

Solutions to Final Step-A | Mathematics

Sets, Relations, Statistics & Mathematical Reasoning

$$1.(A) \quad \text{Mean } (\bar{x}) = \frac{1+(1+d)+\dots+(1+100d)}{101} = \frac{\frac{100}{2}[2+100d]}{101} = 1+50d$$

$$\text{M.D. } (\bar{x}) = \frac{\sum |x_i - \bar{x}|}{n} = \frac{50d + 49d + \dots + d + 0 + d + \dots + 50d}{101} = \frac{2(d+2d+3d+\dots+50d)}{101}$$

$$= \frac{2d(1+2+3+\dots+50)}{101} = \frac{d \times 50 \times 51}{101}$$

$$\text{M.D. } (\bar{x}) = 255 \text{ (given)} \quad \therefore \quad \frac{d \times 50 \times 51}{101} = 255 \quad \Rightarrow \quad d = \frac{101}{10} = 10.1$$

$$2.(C) \quad \text{Here, } P \text{ is False and } Q \text{ is True} \quad \therefore \quad (V_1, V_2) = (F, T).$$

3.(D) Let 5 observations be 1, 2, 6, x and y.

$$\therefore \quad \text{Mean } (\bar{x}) = \frac{1+2+6+x+y}{5} \quad \Rightarrow \quad 5 \times 5 = 9+x+y \quad \Rightarrow \quad x+y=16$$

$$\text{Variance} = \frac{1}{5} \sum x_i^2 - (\bar{x})^2 \Rightarrow 124 = \frac{1}{5} (1+4+36+x^2+y^2) - 25$$

$$\Rightarrow x^2 + y^2 = 704 \Rightarrow x^2 + (16-x)^2 = 704 \quad \{\because x+y=16\}$$

$$\Rightarrow x^2 - 16x - 224 = 0 \Rightarrow x = 8 \pm 12\sqrt{2}$$

$$\therefore x = 8+12\sqrt{2} \text{ and } y = 8-12\sqrt{2}$$

$$\text{M.D. } (\bar{x}) = \frac{1}{5} \sum |x_i - \bar{x}|$$

$$= \frac{1}{5} [|1-5| + |2-5| + |6-5| + |8+12\sqrt{2}-5| + |8-12\sqrt{2}-5|] = \frac{1}{5} [4+3+1+|3+12\sqrt{2}| + |3-12\sqrt{2}|]$$

$$= \frac{1}{5} [8+3+12\sqrt{2}+12\sqrt{2}-3] = \frac{1}{5} [8+24\sqrt{2}] = 8.38$$

4.(D) Contra-positive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$.

Given statement is in the form of $p \rightarrow q$, so its contra-positive is

If the area of a square does not increase four times, then its side is not doubled.

5.(D) Question based on Probability

A = family owning a phone ; B = family owning a car

$$P(A) = \frac{25}{100}, P(B) = \frac{15}{100}$$

$$P(\bar{A} \cap \bar{B}) = \frac{65}{100} \Rightarrow P(\overline{A \cup B}) = \frac{65}{100} \quad \therefore \quad P(A \cup B) = \frac{35}{100}$$

$$\Rightarrow P(A) + P(B) - P(A \cap B) = \frac{35}{100} \Rightarrow P(A \cap B) = \frac{5}{100} \text{ and } n(A \cap B) = 2000$$

$$\Rightarrow \frac{n(A \cap B)}{n(S)} = \frac{5}{100} \Rightarrow n(S) = 40,000$$

$\Rightarrow$ 40,000 families live in the town

$\Rightarrow P(A \cap B) = \frac{5}{100} \Rightarrow$ 5% families own both a car and a phone.

$\Rightarrow P(A \cup B) = \frac{35}{100} \Rightarrow$ 35% families own either a car and a phone.

6.(C) Combined mean $\bar{x} = \frac{n_1 \bar{x}_1 + n_2 \bar{x}_2}{n_1 + n_2}$

Given $n_1 = 70, n_2 = 30, \bar{x}_1 = 54, \bar{x} = 60$

$$\therefore 60 = \frac{70 \times 54 + 30 \times \bar{x}_2}{70 + 30} \Rightarrow \bar{x}_2 = 74$$

$\therefore$ Mean wage of the night shift workers is Rs. 74.

7.(A) Given statement is $p \rightarrow \sim q$ p : It is raining

Its contra-positive is $\sim(\sim q) \rightarrow \sim p$ i.e. $q \rightarrow \sim p$ q : I will come

Thus, contra-positive of given statement is "If I will come, then it is not raining".

8.(D) Given statement is

$$(P \wedge \sim R) \leftrightarrow Q$$

$$Q \leftrightarrow (P \wedge \sim R)$$

Negation of this statement is $\sim(Q \leftrightarrow (P \wedge \sim R))$

9.(C) Not reflection : For $a = \pi/2$ (real number)

$$(a, a) \notin P \text{ as } \sec^2 a - \tan^2 a - \tan^2 a \neq 1$$

Symmetric : Let $(a, b) \in P \Rightarrow \sec^2 a - \tan^2 b = 1$

$$\Rightarrow \sec^2 a - 1 = \tan^2 b \Rightarrow \tan^2 a = \sec^2 b - 1$$

$$\Rightarrow \sec^2 b - \tan^2 a = 1 \Rightarrow (b, a) \in P$$

Transitive : Let $(a, b) \in P$ and $(b, c) \in P$

$$\Rightarrow \sec^2 a - \tan^2 b = 1 \text{ and } \sec^2 b - \tan^2 c = 1 \Rightarrow \sec^2 a - \tan^2 b + \sec^2 b - \tan^2 c = 1 + 1$$

$$\Rightarrow \sec^2 a - \tan^2 c = 1 \Rightarrow (a, c) \in P$$

10.(A) Let the observations in ascending order be $x_1, x_2, x_3, \dots, x_n, x_{n+1}, x_{2n}$

$$\text{Mean} = \frac{x_1 + x_2 + \dots + x_{2n}}{2n} = \bar{x} \quad \dots (i)$$

Median (for even) = $\frac{x_n + x_{n+1}}{2}$ which lies between x_n and x_{n+1} .

So, the observations below the median are $x_1, x_2, \dots, x_n$ and the remaining observations are $x_{n+1}, x_{n+2}, \dots, x_{2n}$.

$$\text{New mean} = \frac{(x_1 + 5) + (x_2 + 5) + \dots + (x_n + 5) + (x_{n+1} - 3) + (x_{n+2} - 3) + \dots + (x_{2n} - 3)}{2n}$$

$$= \frac{x_1 + x_2 + x_3 + \dots + x_{2n} + 5n - 3n}{2n} = \bar{x} + 1 \quad [\text{From (i)}]$$

Thus, new mean increases by 1.

11.(B) $p \rightarrow q$ means if p then q .

Contra-positive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$.

Let p : It rains ; q : I go to school.

So, the given statement in the form of $\sim p \rightarrow q$.

Thus, its contrapositive is $\sim q \rightarrow p$. Hence, contrapositive is "If I do not go to school, it rains".

$$\begin{aligned} 12.(C) \quad \sim(p \vee \sim q) \vee \sim(p \vee q) &= \sim[(p \vee \sim q) \wedge (p \vee q)] \\ &= \sim[p \vee (\sim q \wedge q)] = [p \vee f] \quad \because \quad \sim q \wedge q = f \text{ (fallacy)} \\ &= \sim p \end{aligned}$$

$$13.(D) \quad \text{M.D.} = \frac{\sum |x_i - \bar{X}|}{n} \quad (\text{By defn.})$$

Obviously, new mean $= \bar{X} + 5$

$$\text{New M.D.} = \frac{\sum |(x_i + 5) - (\bar{X} + 5)|}{n} = \text{M.D.}$$

14.(B) $p \Rightarrow (q \vee r)$ is equivalent to $(p \Rightarrow q) \vee (p \Rightarrow r)$ by truth table

p	q	r	$q \vee r$	$p \Rightarrow (q \vee r)$	$p \Rightarrow q$	$p \Rightarrow r$	$(p \Rightarrow q) \vee (p \Rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T
T	F	T	T	T	F	T	T
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	F	F	F	T	T	T	T

Equivalent

$$15.(C) \quad A = \{-2, -1, 0, 1, 2\}$$

$$B = \{(0, 0), (1, 1), (2, 2), (-2, 2)\}$$

$\therefore$ Number of elements in the power set of $R = 2^4$.

$$16.(B) \quad \text{Mean} = \frac{x_1 + x_2 + \dots + x_n}{n} = \bar{x} \quad \text{Variance} = \frac{\sum (x_i - \bar{x})^2}{n} = \sigma^2$$

$$\text{New mean } (\bar{d}) = \frac{d_1 + d_2 + \dots + d_n}{n} = \frac{-x_1 - x_2 - \dots - x_n - na}{n} = -\bar{x} - a$$

$$\text{New variance} = \frac{\sum (d_i - \bar{d})^2}{n} = \frac{\sum \{-x_i - a - (-\bar{x} - a)\}^2}{n} = \frac{\sum (-x_i + \bar{x})^2}{n} = \frac{\sum (x_i - \bar{x})^2}{n} = \sigma^2$$

$\therefore$ Mode of $x_1, x_2, \dots, x_n$ is M

$\therefore$ Mode of $-x_1 - a, -x_2 - a, \dots, -x_n - a$ is $-M - a$

Mode has the highest frequency

$\therefore$ Both the statements are true.

17.(C) Contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$. Given statement is in the form of $\sim p \rightarrow q$. So, its contrapositive form is $\sim q \rightarrow p$. Hence option (C).

$$18.(A) \quad \frac{\sum x_i}{n} = \bar{x} \text{ (mean)} \Rightarrow \sum x_i = n\bar{x}.$$

Sum of all observations (incorrect) = $40 \times 20 = 800$. Sum of all observations (correct) = $800 - 33 + 53 = 820$

$$\therefore \text{Correct mean} = \frac{\text{sum(correct)}}{20} = \frac{820}{20} = 41$$

19.(C) Every element of set A has distinct image in A . So R is a one-one and onto function and hence it has an inverse.

$$20.(A) \quad \frac{\sum x_i}{n} = \bar{x} \text{ (mean)} \Rightarrow \sum x_i = n\bar{x}$$

Sum of observations = $7 \times 5 = 35$ $\therefore$ 5th observation = $35 - (6 + 7 + 8 + 10) = 4$

$$\text{Variance} = \frac{\sum (x_i - \bar{x})^2}{n} = \frac{\sum (x_i - 7)^2}{n} = \frac{1+0+1+9+9}{5} = 4$$

21.(D) Clearly, reflexive as $(a, a) \in R \forall a \in A$.

$(5, 12) \in R$ but, $(12, 5) \notin R$

So not symmetric

$(3, 5) \in R$ and $(5, 12) \in R \Rightarrow (3, 12) \in R$, so transitive.

22.(B)

p	q	$q \rightarrow p$	$p \rightarrow (q \rightarrow p)$	$p \vee q$	$p \rightarrow (p \vee q)$	$p \wedge q$	$p \rightarrow (p \wedge q)$	$p \rightarrow q$	$p \rightarrow (p \rightarrow q)$
T	T	T	T	T	T	T	T	T	T
T	F	T	T	T	T	F	F	F	F
F	T	F	T	T	T	F	T	T	T
F	F	T	T	F	T	F	T	T	T

Clearly, $p \rightarrow (q \rightarrow p)$ and $p \rightarrow (p \vee q)$ are equivalent.

23.(A) (B) $P : 4$ divides 8

$Q : 4$ divides 8^2

But 4 is not a prime $\therefore P \wedge Q \rightarrow R$ is not True.

(C) $Q : 12$ divides 6^2

But 12 is not a prime $\therefore Q \rightarrow R$ is not True.

(D) $Q : 12$ divides 6^2

But 12 is not a prime

$\therefore Q \rightarrow P$ is not divide 6. $\therefore Q \rightarrow P$ is not true

(A) $Q : m$ divides n^2

$R : m$ is a prime

then m divides n

$\therefore Q \wedge R \rightarrow P$ is True.

e.g. 3 divides 12^2 and 3 is a prime then 3 also divides 12.

24.(D) $0 < y < x < 2y \Rightarrow y < x < 2y$

$\Rightarrow y - y < x - y < 2y - y \Rightarrow 0 < x - y < y$

$\therefore x - y < y < x < 2x + y$ (Even)

Median = $\frac{y+x}{2} = 10 \Rightarrow x + y = 20$ (i)

Range = 28 $\Rightarrow 2x + y - (x - y) = 28 \Rightarrow x + 2y = 28$ (ii)

Solve (i) and (ii), $x = 12, y = 8$

$x = 12, y = 8, 2x + y = 32, x - y = 4$

$\therefore \text{Mean} = \frac{12+8+32+4}{4} = \frac{56}{4} = 14$

25.(A) $x^2 - 4xy + 3y^2 = 0 \Rightarrow (x - y)(x - 3y) = 0$

$\Rightarrow x = y, x = 3y, x, y \in N$

$\therefore R = \{(1, 1), (2, 2), (3, 3), \dots, (3, 1), (6, 2), (9, 3), \dots\}$

Hence Reflexive, not symmetric, not Transitive.

26.(B) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

$p \vee \sim p = t$ (Tautology), $p \wedge t = p$

$= (p \vee q) \wedge (\sim q \vee q) \vee (\sim p \wedge q) = (p \vee q) \wedge t \vee (\sim p \wedge q)$

$= (p \vee q) \vee (\sim p \wedge q) = (p \vee q \vee \sim p) \wedge (p \vee q \vee q)$

$= (p \vee \sim p \vee q) \wedge (p \vee q) = (t \vee q) \wedge (p \vee q) = t \wedge (p \vee q) = p \vee q$

27.(B) We know that

$p \vee \sim p = t, p \wedge t = p, \sim(\sim p) = p, \sim(p \wedge q) = \sim p \vee \sim q,$

$p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

$\therefore \sim s \vee (\sim r \vee s) = (\sim s \vee \sim r) \wedge (\sim s \vee s) = \sim(s \wedge r) \wedge t = \sim(s \wedge r)$

Negation of $\sim s \vee (\sim r \wedge s) = \sim(\sim(s \wedge r)) = s \wedge r$

28.(B) $n = 16, \bar{x} = 16, \bar{x} = \frac{\sum x_i}{n}$

$\therefore \sum x_i = n\bar{x} = 16 \times 16 = 256$

$\therefore$ Sum of 16 observations = 256

One observation valued 16 is deleted and 3 new observation valued 3, 4 and 5 are added.

$\therefore$ Sum of 18 observations = $256 - 16 + 3 + 4 + 5 = 252$

Mean of resultant data = $\frac{252}{18} = 14$

29.(C) $(1+x^n) = 1 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n$

$\therefore 4^n - 3n - 1 = (1+3)^n - 3n - 1 = 1 + {}^nC_1 3^1 + {}^nC_2 3^2 + \dots + {}^nC_n 3^n - 3n - 1$
 $= 9({}^nC_2 + {}^nC_3 3 + \dots + {}^nC_n 3^{n-2})$ as ${}^nC_1 = n$

$$\therefore X = \left\{ 9 \left({}^nC_2 + {}^nC_3 3 + \dots + {}^nC_n 3^{n-2} \right) : n \in N \right\} = \{0, 9, 54, \dots\}$$

= a set of some multiples of 9

$$Y = \{0, 9, 18, 27, \dots\} = \text{a set of all multiples of 9}$$

Clearly, $X \subset Y \quad \therefore X \cup Y = Y$

30.(A) $\sum n = \frac{n(n+1)}{2}, \sum n^2 = \frac{n(n+1)(2n+1)}{6}.$

$$\text{Variance} = \frac{\sum x_i^2}{n} - (\bar{x})^2$$

$$\bar{x} = \frac{2+4+\dots+100}{50} = \frac{2(1+2+\dots+50)}{50} = \frac{2 \times 50 \times 51}{50 \times 2} = 51$$

$$\sum x_i^2 = 2^2 + 4^2 + \dots + 100^2 = 2^2 (1^2 + 2^2 + \dots + 50^2) = \frac{2^2 \times 50 \times 51 \times 101}{6}$$

$$\therefore \text{Variance} = \frac{2^2 \times 50 \times 51 \times 101}{50 \times 6} - 51^2 = 3434 - 2601 = 833.$$

31.(D)

p	q	$\sim q$	$p \leftrightarrow \sim q$	$\sim(p \leftrightarrow \sim q)$	$p \leftrightarrow q$
T	T	F	F	T	T
T	F	T	T	F	F
F	T	F	T	F	F
F	F	T	F	T	T

32.(C) **Statement I :** $(p \wedge \sim q) \wedge (\sim p \wedge q) = p \wedge \sim q \wedge \sim p \wedge q$
 $= (p \wedge \sim p) \wedge (\sim q \wedge q) = f \wedge f = f.$ Fallacy.

Statement II : $(p \rightarrow q) \leftrightarrow (\sim q \wedge \sim p)$

i.e. $(p \rightarrow q) \leftrightarrow (p \rightarrow q)$ is always true.

As $\sim q \rightarrow \sim p$ is contrapositive of $p \rightarrow q$

$\therefore$ Tautology.

33.(D) $n(A \times B) = n(A)n(B) = 2 \times 4 = 8$

Total subsets = 2^8

Number of subsets having 3 or more elements

$$= {}^8C_3 + {}^8C_4 + {}^8C_5 + {}^8C_6 + {}^8C_7 + {}^8C_8 = 2^8 - ({}^8C_0 + {}^8C_1 + {}^8C_2) = 256 - (1 + 8 + 28) = 219$$

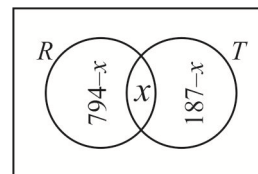
34.(21) $3N = \{x \in N : x \text{ is a multiple of } 3\}$

$7N = \{x \in N : x \text{ is a multiple of } 7\}$

$\therefore 3N \cap 7N = \{x \in N : x \text{ is a multiple of } 3 \text{ and } 7\}$

$\{x \in N : x \text{ is a multiple of } 21\} = 21N$

- 35.(41) Let R be the set of families having a radio and T the set families having a T.V.,
Then $n(R \cup T)$ = The number of families having at least one of the radio and T.V.
 $= 1003 - 63 = 940$
 $n(R) = 794$ and $n(T) = 187$



Let x families have both a radio and a T.V. Then number of families who have only radio $= 794 - x$ and the number of families who have only T.V. $= 187 - x$
From venn diagram, $794 - x + x + 187 - x = 940$
 $\Rightarrow 981 - x = 940$ or $x = 981 - 940 = 41$

- 36.(99) $n((A \times B) \cap (B \times A))$
 $= n((A \cap B) \times (B \cap A)) = n(A \cap B) \cdot n(B \cap A)$
 $= n(A \cap B) \cdot n(A \cap B) = (99)(99) = 99^2$

- 37.(45) Let A denote the set of Americans who like cheese and let B denote the set of Americans who like apples. Let population of America be 100.

Then $n(A) = 63, n(B) = 76$

$$\text{Now, } n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$= 63 + 76 - n(A \cap B)$$

$$\Rightarrow n(A \cap B) = 139 - n(A \cup B)$$

But $n(A \cup B) \leq 100$

$$\therefore 139 - n(A \cup B) \geq 139 - 100 = 39$$

$$\therefore n(A \cap B) \geq 39 \text{ i.e., } 39 \leq n(A \cap B)$$

Again, $A \cap B \subseteq A, A \cap B \subseteq B$

$$\therefore n(A \cap B) \leq n(A) = 63 \text{ and } n(A \cap B) \leq n(B) = 76$$

$$\therefore n(A \cap B) \leq 63$$

$$\text{Then, } 39 \leq n(A \cap B) \leq n(B) \Rightarrow 39 \leq x \leq 63$$

- 38.(6) Since $A \subseteq B$, $\therefore A \cup B = B$

$$\text{So, } n(A \cup B) = n(B) = 6$$

- 39.(6) Number of proper subsets of the set $\{1, 2, 3\} = 2^3 - 2 = 6$

- 40.(2) $n(A) = 4, n(B) = 3$

$$n(A) \times n(B) \times n(C) = n(A \times B \times C)$$

$$4 \times 3 \times n(C) = 24$$

$$\Rightarrow n(C) = \frac{24}{12} = 2$$

- 41.(8) Given set is $\{(a, b) : 2a^2 + 3b^2 = 35, a, b \in \mathbb{Z}\}$

We can see that, $2(\pm 2)^2 + 3(\pm 3)^2 = 35$ and $2(\pm 4)^2 + 3(\pm 1)^2 = 35$

$\therefore (2, 3), (2, -3), (-2, -3), (-2, 3), (4, 1), (4, -1), (-4, -1), (-4, 1)$ are eight elements of the set, so $n = 8$

42.(3) $n(A \cup B) = n(A) + n(B) - n(A \cap B) = 12 + 9 - 4 = 17$

Now, $n((A \cup B)^C) = n(U) - n(A \cup B) = 20 - 17 = 3$

43.(16) Given $n(N) = 12, n(P) = 16, n(H) = 18, n(N \cup P \cup H) = 30$

From, $n(N \cup P \cup H) = n(N) + n(P) + n(H) - n(N \cap P) - n(P \cap H) - n(N \cap H) + n(N \cap P \cap H)$

$\therefore n(N \cap P) + n(P \cap H) + n(N \cap H) = 16$

44.(60) Marks obtained from three subjects out of 300 is $75 + 80 + 85 = 240$. If the marks of another subject is added, then the marks will be ≥ 240 out of 400

$\therefore$ Minimum average marks $= \frac{240}{4} = 60\%$

[When marks in the fourth subject = 0]

45.(50) Sum of 100 items $= 49 \times 100 = 4900$

Sum of items added $= 60 + 70 + 80 = 210$

New sum $= 4900 + 210 - 110 = 5000$

$\therefore$ Correct mean $= \frac{5000}{100} = 50$

46.(57) The total weight of seven student is $55 \times 7 = 385$ kg

The sum of the weights of six students is $52 + 58 + 55 + 53 + 56 + 54 = 328$ kg

Hence, the weight of the seventh student is $385 - 328 = 57$ kg

47.(16) $\sqrt{\frac{\sum_{i=1}^n x_i^2}{n}} \geq \frac{\sum_{i=1}^n x_i}{n} = \sqrt{\frac{400}{n}} \geq \frac{80}{n} \Rightarrow n \geq 16$

Hence, the possible value of n is 18

48.(8) For a moderately skewed distribution, Mode $= 3$ median $- 2$ neab

$\Rightarrow 6\lambda = 3$ median $- 18\lambda \Rightarrow$ median $= 8\lambda$

Sequence & Series

1.(A) Let first term be A and common ratio be R. Given, p th term, $T_p = q$ and q th term, $T_q = p$

$Ar^{p-1} = q$ and $AR^{q-1} = p$... (i)

$\therefore \frac{AR^{p-1}}{AR^{q-1}} = \frac{q}{p} \Rightarrow R^{p-q} = \frac{q}{p} \Rightarrow R = \left(\frac{q}{p}\right)^{\frac{1}{p-q}}$

On putting the value of R in equation (i), we get : $A \cdot \left(\frac{q}{p}\right)^{\frac{p-1}{p-q}} = q \Rightarrow A = q \cdot \left(\frac{p}{q}\right)^{\frac{p-1}{p-q}}$

Now, $(p+q)$ th term, $T_{p+q} = AR^{p+q-1}$

$$= q \cdot \left(\frac{p}{q}\right)^{\frac{p-1}{p-q}} \times \left(\frac{q}{p}\right)^{\frac{p+q-1}{p-q}} = \frac{q^{\frac{1-p-1}{p-q} + \frac{p+q-1}{p-q}}}{p^{\frac{p-1}{p-q} - \frac{p+q-1}{p-q}}} = \frac{q^{\frac{p-q-p+1+p+q-1}{p-q}}}{p^{\frac{p-1-p+q-1-p+1}{p-q}}} = \frac{q^{\frac{p}{p-q}}}{p^{\frac{q}{p-q}}} = \left(\frac{q^p}{p^q}\right)^{\frac{1}{p-q}} = \left(\frac{q^p}{p^q}\right)^{\frac{1}{p-q}}$$

2.(A)

3.(D)

4.(B) $\therefore a_1, a_2, a_3, a_4$ are in HP $\therefore \frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \frac{1}{a_4}$ are in AP

$$\Rightarrow \frac{1}{a_2} - \frac{1}{a_1} = \frac{1}{a_3} - \frac{1}{a_2} = \frac{1}{a_4} - \frac{1}{a_3} = \lambda \text{ (say)}$$

$$\Rightarrow \frac{a_1 - a_2}{a_1 a_2} = \frac{a_2 - a_3}{a_2 a_3} = \frac{a_3 - a_4}{a_3 a_4} = \frac{a_1 - a_4}{a_1 a_2 + a_2 a_3 + a_3 a_4} \quad \text{or} \quad \frac{a_1 - a_2}{a_1 a_2} = \frac{(a_1 - a_4)}{\sum_{r=1}^3 a_r a_{r+1}}$$

$$\therefore \frac{1}{a_1 a_4} \sum_{r=1}^3 a_r a_{r+1} = \frac{1}{a_1 a_4} \times \frac{a_1 a_2 (a_1 - a_4)}{(a_1 - a_2)} = \frac{\frac{1}{a_4} - \frac{1}{a_1}}{\frac{1}{a_2} - \frac{1}{a_1}} = \frac{3\lambda}{\lambda} = 3. \text{ Hence, 3 is a root of } x^2 + 2x - 15 = 0$$

5.(C) $\therefore f(k) = \sum_{r=1}^n a_r - a_k$

$$\sum_{r=1}^n a_r = a_k + f(k) \Rightarrow a_1 + f(1) = a_2 + f(2) = a_3 + f(3) = \dots = a_n + f(n)$$

Given $a_1, a_2, a_3, \dots, a_n$ are in HP.

$\therefore \frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \dots, \frac{1}{a_n}$ are in AP. $\Rightarrow \frac{a_1 + f(1)}{a_1}, \frac{a_2 + f(2)}{a_2}, \frac{a_3 + f(3)}{a_3}, \dots, \frac{a_n + f(n)}{a_n}$ are also in AP.

Subtracting 1 in each term, then : $\frac{f(1)}{a_1}, \frac{f(2)}{a_2}, \frac{f(3)}{a_3}, \dots, \frac{f(n)}{a_n}$ are also in AP.

$\therefore \frac{a_1}{f(1)}, \frac{a_2}{f(2)}, \frac{a_3}{f(3)}, \dots, \frac{a_n}{f(n)}$ are in HP.

6.(A) $\alpha = a_2 + a_4 + a_6 + \dots + a_{200}$ & $\beta = a_1 + a_3 + a_5 + \dots + a_{199}$

Since, $\alpha = (a_1 r + a_3 r + a_5 r + \dots + a_{199} r) = r \beta \therefore r = \alpha / \beta$

7.(D) $(1 + 2 + 2^2 + 2^3 + \dots + 2^{n-1})^2 = (1^2 + 2^2 + 2^4 + 2^6 + \dots + 2^{2n-2}) + 2P$ (where P is sum of products taken two at a time)

$$\therefore P = \frac{(2^n - 1)^2 - \frac{1^2 \cdot (2^{2n} - 1)}{(2^2 - 1)}}{2} = \frac{1}{3} \cdot 2^{2n} - 2^n + \frac{2}{3}$$

8.(C) Let d be the common difference $\therefore a_7 = 9 \therefore a_1 + 6d = 9$

$$\text{Let } D = a_1 a_2 a_7 = (9 - 6d)(9 - 5d)9 = 270 \left\{ \left(d - \frac{33}{20} \right)^2 - \frac{9}{400} \right\}$$

For least value of D ; $d - \frac{33}{20} = 0 \therefore d = 33/20$

$$\begin{aligned}
 9.(A) \quad L &= \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{n^4} \sum_{r=1}^n r(n-r+1)(n-r+2) = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{n^4} \sum_{r=1}^n \{r^3 - (2n+3)r^2 + (n+1)(n+2)r\} \\
 &= \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{n^4} \left\{ \sum n^3 - (2n+3) \sum n^2 + (n+1)(n+2) \sum n \right\} \\
 &= \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{n^4} \left\{ \left(\frac{n(n+1)}{2} \right)^2 - \frac{(2n+3)n(n+1)(2n+1)}{6} + \frac{(n+1)(n+2)n(n+1)}{2} \right\} \\
 &= \frac{1}{2} \lim_{n \rightarrow \infty} \left\{ \left(\frac{1 \cdot \left(1 + \frac{1}{n}\right)}{2} \right)^2 - \frac{\left(2 + \frac{3}{n}\right) 1 \left(1 + \frac{1}{n}\right) \left(2 + \frac{1}{n}\right)}{6} + \frac{\left(1 + \frac{1}{n}\right)^2 1 \left(1 + \frac{2}{n}\right)}{2} \right\} \\
 &= \frac{1}{2} \left\{ \frac{1}{4} - \frac{4}{6} + \frac{1}{2} \right\} = \frac{1}{8} - \frac{1}{3} + \frac{1}{4} = \frac{3-8+6}{24} = \frac{1}{24}
 \end{aligned}$$

- 10.(A) Given, $a + b + c = 25$ (i); $2a = 2 + b$ (ii); $c^2 = 18b$ (iii)
Substituting the values of a and b in terms of c from equations (ii) and (iii) in equation (i), then

$$\frac{1}{2} \left(2 + \frac{c^2}{18} \right) + \frac{c^2}{18} + c = 25 \Rightarrow c^2 + 12c - 288 = 0 \quad \therefore c = 12, -24$$

$\therefore a, b, c$ are between 2 and 18 ($\therefore c \neq -24$). Hence, $c = 12$

From equation (iii), we get $b = \frac{(12)^2}{18} = 8$ & from equation (ii), we get $a = \frac{2+b}{2} = \frac{10}{2} = 5$

$\therefore a = 5, b = 8, c = 12$. Then $G = 12, L = 5$

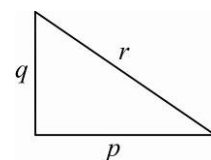
- 11.(D) For a GP product of equidistant terms from beginning and end is constant,

$$\therefore t_1 t_{25} = t_5 t_{21} = t_{10} t_{16} \quad \therefore t_1 t_{25} = 512^{1/3} = 8 \quad \therefore t_{13} = \sqrt{t_1 t_{25}} = \sqrt{8}$$

- 12.(A) $(1+3+5+7+\dots+(2P-1)) + (1+3+5+\dots+(2a-1)) = (1+3+\dots+(2r-1)) \Rightarrow p^2 + q^2 = r^2$

$$p = m^2 - n^2, q = 2mn, r = m^2 + n^2$$

$$\text{Take } m = 3, n = 1 \quad \therefore p = 8, q = 6, r = 10, p + q + r = 24$$



- 13.(B) $f(n) = \lfloor n \rfloor$ & $r f(r) = r(\lfloor r \rfloor)$

$$\sum_{r=1}^{20} r \lfloor r \rfloor = \sum_{r=1}^{20} ((r+1)-1) \lfloor r \rfloor = \sum_{r=1}^{20} (\lfloor r+1 \rfloor - \lfloor r \rfloor) = \lfloor 21 \rfloor - \lfloor 1 \rfloor = f(21) - 1 \Rightarrow x = 21$$

- 14.(D) Coefficient of $(x^0) = (-1)^{(\text{No. of roots})} \times (\text{product of roots})$

$$15.(A) \quad x = \frac{1}{1-a} \quad y = \frac{1}{1-b} \quad z = \frac{1}{1-\frac{a}{b}}$$

$$a = \frac{x-1}{x} \quad b = \frac{y-1}{y} \quad \frac{a}{b} = \frac{y(x-1)}{x(y-1)}$$

$$z = \frac{1}{1-\frac{y(x-1)}{x(y-1)}} \Rightarrow z = \frac{xy-x}{y-x} \Rightarrow zy-zx = xy-x \Rightarrow xy+xz = x+yz \Rightarrow x(y+z) = x+yz$$

16.(C) Series is $(1.05) + (1.05)^2 + (1.05)^3 + (1.05)^4 + \dots (1.05)^{49}$

$$= (1.05) \left[\frac{(1.05)^{50} - 1}{1.05 - 1} \right] = \frac{(1.05)^{50} - 1.05}{0.05} = \frac{11.658 - 1.05}{0.05} = \frac{10.608}{0.05} = 212.16$$

17.(D) $P_1 \equiv 17, 21, 25, \dots, 417$; $P_2 \equiv 16, 21, 26, \dots, 466$

General term for $P_1 = 17 + 4(n-1) = 4n + 13$; General term for $P_2 = 16 + 5(k-1) = 5k + 11$

$$\text{Now } 4n + 13 = 5k + 11 \Rightarrow n = \frac{5k - 2}{4} = k + \frac{k - 2}{4}$$

As k and n both are integers $\frac{k - 2}{4} = I$

$$k = 4I + 2 \quad \text{and} \quad 0 \leq x \leq 111, 0 \leq k \leq 91 \quad \therefore k \in \{2, 6, 10, 14, \dots, 86\}$$

We can't include $k \in \{90\}$ as for this n exceeds 111.

$\therefore k$ has 22 values $\Rightarrow n$ has 22 values.

$$18.(B) \quad t_r^2 = 2^r + 2^{-r} + 2 \Rightarrow \sum_{r=1}^{10} t_r^2 = \sum_{r=1}^{10} 2^r + \sum_{r=1}^{10} 2^{-r} + 20 = \frac{2(2^{10} - 1)}{2 - 1} + 2^{-1} \frac{\left(\frac{1}{2^{10}} - 1\right)}{\frac{1}{2} - 1} + 20 = \frac{2^{21} - 1}{2^{10}} + 19$$

$$19.(A) \quad T_r = \frac{1}{\sqrt{4r-1} + \sqrt{4r+3}} = \frac{1}{4} (\sqrt{4r+3} - \sqrt{4r-1}) \Rightarrow \sum_{r=1}^n T_r = \frac{1}{4} (\sqrt{4n+3} - \sqrt{3})$$

$$20.(A) \quad \text{As } A.M. > G.M. \Rightarrow \frac{a+b+c}{3} > (abc)^{1/3} \text{ and } \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} > \left(\frac{1}{abc}\right)^{1/3} \quad (\text{for distinct +ve numbers})$$

$$\Rightarrow a+b+c > 3.(abc)^{1/3} \text{ and } \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) > \frac{3}{(abc)^{1/3}} \Rightarrow (a+b+c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) > 9$$

21.(D) Lengths of line segments on one side of the diagonal are $\sqrt{2}, 2\sqrt{2}, 3\sqrt{2}, \dots, (n-1)\sqrt{2}$

$$\text{So required sum} = 2 \left[\sqrt{2} + 2\sqrt{2} + 3\sqrt{2} + \dots + (n-1)\sqrt{2} \right] + n\sqrt{2} = 2\sqrt{2} \frac{n(n-1)}{2} + n\sqrt{2} = n^2 \sqrt{2}$$

22.(A) $S = (1)(2003) + 2(2002) + 3(2001) + \dots$ and $k = 1^2 + 2^2 + 3^2 + \dots + 2003^2$

Adding, we get : $S + k = 2004 [1 + 2 + 3 + \dots + 2003]$

$$\Rightarrow (2003)(334)(x) + (2003)(4007)(334) = (2004)(1002)(2003) \Rightarrow x = 2005$$

23.(A) In a ΔABC , sum of any two sides is always greater than third side

$$\therefore a+b-c > 0, b+c-a > 0 \text{ and } c+a-b > 0$$

For three non zero positive numbers $A.M. \geq G.M$

$$\Rightarrow \left\{ \frac{(a+b-c) + (b+c-a) + (c+a-b)}{3} \right\} \geq \{(a+b-c)(b+c-a)(c+a-b)\}^{1/3}$$

$$\Rightarrow \left(\frac{a+b+c}{3} \right) \geq \{(a+b-c)(b+c-a)(c+a-b)\}^{1/3} \Rightarrow (a+b+c)^3 \geq 27(a+b-c)(b+c-a)(c+a-b)$$

24.(B) $a_1, a_2, \dots, a_n, \dots$ is an AP, $b_1, b_2, \dots, b_n, \dots$ is a GP and $a_1 = b_1 = 1, a_9 = b_9 = b$

$$\sum_{r=1}^9 a_r = 369 \Rightarrow \frac{9}{2}(1+b) = 369 \Rightarrow b = 81$$

$$b_n = (1)r^{n-1} \Rightarrow b_9 = r^8 = 81 \Rightarrow r = \sqrt[3]{3} \Rightarrow b_7 = r^6 = 3^3 = 27$$

25.(D) Given $x + y + z = 12 \Rightarrow 3\left(\frac{x}{3}\right) + 4\left(\frac{y}{4}\right) + 5\left(\frac{z}{5}\right) = 12$

For $\frac{x}{3}, \frac{x}{3}, \frac{x}{3}, \frac{y}{4}, \dots, \frac{z}{5}$ (12 numbers)

AM $\geq$ GM

$$\Rightarrow \frac{3 \cdot \frac{x}{3} + 4 \cdot \frac{y}{4} + 5 \cdot \frac{z}{5}}{12} \geq \left[\left(\frac{x}{3}\right)^3 \left(\frac{y}{4}\right)^4 \left(\frac{z}{5}\right)^5 \right]^{1/12} \Rightarrow \frac{x+y+z}{12} \geq \left[\frac{x^3 y^4 z^5}{3^3 \cdot 4^4 \cdot 5^5} \right]$$

Since $x + y + z = 12$ and $x^3 y^4 z^5 = (0.1)(600)^3$

$$\therefore 1 \geq \left[\frac{0.1 \times (600)^3}{(3 \times 4 \times 5)^3 \times (4 \times 5^2)} \right] \Rightarrow 1 \geq \left[\frac{0.1 \times (600)^3}{(60)^3 \times 100} \right]^{1/12} \Rightarrow 1 \geq 1 \Rightarrow \text{AM} = \text{GM}$$

$\therefore$ All the terms are equal

$$\Rightarrow \frac{x}{3} = \frac{y}{4} = \frac{z}{5} = \lambda \text{ (let)} \Rightarrow x = 3\lambda, y = 4\lambda, z = 5\lambda$$

But $x + y + z = 12 \Rightarrow \lambda = 1 \therefore x = 3, y = 4, z = 5 \therefore x^3 + y^3 + z^3 = 3^3 + 4^3 + 5^3 = 216.$

26.(A) If $a_1, a_2, a_3, \dots, a_n$ are in A.P., then the sum of the terms equidistant from the beginning and the end is equal to the sum of first and last terms.

$$\text{i.e. } a_1 + a_n = a_2 + a_{n-1} = a_3 + a_{n-2} = \dots \text{ and } S_n = \frac{n}{2}[a_1 + a_n] \therefore S_{17} = \frac{17}{2}[a_1 + a_{17}] \dots (i)$$

$$\text{Given } a_3 + a_7 + a_{11} + a_{15} = 72 \Rightarrow (a_3 + a_{15}) + (a_7 + a_{11}) = 72 \dots (ii)$$

But $a_1 + a_{17} = a_3 + a_{15} = a_7 + a_{11}$

$$\text{By (ii), } 2(a_1 + a_{17}) = 72 \Rightarrow a_1 + a_{17} = 36 \quad \text{By (i), } S_{17} = \frac{17}{2} \times 36 = 17 \times 18 = 306.$$

27.(B) $(r^2 + 1) \times r! = (r^2 - 1 + 2) \times r!$

$$= [(r+1)(r-1) + 2]r! = (r+1)!(r+2-3) + 2r! = [(r+2)! - 3(r+1)!] + 2r! = [(r+2)! - (r+1)!] - 2[(r+1)! - r!]$$

$$\therefore \text{Req. sum} = \sum_{r=1}^{10} [(r+2)! - (r+1)!] - 2 \sum_{r=1}^{10} [(r+1)! - r!] = 10(11!)$$

28.(B) $\sum_{r=16}^{30} (r+2)(r-3) = \sum_{r=16}^{30} (r^2 - r - 6)$

$$\sum_{r=16}^{30} r^2 - \sum_{r=16}^{30} r - \sum_{r=16}^{30} 6 = \left(\sum_{r=1}^{30} r^2 - \sum_{r=1}^{15} r^2 \right) - \left(\sum_{r=1}^{30} r - \sum_{r=1}^{15} r \right) - 6 \times 15$$

$$= \left(\frac{30 \times 31 \times 61}{6} - \frac{15 \times 16 \times 31}{6} \right) - \left(\frac{30 \times 31}{2} - \frac{15 \times 16}{2} \right) - 90 = (9455 - 1240) - (465 - 120) - 90 = 7780$$

29.(C) A.P. $a, a+d, a+2d, \dots$

$\therefore$ All terms are positive integers

$\therefore a$ and b are positive integers

Given $200 < S_9 < 220$

$$200 < \frac{9}{2}(2a+8d) < 220$$

$$200 < 9(a+4d) < 220$$

$$\therefore a_2 = 12 \Rightarrow a+d = 12$$

$$200 < 9(a+4d) < 220$$

$$\frac{92}{27} < d < \frac{112}{27} \Rightarrow 3.4 < d < 4.1 \Rightarrow d = 4$$

But $a+d = 12 \Rightarrow a = 8 \therefore a_4 = a+3d = 8+12 = 20$

30.(A) $t_n = \frac{2n+1}{1^2+2^2+3^2+\dots+n^2} = \frac{6(2n+1)}{n(n+1)(2n+1)} = \frac{6}{n(n+1)} = 6\left[\frac{1}{n} - \frac{1}{n+1}\right]$

$$\therefore S_{20} = 6\left[\left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) + \dots + \left(\frac{1}{20} - \frac{1}{21}\right)\right] = 6\left(1 - \frac{1}{21}\right) = \frac{120}{21} = \frac{k}{21} \text{ (given)}$$

$$\therefore k = 120$$

31.(C) Given $\frac{a+ar+ar^2+ar^3+ar^4}{\frac{1}{a}+\frac{1}{ar}+\frac{1}{ar^2}+\frac{1}{ar^3}+\frac{1}{ar^4}} = \frac{49}{1}$

$$\Rightarrow \frac{a(r^5-1)}{(r-1)} \div \frac{1\left(1-\frac{1}{r^5}\right)}{a\left(1-\frac{1}{r}\right)} = \frac{49}{1} \Rightarrow \frac{a^2(r^5-1) \cdot r^5 \cdot (r-1)}{(r^5-1) \cdot (r-1) \cdot r} = \frac{49}{1} \Rightarrow a^2 r^4 = 49 \Rightarrow ar^2 = 7$$

But $a+ar^2 = 35 \therefore a = 28$

32.(B) First A.P. $3+7+11+15+19+23+27+31+\dots$

Second A.P. $1+6+11+16+21+26+31+\dots$

Common difference of first A.P. is 4 and common difference of second A.P. is 5.

$$\therefore \text{LCM}(4 \text{ and } 5) = 20$$

So, common terms of both APs will form A.P. with common difference 20 whose first term is 11.

$11+31+51+\dots$

$$\therefore S_{20} = \frac{20}{2} [2 \times 11 + (20-1)20] = 4020$$

33.(C) $G = \sqrt{ab}, M = \frac{1}{2}\left(\frac{1}{a} + \frac{1}{b}\right) = \frac{a+b}{2ab}$

$$\text{Given } \frac{1}{M} : G = 4 : 5 \Rightarrow \frac{2ab}{a+b} \div \frac{1}{\sqrt{ab}} = \frac{4}{5} \Rightarrow \frac{a+b}{2\sqrt{ab}} = \frac{5}{4}$$

By componendo and dividendo

$$\frac{a+b+2\sqrt{ab}}{a+b-2\sqrt{ab}} = \frac{5+4}{5-4} \Rightarrow \frac{(\sqrt{b}+\sqrt{a})^2}{(\sqrt{b}-\sqrt{a})^2} = \frac{9}{1} \Rightarrow \frac{\sqrt{b}+\sqrt{a}}{\sqrt{b}-\sqrt{a}} = \frac{3}{1}$$

Again, by componendo and dividendo

$$\frac{(\sqrt{b}+\sqrt{a})+(\sqrt{b}-\sqrt{a})}{(\sqrt{b}+\sqrt{a})-(\sqrt{b}-\sqrt{a})} = \frac{3+1}{3-1} \Rightarrow \frac{\sqrt{b}}{\sqrt{a}} = \frac{2}{1} \Rightarrow \frac{b}{a} = \frac{4}{1} \Rightarrow \frac{a}{b} = \frac{1}{4}$$

34.(D) $1 - 2\left(\frac{1}{3} + \frac{1}{3^2} + \dots + \frac{1}{3^{n-1}}\right) < \frac{1}{100}$

$$1 - \frac{2\left(\frac{1 - \frac{1}{3^{n-1}}}{1 - \frac{1}{3}}\right)}{1 - \frac{1}{3}} < \frac{1}{100} \Rightarrow 1 - \left(1 - \frac{1}{3^{n-1}}\right) < \frac{1}{100} \Rightarrow \frac{1}{3^{n-1}} < \frac{1}{100} \Rightarrow 100 < 3^{n-1}$$

Latest positive integer $n = 6$

35.(A) Let the number of terms in A.P. be $2n$ with common difference d

$$a_1 + a_3 + a_5 + \dots + a_{2n-1} = 24$$

$$\Rightarrow \frac{n}{2}[2a_1 + (n-1)2d] = 24 \Rightarrow n[a_1 + (n-1)d] = 24 \quad \dots\dots (i)$$

$$a_2 + a_4 + a_6 + \dots + a_{2n} = 30$$

$$\Rightarrow \frac{n}{2}[2a_2 + (n-1)2d] = 30 \Rightarrow n[a_2 + (n-1)d] = 30 \quad \dots\dots (ii)$$

$$(ii) - (i) \Rightarrow nd = 6 \quad \dots\dots (iii)$$

$$\text{Also, } a_{2n} - a_1 = \frac{21}{2} \Rightarrow (2n-1)d = \frac{21}{2} \quad \dots\dots (iv)$$

$$(iv) \div (iii) \Rightarrow n = 4$$

36.(B) $\frac{\frac{p}{2}[2a_1 + (p-1)d]}{\frac{q}{2}[2a_1 + (q-1)d]} = \frac{p^3}{q^3}$

$$\Rightarrow \frac{2a_1 + (p-1)d}{2a_1 + (q-1)d} = \frac{p^2}{q^2} \quad \dots\dots (i)$$

$$\frac{a_6}{a_{21}} = \frac{a_1 + 5d}{a_1 + 20d} = \frac{2a_1 + 10d}{2a_1 + 40d} \quad \dots\dots (ii)$$

Equating (i) and (ii), we get :

$$p-1 = 10 \Rightarrow p = 11 \text{ and } q-1 = 40 \Rightarrow q = 41 \quad \therefore \frac{a_6}{a_{21}} = \frac{11^2}{41^2} = \frac{121}{1681}$$

37.(C) $t_n = \frac{1}{1+2+3+\dots+n} = \frac{2}{n(n+1)} = 2\left[\frac{1}{n} - \frac{1}{n+1}\right] \therefore S_{10} = 2\left[\left(\frac{1}{1} - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \dots + \left(\frac{1}{10} - \frac{1}{11}\right)\right] = 2\left[1 - \frac{1}{11}\right] = \frac{20}{11}$

38.(B) $S_n = 2n + 3n^2 \quad \therefore \quad a = 5, d = 2 \times 3 = 6$

2nd A.P. $a = 5, d = 12 \quad \therefore \quad S_n = \frac{n}{2} [2 \times 5 + (n-1)12] = 6n^2 - n$

39.(C) $t_n = \frac{2n+1}{1^2 + 2^2 + \dots + n^2} = \frac{6(2n+1)}{n(n+1)(2n+1)} = \frac{6}{n(n+1)} = 6 \left[\frac{1}{n} - \frac{1}{n+1} \right]$

$\therefore \quad S_{11} = 6 \left[\left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \dots + \left(\frac{1}{11} - \frac{1}{12} \right) \right] = 6 \left[1 - \frac{1}{12} \right] = \frac{11}{2}$

40.(C) $\sum n^3 = \left[\frac{n(n+1)}{2} \right]^2$

$\therefore \quad \text{Sum} = 2^2 [1^2 + 2 \cdot 2^2 + 3 \cdot 3^2 + \dots + 10 \cdot 10^2] = 4 [1^2 + 2^3 + 3^3 + \dots + 10^3] = 4 \times \left(\frac{10 \times 11}{2} \right)^2 = 12100$

41.(C) $a = 10, S_3 = 39 \Rightarrow \frac{3}{2} [2 \times 10 + 2d] = 39 \Rightarrow d = 3$

Let the last four terms of AP be $a_{n-3}, a_{n-2}, a_{n-1}, a_n \quad \therefore \quad a_{n-3} + a_{n-2} + a_{n-1} + a_n = 178$

$\Rightarrow \quad \frac{4}{2} [2a_{n-3} + 3d] = 178 \Rightarrow 2 [2\{a + (n-4)d\} + 3 \times 3] = 178$

$\Rightarrow \quad 2 [2\{10 + (n-4)3\} + 9] = 178 \Rightarrow n = 14 \quad \therefore \quad \text{Number of terms in AP} = 14 (\text{even})$

Median = $\frac{7\text{th term} + 8\text{th term}}{2} = \frac{a_7 + a_8}{2} = \frac{a + 6d + a + 7d}{2} = \frac{59}{2} = 29.5$

42.(A) $t_n = \frac{1}{n(n+1)(n+2)(n+3)} = \frac{1}{3} \left[\frac{(n+3) - n}{n(n+1)(n+2)(n+3)} \right] = \frac{1}{3} \left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)} \right]$

$\therefore \quad \text{Required sum} = \frac{1}{3} \sum_{n=1}^5 \left[\frac{1}{n(n+1)(n+2)} - \frac{1}{(n+1)(n+2)(n+3)} \right] = \frac{1}{3} \left(\frac{1}{1.2.3} - \frac{1}{6.7.8} \right) = \frac{1}{3} \times \frac{55}{336}$

43.(B) Given $a_3 + a_4 = 60 \quad (\text{G.P.})$

$ar^2 + ar^3 = 60$

$ar(r + r^2) = 60 \quad \dots\dots (i)$

and $a_1 \cdot a_2 \cdot a_3 = 1000$

$a \cdot ar \cdot ar^2 = 1000$

$(ar)^3 = (10)^3 \Rightarrow ar = 10$

From (i), $10(r + r^2) = 60$

$r^2 + r - 6 = 0$

$(r+3)(r-2) = 0$

$r = -3, r = 2$

When $r = -3; a = -10/3 < 0$ (rejected) $\therefore a > 0$

When $r = 2; a = 5 \quad \therefore \quad a_7 = ar^6 = 5(2)^6 = 320$

$$44.(C) \quad a_4 - a_7 + a_{10} = m \quad \Rightarrow \quad a + 3d - (a + 6d) + a + 9d = m \quad \Rightarrow \quad a + 6d = m$$

$$S_{13} = \frac{13}{2} [2a + 12d] = 13(a + 6d) = 13m$$

$$45.(A) \quad 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\left(\frac{8}{5}\right)^2 + \left(\frac{12}{5}\right)^2 + \left(\frac{16}{5}\right)^2 + \dots \text{ to ten terms}$$

$$= \frac{4^2}{5^2} (2^2 + 3^2 + 4^2 + \dots + 11^2) = \frac{16}{25} (1^2 + 2^2 + 3^2 + \dots + 11^2 - 1^2) = \frac{16}{25} \left[\frac{11 \times 12 \times 23}{6} \right] = \frac{16}{25} \times 505 = \frac{16}{5} \times 101$$

$$\text{Given } \frac{16}{5} \times 101 = \frac{16}{5} m \quad \Rightarrow \quad m = 101$$

46.(4) Let the series has $2n$ terms and the series is $a, a + d, a + 2d, \dots, a + (2n-1)d$ according to the given conditions

We have

$$[a + (a + 2d) + (a + 4d) + \dots + (a + (2n-2)d)] = 24$$

$$\Rightarrow \quad \frac{n}{2} [2a + (n-1)2d] = 24$$

$$\Rightarrow \quad n[a + (n-1)d] = 24$$

$$\text{Also } [(a + d) + (a + 3d) + \dots + (a + (2n-1)d)] = 30$$

$$\Rightarrow \quad \frac{n}{2} [2(a + d) + (n-1)2d] = 30$$

$$\Rightarrow \quad n[(a + d) + (n-1)d] = 30$$

Also last term exceeds the first by $21/2$

$$\Rightarrow \quad a + (2n-1)d - a = 21/2$$

$$\Rightarrow \quad (2n-1)d = 21/2$$

Now subtracting (1) from (2)

$$nd = 6$$

Dividing (3) by (4)

$$\Rightarrow \quad \frac{2n-1}{n} = \frac{21}{12}$$

$$\Rightarrow \quad n = 4$$

47.(6) Let a be the first term, r the common ratio and n be the numbers of terms of the given G.P. Then,

$$a_1 + a_n = 66 \Rightarrow a + ar^{n-1} = 66$$

$$a_2 \cdot a_{n-1} = 128$$

$$\Rightarrow \quad ar \cdot ar^{n-2} = 128 \Rightarrow a^2 r^{n-1} = 128 \Rightarrow a \cdot (ar^{n-1}) = 128 = ar^{n-1} = \frac{128}{a}$$

Putting this value of ar^{n-1} in (1), we get $2 + 2 \cdot r^{n-1} = 66 \Rightarrow r^{n-1} = 32$

Putting $a = 64$ in (1), we get $64 + 64r^{n-1} = 66 \Rightarrow r^{n-1} = \frac{1}{32}$

We reject the second value as the G.P. is an increasing G.P. and therefore, $r > 1$.

Given, $S_n = 126$

$$\Rightarrow 2\left(\frac{r^n - 1}{r - 1}\right) = 126 \Rightarrow \frac{r^{n-1} \cdot r - 1}{r - 1} = 63 \Rightarrow \frac{32r - 1}{r - 1} = 63 \Rightarrow r = 2$$

$$\therefore r^{n-1} \cdot 32 \Rightarrow 2^{n-1} = 32 = 2^5 \Rightarrow n - 1 = 5 \Rightarrow n = 6$$

48.(3) Let $1/a = A, 1/b = A + D, 1/c = A + 2D, 1/d = A + 3D$ Thus, $a^{-2} - d^{-2} = A^2 - (A + 3D)^2 = -3D(2A + 3D)$ and $b^{-2} - c^{-2} = -D(2A + 3D)$

$\therefore$ Required ratio is 3.

49.(10) $T_n = \frac{(n+1)^2 - n}{n(n+1)} = 1 + \left(\frac{1}{n} - \frac{1}{n+1}\right)$

$$\therefore S_{10} = 10 + \left(1 - \frac{1}{11}\right) = \frac{120}{11}$$

50.(2) Given $b^2 = ac$ and $x = \frac{a+b}{2}, y = \frac{b+c}{2}$

$$\Rightarrow \frac{a}{x} + \frac{c}{y} = \frac{2a}{a+b} + \frac{2c}{b+c} = \frac{2a(b+c) + 2c(a+b)}{(a+b)(b+c)} = 2 \frac{2ac + ab + bc}{ab + ac + b^2 + bc} = 2 \frac{2ac + ab + bc}{2ac + ab + bc} = 2$$

51.(6) Since A.M. $\geq$ G.M.

$$\frac{a}{b} + \frac{b}{a} + \frac{b}{c} + \frac{c}{b} + \frac{c}{a} \geq 6 \Rightarrow \frac{b+c}{a} + \frac{b+a}{b} + \frac{a+b}{c} \geq 6$$

Hence, the least value is 6.

52.(3) Give expression is $\frac{1}{2} \sum \frac{2a}{b+c-a}$

$$= \frac{1}{2} \sum \left(\frac{2a}{b+c-a} + 1 \right) - \frac{3}{2}$$

$$= \frac{1}{2} (a+b+c) \sum \left(\frac{1}{b+c-a} \right) - \frac{3}{2}$$

Now, as $(a+b+c) = \sum (b+c-a)$

Applying A.M. $\geq$ H.M.

Minimum value of the expression $= \frac{1}{2} \times 9 - \frac{3}{2} = 3$

53.(10) $\sum_{i=1}^n \sum_{j=1}^i \sum_{k=1}^j 1$

$$= \sum_{i=1}^n \sum_{j=1}^i j = \sum_{i=1}^n \frac{i(i+1)}{2} = \frac{1}{2} \sum_{i=1}^n (i^2 + i) = \frac{n(n+1)(n+2)}{6} = 220$$

$$\therefore n = 10$$

$$54.(1) \quad S_n = 1 + 3 + 7 + 13 + 21 + \dots + t_n$$

$$S_n = 1 + 3 + 7 + 13 + \dots + t_{n-1} + t_n$$

Subtracting

$$0 = 1 + 2 + 4 + 6 + 8 + \dots + (t_n - t_{n-1}) - t_n$$

$$t_n = 1 + 2 + 4 + 6 + 8 + \dots + (t_n - t_{n-1})$$

$$= 1 + \frac{n-1}{2} [2 \times 2 + (n-2) \times 2]$$

$$= 1 + (n-1)n = n^2 - n + 1$$

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 - n + 1} - n)$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 - n + 1 - n^2}{\sqrt{n^2 - n + 1} + n}$$

$$= -\frac{1}{2}$$

$$55.(1) \quad U_n = \frac{1}{(n+2)(n+3)}$$

$$= \left(\frac{1}{n+2} - \frac{1}{n+3} \right)$$

$$S_n = \sum_{n=1}^n \left(\frac{1}{n+2} - \frac{1}{n+3} \right)$$

$$= \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \dots + \left(\frac{1}{n+2} - \frac{1}{n+3} \right)$$

$$= \frac{1}{3} - \frac{1}{n+3}$$

$$\therefore \lim_{n \rightarrow \infty} S_n = \frac{1}{3}$$

56.(7) There are 7 solutions (1, 2, 4, 8, 16), (2, 4, 8, 16, 32), (3, 6, 12, 24, 48), (4, 8, 16, 32, 64), (6, 12, 24, 48, 96) and (5, 10, 20, 40, 80), (1, 3, 9, 27, 81).

57.(0) Given that a, b, c are in G.P.,

$$\Rightarrow b^2 = ac$$

Also, given that $a-b, c-a, b-c$ are in H.P.

$$\frac{2}{c-a} = \frac{1}{a-b} + \frac{1}{b-c}$$

$$\Rightarrow \frac{2}{c-a} = \frac{a-c}{(a-b)(b-c)}$$

$$\Rightarrow 2(a-b)(b-c) = -(a-c)^2$$

$$\Rightarrow -(a-c)^2 = 2(ab - 2b^2 + bc)$$

$$\Rightarrow -(a-c)^2 = 2b(a - 2\sqrt{ac} + c)$$

$$\Rightarrow \left\{ (\sqrt{a} - \sqrt{c})(\sqrt{a} + \sqrt{c}) \right\}^2 = 2b(\sqrt{a} - \sqrt{c})^2$$

$$\Rightarrow 2b = -\{a + 2\sqrt{ac} + c\}$$

$$\Rightarrow 2b = -a - 2b - c$$

$$\Rightarrow a + 4b + c = 0$$

$$58.(2) \quad T_n = \frac{1+2+\dots+n}{1^3+2^3+\dots+n^3}$$

$$= \frac{\frac{n(n+1)}{2}}{\frac{n^2(n+1)^2}{2^2}} = \frac{2}{n(n+1)} = 2\left(\frac{1}{n} - \frac{1}{n+1}\right)$$

$$S_n = 2\left[\frac{1}{1} - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{n} - \frac{1}{n+1}\right]$$

$$= 2\left[1 - \frac{1}{n+1}\right] = 2 - \frac{2}{n+1} < 2$$

$$59.(5) \quad \text{We have } x + 2y + 3z + 4w = 50$$

Using the fact A.M. $\geq$ G.M., we get

$$\frac{2\left(\frac{x}{2}\right) + 4\left(\frac{y}{2}\right) + 3\left(\frac{z}{1}\right) + 1\left(\frac{4w}{1}\right)}{2+4+3+1} = \frac{50}{10} \geq \left[\left(\frac{x}{2}\right)^2 \left(\frac{y}{2}\right)^4 (z)^3 (4w)\right]^{1/10}$$

$$\Rightarrow 5 \geq \left[\left(\frac{x^2}{2^2}\right)(z)^3(2^2w)\right]^{1/10}$$

$$\Rightarrow 5 \geq \left(\frac{x^2y^4z^3w}{16}\right)^{1/10}$$

$$60.(3) \quad \text{Since } f(0) = f(1) = f(2)$$

$$\Rightarrow f(x) = 0 \quad \forall x \in R$$

$$\Rightarrow a = 2 \text{ and } b = 3$$

$$\text{Now, } \frac{1}{10} \sum_{r=4}^8 A_r H_{11-r} = \frac{1}{10} (A_4 \cdot H_7 + A_5 \cdot H_6 + \dots + A_9 \cdot H_3)$$

$$= \frac{1}{10} (6 \times 5) = 3$$

Quadratic Equations & Inequalities

1.(ABC) We have $D = b^2 - 4ac$. If $D > 0$, then the roots of the equation are given by $x = \frac{-b \pm \sqrt{D}}{2a}$

As $D = b^2 - 4ac < b^2$ ($a > 0, c > 0$), it follows that the roots of the quadratic equation are negative.

In case $D < 0$, then the roots of the equation are given by $x = \frac{-b \pm i\sqrt{-D}}{2a}$ which have negative real parts. (Both a and b are positive)

2.(C) $x^2 - (a-3)x + a = 0$

Atleast one root +ve $\Rightarrow$ either one is +ive or both are +ive

$\Rightarrow$ when both roots are +ive, Sum of roots $> 0 \Rightarrow a-3 > 0 \Rightarrow a > 3$

and when one root is +ive, Product of roots $< 0 \Rightarrow a < 0$

$\Rightarrow$ Take the union combination to get : $a \in (-\infty, 0) \cup (3, \infty)$ (i)

But $D \geq 0 \Rightarrow (a-3)^2 - 4a \geq 0 \Rightarrow a^2 - 10a + 9 \geq 0 \Rightarrow (a-9)(a-1) \geq 0$ (ii)

Combined solution : $a \in (-\infty, 0) \cup (0, 1] \cup [9, \infty)$

3.(C) Let $f(x) = 9^x - 3^x + 1$ (i)

Further, Let $3^x = t$

Replace ($3^x = t$) in (i), to get :

$$f(t) = t^2 - t + 1$$
 (ii)

We know that 3^x can take only positive value $x \in R$ (iii)

From (ii) and (iii),

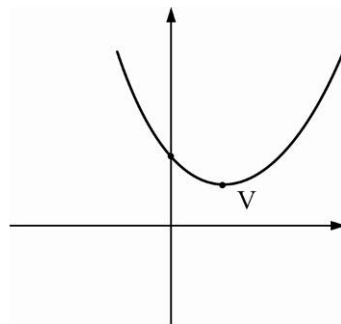
$$\Rightarrow t = 3^x > 0 \quad x \in R$$

We need to find all values $f(t)$ can take values $t > 0$.

$\Rightarrow$ To find all values $t^2 - t + 1$ can take $t > 0$.

Draw graph of $f(t)$ as shown $V \equiv \left(\frac{-b}{2a}, \frac{-D}{4a}\right) \equiv \left(\frac{1}{2}, \frac{3}{4}\right)$

From graph, $f(t)$ can take values $\left[\frac{3}{4}, \infty\right) \forall t > 0$



Alternate Method :

Let $f(x) = 9^x - 3^x + 1$

Expressing in perfect square form, $f(x) = \left(3^{2x} - 2 \cdot 3^x \cdot \left(\frac{1}{2}\right) + \frac{1}{4}\right) + \frac{3}{4}$; $f(x) = \left(3^x - \frac{1}{2}\right)^2 + \frac{3}{4}$

If we increase x to infinity, $f(x)$ also increases to infinity. $\Rightarrow$ Maximum value of $f(x) = \infty$.

Minimum value of $f(x) = \text{Min} \left[\left(3^x - \frac{1}{2}\right)^2 \right] + \frac{3}{4} = 0 + \frac{3}{4} \left\{ \because \left(3^x - \frac{1}{2}\right)^2 = 0 \text{ at } x = \frac{-\log 2}{\log 3} \right\} = \frac{3}{4}$

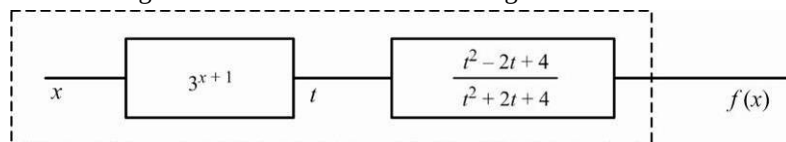
Minimum value of $f(x) = \frac{3}{4}$, Maximum value is ∞ and in between $f(x)$ can possess all values.

$$\Rightarrow f(x) \in \left[\frac{3}{4}, \infty\right)$$

4.(D) Let $f(x) = \frac{9 \cdot 3^{2x} - 6 \cdot 3^x + 4}{9 \cdot 3^{2x} + 6 \cdot 3^x + 4} \Rightarrow \frac{3^{2x+2} - 2 \cdot 3^{x+1} + 4}{3^{2x+2} + 2 \cdot 3^{x+1} + 4}$

Put $3^{x+1} = t$ to get : $f(t) = \frac{t^2 - 2t + 4}{t^2 + 2t + 4}$

We can consider the given function as shown in the diagram



$\Rightarrow$ 2nd block in the above diagram is getting only positive values as input. This is because $t = 3^{x+1} > 0, x \in R$.

$\Rightarrow$ We need to find the range of $f(t) = \frac{t^2 - 2t + 4}{t^2 + 2t + 4} t > 0$

Let $f(t) = y = \frac{t^2 - 2t + 4}{t^2 + 2t + 4} \Rightarrow y(t^2 + 2t + 4) = t^2 - 2t + 4$

$\Rightarrow (1-y)t^2 - 2(1+y)t + 4(1-y) = 0 \quad \forall t > 0$

We have to find all values of y such that the above equation has positive roots.

[If α, β are roots of $px^2 + qx + r = 0$, then both α and β are positive if :

$p f(0) > 0, x_v = x\text{-coordinate of vertex} > 0 \text{ and } D \geq 0$

$(1-y)\{4(1-y)\} > 0 \Rightarrow 4(1-y)^2 > 0 \Rightarrow y \in R - \{1\} \quad \dots\dots (i)$

$x_v = \frac{2(1+y)}{2(1-y)} > 0 \Rightarrow \frac{y+1}{y-1} < 0 \Rightarrow y \in (-1, 1) \quad \dots\dots (ii)$

$D \geq 0 \Rightarrow 4(1+y)^2 - 16(1-y)^2 \geq 0$

$\Rightarrow [1+y]^2 - [2(1-y)]^2 \geq 0 \Rightarrow (1+y+2-2y)(1+y-2+2y) \geq 0$

$\Rightarrow (3-y)(3y-1) \geq 0 \Rightarrow (y-3)(3y-1) \leq 0 \Rightarrow y \in \left[\frac{1}{3}, 3\right] \quad \dots\dots (iii)$

Combining (i), (ii) and (iii), $y \in \left[\frac{1}{3}, 1\right) \Rightarrow f(x) \in \left[\frac{1}{3}, 1\right)$.

5.(BC) Let $f(x) = (x-a)(x-c) + \lambda(x-b)(x-d)$

So, $f(a) = \lambda(a-b)(a-d)$ and $f(c) = \lambda(c-b)(c-d)$

$\Rightarrow f(a)f(c) = \lambda^2(a-b)(a-d)(c-b)(c-d) < 0 \quad [\text{Using } a < b < c < d]$

Also, $f(b) = (b-a)(b-c)$ and $f(d) = (d-a)(d-c)$

$\Rightarrow f(b)f(d) = (b-a)(b-c)(d-a)(d-c) < 0 \quad [\text{Using } a < b < c < d]$

$\Rightarrow$ Hence given equations has one real root between a and c and other real root between b and d .

6.(BC) The given equation is, $\pi^e(x-\pi)(x-\pi-e) + e^\pi(x-e)(x-\pi-e) + (\pi^\pi + e^e)(x-e)(x-\pi) = 0$

Let $f(x) = \pi^e(x-\pi)(x-\pi-e) + e^\pi(x-e)(x-\pi-e) + (\pi^\pi + e^e)(x-e)(x-\pi)$

Then $f(e) = \pi^e(e-\pi)(-\pi) > 0 \quad [\text{Using } e < \pi] \quad \dots\dots (i)$

and $f(\pi) = e^\pi (\pi - e) (-e) < 0$ (ii)

Combining (i) and (ii) to get : $f(e)f(\pi) < 0 \Rightarrow$ Equation $f(x) = 0$ has a real root in (e, π) .

Also $f(\pi + e) = (\pi^\pi + e^e)(\pi)(e) > 0$ (iii)

Combining (i) and (iii) to get $f(\pi)f(\pi + e) < 0 \Rightarrow f(x) = 0$ has a real root in $(\pi, \pi + e)$.

$f(x) = 0$ has a real roots in (e, π) and other in $(\pi, \pi + e)$

Using diagram, equation $f(x) = 0$ has two real roots in $(\pi - e, \pi + e)$

7.(A) $0 < a < b < c, \alpha + \beta = -\frac{b}{a}$ and $\alpha\beta = \frac{c}{a}$

For non real complex roots $b^2 - 4ac < 0$

$$\Rightarrow \frac{b^2}{a^2} - \frac{4c}{a} < 0 \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta < 0 \Rightarrow (\alpha - \beta)^2 < 0$$

$$\therefore 0 < a < b < c$$

$\therefore$ Roots are conjugate, then $|\alpha| = |\beta|$

But $\alpha = \frac{c}{a}$

$$|\alpha\beta| = \left| \frac{c}{a} \right| > 1 \quad \left(\because a < c, \therefore \frac{c}{a} > 1 \right) \Rightarrow |\alpha||\beta| = 1$$

$$|\alpha^2| > 1 \text{ or } |\alpha| > 1$$

8.(A) If roots is $\alpha, \frac{1}{\alpha}$

$$\therefore \alpha \cdot \frac{1}{\alpha} = \frac{c}{a} \Rightarrow c = a$$

$$\text{Since, } (|a| - |b|)^2 \geq 0 \text{ or } |a| \geq |b|$$

$$\text{But } a = c, \therefore |c| \geq |b|$$

9.(B) $y = \frac{x+a}{x^2+bx+c^2} \Rightarrow yx^2 + (by-1)x + c^2y - a = 0$

As $x \in R, D \geq 0$

$$\Rightarrow (by-1)^2 - 4y(c^2y-a) \geq 0 \Rightarrow (b^2-4c^2)y^2 + (4a-2b)y + 1 \geq 0$$

$$\text{Now, } b^2 - 4c^2 > 0 \text{ (given) and } D = (4a-2b)^2 - 4(b^2-4c^2) = 4[(2a-b)^2 - b^2 + 4c^2]$$

$$D = 16[a^2 + c^2 - ab] < 0 \text{ (given)}$$

Hence, y can take all values $\Rightarrow y \in R$

10.(C) $y = \frac{ax^2+x-2}{a+x-2x^2} \Rightarrow (a+2y)x^2 + (1-y)x - 2 - ay = 0$

As $x \in R, D \geq 0$

$$\Rightarrow (y-1)^2 - 4(a+2y)(-2-ay) \geq 0 \Rightarrow (8a+1)y^2 + (4a^2+14)y + (8a+1) \geq 0$$

Now, y will take all real values if this inequality is true for all values of y .

$$\Rightarrow 8a + 1 > 0 \Rightarrow a > -\frac{1}{8} \quad \dots\text{(i)}$$

$$\begin{aligned} \text{and } D \leq 0 &\Rightarrow (4a^2 + 14)^2 - 4(8a + 1)^2 \leq 0 \Rightarrow (a^2 + 4a + 4)^2 (a^2 - 4a + 3) \leq 0 \\ &\Rightarrow a^2 - 4a + 3 \leq 0 \quad [\because a^2 + 4a + 4 \geq 0] \Rightarrow (a - 3)(a - 1) \leq 0 \quad \dots\text{(ii)} \end{aligned}$$

Combined solution of (i) and (ii) : $a \in [1, 3]$

11.(B) Here, $f(x) = 4x^2 - 2(1 + 2 + 3 + 4)x + 30$

$$= 4x^2 - 20x + 30 = 4\left(x^2 - 5x\right) + 30 = 4\left\{x^2 - 5x + \frac{25}{4}\right\} + 30 - 25 = 4\left(x - \frac{5}{2}\right)^2 + 5$$

Now, $f(x)$ is minimum

when $\left(x - \frac{5}{2}\right)^2$ is minimum, i.e., when $\left(x - \frac{5}{2}\right)^2 = 0$, i.e., when $x = \frac{5}{2}$.

12.(B) Since, $x^4 + 20 + \sqrt{x^4 + 20} = 22 + 20$

Let $\sqrt{x^4 + 20} = y \quad \therefore y^2 + y - 42 = 0 \Rightarrow y^2 + y - 42 = 0$

$$\Rightarrow (y - 6)(y + 7) = 0 \Rightarrow y = 6 \quad (\because y \neq -7)$$

$$\Rightarrow \sqrt{x^4 + 20} = 6 \Rightarrow x^4 + 20 = 36$$

$$\Rightarrow x^4 = 16 \Rightarrow x = \pm 2$$

Hence, the number of real roots of the equation is 2.

13.(B) Remainders of $f(x)$ when divided by $x - 1, x - 2, x - 3$ are $f(1), f(2), f(3)$ respectively.

Then $f(1) = 3, f(2) = 7, f(3) = 13$

Since $(x - 1)(x - 2)(x - 3)$ is a third degree expression, the remainder of $f(x)$ when divided by

$(x - 1)(x - 2)(x - 3)$ is of second degree $\therefore f(x) = q(x - 1)(x - 2)(x - 3) + ax^2 + bx + c$

$$f(1) = 3 \Rightarrow a + b + c = 3 \quad \dots\text{(i)}, \quad f(2) = 7 \Rightarrow 4a + 2b + c = 7 \quad \dots\text{(ii)}$$

$$f(3) = 13 \Rightarrow 9a + 3b + c = 13 \quad \dots\text{(iii)}$$

$$(ii) - (i) \Rightarrow 3a + b = 4 \quad \dots\text{(iv)}$$

$$(iii) - (ii) \Rightarrow 5a + b = 6 \quad \dots\text{(v)}$$

$$(v) - (iv) \Rightarrow 2a = 2 \Rightarrow a = 1 \quad \therefore b = 1, c = 1 \quad \therefore \text{Remainder} = x^2 + x + 1$$

14.(C) $0 < \alpha < \beta < \gamma < \frac{\pi}{2} \Rightarrow \sin \alpha < \sin \beta < \sin \gamma$

Now, the given equation is $(x - \sin \beta)(x - \sin \gamma) + (x - \sin \alpha)(x - \sin \gamma) + (x - \sin \alpha)(x - \sin \beta) = 0$

Let $f(x) = (x - \sin \beta)(x - \sin \gamma) + (x - \sin \alpha)(x - \sin \gamma) + (x - \sin \alpha)(x - \sin \beta) = 0$

$$\Rightarrow f(\sin \alpha) = (\sin \alpha - \sin \beta)(\sin \alpha - \sin \gamma) > 0 \Rightarrow f(\sin \beta) = (\sin \beta - \sin \alpha)(\sin \beta - \sin \gamma) < 0$$

$$\Rightarrow f(\sin \gamma) = (\sin \gamma - \sin \alpha)(\sin \gamma - \sin \beta) > 0$$

Hence, equation $f(x) = 0$ has 1 root between $\sin \alpha$ and $\sin \beta$ and other root between $\sin \beta$ and $\sin \gamma$.

15.(C) α, β are roots of the equation $(x-a)(x-b)+c=0$ or $x^2-(a+b)x+ab+c=0$

$\Rightarrow \alpha + \beta = a + b \quad \dots\text{(i)}$

and $\alpha\beta = ab + c \quad \dots\text{(ii)}$

Let p, q be roots of the equation $(x-c-\alpha)(x-c-\beta)=c$

or $x^2-(\alpha+\beta+2c)x+c^2+c(\alpha+\beta-1)+\alpha\beta=0$

$\Rightarrow p+q=\alpha+\beta+2c \quad \dots\text{(iii)}$

and $pq=c^2+c(\alpha+\beta-1)+\alpha\beta \quad \dots\text{(iv)}$

Putting value of $\alpha + \beta$ from (i) into (iii), (iv) and value of $\alpha\beta$ from (ii) into (iv), we get :

$p+q=a+b+2c=(a+c)+(b+c) \quad \text{and} \quad pq=c^2+c(a+b-1)+ab+c=(a+c) \cdot (b+c)$

$\Rightarrow p=a+c \quad \text{and} \quad q=b+c$

16.(A) Consider $x^2+4x+3=(x+1)(x+3) \Rightarrow x^2+4x+3 \geq 0$ for $x \in (-\infty, -3] \cup [-1, \infty)$ and $x^2+4x+3 < 0$ for $x \in (-3, -1)$

Let us take $x \in (-\infty, -3] \cup [-1, \infty)$.

Then $|x^2+4x+3|+(2x+5)=0$ gives $(x^2+4x+3)+(2x+5)=0$

or $x^2+6x+8=0 \Rightarrow x=\frac{-6 \pm 2}{2} = -2, -4$

Reject $x=-2$ So, $x=-4$ Now take $x \in (-3, -1)$

Then $|x^2+4x+3|+(2x+5)=0$ gives $-(x^2+4x+3)+2x+5=0$

or $-x^2-2x+2=0$ or $x^2+2x-2=0$ or $x=\frac{-2 \pm 2\sqrt{3}}{2} = -1 \pm \sqrt{3}$

Reject $x=-1+\sqrt{3}$ So, $x=-1-\sqrt{3}$

17.(C) $\alpha + \beta = -b$ and $\alpha\beta = 1$

Now, for the required equation sum of roots $= -\left(\alpha + \beta + \frac{1}{\alpha} + \frac{1}{\beta}\right) = -\left(\alpha + \beta + \frac{\alpha + \beta}{\alpha\beta}\right) = -\left[-b + \frac{-b}{1}\right] = 2b$

Product of roots $= -\left(\alpha + \frac{1}{\beta}\right) \times \left(-\left[\beta + \frac{1}{\alpha}\right]\right) = \alpha\beta + \frac{1}{\alpha\beta} + 2 = 1 + \frac{1}{1} + 2 = 4$

So, the required equation is $x^2 - 2bx + 4 = 0$

18.(A) $x = \sqrt{7+4\sqrt{3}} = \sqrt{(2+\sqrt{3})^2} = 2+\sqrt{3}$

$x + \frac{1}{x} = (2+\sqrt{3}) + \frac{1}{(2+\sqrt{3})} = (2+\sqrt{3}) + (2-\sqrt{3}) = 4$

$$19.(B) \quad \frac{1}{|x|-3} - \frac{1}{2} < 0 \Rightarrow \frac{2-|x|+3}{|x|-3} < 0$$

$$\frac{5-|x|}{|x|-3} < 0 \Rightarrow \frac{|x|-5}{|x|-3} > 0$$

$$\Rightarrow |x| > 5 \text{ or } |x| < 3$$

$$\Rightarrow (x > 5 \text{ or } x < -5) \text{ or } (-3 < x < 3)$$

$$x \in (-\infty, -5) \cup (5, \infty) \text{ or } x \in (-3, 3) \Rightarrow x \in (-\infty, -5) \cup (-3, 3) \cup (5, \infty)$$

$$20.(C) \quad \left| \frac{x^2 - 5x + 4}{x^2 - 4} \right| \leq 1 \text{ and } x \neq \pm 2$$

$$|x^2 - 5x + 4| \leq |x^2 - 4|$$

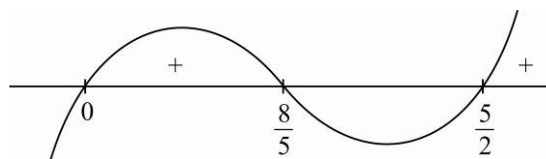
$$(x^2 - 5x + 4)^2 \leq (x^2 - 4)^2$$

$$(x^2 - 5x + 4 - x^2 + 4)(x^2 - 5x + 4 + x^2 - 4) \leq 0$$

$$(8 - 5x)(2x^2 - 5x) \leq 0$$

$$(5x - 8)(2x^2 - 5x) \geq 0$$

$$x(5x - 8)(2x - 5) \geq 0 \Rightarrow x \in \left[0, \frac{8}{5}\right] \cup \left[\frac{5}{2}, \infty\right)$$



$$21.(A) \quad \alpha - \beta = 1 \Rightarrow p^2 - 4q = 1 \Rightarrow p^2 - 4q + 4q^2 = 1 + 4q^2 \Rightarrow p^2 + 4q^2 = (1 + 2q)^2$$

$$22.(C) \quad 3^{2(x^2-x-6)} - 2 \cdot 3^{(x^2-x-6)} + 1 = 0$$

$$\Rightarrow 3^{(x^2-x-6)} = 1 \Rightarrow x^2 - x - 6 = 0$$

$$\Rightarrow (x-3)(x+2) = 0 \Rightarrow x = -2, 3$$

$$23.(C) \quad x^3 - px^2 + qx - r = 0$$

α, β, γ are roots of equation

$$\sum \alpha^2 \beta = \alpha^2 \beta + \alpha^2 \gamma + \beta^2 \alpha + \beta^2 \gamma + \gamma^2 \alpha + \gamma^2 \beta$$

$$= (\alpha^2 \beta + \alpha^2 \gamma + \alpha \beta \gamma) + (\beta^2 \alpha + \beta^2 \gamma + \alpha \beta \gamma) + (\gamma^2 \alpha + \gamma^2 \beta + \alpha \beta \gamma) - 3\alpha \beta \gamma$$

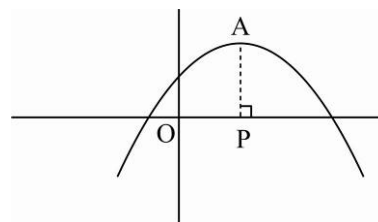
$$= \alpha(\alpha\beta + \beta\gamma + \alpha\gamma) + \beta(\beta\alpha + \beta\gamma + \alpha\gamma) + \gamma(\alpha\beta + \beta\gamma + \alpha\gamma) - 3\alpha\beta\gamma$$

$$= (\alpha\beta + \beta\gamma + \gamma\alpha)(\alpha + \beta + \gamma) - 3\alpha\beta\gamma = qp - 3r$$

$$24.(C) \quad \text{Equation of curve } y = ax^2 + bx + c$$

$$AP = \frac{-(b^2 - 4ac)}{4a} = 3 \Rightarrow \frac{-2b}{4a} = 3$$

$$\text{So } \frac{-b}{2a} = 3 \Rightarrow OP = 3$$



25.(C) $p=13, q=(-2)(-15)=30; \quad x^2+px+q=0 \Rightarrow x^2+13x+30=0 \Rightarrow x=-3,-10$

26.(D) Since roots are real, we have $\cos p - 1 \neq 0$ and discriminant $\geq 0 \therefore \cos^2 p - 4(\cos p - 1)\sin p \geq 0$

$$\Rightarrow \cos^2 p + 4(1 - \cos p)\sin p \geq 0 \Rightarrow \sin p \geq 0 \quad (\because \cos p - 1 \neq 0, -1 \leq \cos p \leq 1 \Rightarrow 1 - \cos p > 0)$$

$$\Rightarrow p \in [0, \pi] \Rightarrow p \in (0, \pi) \quad (\because \cos p \neq 1)$$

27.(B) We have $(b-c)x^2 + (c-a)x + (a-b) = 0 \quad \dots(i)$

a, b, c are in A.P. Let common difference be k .

$$\therefore (i) \Rightarrow -kx^2 + 2kx - k = 0 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow x = 1, 1$$

$$\therefore 1 \text{ is a root of } 2(c+a)x^2 + (b+c)x = 0 \Rightarrow 2(c+a) + (b+c) = 0 \Rightarrow 2a + b + 3c = 0$$

$$a, b, c \text{ are in A.P.} \Rightarrow a - 2b + c = 0 \quad \therefore \frac{a}{1+6} = \frac{b}{3-2} = \frac{c}{-4-1} \Rightarrow \frac{a}{7} = \frac{b}{1} = \frac{c}{-5} = \lambda \text{ (say)}$$

$$\Rightarrow a = 7\lambda, b = \lambda, c = -5\lambda \Rightarrow a^2 = 49\lambda^2, b^2 = \lambda^2, c^2 = 25\lambda^2 \Rightarrow a^2 + b^2 = 2c^2 \Rightarrow a^2, c^2, b^2 \text{ are in A.P.}$$

28.(B) α, β are roots of $x^2 + px + q = 0 \Rightarrow \alpha + \beta = -p$ and $\alpha\beta = q$

$$\alpha, \beta \text{ are also roots of } x^{2n} + p^n x^n + q^n = 0 \Rightarrow \alpha^{2n} + p^n \alpha^n + q^n = 0 \quad \dots(i)$$

$$\beta^{2n} + p^n \beta^n + q^n = 0 \quad \dots(ii)$$

$$(i) - (ii) \Rightarrow \alpha^{2n} - \beta^{2n} + p^n(\alpha^n - \beta^n) = 0 \Rightarrow (\alpha^n - \beta^n)(\alpha^n + \beta^n + p^n) = 0 \Rightarrow \alpha^n + \beta^n = -p^n \quad \dots(iii)$$

$$\frac{\alpha}{\beta} \text{ and } \frac{\beta}{\alpha} \text{ are roots of } x^n + 1 + (x+1)^n = 0 \Rightarrow \frac{\alpha^n}{\beta^n} + 1 + \left(\frac{\alpha}{\beta} + 1\right)^n = 0 \Rightarrow \alpha^n + \beta^n + (\alpha + \beta)^n = 0$$

$$\Rightarrow -p^n + (-p)^n = 0 \Rightarrow p^n[-1 + (-1)^n] = 0 \text{ This is possible only when } n \text{ is even.}$$

29.(A) Given, $\log_{10}(x^3 + y^3) - \log_{10}(x^2 + y^2 - xy) \leq 2$

$$\Rightarrow \log_{10} \frac{(x^3 + y^3)}{x^2 + y^2 - xy} \leq 2 \Rightarrow \log_{10}(x + y) \leq 2 \Rightarrow x + y \leq 100$$

Using $AM \geq GM$

$$\therefore \frac{x+y}{2} \geq \sqrt{xy} \Rightarrow \sqrt{xy} \leq \frac{x+y}{2} \leq \frac{100}{2} \Rightarrow xy \leq 2500$$

30.(D) $x^{12} - x^9 + x^4 - x + 1 > 0$, three cases arise

Case 1: When $x \leq 0$

$$x^{12} > 0, -x^9 > 0, x^4 > 0, -x > 0 \Rightarrow x^{12} - x^9 + x^4 - x + 1 > 0, \forall x \leq 0 \quad \dots(i)$$

Case 2: When $0 < x \leq 1$

$$x^9 < x^4, x < 1 \Rightarrow -x^9 + x^4 > 0 \text{ and } 1 - x > 0$$

$$\therefore x^{12} - x^9 + x^4 - x + 1 > 0, \forall 0 < x \leq 1 \quad \dots(ii)$$

Case 3: When $x > 1$

$$x^{12} > x^9, x^4 > x \Rightarrow x^{12} - x^9 + x^4 - x + 1 > 0, \forall x > 1 \quad \dots(iii)$$

$\therefore$ From equations (i), (ii) and (iii) the above equation holds for $x \in R$.

$$31.(A) \text{ Product} = \frac{1}{\sqrt{\alpha}} + \frac{1}{\sqrt{\beta}} = \frac{1}{a} \Rightarrow \sqrt{\alpha}\sqrt{\beta} = a$$

$$\text{Sum} = \frac{1}{\sqrt{\alpha}} \frac{1}{\sqrt{\beta}} = -\frac{b}{a} \Rightarrow \frac{\sqrt{\alpha} + \sqrt{\beta}}{\sqrt{\alpha}\sqrt{\beta}} = -\frac{b}{a} \Rightarrow \sqrt{\alpha} + \sqrt{\beta} = -b$$

Let α' and β' be the roots of $x^2 + (b^3 - 3ab)x + a^3 = 0$

$$\text{Sum} = \alpha' + \beta' = 3ab - b^3 \text{ and product} = \alpha'\beta' = a^3 = (\sqrt{\alpha}\sqrt{\beta})^3 \therefore \alpha'\beta' = \alpha^{3/2}\beta^{3/2}$$

$$\text{Now, } \alpha^{3/2} + \beta^{3/2} = (\sqrt{\alpha})^3 + (\sqrt{\beta})^3 = (\sqrt{\alpha} + \sqrt{\beta})^3 - 3\sqrt{\alpha}\sqrt{\beta}(\sqrt{\alpha} + \sqrt{\beta}) = -b^3 - 3a(-b) = 3ab - b^3$$

$$\therefore \text{Roots are } \alpha^{3/2} \text{ and } \beta^{3/2}$$

$$32.(B) x^2 + |2x - 3| - 4 = 0$$

$$(i) \text{ When } 2x - 3 \geq 0 \Rightarrow x \geq 3/2$$

$$x^2 + 2x - 3 - 4 = 0 \Rightarrow (x+1)^2 = 8 \Rightarrow x = 2\sqrt{2} - 1 \text{ as } x \geq 3/2$$

$$(ii) \text{ When } 2x - 3 < 0 \Rightarrow x < 3/2$$

$$x^2 - 2x + 3 - 4 = 0 \Rightarrow (x-1)^2 = 2 \Rightarrow x = 1 - \sqrt{2} \text{ as } x < 3/2$$

$$\therefore \text{Sum of roots} = (2\sqrt{2} - 1) + (1 - \sqrt{2}) = \sqrt{2}$$

$$33.(A) \sqrt{3x^2 + x + 5} = x - 3 \dots\dots (i)$$

On squaring, $3x^2 + x + 5 = (x-3)^2 \Rightarrow (2x-1)(x+4) = 0 \Rightarrow x = 1/2, -4$. They don't satisfy (i). Hence solution

$$34.(D) f(x) = (x+b)^2 + 2c^2 - b^2 \Rightarrow \min f(x) = 2c^2 - b^2$$

$$g(x) = -(x+c)^2 + b^2 + c^2 \Rightarrow \max g(x) = b^2 + c^2$$

$$\min f(x) > \max g(x)$$

$$2c^2 - b^2 > b^2 + c^2 \Rightarrow c^2 > 2b^2 \Rightarrow \left|\frac{c}{b}\right|^2 > 2 \Rightarrow \left|\frac{c}{b}\right| > \sqrt{2}$$

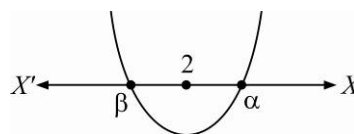
35.(C) Let two roots be α and β

Given $\alpha > 2, \beta < 2$

$$\text{Let } f(x) = x^2 - (a+1)x + a^2 + a - 8$$

$$\therefore f(\alpha) < 0 \Rightarrow 4 - (a+1)2 + a^2 + a - 8 < 0$$

$$\Rightarrow (a-3)(a+2) < 0 \Rightarrow -2 < a < 3$$



$$36.(C) \alpha + \beta = -p, \alpha\beta = \frac{3p}{4}$$

$$|\alpha - \beta|^2 = 10 \Rightarrow (\alpha + \beta)^2 - 4\alpha\beta = 10$$

$$\Rightarrow p^2 - 3p - 10 = 0 \Rightarrow (p-5)(p+2) = 0 \Rightarrow p = 5, -2$$

37.(D) Let α be a common root.

$$\therefore \alpha^2 + b\alpha - 1 = 0 \text{ and } \alpha^2 + \alpha + b = 0$$

$$\text{By cross-multiplication } \frac{\alpha^2}{b^2 + 1} = \frac{\alpha}{-1 - b} = \frac{1}{1 - b} \therefore \alpha = \frac{b^2 + 1}{-(1 + b)} \text{ and } \alpha = \frac{-(1 + b)}{1 - b}$$

$$\Rightarrow \frac{b^2+1}{-(1+b)} = \frac{-(1+b)}{1-b} \Rightarrow (b^2+1)(1-b) = (1+b)^2$$

$$\Rightarrow b(b^2+3) = 0 \Rightarrow b = 0 \text{ or } b^2 = -3.$$

$$\text{When } b = 0, \text{ then } \alpha^2 - 1 = 0 \Rightarrow \alpha = \pm 1 \text{ and } \alpha^2 + a = 0 \Rightarrow a = 0, -1$$

$$\therefore \text{Common root is } -1, \text{ which is rejected. So, } b^2 = -3 \Rightarrow b = \pm\sqrt{3}i \Rightarrow |b| = \sqrt{3}.$$

38.(A) $\sqrt{2x+1} = \sqrt{2x-1} + 1$

Squaring, $2x+1 = 2x-1+1+2\sqrt{2x-1}$

$$2\sqrt{2x-1} = 1 \Rightarrow 4(2x-1) = 1 \Rightarrow x = 5/8 \quad \therefore \sqrt{4x^2-1} = \sqrt{4\left(\frac{5}{8}\right)^2-1} = \sqrt{\frac{9}{16}} = \frac{3}{4}.$$

39.(B) In a polynomial equation, imaginary roots are always in pair.

$$2x^3 - 9x^2 + kx - 13 = 0$$

If $a = 2+3i$, then $\beta = 2-3i$ (conjugate pair)

$$\text{Now } \alpha + \beta + \gamma = \frac{-(-9)}{2} \Rightarrow 2+3i+2-3i+\gamma = \frac{9}{2} \Rightarrow \gamma = \frac{1}{2} \quad \therefore \text{Real root is } 1/2.$$

40.(C) $x^4 + x^2 + 1 = (x^4 + 2x^2 + 1) - x^2 = (x^2 + 1)^2 - x^2 = (x^2 + 1 + x)(x^2 + 1 - x)$

$$x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} > 0$$

$$\text{Now, } (a-1)(x^4 + x^2 + 1) + (a+1)(x^2 + x + 1)^2 = 0$$

$$\Rightarrow (x^2 + 1 + x) \left[(a-1)(x^2 + 1 - x) + (a+1)(x^2 + x + 1) \right] = 0$$

$$\text{Since } x^2 + x + 1 > 0$$

$$\therefore (a-1)(x^2 + 1 - x) + (a+1)(x^2 + x + 1) = 0 \Rightarrow ax^2 + x + a = 0$$

For two distinct roots $a \neq 0$ and discriminant > 0

$$\therefore 1^2 - 4a^2 > 0 \Rightarrow 4a^2 < 0 \Rightarrow a^2 < \frac{1}{4} \Rightarrow |a| < \frac{1}{2} \quad \therefore -\frac{1}{2} < a < \frac{1}{2} \text{ and } a \neq 0$$

$$\text{So, set of all values of } a \text{ is } \left(-\frac{1}{2}, 0\right) \cup \left(0, \frac{1}{2}\right).$$

41.(B) For $\ln k$, $k > 0$ and $e^{4\ln k} = e^{\ln k^4} = k^4$

$$x^2 - 4\sqrt{2}kx + 2k^4 - 1 = 0 ; \alpha + \beta = 4\sqrt{2}k, \alpha\beta = 2k^4 - 1$$

$$\text{Given } \alpha^2 + \beta^2 = 66 \Rightarrow (\alpha + \beta)^2 - 2\alpha\beta = 66$$

$$\Rightarrow 4(\sqrt{2}k)^2 - 2(2k^4 - 1) = 66 \Rightarrow (k^2 - 4)^2 = 0 \Rightarrow k^2 = 4 \Rightarrow k = 2, \text{ as } k > 0$$

$$\therefore \alpha + \beta = 8\sqrt{2}, \alpha\beta = 31$$

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = 8(\sqrt{2})^3 - 3 \times 31 \times 8\sqrt{2} = 8\sqrt{2}(128 - 93) = 280\sqrt{2}$$

42.(D) $f(-1) f(2) < 0$

$$[1 + (3 - 2a)(-1) + a][4 + 2(3 - 2a) + a] < 0$$

$$(3a - 2)(10 - 3a) < 0 \Rightarrow (3a - 2)(3a - 10) > 0$$

$$a > \frac{10}{3} \text{ or } a < \frac{2}{3} \quad \text{For } f(-1) f(2) = 0$$

Case - 1

$$f(-1) = 0$$

$$a = \frac{2}{3}$$

Equation becomes

$$x^2 + \frac{5x}{3} + \frac{2}{3} = 0$$

$$3x^2 + 5x + 2 = 0$$

$$(x + 1)(3x + 2) = 0$$

$$x = -1, -\frac{2}{3}$$



$-2/3$ lies in $(-1, 2) \therefore a = 2/3$ also

Case-2

$$f(2) = 0$$

$$a = \frac{10}{3}$$

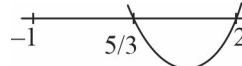
Equation becomes

$$x^2 - \frac{11x}{3} + \frac{10}{3} = 0$$

$$3x^2 - 11x + 10 = 0$$

$$(x - 2)(3x - 5) = 0$$

$$x = 2, \frac{5}{3}$$



$5/3$ lies in $(-1, 2) \therefore a = 10/3$ also.

43.(D) $\alpha - 2 = \frac{1}{\alpha} \Rightarrow \alpha^2 - 2\alpha = 1 \Rightarrow \alpha^2 - 2\alpha - 1 = 0$

$$x^2 - 2x - 1 = 0$$

$$b = -2, c = -1$$

44.(C) $\alpha^2 + p\alpha + q = 0, \Rightarrow \alpha^2 + p\alpha = -q$

$$(x - r)(x - \delta) = x^2 + px - r \text{ as } \gamma, \delta \text{ are root of } x^2 + px - r = 0$$

$$(\alpha - r)(\alpha - \delta) = \underbrace{\alpha^2 + p\alpha}_{=-q} - r = -q - r = -(q + r)$$

45.(A) $y = x^2 - 3x + 5 \quad D < 0$

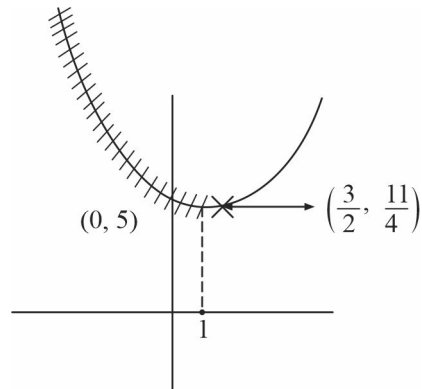
Coordinates of vertex

$$\left(\frac{-b}{2a}, \frac{-D}{4a} \right) = \left(\frac{3}{2}, \frac{11}{4} \right)$$

Clearly from graph for $x \in [-4, 1]$, y is decreasing

$$\therefore y_{\min} \text{ at } x = 1$$

$$y_{\min} = 1 - 3 + 5 = 3$$



46.(C) Let $f(x) = ax^2 + bx + c = 0$

Now as $a + b + c < 0 \Rightarrow f(1) < 0 \dots\dots(i)$

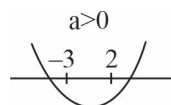
$a - b + c < 0 \Rightarrow f(-1) < 0 \dots\dots(ii)$

$c > 0 \Rightarrow f(0) > 0 \dots\dots(iii)$

Now from (i), (ii) and (iii), we can say that one of the root lies between $(-1, 0)$ and other root lies between $(0, 1)$.

$\Rightarrow [\alpha] + [\beta] = -1 + 0 = -1.$

47.(B) Case 1



$f(-3) < 0, f(2) < 0$

Clearly $f(1)$ & $f(-1)$

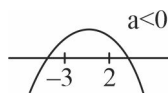
Would both be negative

$a + |b| + c < 0$

$a > 0$

Combining both cases. $a(a + |b| + c) < 0$

Case 2



$f(-3) > 0, f(2) > 0$

$f(1)$ & $f(-1)$ both would be positive

$f(-3) \text{ \& } f(s) > 0$

$\therefore a + |b| + c > 0$

$a < 0$

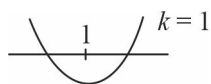
48.(3) $x \in [1, 2]$ is also solution of $x^2 - mx + 1 < 0$

$f(1) < 0 \quad 1 - m + 1 < 0 \Rightarrow 2 < m \dots(1)$

& $f(2) < 0 \quad \& \quad 4 - 2m + 1 < 0 \quad 2m > 5 \quad m > \frac{5}{2} \dots(2)$

Using (1) & (2), least value of $m = 3$

49.(5) $A f(k) < 0 \quad (a^2 + a + 2)[a^2 + a + 2 - a - 4 - 7] < 0$



As $a^2 + a + 2$ is always +ve $(a^2 - 9) < 0$

$-3 < a < 3 \quad a = -2, -1, 0, 1, 2$

50.(5) $D > 0$

$4p^2 - 4(3p + 4) > 0$

$p^2 - 3p - 4 > 0$

$(p - 4)(p + 1) > 0$

$\frac{+}{-1} \quad \frac{-}{4} \quad \frac{+}{p} \quad p < -1 \text{ or } p > 4$

51.(3) $f(x) = (\lambda^2 + \lambda - 2)x^2 + (\lambda + 2)x - 1 < 0$

Case 1 If $f(x)$ is quadratic

If $\lambda^2 + \lambda - 2 \neq 0$

Then graph of $f(x)$ is



$A < 0, D < 0$

$(\lambda + 2)(\lambda - 1) < 0$

Case 2 : If $f(x)$ is not a quadratic

Then $f(x) < 0$ if $\lambda^2 + \lambda - 2 = 0$

$(\lambda + 2)(\lambda - 1) = 0$

$\lambda = -2, \lambda = 1$

for $\lambda = -2$

$f(x) = -1 < 0$

$\lambda = 1$

$f(x) = 3x - 1 < 0$

55.(6) Let α be the common +ve root ($\alpha > 0$)

$$\therefore \alpha^2 + a\alpha + 12 = 0 \dots(1)$$

$$\alpha^2 + b\alpha + 15 = 0 \dots(2)$$

$$\alpha^2 + a\alpha + b\alpha + 36 = 0 \dots(3)$$

$$(3)-(2) \quad a\alpha + 21 = 0$$

$$a\alpha = -21$$

Putting in equation (1)

$$\alpha^2 - 21 + 12 = 0 \Rightarrow \alpha^2 - 9 = 0$$

$$\alpha = 3, -3 \text{ (rejected as } \alpha > 0 \text{)}$$

$$\therefore \alpha = 3$$

$$a = -7$$

Using equation (2)

$$9 + 3b + 15 = 0$$

$$3b = -24 \Rightarrow b = -8$$

$$b - 2a = -8 + 14 = 6$$

56.(1) $x^2 - x + 1 = 0$

$\alpha, \beta = -\omega, -\omega^2$ where ω is complex cube root of unity.

$$\alpha^{2020} + \beta^{2020} = \omega^{2020} + \omega^{4040} = \omega + \omega^2 = -1$$

57.(5) $\alpha, \beta, \gamma, \delta$ are in G.P. with $r > 1$

$$\beta = \alpha r$$

$$\gamma = \alpha r^2$$

$$\delta = \alpha r^3$$

$$\alpha + \beta = 4 \Rightarrow \alpha + \alpha r = 4 \dots(1)$$

$$\gamma + \delta = 36 \Rightarrow \alpha r^2 + \alpha r^3 = 36 \dots(2)$$

Dividing (1) & (2) we get $r^2 = 9 \Rightarrow r = 3$

$$\alpha + \beta = 4 \Rightarrow \alpha + \alpha r = 4 \Rightarrow \alpha(1 + r) = 4$$

$$\alpha = 1$$

$$A = \alpha\beta = \alpha^2 r = 3$$

$$B = \gamma\delta = \alpha^2 r^5 = r^5 = 3^5$$

58.(2) $3x^2 - 17x + 10 = 0$

$$3x^2 - 15x - 2x + 10 = 0$$

$$3x(x-5) - 2(x-5) = 0 \Rightarrow x = 5 \text{ or } 2/3$$

$\therefore$ Common root is either 5 or $2/3$.

Case 1 : If 5 is common root then $x^2 - 5x + m = 0$

$$5^2 - 5 \times 5 + m = 0$$

$$m = 0$$

Case 2 : If $2/3$ is common root. $\left(\frac{2}{3}\right)^2 - 5\left(\frac{2}{3}\right) + m = 0$

$$\frac{4}{9} - \frac{10}{3} + m = 0 ; \quad m = \frac{10}{3} - \frac{4}{9} = \frac{30}{9} - \frac{4}{9} = \frac{26}{9}$$

59.(1) $\frac{1}{r+s} = \frac{r+s}{rs} \Rightarrow (r+s)^2 = rs$

$$r^2 + s^2 + rs = 0$$

$$\left(\frac{r}{s}\right)^2 + \left(\frac{r}{s}\right) + 1 = 0$$

Let $\frac{r}{s} = t \quad \therefore \quad t^2 + t + 1 = 0$

$t = \omega, \omega^2$ where ω is complex cube root of unity.

$$\left(\frac{r}{s}\right)^3 = t^3 = 1$$

60.(2) $f(x) = 3x^2 + kx + 3$

$$g(x) = x^2 + kx - 1$$

$$f(x)g(x) = 0$$

$$D_1 = k^2 - 36 \text{ which may or may not be positive.}$$

$\therefore f(x)$ can have 2 or 0 number of real roots.

$$D_2 = k^2 + 4 \quad D_2 > 0$$

$g(x)$ has always 2 real roots.

$\therefore f(x) \cdot g(x) = 0$ has atleast 2 real roots.

61.(3) $4x^2 - 16x + \lambda = 0, \lambda \in R$

I: $f(1)f(2) < 0$

$$[4 - 16 + \lambda][16 - 32 + \lambda] < 0$$

$$(\lambda - 12)(\lambda - 16) < 0$$

Combined solution: $\lambda \in (12, 16)$

Integral values of $\lambda = 13, 14, 15$

II: $f(2)f(3) < 0$

$$[16 - 32 + \lambda][36 - 48 + \lambda] < 0$$

$$(\lambda - 16)(\lambda - 12) < 0$$

62.(9) $2x^2 + 3x + 4 = 0, D = 9 - 4 \times 2 \times 4 = -23 < 0$

Roots are imaginary and imaginary roots are in pair, so, both roots are common.

$$\frac{a}{2} = \frac{b}{3} = \frac{c}{4} \Rightarrow a : b : c = 2 : 3 : 4$$

Straight Line

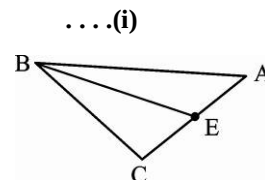
1.(B) Let E is the mid-point of AC. $\Rightarrow E \equiv \left(4, \frac{5}{2}\right)$

Equation of median through $B(-4, 9)$ and $E\left(4, \frac{5}{2}\right)$ using : $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

$(x_1, y_1) \equiv B(-4, 9)$ and $(x_2, y_2) \equiv E\left(4, \frac{5}{2}\right)$

Substituting the value of (x_1, y_1) and (x_2, y_2) in (i), we get : $y - 9 = \frac{5/2 - 9}{4 - (-4)}(x + 4)$

Simplify to get : $13x + 16y = 92$ is the required equation.



2.(BC) Let $P(t, 4 - t)$ be an arbitrary point on the line $x + y = 4$

Distance of P from $4x + 3y - 10 = 0$ is unity.

$\Rightarrow \frac{|4t + 3(4 - t) - 10|}{\sqrt{16 + 9}} = 1 \Rightarrow |t + 2| = 5$

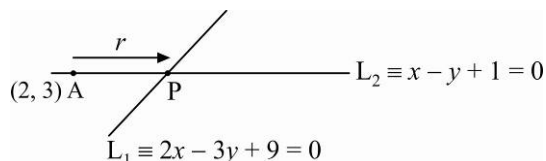
$\Rightarrow t = -2 \pm 5 = -7, 3 \Rightarrow$ Points are $(-7, 11)$ and $(3, 1)$

Draw the diagram yourself.

3.(A) $P \equiv \left[2 + \frac{r}{\sqrt{2}}, 3 + \frac{r}{\sqrt{2}}\right]$

Since, P lies on $2x - 3y + 9 = 0$

$\Rightarrow 2\left[2 + \frac{r}{\sqrt{2}}\right] - 3\left[3 + \frac{r}{\sqrt{2}}\right] + 9 = 0 \Rightarrow r = 4\sqrt{2}$



4.(C) Since, the given lines are concurrent.

$\therefore \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \Rightarrow a^3 + b^3 + c^3 - 3abc = 0 \Rightarrow (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 0$

$\Rightarrow \frac{(a + b + c)}{2} \{(a - b)^2 + (b - c)^2 + (c - a)^2\} = 0 \Rightarrow a + b + c = 0$ (As $a \neq b \neq c$)

5.(A) Let equation of line is $\frac{x - x_1}{\cos \theta} = \frac{y - y_1}{\sin \theta} = r \Rightarrow \frac{x - 1}{\cos \theta} = \frac{y - 2}{\sin \theta} = \frac{\sqrt{6}}{3} \Rightarrow x = \frac{\sqrt{6}}{3} \cos \theta + 1, y = \frac{\sqrt{6}}{3} \sin \theta + 2$

It satisfies $x + y = 4$

$\therefore \frac{\sqrt{6}}{3} \cos \theta + 1 + \frac{\sqrt{6}}{3} \sin \theta + 2 = 4 \Rightarrow \frac{\sqrt{6}}{3} \cos \theta + \frac{\sqrt{6}}{3} \sin \theta = 1$

$\Rightarrow \theta = 15^\circ$

or $\sin 2\theta = \sin(180^\circ - 30^\circ) \Rightarrow 2\theta = 150^\circ \Rightarrow \theta = 75^\circ$

6.(A) Given, equation of line is $\sqrt{3}x + y + 2 = 0$

$$\Rightarrow \sqrt{3}x + y = -2 \Rightarrow -\sqrt{3}x - y = 2$$

On dividing above equation by $\sqrt{(\text{coefficient of } x)^2 + (\text{coefficient of } y)^2}$

$$\text{i.e., } \sqrt{(-3)^2 + (-1)^2} = \sqrt{3+1} = \sqrt{4} = 2 \quad \Rightarrow \quad \frac{-\sqrt{3}}{2}x - \frac{1}{2}y = \frac{2}{2}$$

$$\Rightarrow -\cos 30^\circ x - \sin 30^\circ y = 1 \quad (\because \text{For convert in the form of } x \cos \alpha + y \sin \alpha = p)$$

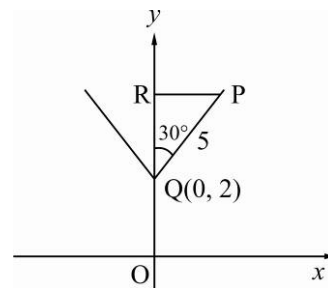
$$\Rightarrow \cos(180^\circ + 30^\circ)x + \sin(180^\circ + 30^\circ)y = 1 \quad (\because \cos \theta \text{ and } \sin \theta \text{ both are negative in third quadrant})$$

$$\Rightarrow (\cos 210^\circ)x + (\sin 210^\circ)y = 1$$

On comparing with $x \cos \theta + y \sin \theta = p$, we get : $\theta = 210^\circ$ and $p = 1$

7.(B) Given that, $y = \sqrt{3}|x| + 2$ and $PQ = 5$, so $QR = \frac{5\sqrt{3}}{2}$

So, coordinates of R are $\left(0, 2 + \frac{5\sqrt{3}}{2}\right)$ or $\left(0, \frac{4+5\sqrt{3}}{2}\right)$



8.(C) Each side of square is 5 units, distance between given lines is 5 units,

$$\text{i.e., } \left| \frac{k_1 - k_2}{5} \right| = 5 \quad \Rightarrow \quad |k_1 - k_2| = 25$$

9.(B) Since, t_1, t_2 are the roots of the equation $t^2 + \lambda t + 1 = 0$, $\therefore t_1 + t_2 = -\lambda, t_1 t_2 = 1$

The equation of a line passing through $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$ is

$$y - 2at_2 = \frac{2}{t_1 + t_2} (x - at_2^2) \Rightarrow y - 2at_2 = -\frac{2}{\lambda} (x - at_2^2)$$

$$\Rightarrow \lambda y - 2a\lambda t_2 = -2x + 2at_2^2 \Rightarrow \lambda y + 2x = 2a(\lambda t_2 + t_2^2) \Rightarrow \lambda y + 2x = 2a(-1) \Rightarrow 2(x + a) + \lambda y = 0$$

10.(D) If the line meets the x and y -axes at A and B, then $A \equiv \left(-\frac{c}{a}, 0\right), B \equiv \left(0, -\frac{c}{b}\right)$. The line will pass through the first quadrant, if $-\frac{c}{a} > 0$ and $-\frac{c}{b} > 0 \Rightarrow ac < 0$ and $bc < 0$

11.(C) $ax + by = 1$ will be one of the bisectors of the given line. Equation of bisectors of the given lines are :

$$\frac{3x + 4y - 5}{5} = \pm \left(\frac{5x - 12y - 10}{13} \right)$$

$$\Rightarrow 64x - 8y = 115 \text{ or } 14x + 112y = 15 \quad \Rightarrow \quad a = \frac{64}{115}, b = -\frac{8}{115} \text{ or } a = \frac{14}{15}, b = \frac{12}{115}$$

12.(C) As $x + y = |a|$ and $ax - y = 1$, intersect in first quadrant.

$\therefore$ x and y -intercepts are positive.

$$\therefore x = \frac{1 + |a|}{1 + a} \geq 0 \quad \text{and} \quad y = \frac{a|a| - 1}{a + 1} \geq 0$$

$$\Rightarrow 1 + a \geq 0 \text{ and } a|a| - 1 \geq 0 \quad \Rightarrow \quad a \geq -1 \text{ and } a|a| \geq 1 \quad \dots \text{(i)}$$

$$\text{If } -1 \leq a < 0 \quad \Rightarrow \quad -a^2 > 1 \quad [\text{Not possible}]$$

$$\text{If } a \geq 0 \Rightarrow a^2 \geq 1 \quad \Rightarrow \quad a \geq 1 \quad \dots \text{(ii)}$$

$$\therefore a > 1 \text{ or } a \in [1, \infty)$$

- 13.(A) The point (a, a^2) lies in the sector bounded by the lines $x - 2y = 0$ and $3x - y = 0$

$$\therefore (a - 2a^2)(3a - a^2) < 0 \Rightarrow \left[a - \frac{1}{2}\right](a - 3) < 0 \therefore a \in \left(\frac{1}{2}, 3\right)$$

- 14.(B) Equation of line: $\frac{x}{a} + \frac{y}{b} = 1$

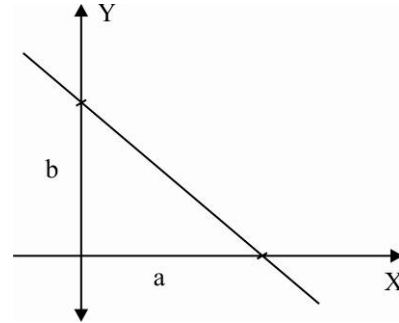
It passes through, (α, β)

$$\Rightarrow \frac{\alpha}{a} + \frac{\beta}{b} = 1 \quad \text{or} \quad b = \frac{\beta}{1 - \frac{\alpha}{a}}$$

$$S = \frac{1}{2}ab = \frac{1}{2} \frac{a\beta}{1 - \frac{\alpha}{a}} = \frac{1}{2} \frac{\beta}{\frac{1}{a} \left(1 - \frac{\alpha}{a}\right)}; \quad S = \frac{\beta/2}{\frac{-\alpha}{a^2} + \frac{1}{a}}$$

Let $\frac{1}{a} = t$ to get $S = \frac{\beta/2}{-\alpha t^2 + t}$

S is minimum when $-\alpha t^2 + t$ is maximum which occurs at $t = \frac{1}{2\alpha}$ and corresponding area is $\frac{\beta/2}{-\alpha \left(\frac{1}{4\alpha^2}\right) + \frac{1}{2\alpha}} = 2\alpha\beta$



- 15.(C) Let equal sides are along X-axis and Y-axis.

Equation of PQ is:

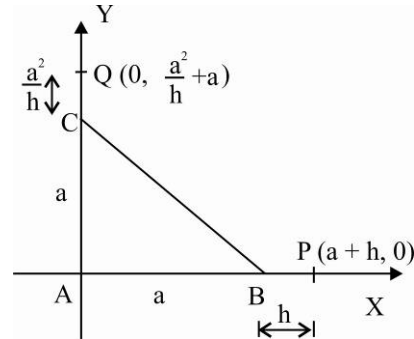
$$\frac{x}{a+h} + \frac{y}{\frac{a^2}{h} + a} = 1$$

$$\Rightarrow \frac{x}{a+h} + \frac{yh}{a(a+h)} = 1$$

$$\Rightarrow ax + yh = a^2 + ah$$

$$\Rightarrow (ax - a^2) + h(y - a) = 0$$

$\Rightarrow$ Family of lines passing through the point of intersection of $x = a$ and $y = a$.



- 16.(B) $\frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{c_1}{b_1} = \frac{c_2}{b_2}$. Both lines have same Y-intercept. Hence given family is family of concurrent lines having the same Y-intercept.

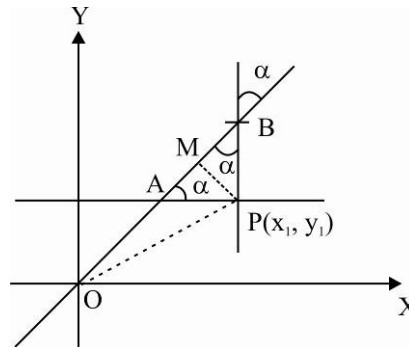
- 17.(A) $OP^2 = OM^2 + PM^2$

$$x_1^2 + y_1^2 = \left(1 + \frac{1}{2}\right)^2 + \left(\frac{1}{2} \tan \alpha\right)^2$$

$$\Rightarrow x_1^2 + y_1^2 = \frac{9 + \tan^2 \alpha}{4}$$

Replace x_1, y_1 by x, y

To get: $x^2 + y^2 = \frac{9 + \tan^2 \alpha}{4}$



$$18.(C) \quad (4, 1) \xrightarrow{1} (1, 4) \xrightarrow{2} (3, 4) \xrightarrow{3} \left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$$

$$\therefore B \equiv (5 \cos(45^\circ + \alpha), 5 \sin(45^\circ + \alpha)) \equiv \left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right), \quad \cos \alpha = \frac{3}{5}$$

$$19.(A) \quad y = mx \text{ cuts the curve } x^3 + xy^2 + 2x^2 + 2y^2 + 3x + 1 = 0$$

$$\therefore x^3 + x(mx)^2 + 2x^2 + 2(mx)^2 + 3x + 1 = 0 \Rightarrow (1+m^2)x^3 + (2+2m^2)x^2 + 3x + 1 = 0$$

Roots are in H.P.

$$\therefore \text{Replace } x \text{ by } \frac{1}{t} \text{ to get,}$$

$$(1+m^2)\frac{1}{t^3} + 2(1+m^2)\frac{1}{t^2} + \frac{3}{t} + 1 = 0 \quad \text{or} \quad t^3 + 3t^2 + 2(1+m^2)t + (1+m^2) = 0$$

Now, roots are in A.P.

Assume roots are $a-d, a, a+d$

$$3a = \frac{-3}{1} \Rightarrow a = -1$$

$$\therefore a^3 + 3a^2 + 2(1+m^2)a + (1+m^2) = 0 \Rightarrow -1 + 3 - 2(1+m^2) + (1+m^2) = 0 \Rightarrow m = \pm 1$$

$$20.(A) \quad \text{Distance between } || \text{ lines} = \frac{\left|4 - \frac{5}{2}\right|}{\sqrt{2}} = \frac{3}{2\sqrt{2}}$$

Hence, no point lies on $x+y=4$ which is at a distance

of 1 unit from $x+y=\frac{5}{2}$

$$21.(B) \quad \text{Area of trapezium} = \frac{1}{2} \left[\frac{k}{4} + \frac{2+k}{4} \right] \times 2 = \frac{1}{2} \times 2 \times 3$$

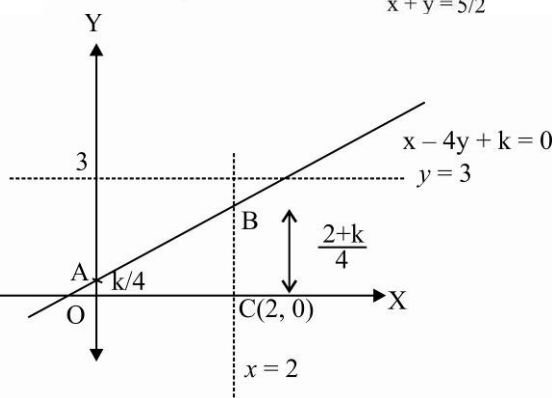
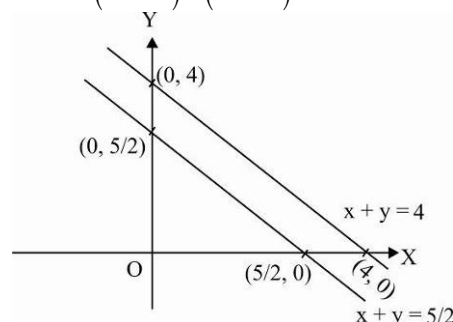
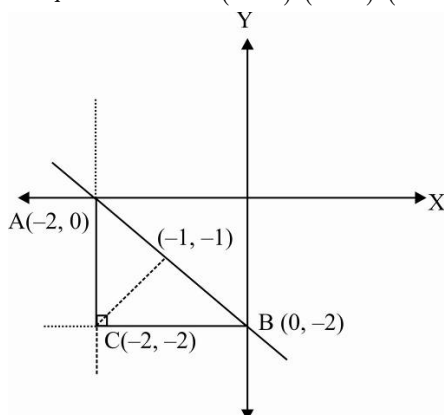
$$\Rightarrow k = 5$$

Equation of line is :

$$x - 4y + 5 = 0$$

$$22.(B) \quad xy + 2x + 2y + 4 = 0 \Rightarrow (y+2)(x+2) = 0$$

$(-1, -1)$ is equidistant from $(-2, 0), (0, -2), (-2, -2)$



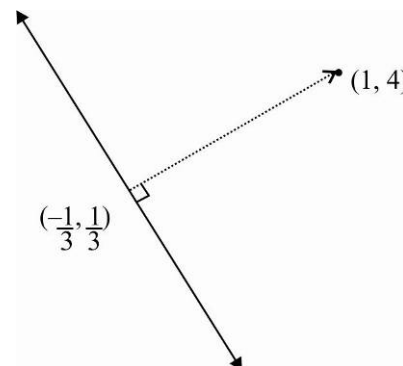
23.(B) $(x+y) + k(2x-y+1) = 0$

Solve $x+y=0$ and $2x-y+1=0 \Rightarrow x = -\frac{1}{3}, y = \frac{1}{3}$

Line through $\left(-\frac{1}{3}, \frac{1}{3}\right)$ at a maximum distance $(1, 4)$ is perpendicular

to line joining $\left(-\frac{1}{3}, \frac{1}{3}\right)$ and $(1, 4)$ with slope $\frac{11}{4}$.

$$\therefore \frac{y - \frac{1}{3}}{x + \frac{1}{3}} = -\frac{4}{11} \Rightarrow 11(3y-1) + 4(3x+1) = 0 \Rightarrow 12x + 33y - 7 = 0$$

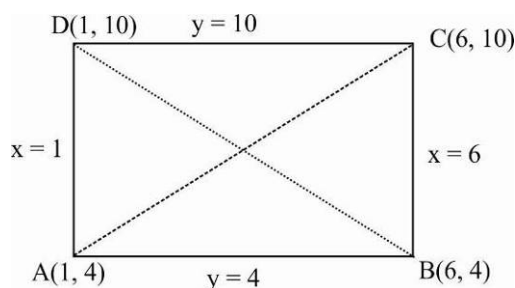


24.(C) Equation of AC is: $\frac{y-4}{x-1} = \frac{10-4}{6-1}$

$$\Rightarrow 6x - 5y + 14 = 0$$

Equation of BD is: $\frac{y-4}{x-6} = \frac{10-4}{1-6}$

$$\Rightarrow 6x + 5y = 56$$



25.(B) $-x+2y+4=0$ and $-2x+y+4=0$ intersect at $\left(\frac{4}{3}, -\frac{4}{3}\right)$

$$a_1a_2 + b_1b_2 = (-1)(-2) + (2)(1) > 0$$

Acute $\angle$ bisector is: $\frac{-x+2y+4}{\sqrt{5}} = -\left(\frac{-2x+y+4}{\sqrt{5}}\right)$

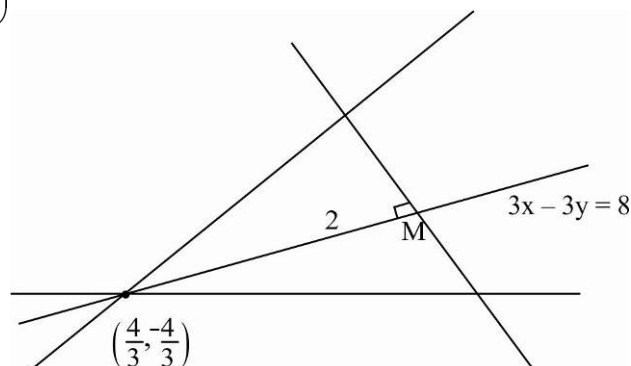
$$\Rightarrow -x+2y+4-2x+y+4=0$$

$$\Rightarrow 3x-3y-8=0$$

Slope $= m = \tan \theta = 1$

$$\sin \theta = \frac{1}{\sqrt{2}}, \cos \theta = \frac{1}{\sqrt{2}}$$

$$M \equiv \left[\frac{4}{3} \pm 2\left(\frac{1}{\sqrt{2}}\right), -\frac{4}{3} \pm 2\left(\frac{1}{\sqrt{2}}\right) \right] \equiv \left[\frac{4}{3} \pm \sqrt{2}, -\frac{4}{3} \pm \sqrt{2} \right]$$

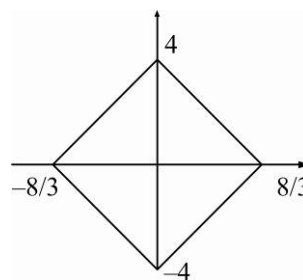


26.(C) $f(x+y) = f(x) + f(y)$

$$\Rightarrow f(x) = a^x \text{ where } f(1) = 2 \Rightarrow f(x) = 2^x$$

Area $= 4 \times \left(\frac{1}{2} \times 4 \times \frac{8}{3}\right) = \frac{64}{3}$ square units

$$= \frac{f(6)}{3}$$



27.(C) $B' \equiv (0 + 2 \cos 30^\circ, 0 + 2 \sin 30^\circ) \equiv (\sqrt{3}, 1)$

$D' \equiv (0 + 2 \cos 120^\circ, 0 + 2 \sin 120^\circ) \equiv (-1, \sqrt{3})$

$$\frac{y-1}{x-\sqrt{3}} = \frac{\sqrt{3}-1}{-1-\sqrt{3}} = \frac{-(\sqrt{3}-1)^2}{(\sqrt{3}+1)(\sqrt{3}-1)}$$

$$\Rightarrow \frac{y-1}{x-\sqrt{3}} = -\frac{(4-2\sqrt{3})}{2} = \sqrt{3}-2 \Rightarrow (2-\sqrt{3})x + y = 2(\sqrt{3}-1)$$

28.(A) $\frac{x}{t^3} = \frac{y}{t^2-3} = \frac{1}{t-1} \Rightarrow x = \frac{t^3}{t-1}, y = \frac{t^2-3}{t-1}$

Let equation of line is : $\ell x + m y + n = 0$

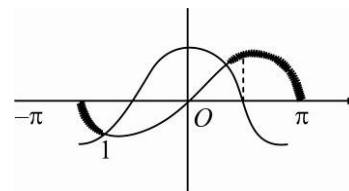
$$\Rightarrow \ell \left(\frac{t^3}{t-1} \right) + m \left(\frac{t^2-3}{t-1} \right) + n = 0 \Rightarrow \ell t^3 + m t^2 + n t + (-3m-n) = 0$$

$$\Rightarrow a+b+c = \frac{-m}{\ell}, ab+bc+ca = \frac{n}{\ell}, abc = \frac{-(3m+n)}{\ell}$$

$$abc = -3(a+b+c) + (ab+bc+ca) \therefore abc + 3(a+b+c) = ab+bc+ca$$

29.(A) $3\left(\frac{\sin \alpha}{3} - 1\right) - 2\left(\frac{\cos \alpha}{2} - 1\right) + 1$ and $3(1) - 2(1) + 1$ have same sign i.e., $\sin \alpha - \cos \alpha > 0 \Rightarrow \sin \alpha > \cos \alpha$

$$\alpha \in \left[-\pi, \frac{-3\pi}{4}\right) \cup \left(\frac{\pi}{4}, \pi\right]$$



30.(B) $x \cos \alpha + y \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} = 0$

$$p_1 = \left| m^2 \cos \alpha + 2m \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} \right| = \left| \frac{(m \cos \alpha + \sin \alpha)^2}{\cos \alpha} \right|, p_2 = \left| mm' \cos \alpha + (m+m') \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} \right|$$

$$p_2 = \left| \frac{(m \cos \alpha + \sin \alpha)(m' \cos \alpha + \sin \alpha)}{\cos \alpha} \right|, p_3 = \left| m'^2 \cos \alpha + 2m' \sin \alpha + \frac{\sin^2 \alpha}{\cos \alpha} \right| = \left| \frac{(m' \cos \alpha + \sin \alpha)^2}{\cos \alpha} \right|$$

$$p_1 p_3 = \frac{(m \cos \alpha + \sin \alpha)^2 (m' \cos \alpha + \sin \alpha)^2}{\cos^2 \alpha} = p_2^2$$

31.(D) $OA = OB = OC$

$$x_1^2 (1 + \tan^2 \alpha) = x_2^2 (1 + \tan^2 \beta) = x_3^2 (1 + \tan^2 \gamma) = R^2$$

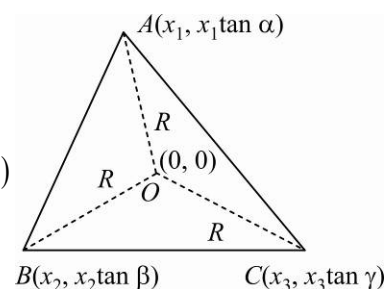
$$x_1 = R \cos \alpha, x_2 = R \cos \beta, x_3 = R \cos \gamma$$

Co-ordinates of Δ are : $(R \cos \alpha, R \sin \alpha), (R \cos \beta, R \sin \beta), (R \cos \gamma, R \sin \gamma)$

Centroid divides orthocentre and circumcentre in the ratio of 2 : 1 internally

$$\left(\frac{\sum R \cos \alpha}{3}, \frac{\sum R \sin \alpha}{3} \right) \equiv \left[\frac{2(0)+1(a)}{3}, \frac{2(0)+1(b)}{3} \right]$$

$$\frac{a}{b} = \frac{\cos \alpha + \cos \beta + \cos \gamma}{\sin \alpha + \sin \beta + \sin \gamma}$$



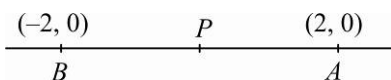
32.(B) Let $P(x, y)$ be any point and $A(2, 0)$ and $B(-2, 0)$ are fixed point

$\sqrt{(x-2)^2 + (y-0)^2}$ is distance of point $P(x, y)$ from A

Similarly, $\sqrt{(x+2)^2 + (y-0)^2}$ is distance of point $P(x, y)$ from B

Hence, $\sqrt{(x-2)^2 + (y-0)^2} + \sqrt{(x+2)^2 + (y-0)^2} = 4$ can be written as $AP + BP = AB$

P lies on the line segment AB



33.(B) Point P lies on the origin side of $x - y - 1 = 0$ and non-origin side of $x + y - 1 = 0$ or vice-versa.

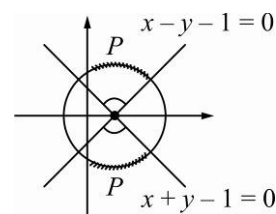
i.e., $\cos\theta - \sin\theta - 1 < 0$, $\cos\theta + \sin\theta - 1 > 0$

or $\cos\theta - \sin\theta - 1 > 0$, $\cos\theta + \sin\theta - 1 < 0$

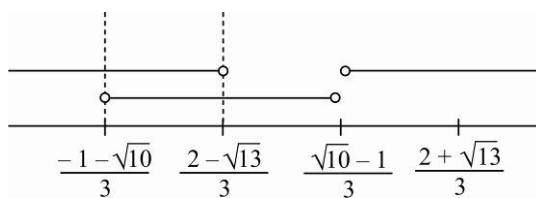
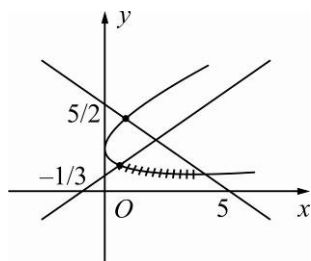
i.e., $(\cos\theta - \sin\theta - 1)(\cos\theta + \sin\theta - 1) < 0 \Rightarrow (\cos\theta - 1)^2 - \sin^2\theta < 0$

$\therefore \cos^2\theta + 1 - 2\cos\theta - 1 + \cos^2\theta < 0$

$\Rightarrow \cos\theta(\cos\theta - 1) < 0 \Rightarrow \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



34.(A)



$(\infty^2, \infty + 1)$ lies in the origin side of $3x - 4y + 1 = 0$ and $x + 2y - 5 = 0$

i.e., $3\infty^2 - 4(\infty + 1) + 1 > 0$ and $3\infty^2 + 2(\infty + 1) - 5 < 0$

$\Rightarrow 3\infty^2 - 4\infty - 3 > 0$ and $3\infty^2 + 2\infty - 3 < 0$

$\Rightarrow \left(\alpha - \frac{2+\sqrt{13}}{3}\right)\left(\alpha - \frac{2-\sqrt{13}}{3}\right) > 0$ and $\left(\alpha - \frac{-1+\sqrt{10}}{3}\right)\left(\alpha - \frac{-1-\sqrt{10}}{3}\right) < 0$

$\Rightarrow \alpha \in \left(\frac{-1-\sqrt{10}}{3}, \frac{2-\sqrt{13}}{3}\right) \sim (-1.3, -0.5) \Rightarrow [\alpha] = -1$

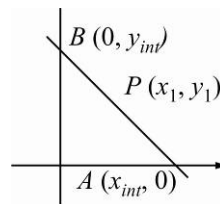
35.(B) Family of lines : $\left(\frac{x}{3} + \frac{y}{4} - 1\right) + k\left(\frac{x}{4} + \frac{y}{3} - 1\right) = 0$

$\left(\frac{1}{3} + \frac{k}{4}\right)x + \left(\frac{1}{4} + \frac{k}{3}\right)y = 1 + k$

$x\text{-int.} = \frac{1+k}{\frac{1}{3} + \frac{k}{4}}, y\text{-int.} = \frac{1+k}{\frac{1}{4} + \frac{k}{3}}$

Let mid-point be (x_1, y_1)

$\Rightarrow 2x_1 = \frac{12(1+k)}{3k+4}, 2y_1 = \frac{12(1+k)}{4k+3} \Rightarrow \frac{y_1}{x_1} = \frac{3k+4}{4k+3} \text{ or } k = \frac{4x_1-3y_1}{4y_1-3x_1}$



Now, $(3k + 4)x_1 = 6(1 + k)$ or $\left(\frac{12x_1 - 9y_1}{4y_1 - 3x_1} + 4\right)x_1 = 6\left(1 + \frac{4x_1 - 3y_1}{4y_1 - 3x_1}\right) \Rightarrow 7x_1y_1 = 6(y_1 + x_1)$

Replace (x_1, y_1) by (x, y)

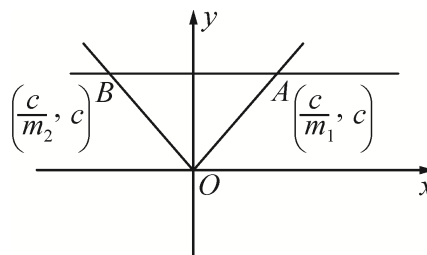
$$2xy = \frac{12}{7}(x + y) \Rightarrow k = \frac{12}{7}$$

Hence, $343k = 588$

36.(B) $m_1 + m_2 = -(2 + \sqrt{3}), m_1m_2 = \sqrt{3} - 1$

$$\begin{aligned} \text{Area} &= \frac{1}{2} \times c \times \left[\frac{c}{m_1} - \frac{c}{m_2} \right] \\ &= \frac{1}{2} c^2 \left| \frac{m_2 - m_1}{m_1 m_2} \right| = \frac{1}{2} c^2 \left| \frac{\sqrt{(m_1 + m_2)^2 - 4m_1 m_2}}{m_1 m_2} \right| \\ &= \frac{1}{2} c^2 \left| \frac{\sqrt{7 + 4\sqrt{3} - 4\sqrt{3} + 4}}{\sqrt{3} - 1} \right| = \frac{1}{2} c^2 \left| \frac{\sqrt{11} [\sqrt{3} + 1]}{2} \right| = \frac{1}{4} c^2 [\sqrt{33} + \sqrt{11}] \\ a &= \frac{\sqrt{33}}{4}, b = \frac{\sqrt{11}}{4} \end{aligned}$$

$$2008(a^2 + b^2) = \frac{2008}{16}(33 + 11) = 5522$$



37.(B) Point B is the orthocentre which has integral co-ordinates. Also, centroid is a rational point. As centroid divides orthocentre and circumcentre in the ratio of 2:1, hence circumcentre is also a rational point.

$$\begin{aligned} \text{Incentre} &\equiv \left[\frac{ax_1 + ax_3 + \sqrt{2}ax_2}{a + a + \sqrt{2}a}, \frac{ay_1 + ay_3 + \sqrt{2}ay_2}{a + a + \sqrt{2}a} \right] \\ &\equiv \left[\frac{x_1 + \sqrt{2}x_2 + x_3}{2 + \sqrt{2}}, \frac{y_1 + \sqrt{2}y_2 + y_3}{2 + \sqrt{2}} \right] \end{aligned}$$

Hence, incentre is not a rational point.

38.(C) Line passing through $A(2, 1)$ and $\parallel$ to L is

$$\begin{aligned} x - y = 1 &\Rightarrow m = 1 \Rightarrow \tan \theta = 1 \\ \Rightarrow \theta &= \pi/4 \end{aligned}$$

Point Q is obtained by

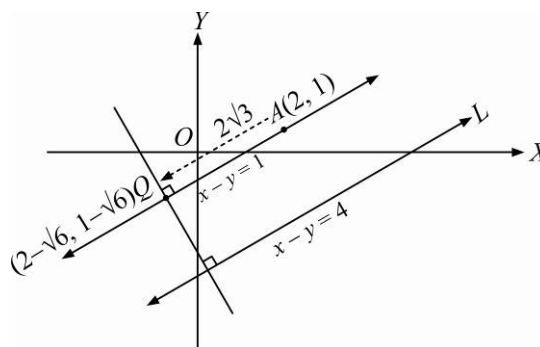
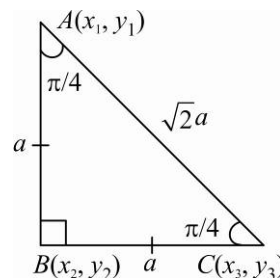
$$\frac{x-2}{\cos \frac{\pi}{4}} = \frac{y-1}{\sin \frac{\pi}{4}} = -2\sqrt{3}$$

$$x - 2 = \frac{-2\sqrt{3}}{\sqrt{2}} = -\sqrt{6} \Rightarrow x = 2 - \sqrt{6}$$

$$y - 1 = \frac{-2\sqrt{3}}{\sqrt{2}} = -\sqrt{6} \Rightarrow y = 1 - \sqrt{6}$$

$$\therefore Q(2 - \sqrt{6}, 1 - \sqrt{6})$$

$\therefore$ Equation of line passing through Q and perp. to L is $x + y = 3 - 2\sqrt{6}$.



39.(A) Take a point $A(-1, 3)$ on the reflected ray. So, its image A' is given by

$$\frac{x+1}{7} = \frac{y-3}{-1} = \frac{-2(-7-3+1)}{7^2+1} = \frac{9}{25}$$

$$\frac{x+1}{7} = \frac{9}{25} \Rightarrow x = \frac{38}{25}$$

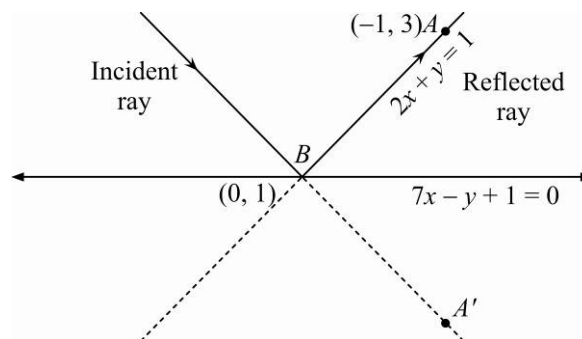
and $\frac{y-3}{-1} = \frac{9}{25} \Rightarrow y = \frac{66}{25}$

$$\therefore A' \left(\frac{38}{25}, \frac{66}{25} \right)$$

Image A' lies on incident ray (extended).

$\therefore$ Eq. of incident ray L (i.e. BA') is

$$y-1 = \frac{\frac{66}{25}-1}{\frac{38}{25}-0}(x-0) \Rightarrow 41x-38y+38=0$$



40.(C) Given $4x+3y-10=0 \Rightarrow 8x+6y-20=0$ and $8x+6y+5=0$ are parallel.

Putting $(0, 0)$ in them gives $-ve$ and $+ve$ signs. So, origin lies between them.

$OA/OB = OP/OQ$ (similarly)

$$\frac{OA}{OB} = \frac{\left| \frac{-20}{\sqrt{8^2+6^2}} \right|}{\left| \frac{5}{\sqrt{8^2+6^2}} \right|} = \frac{|-20|}{5} = \frac{4}{1}.$$

41.(D) Suppose $A\left(0, \frac{8}{3}\right)$, $B(1, 3)$ and $C(82, 30)$

$$\text{Slope of } AB = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3 - 8/3}{1 - 0} = \frac{1}{3}$$

$$\text{Slope of } BC = \frac{30 - 3}{82 - 1} = \frac{1}{3} \quad \therefore \text{Slope of } AB = \text{Slope of } AC$$

So, given points lie on a straight line.

42.(A) Let the line L be $\frac{x}{a} + \frac{y}{b} = 1$

Given $P(1, 2)$ is mid-point of AB .

$$\therefore \frac{a+0}{2} = 1, \frac{0+b}{2} = 2 \Rightarrow a = 2, b = 4$$

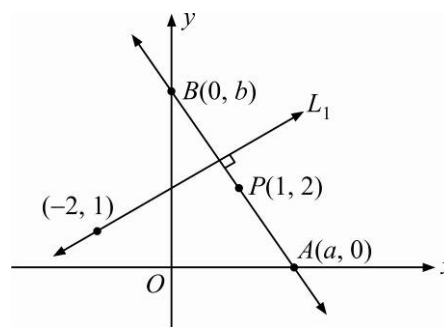
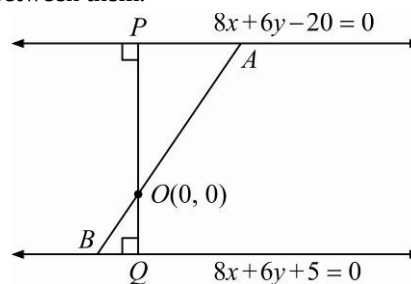
$$\therefore L \text{ is } \frac{x}{2} + \frac{y}{4} = 1 \Rightarrow 2x + y = 4 \quad \dots\dots (i)$$

$$\because L_1 \perp L \quad \therefore \text{Slope of } L_1 = 1/2$$

$$\text{Equation of } L_1 \text{ is } y-1 = \frac{1}{2}(x+2)$$

$$x-2y = -4 \quad \dots\dots (ii)$$

$$\text{Solving (i) and (ii), we get : } x = \frac{4}{5}, y = \frac{12}{5}. \text{ Required point } C\left(\frac{4}{5}, \frac{12}{5}\right)$$



- 43.(B) Clearly, there are two possible lines passing through $(3, -2)$ which can make 60° with the line $\sqrt{3}x + y = 1$.

Let slope of the required line be m .

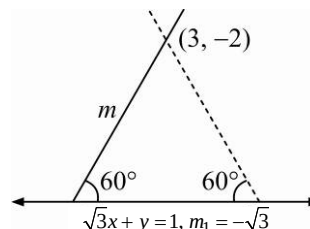
$$\therefore \tan 60^\circ = \left| \frac{m_2 - m_1}{1 + m_2 \cdot m_1} \right| \Rightarrow \sqrt{3} = \left| \frac{m + \sqrt{3}}{1 - \sqrt{3}m} \right| \Rightarrow \frac{m + \sqrt{3}}{1 - \sqrt{3}m} = \pm \sqrt{3}$$

Take +ve, $m + \sqrt{3} = \sqrt{3}(1 - \sqrt{3}m)$

$$m + \sqrt{3} = \sqrt{3} - 3m \Rightarrow m = 0$$

Take -ve, $m + \sqrt{3} = -\sqrt{3}(1 - \sqrt{3}m)$

$$m + \sqrt{3} = -\sqrt{3} + 3m \Rightarrow m = \sqrt{3}$$



So, equation of line with $m = 0$ and passing point $(3, -2)$ is $y + 2 = 0(x + 3) \Rightarrow y + 2 = 0$

This line is parallel to x -axis which does not intersect the x -axis.

Equation of line with $m = \sqrt{3}$ and passing point $(3, -2)$ is $y + 2 = \sqrt{3}(x - 3) \Rightarrow y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

This is the required line L which intersects the x -axis.

- 44.(B) In the equilateral triangle, incentre and circumcentre are the same.

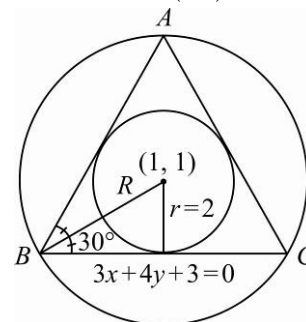
Incentre of an equilateral triangle is $(1, 1)$, so the circumcentre of circumcircle of the triangle is $(1, 1)$

$$r = \left| \frac{3 \cdot 1 + 4 \cdot 1 + 3}{\sqrt{3^2 + 4^2}} \right| = \left| \frac{3 + 4 + 3}{5} \right| = 2$$

$$\sin 30^\circ = \frac{r}{R} = \frac{2}{R} \Rightarrow R = 4$$

$\therefore$ Equation of circumcircle with centre $(1, 1)$ and $R = 4$ is

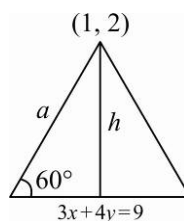
$$(x - 1)^2 + (y - 1)^2 = 4^2 \Rightarrow x^2 + y^2 - 2x - 2y - 14 = 0$$



- 45.(B) Let each side of an equilateral triangle be a .

$$h = \left| \frac{3 \cdot 1 + 4 \cdot 2 - 9}{\sqrt{3^2 + 4^2}} \right| = \frac{2}{5}$$

$$\sin 60^\circ = \frac{h}{a} \Rightarrow a = \frac{4}{5\sqrt{3}} = \frac{4\sqrt{3}}{15}$$



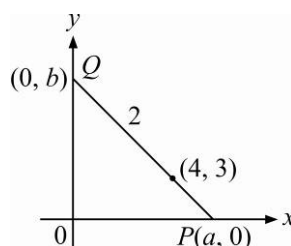
- 46.(C) Let $\frac{x}{a} + \frac{y}{b} = 1$

$$\text{Given } \frac{PA}{AQ} = \frac{1}{2}$$

$$\therefore 4 = \frac{2 \cdot a + 1 \cdot 0}{2 + 1} \Rightarrow a = 6$$

$$\text{and } 3 = \frac{2 \cdot 0 + 1 \cdot b}{2 + 1} \Rightarrow b = 9$$

$$\therefore \frac{x}{6} + \frac{y}{9} = 1 \Rightarrow 3x + 2y = 18$$



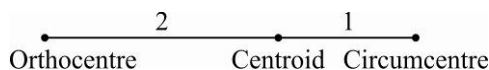
- 47.(D) In a triangle, orthocentre, centroid and circumcentre are collinear and centroid divides it in the ratio of 2 : 1.

Circumcentre (0, 0), centroid $\left(\frac{(a+1)^2}{2}, \frac{(a-1)^2}{2}\right)$ and let its orthocentre be (x, y)

$$\frac{(a+1)^2}{2} = \frac{1 \cdot x + 2 \cdot 0}{3} \Rightarrow x = \frac{3}{2}(a+1)^2$$

$$\frac{(a-1)^2}{2} = \frac{1 \cdot y + 2 \cdot 0}{3} \Rightarrow y = \frac{3}{2}(a-1)^2$$

Clearly, orthocentre $\left(\frac{3}{2}(a+1)^2, \frac{3}{2}(a-1)^2\right)$ satisfied the option (D).



- 48.(B) Let the line perpendicular to $5x - y = 1$ be $L : x + 5y = \lambda$ which intersect x-axis at A and y-axis at B.

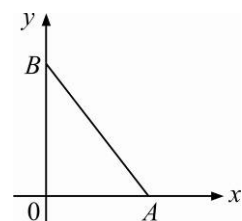
$$OA = \lambda, OB = \lambda/5$$

$$\text{Area } (AOB) = 5 \Rightarrow \frac{1}{2}(OA)(OB) = 5$$

$$\Rightarrow \lambda^2 = 50 \Rightarrow \pm 5\sqrt{2}$$

$$\therefore L \text{ is } x + 5y = \pm 5\sqrt{2} \quad \dots\dots (i)$$

$$\therefore \text{Distance of } x + 5y = 0 \text{ from (i)} = \frac{5\sqrt{2}}{\sqrt{1+25}} = \frac{5}{\sqrt{13}}$$

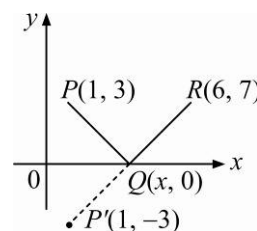


- 49.(D) Image of $P(1, 3)$ in x-axis is $P'(1, -3)$.

Here reflected ray also passes through P' .

$$\text{Equation of } RP' \text{ is } y + 3 = 2(x - 1)$$

As it crosses x-axis, put $y = 0$ gives $x = 5/2$



- 50.(A) $l_1 : x - 3y = p \Rightarrow m_1 = \frac{1}{3}; l_2 : ax + 2y = q \Rightarrow m_2 = -\frac{a}{2}$

$$l_3 : ax + y = r \Rightarrow m_3 = -a$$

To form a right-angled triangle $m_1 m_2 = -1, m_2 m_3 = -1$ or $m_1 m_3 = -1$

$$\Rightarrow \frac{1}{3} \cdot \left(-\frac{a}{2}\right) = -1, \left(-\frac{a}{2}\right)(-a) = -1 \text{ or } \frac{1}{3}(-a) = -1 \Rightarrow a = 6, a^2 = -2 \text{ (rejected) or } a = 3$$

$$\therefore (a-6)(a-3) = 0 \Rightarrow a^2 - 9a + 18 = 0$$

- 51.(D) $3x + 4y = 12 \Rightarrow \frac{x}{4} + \frac{y}{3} = 1$

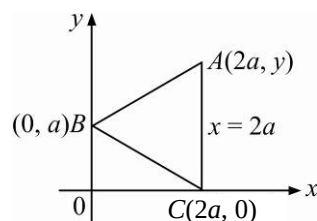
x-intercept = 4, y-intercept = 3

For new line x-intercept = 8, y-intercept = 3/2. Equation of L is $\frac{x}{8} + \frac{y}{3/2} = 1$, so slope = $-\frac{3}{16}$

- 52.(B) In $\triangle ABC$, $AB = AC$

$$\Rightarrow (2a)^2 + (y-a)^2 = y^2 \Rightarrow y = \frac{5a}{2}$$

$$\text{Area} = \frac{1}{2} \times \frac{5a}{2} \times 2a = \frac{5a^2}{2}$$



- 53.(A) θ_1 and θ_2 are not dependent on c_1, c_2 and c_3 .

- 54.(3) Perpendicular distance of (2, 3) from line $3x + 4y - 5 = 0$,

$$P_1 = \frac{|3(2) + 4(3) - 5|}{5} = \frac{13}{5}$$

$$\therefore 1 + \sin^2 \theta < P_1 \forall \theta \in R \Rightarrow n_1 = 0$$

Perpendicular distance of (1, 3) from line $3x + 4y - 5 = 0$,

$$P_2 = \frac{|3 + 12 - 5|}{5} = 2$$

$$\therefore \sec^2 \theta + 2 \operatorname{cosec}^2 \theta \geq 3 > P_2 \forall \theta \in R$$

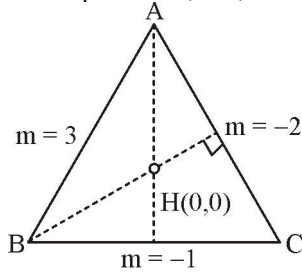
$$\Rightarrow n_2 = 2$$

$$\therefore \text{Equation becomes, } 2x^2 - 6x = 0$$

$$\Rightarrow x = 0, 3$$

Hence, sum of roots = 3

- 55.(7) Let slopes of BC, CA, AB be -1, -2, 3 respectively



$$\Rightarrow \text{Slope of } AH = 1 \Rightarrow A = (x_1, x_1)$$

$$\text{Slope of } BH = \frac{1}{2} \Rightarrow B = \left(x_2, \frac{1}{2}x_2\right)$$

$$\text{Slope of } CH = -\frac{1}{3} \Rightarrow C = \left(x_3, -\frac{x_3}{3}\right)$$

$$\therefore \text{Slope of } BC = \frac{\frac{x_2}{2} + \frac{x_3}{3}}{x_2 - x_3} = \frac{3x_2 + 2x_3}{6(x_2 - x_3)}$$

$$= -1 \Rightarrow 9x_2 = 4x_3$$

$$\therefore \text{Slope of } CA = \frac{x_1 + \frac{x_3}{3}}{x_1 - x_3} = -2$$

$$= \frac{3x_1 + x_3}{3(x_1 - x_3)} \Rightarrow 9x_1 = 5x_3$$

$$G \equiv (h, k) = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{x_1 + \frac{x_2}{2} - \frac{x_3}{3}}{3}\right)$$

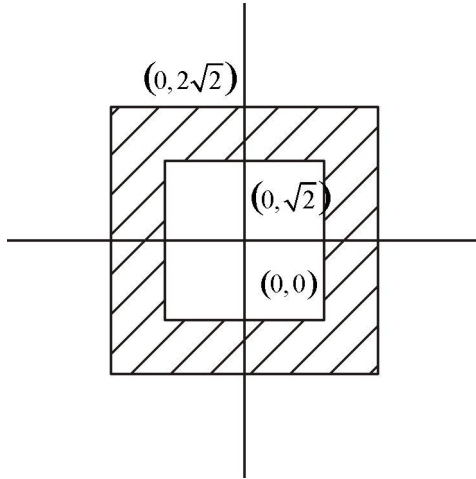
$$3h = x_3 + \frac{5x_3}{9} + \frac{4x_3}{9} = 2x_3 \quad \dots(i)$$

$$3k = \frac{5x_3}{9} + \frac{2x_3}{9} - \frac{x_3}{3} = \frac{4x_3}{9} \quad \dots(ii)$$

From equation (i) and (ii), we get

$$\frac{k}{h} = \frac{2}{9} \Rightarrow y = \frac{2}{9}x; \quad b - a = 9 - 2 = 7$$

56.(6) $\frac{|x+y|}{\sqrt{2}} + \frac{|y-x|}{\sqrt{2}} \in [2, 4]$



If $y \leq x$ and $y + x \geq 0$

$$\Rightarrow y \in [2\sqrt{2}, 2\sqrt{2}]$$

If $y \leq x$ and $y + x \leq 0$

$$\Rightarrow y \in [-2\sqrt{2}, -\sqrt{2}]$$

If $y \geq x$ and $y + x \leq 0$

$$\Rightarrow x \in [-2\sqrt{2}, -\sqrt{2}]$$

If $y \leq x$ and $y + x \geq 0$

$$\Rightarrow x \in [\sqrt{2}, 2\sqrt{2}]$$

Area of region

$$= (4\sqrt{2})^2 - (2\sqrt{2})^2 = 32 - 8 = 24$$

57.(3) Q is mid-point of P_1 & P_2

$$Q = (h, k)$$

$$P_1 = (h + r \cos \theta, k + r \sin \theta)$$

$$P_2 = (h - r \cos \theta, k - r \sin \theta)$$

$$k + r \sin \theta = 3(h + r \cos \theta) + 1 \text{ \& } k - r \sin \theta = 3 - 2(h - r \cos \theta)$$

$$r(\sin \theta - 3 \cos \theta) = 3h - k + 1 \text{ \& } (2 \cos \theta + \sin \theta)r = 2h + k - 3$$

$$r = \frac{3h - k + 1}{\sin \theta - 3 \cos \theta} \quad \dots(i)$$

$$r = \frac{2h + k - 3}{2 \cos \theta + \sin \theta} \quad \dots(ii)$$

From (i) & (ii)

$$(3h - k + 1)(2 \cos \theta + \sin \theta) = (2h + k - 3)(\sin \theta - 3 \cos \theta)$$

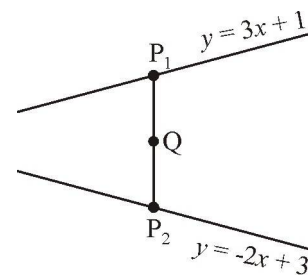
$$(6 \cos \theta + 3 \sin \theta)h - k(2 \cos \theta + \sin \theta) + 2 \cos \theta + \sin \theta = 2(\sin \theta - 3 \cos \theta)h + k(\sin \theta - 3 \cos \theta) - 3(\sin \theta - 3 \cos \theta)$$

Local of θ

$$\Rightarrow (\sin \theta - 12 \cos \theta)x - 4(2 \sin \theta - \cos \theta) + 4 \sin \theta + 7 \cos \theta = 0$$

$$\tan \theta = \frac{1}{2} = \frac{p}{q}$$

$$p + q = 3$$

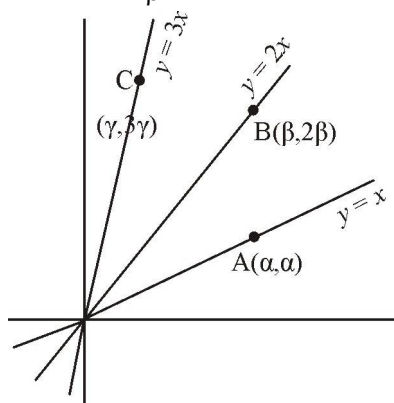


58.(0) Slope of $AB = \frac{\alpha}{\alpha-1} = \frac{2\beta}{\beta-1}$

$$\Rightarrow \alpha = \frac{2\beta}{\beta+1}$$

Slope of $BC = \frac{2\beta+1}{\beta} = \frac{3\gamma+1}{\gamma}$

$$\Rightarrow \gamma = \frac{\beta}{1-\beta}$$



Equation of AC is

$$y - \frac{2\beta}{\beta+1} = \left(\frac{\frac{3\beta}{1-\beta} - \frac{2\beta}{\beta+1}}{\frac{\beta}{1-\beta} - \frac{2\beta}{\beta+1}} \right) \left(x - \frac{2\beta}{\beta+1} \right)$$

$$\Rightarrow (x+y) - \beta(3y-5x+4) = 0$$

which passes through fixed point $\left(\frac{1}{2}, -\frac{1}{2} \right)$

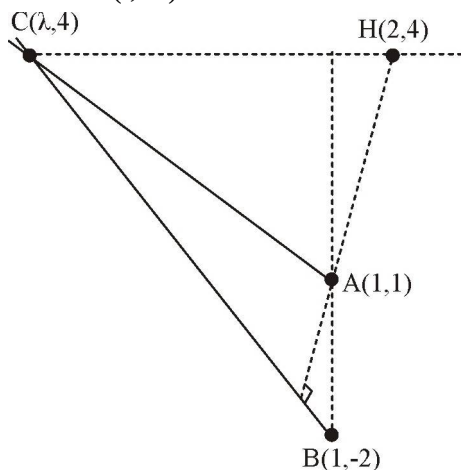
$$\therefore h+k=0$$

59.(14) $a+c-2b=0$

$$\Rightarrow ax+by+c=0 \text{ passes through fixed}$$

Point (1,-2)

$$\Rightarrow B \equiv (1, -2)$$



$$\text{Slope of } BC = -\frac{2-1}{4-1} = \frac{4+2}{\lambda-1}$$

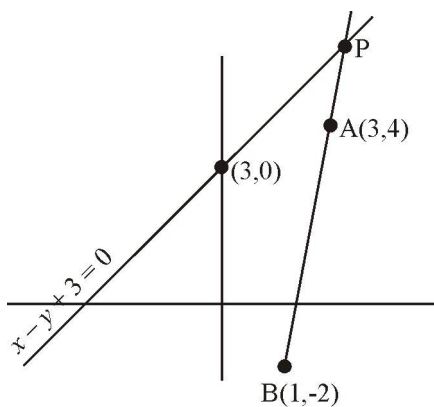
$$\lambda - 1 = -18 \Rightarrow \lambda = -17$$

$$C \equiv (-17, 4) \equiv (h, k)$$

$$\therefore 2h + 12k = -34 + 48 = 14$$

60.(5) $(3x + 7y + 11)\sec\theta + (5x - 3y - 11) = 0$

$\cos\theta = 0$ passes through intersection of $3x + 7y + 11 = 0$ and $5x - 3y - 11 = 0 \forall$ permissible values of θ given by $(1, -2)$

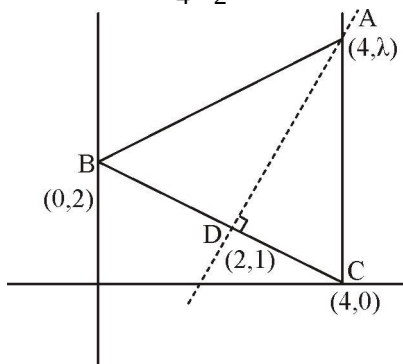


$$\Rightarrow B \equiv (1, -2)$$

It is clear that $|PA - PB| \leq AB$

$$\Rightarrow |PA - PB|_{\max} = AB = \sqrt{6^2 + 2^2} = 2\sqrt{10} \Rightarrow n = 5$$

61.(8) Slope of $AD = \frac{\lambda - 1}{4 - 2} = 2 \Rightarrow \lambda = 5$



$\therefore$ Equation of AB is

$$y - 2 = \frac{5-2}{4-0}(x-0)$$

$$4y - 8 = 3x$$

$$y = \frac{3}{4}x + 2$$

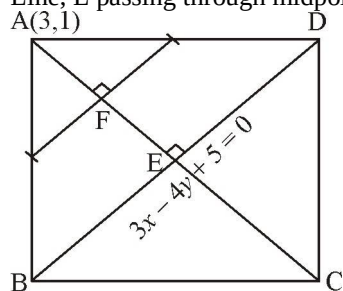
$$\therefore 8m + c = 8\left(\frac{3}{4} + 2\right) = 8$$

62.(1) Point E

$$\frac{x-3}{3} = \frac{y-1}{-4} = -\frac{9-4+5}{25}$$

$$= -\frac{2}{5} \Rightarrow E \equiv \left(\frac{9}{5}, \frac{13}{5} \right)$$

Line, L passing through midpoint of sides, AB and AD passes through F.



$$F \equiv \left(\frac{3 + \frac{9}{5}}{2}, \frac{\frac{13}{5} + 1}{2} \right) \equiv \left(\frac{12}{5}, \frac{9}{5} \right)$$

Equation of line L is

$$3x - 4y = \frac{36}{5} - \frac{36}{5} = 0$$

$$\Rightarrow 3x - 4y = 0 \Rightarrow |a + b + c| = |3 - 4| = 1$$

63.(4) Area = Mod of

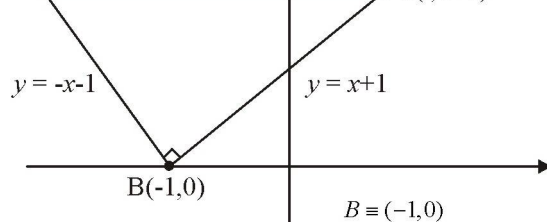
$$\frac{1}{2} \begin{vmatrix} \alpha & 2\alpha + 3 & 1 \\ 1 & 2 & 1 \\ 2 & 3 & 1 \end{vmatrix} = \frac{1}{2} |-\alpha + 2\alpha + 3 - 1|$$

$$\Rightarrow \alpha \in (-8, -6] \cup [2, 4)$$

$$\Rightarrow (\alpha, \beta) = (-7, -11), (-6, -9), (2, 7), (3, 9)$$

Number of possible co-ordinates of A = 4

64.(0) A(m, -m-1) C(l, l+1)



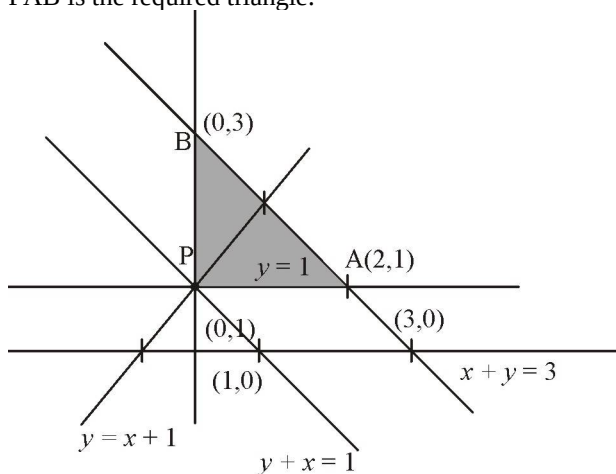
$$\frac{\alpha - 0}{0 - (-1)} = 2 \Rightarrow \alpha = 2$$

$$\text{Centroid} \equiv \left(\frac{-1}{3}, \frac{4}{3} \right)$$

$$\frac{l - m - 1}{3} = \frac{-1}{3} \text{ \& } \frac{-m - 1 + 0 + l + 1}{3} = \frac{4}{3}$$

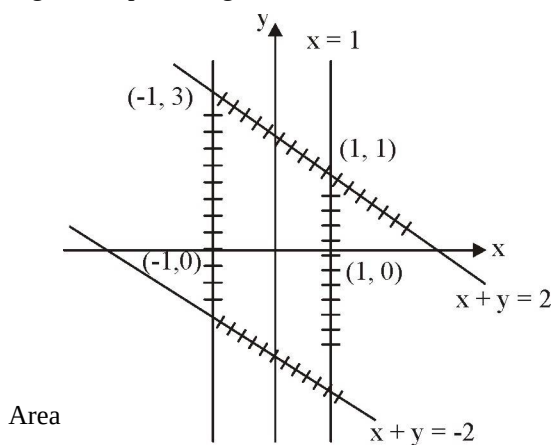
$$l + m = 0 \quad l - m = 4 \\ l = 2, m = -2$$

65.(2) PAB is the required triangle.

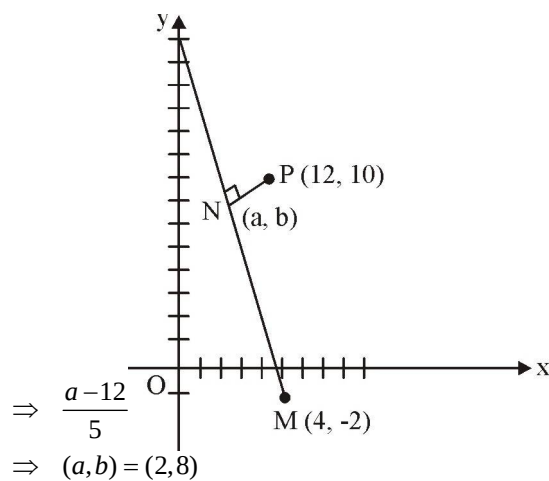


$$\therefore \text{Area of } \triangle PAB = \frac{1}{2} \times 2 \times 2 = 2$$

66.(8) Figure is a parallelogram



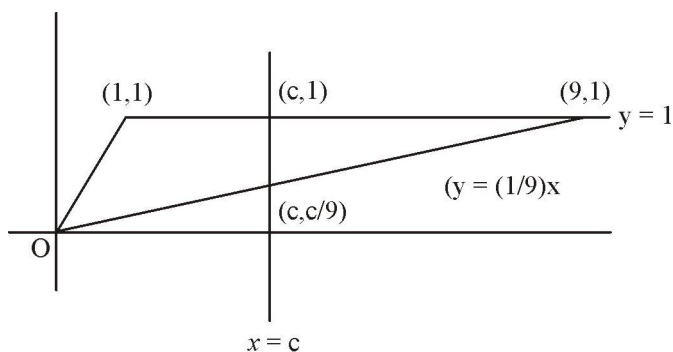
67.(10) (a, b) is the foot of $\perp$ ar of $(12, 10)$ on the line $y = 5x + 18$



$$\Rightarrow \frac{a-12}{5}$$

$$\Rightarrow (a, b) = (2, 8)$$

68.(3)



$$2 \times \frac{1}{2} \left(1 - \frac{c}{9}\right) (9 - c) = \frac{1}{2} \times 8 \times 1$$

$$\Rightarrow c = 3 \quad \text{or} \quad c = 15 \text{ which is not possible}$$

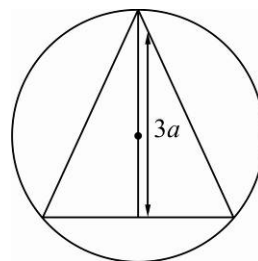
Circle

1.(C) Let equation of circle be $x^2 + y^2 = r^2$

Since, in an equilateral triangle, the centroid coincides with the centre of the circle.

$$\therefore \text{Radius of circle, } r = \frac{2}{3}(3a) = 2a$$

$$\text{On putting } r = 2a \text{ in equation (i), we get : } x^2 + y^2 = (2a)^2 \Rightarrow x^2 + y^2 = 4a^2$$



2.(B) Since, the given circles cut each other orthogonally, $\therefore g_1 g_2 + a^2 = 0 \quad \dots \text{(i)}$

If $lm + my = 1$ is a common tangent of these circles, then

$$\frac{-lg_1 - 1}{\sqrt{l^2 + m^2}} = \pm \sqrt{g_1^2 + a^2} \Rightarrow (lg_1 + 1)^2 = (l^2 + m^2)(g_1^2 + a^2)$$

$$\Rightarrow m^2 g_1^2 - 2lg_1 + a^2(l^2 + m^2) - 1 = 0 \quad \text{Similarly,} \quad m^2 g_2^2 - 2lg_2 + a^2(l^2 + m^2) - 1 = 0$$

So that g_1 and g_2 are the roots of the equation

$$m^2 g^2 - 2lg + a^2(l^2 + m^2) - 1 = 0 \Rightarrow g_1 g_2 = \frac{a^2(l^2 + m^2) - 1}{m^2} = -a^2 \quad [\text{From equation (i)}]$$

$$\Rightarrow a^2(l^2 + m^2) = 1 - a^2 m^2 \quad \dots \text{(ii)}$$

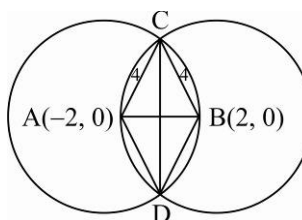
$$\text{Now, } p_1 p_2 = \frac{|ma - 1|}{\sqrt{l^2 + m^2}} \cdot \frac{|-ma - 1|}{\sqrt{l^2 + m^2}} = \frac{|1 - m^2 a^2|}{l^2 + m^2} = a^2 \quad [\text{From equation (ii)}]$$

3.(A) Circles with centres (2, 0) and (-2, 0) each with radius 4.

$\Rightarrow$ y-axis is their common chord.

$\triangle ABC$ is equilateral.

$$\text{Hence, area of ADBC is } \frac{2\sqrt{3}}{4}(4)^2 = 8\sqrt{3}$$

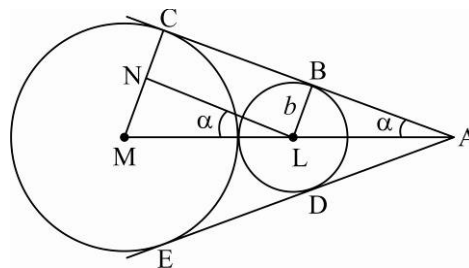


4.(D) From $\triangle MLN$,

$$\sin \alpha = \frac{a-b}{a+b}$$

$$\therefore \alpha = \sin^{-1} \left(\frac{a-b}{a+b} \right)$$

$$\therefore \text{Angle between AB and AD} = 2\alpha = 2 \sin^{-1} \left(\frac{a-b}{a+b} \right)$$



5.(B) Let OA and OB be the tangents from the origin to the given circle with centre $C(-3, 5)$ and radius $\sqrt{9+25-c} = \sqrt{34-c}$

Then, area of the quadrilateral, $OACB = 2 \times \text{Area of the } \triangle OAC = 2 \times \left(\frac{1}{2} \right) \times OA \times AC$

Now, OA = length of the tangent from the origin to the given circle = $\sqrt{c}$ and AC = Radius of the circle = $\sqrt{38-c}$

So, that $\sqrt{c}\sqrt{34-c} = 8$

$$[\text{Given}] \Rightarrow c(34-c) = 64 \Rightarrow c^2 - 34c + 64 = 0$$

6.(B) Let the centroid of the triangle be (h, k) . Then, $h = \frac{0+5\cos\theta+5}{3}$

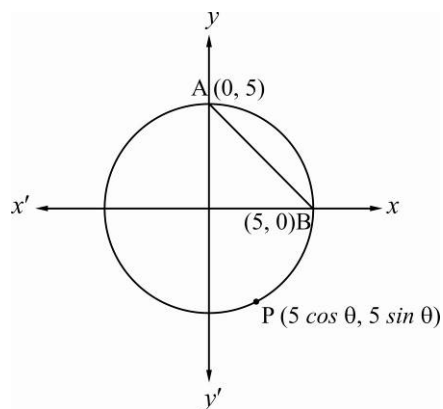
$$\text{and } k = \frac{5+5\sin\theta+0}{3} \Rightarrow 3h = 5(1+\cos\theta) \text{ and } 3k = 5(1+\sin\theta)$$

$$\Rightarrow \frac{(3h-5)}{5} = \cos\theta \text{ and } \frac{(3k-5)}{5} = \sin\theta$$

$$\Rightarrow \frac{9h^2+25-30h}{25} + \frac{9k^2+25-30k}{25} = 1$$

$$\Rightarrow 9h^2+9k^2+50-30h-30k-25=0$$

$$\Rightarrow 9(x^2+y^2)-30x-30y+25=0 \quad \text{Hence, it represents a circle}$$



7.(B) For the circle to have minimum radius the line joining $(1, 0)$ and $(0, 1)$ must be the diameter

Hence, the equation is $(x-1)(x-0) + (y-1)(y-0) = 0$

$$x^2 + y^2 - x - y = 0$$

8.(B) Since (a, b) and (b, a) are the mirror images of each other in the line $y = x$.

Hence $(a-b)^2 + (b-a)^2 = 4c^2$

$$(a-b)^2 = 2c^2 \Rightarrow a = b \pm \sqrt{2}c$$

9.(B) Since, $4\ell^2 - 5m^2 + 6\ell + 1 = 0$

$$\Rightarrow 4\ell^2 + 6\ell + 1 = 5m^2 \Rightarrow 9\ell^2 + 6\ell + 1 = 5(\ell^2 + m^2)$$

$$\Rightarrow \frac{(3\ell+1)^2}{\ell^2+m^2} = 5 \Rightarrow \left| \frac{3\ell+1}{\sqrt{\ell^2+m^2}} \right| = \sqrt{5}$$

Hence the line $\ell x + my + 1 = 0$ touches a circle with centre at $(3, 0)$ and radius as $\sqrt{5}$

10.(A) $(1 + \alpha x)^n = 1 + 8x + 24x^2 + \dots$

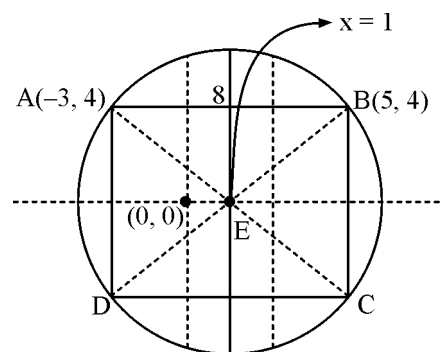
$$PA \cdot PB = |\alpha^2 = n^2 - 4|$$

$$\text{Since } nc_1 \cdot \alpha = 8 \Rightarrow n \alpha = 8$$

$$nc_2 \cdot \alpha^2 = 24 \Rightarrow n \frac{(n-1)}{2} \cdot \alpha^2 = 24$$

$$\text{Solving we get } n = 4; \alpha = 2 \Rightarrow PA \cdot PB = 4$$

- 11.(D) The centre lies at the point of intersection of the lines $x = 1$ and $4y = x + 7$. Which is $(1, 2)$
 $\Rightarrow EB = 5$ Hence length of diagonal = 10, $AD = 6$
 Hence the area of the rectangle = 48sq units



- 12.(A) Centre of the first circle $\equiv (-a, 0)$ Radius $= \sqrt{a^2 - c}$
 Centre of the second circle $\equiv (0, -b)$ Radius $= \sqrt{b^2 - c}$
 $\Rightarrow \sqrt{a^2 + b^2} = \sqrt{a^2 - c} + \sqrt{b^2 - c} \Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c}$

- 13.(B) $C_1 \equiv (0, 0); r_1 = 2; C_2 \equiv (2\lambda, 0); r_2 = \sqrt{4\lambda^2 - 9}$

Now since these circles have two common tangents $|r_1 - r_2| < C_1C_2 < r_1 + r_2$

- 14.(A) The circle $x^2 + y^2 = 1$ will intersect the line $y = mx + 2$ in distinct or coincident points if the length of perpendicular from $(0, 0)$ down upon the straight line is less than or equal to the radius

$$\Rightarrow \frac{2}{\sqrt{1+m^2}} \leq 1 \Rightarrow m^2 + 1 \geq 4$$

$$\Rightarrow m^2 \geq 3 \Rightarrow m \in (-\infty, -\sqrt{3}] \cup [\sqrt{3}, \infty)$$

- 15.(A) $S_1 \equiv x^2 + y^2 - \frac{48}{15}x + \frac{64}{15}y = 0$

Let (h, k) be a point on circle S_1 from which tangents are drawn to two circles

$$S_2 \left(x^2 + y^2 - \frac{24}{5}x + \frac{32}{5}y + 15 = 0 \right) \text{ and } S_3 \left(x^2 + y^2 - \frac{48}{5}x + \frac{64}{5}y + 60 = 0 \right)$$

$$\ell_1 = \sqrt{S_2}_{(h,k)} = \sqrt{h^2 + k^2 - \frac{24}{5}h + \frac{32}{5}k + 15} \text{ and } \ell_2 = \sqrt{S_3}_{(h,k)} = \sqrt{h^2 + k^2 - \frac{48}{5}h + \frac{64}{5}k + 60}$$

$$\text{We have, } h^2 + k^2 - \frac{48}{15}h + \frac{64}{15}k = 0$$

$$\ell_1 = \sqrt{\frac{48}{15}h - \frac{64}{15}k - \frac{24}{5}h + \frac{32}{5}k + 15} = \sqrt{-\frac{24}{15}h + \frac{32}{15}k + 15}$$

$$\ell_2 = \sqrt{\frac{48}{15}h - \frac{64}{15}k - \frac{48}{5}h + \frac{64}{5}k + 60} = \sqrt{4\left(-\frac{24}{15}h\right) + 4\left(\frac{32}{15}k\right) + 4(15)}$$

$$\frac{\ell_1}{\ell_2} = \frac{1}{2}$$

- 16.(A) Let equation of circle be $x^2 + y^2 + 2gx + 2fy + c = 0$

According to question:

$$g(-3) + f(-6) = C + 14$$

$$\dots\dots (i) \text{ and } g(-1) + f(-4) = C + 8$$

$$\dots\dots (ii)$$

$$g(2) + f(-6) = C + 9$$

$$\dots\dots (iii)$$

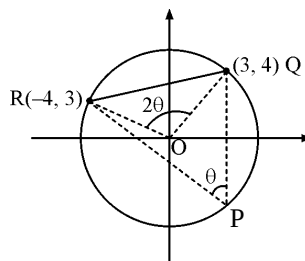
Solve to get $g = -1, f = -2, C = 1$

Hence, circle is : $x^2 + y^2 - 2x - 4y + 1 = 0$

17.(C) $\text{Slope}(OQ) = \frac{4}{3}, \text{slope}(OR) = \frac{3}{-4}$

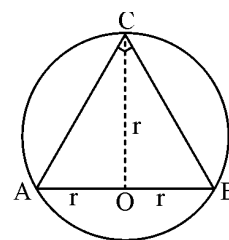
$\text{Slope}(OQ) \times \text{slope}(OR) = -1$

$\Rightarrow \angle QOR = \frac{\pi}{2} = 2\theta \Rightarrow \theta = \frac{\pi}{4}$



18.(A) $ar(\triangle ABC)$ is maximum when height OC is maximum i.e. r of $\triangle ABC$ is isosceles.

Perimeter is also maximum for this position of C .

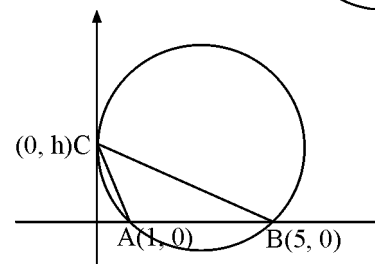


19.(A) Family of circles : $(x - 0)^2 + (y - h)^2 + kx = 0$
As it passes through $(1, 0)$, hence $1 + h^2 + k = 0$ or $k = -(1 + h^2)$

Required circle is : $x^2 + (y - h)^2 - (1 + h^2)x = 0$

It also passes through $(5, 0)$ hence

$25 + h^2 - 5(1 + h^2) = 0 \Rightarrow h = \pm\sqrt{5}$



20.(C) $(x-2)^2 + (y-1)^2 = r^2$ be the circle $x^2 + y^2 - 4x - 2y + 5 - r^2 = 0$ and $x^2 + y^2 - 2x - 6y + 6 = 0$

Common chord of circles is $2x - 4y + 1 + r^2 = 0$

It passes through $(1, 3)$ hence $2 - 12 + 1 + r^2 = 0 \Rightarrow r = 3$

21.(A) $\frac{PA}{PB} = k$ (constant)

A and B are two fixed points.

Let $A(a, b)$, $B(c, d)$ and $P(x_1, y_1)$

$(x_1 - a)^2 + (y_1 - b)^2 = k^2[(x_1 - c)^2 + (y_1 - d)^2]$

Replace $x_1 \rightarrow x, y_1 \rightarrow y$ to get the locus of P is : $[(x - a)^2 - (kx - kc)^2] + [(y - b)^2 - (ky - kd)^2] = 0$

Locus of P is a circle if $k \neq 1$

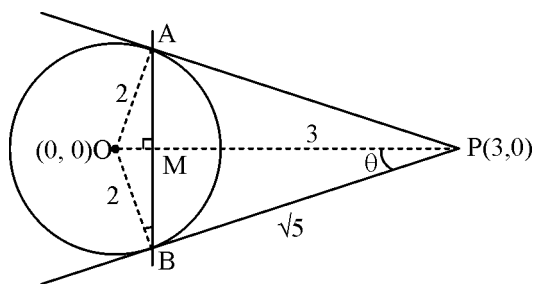
As for $k = 1$, coeff. of x^2 and coeff. of y^2 become zero.

22.(C) $\tan \theta = \frac{2}{\sqrt{5}}$

$ar(\triangle PAB) = \frac{1}{2}(AB)(PM)$

$= \frac{1}{2}(2\sqrt{5} \sin \theta)(\sqrt{5} \cos \theta)$

$= \frac{5}{2} \frac{\tan \theta}{1 + \tan^2 \theta} = \frac{5(2/\sqrt{5})}{1 + \frac{4}{5}} = \frac{10}{9}\sqrt{5}$ sq. units



23.(C) $\perp_r$ Distance of origin from AB is : $\frac{15}{5} = 3$

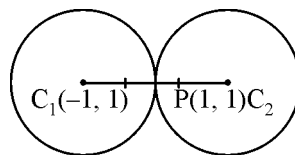
$AB = 2\sqrt{6^2 - 3^2} = 6\sqrt{3}$

24.(A) $S_1 \equiv x^2 + y^2 + 2x - 2y + 1 = 0$, $C_1 \equiv (-1, 1)$, $r_1 = 1$

$S_2 \equiv x^2 + y^2 - 2x - 2y + 1 = 0$, $(1, 1)$, $r_2 = 1$

P divides C_1C_2 in the ratio 1 : 1 externally.

Hence, $P \equiv \left[\frac{-1+1}{2}, \frac{1+1}{2} \right] \equiv (0, 1)$



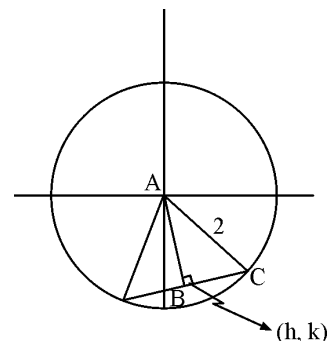
25.(B) Let the mid-point be (h, k) The line joining the centre to (h, k) will be perpendicular to the chord

$$AB = 2\sin 45^\circ$$

$$\Rightarrow \sqrt{h^2 + k^2} = \sqrt{2}$$

$$\Rightarrow h^2 + k^2 = 2$$

Hence the locus is $x^2 + y^2 = 2$



26.(B) Since the line $ax + by + c = 0$ is normal to the circle $x^2 + y^2 + 2gx + 2fy + d = 0$, so it will pass through its centre.

$$\Rightarrow -ag - bf + c = 0$$

$$\Rightarrow ag + bf - c = 0$$

27.(B) Circle touching the line $ax + by + c = 0$ at (x_1, y_1) is $(x - x_1)^2 + (y - y_1)^2 + \lambda(ax + by + c) = 0$.

$\therefore$ Circle touching y-axis (i.e. $x = 0$) at $(0, 2)$ is $x^2 + (y - 2)^2 + \lambda x = 0$.

It passes through $(-2, 4)$

$$\therefore 4 + 4 + \lambda(-2) = 0 \Rightarrow \lambda = 4 \quad \therefore (x + 2)^2 + (y - 2)^2 = 2^2 \Rightarrow \text{Centre } (-2, 2)$$

To be diameter, line must pass through $(-2, 2)$

Clearly, option (B) satisfies it.

28.(B) Eq. of tangent at (x_1, y_1) to circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y - y_1) + c = 0$

Point of intersection is $x = \frac{4}{3}$, $y = \frac{1}{3}$.

$$\therefore \text{Centre } \left(\frac{4}{3}, \frac{1}{3} \right) \text{ and } r = \sqrt{\left(\frac{4}{3} - 1 \right)^2 + \left(\frac{1}{3} - 1 \right)^2} = \frac{\sqrt{17}}{3}$$

$$\therefore \text{Circle is } \left(x - \frac{4}{3} \right)^2 + \left(y - \frac{1}{3} \right)^2 = \frac{17}{9}$$

$$x^2 + y^2 - \frac{8}{3}x - \frac{2}{3}y = 0$$

$$\text{Eq. of tangent at } (1, -1) \text{ is } x \cdot 1 + y(-1) - \frac{4}{3}(x+1) - \frac{1}{3}(y-1) = 0$$

$$x + 4y + 3 = 0$$

29.(B) Equation of circle with end points (x_1, y_1) and (x_2, y_2) of a diameter is $(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$

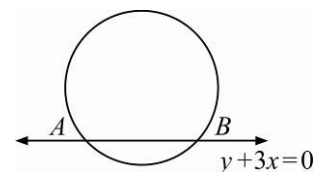
Points of intersections $x^2 + y^2 - 30x = 0$ and $y = -3x$

$$x^2 + 9x^2 - 30x = 0 \Rightarrow x = 0, x = 3$$

$A(O, 0)$ and $B(3, -9)$

$\therefore$ Equation of circle with diameter AB is $(x-0)(x-3) + (y-0)(y+9) = 0$

$$x^2 + y^2 - 3x + 9y = 0$$



30.(B) Given $y-6 = x^2$

$$\frac{dy}{dx} = 2x \Rightarrow \left(\frac{dy}{dx}\right)_{(2,10)} = 4$$

Equation of tangent at $(2, 10)$ is $y-10 = 4(x-2)$

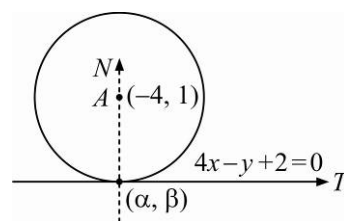
$$y-10 = 4(x-2) \Rightarrow 4x-y+2 = 0 \dots (i)$$

Let this tangent touches the circle at (α, β) .

$\therefore$ Its normal at (α, β) will pass through the centre $A(-4, 1)$

$\therefore$ Normal is $x+4y = 0 \dots (ii)$

Solving (i) and (ii), we get : $x = -\frac{8}{17}, y = \frac{2}{17}$. $(\alpha, \beta) = \left(-\frac{8}{17}, \frac{2}{17}\right)$



31.(C) The circle passes through the point $(-1, 0)$ and touches y -axis at $(0, 2)$.

So, the centre of the circle must lie in the second quadrant.

Let the centre of the circle be $(-a, 2)$, so its radius is a .

Equation of the circle is $(x+a)^2 + (y-2)^2 = a^2$

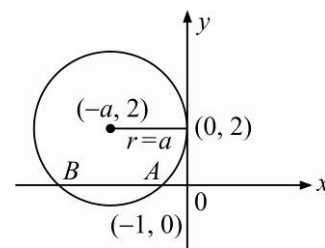
It passes through $A(-1, 0)$, so $(-1+a)^2 + (-2)^2 = a^2 \Rightarrow a = 5/2$

This is the equation of the circle. It intersects x -axis, so put $y = 0$

$$(x+5/2)^2 + 4 = (5/2)^2 \Rightarrow x^2 + 5x + 4 = 0$$

$$(x+4)(x+1) = 0 \Rightarrow x = -1, x = -4$$

$\therefore A(-1, 0)$ and $B(-4, 0)$. Thus, the length of the chord $AB = 3$.



32.(D) $x^2 + y^2 - 6x - 10y + p = 0 \dots (i)$

Centre $(3, 5)$ and $r = \sqrt{9+25-p}$

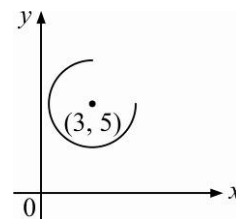
$$\therefore 1+16-6-40+p < 0 \Rightarrow p < 29$$

Circle does not touch the axes, so

Point $(1, 4)$ lies inside (i)

$$r < 3 \Rightarrow \sqrt{9+25-p} < 3 \Rightarrow 9+25-p < 9 \Rightarrow 25 < p$$

$$\therefore p \in (25, 29)$$



33.(B) Two circles intersect if

$$|r_1 - r_2| < C_1 C_2 < r_1 + r_2, C_1(2, 2),$$

$$r_1 = \sqrt{2}; C_2(5, 5), r_2 = \sqrt{50 - \lambda}, C_1 C_2 = 3\sqrt{2}.$$

$$|\sqrt{2} - \sqrt{50 - \lambda}| < 3\sqrt{2} < \sqrt{2} + \sqrt{50 + \lambda}$$

Now $3\sqrt{2} < \sqrt{2} + \sqrt{50 - \lambda}$

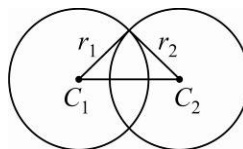
$$\Rightarrow 2\sqrt{2} < \sqrt{50 - \lambda} \Rightarrow 8 < 50 - \lambda \Rightarrow \lambda < 42 \quad \dots\dots (i)$$

$$|\sqrt{2} - \sqrt{50 - \lambda}| < 3\sqrt{2} \Rightarrow \sqrt{50 - \lambda} < 4\sqrt{2} \Rightarrow 50 - \lambda < 32 \Rightarrow 18 < \lambda \quad \dots\dots (ii)$$

Also, $\sqrt{2} - \sqrt{50 - \lambda} < 3\sqrt{2} \Rightarrow -2\sqrt{2} < \sqrt{50 - \lambda}$

For $\lambda < 50 \quad \dots\dots (iii)$

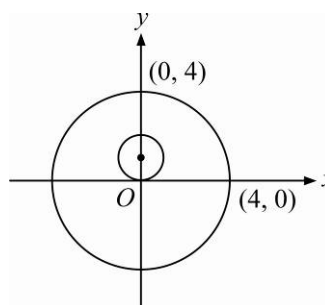
From (i), (ii) and (iii), $18 < \lambda < 42 \Rightarrow \lambda \in (18, 42)$



34.(B) $x^2 + y^2 = 4^2$

$$x^2 + (y - 1)^2 = 1^2$$

Clearly, no common tangent



35.(A) $x^2 + y^2 = 4^2 \Rightarrow$ centre $(0, 0), r = 4$

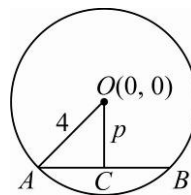
Chord AB is $3x + y + 5 = 0 \quad \dots\dots (i)$

$$\therefore p = \frac{5}{\sqrt{9+1}} = \frac{\sqrt{5}}{2}$$

$$\therefore AC^2 = 4^2 - \frac{5}{2} = \frac{27}{2}$$

Normal of AB through $(0, 0)$ is $x - 3y = 0 \quad \dots\dots (ii)$

Solving (i) and (ii), we get $C\left(-\frac{3}{2}, -\frac{1}{2}\right)$



$$\therefore \text{Circle with centre } C \text{ and radius } AC \text{ is } \left(x + \frac{3}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \frac{27}{2} \Rightarrow x^2 + y^2 + 3x + y - 11 = 0$$

36.(D) Diameter intersect at centre

Solve : $5x + 8y = 13, 4x - y = 3 \quad \therefore x = 1, y = 1$, so centre $(1, 1)$

But centre is $(a^2 - 7a + 11, a^2 - 6a + 6) \quad \therefore a^2 - 7a + 11 = 1; a^2 - 6a + 6 = 1$

$$\Rightarrow a^2 - 7a + 10 = 0; a^2 - 6a + 5 = 0 \Rightarrow (a - 5)(a - 2) = 0; (a - 5)(a - 1) = 0$$

$$\Rightarrow a = 5, 2; a = 5, a = 1 \quad \therefore a = 5$$

$$r = \sqrt{1^2 + 1^2 - (b^3 + 1)} = \sqrt{1 - b^3}$$

$$1 - b^3 > 0 \Rightarrow 1 > b^3 \Rightarrow b \in (-\infty, 1)$$

37.(A) Tangent at $(1, -1)$ is $x(1) + y(-1) + 2(x+1) - 3(y-1) - 12 = 0 \Rightarrow 3x - 4y - 7 = 0 \dots\dots (i)$

Note : Circle touching the line $ax + by + c = 0$ at (x_1, y_1) is $(x - x_1)^2 + (y - y_1)^2 + \lambda(ax + by + c) = 0$

$\therefore$ Equation of circle touching (i) at $(1, -1)$ is $(x - 1)^2 + (y + 1)^2 + \lambda(3x - 4y - 7) = 0 \dots\dots (ii)$

It passes through $(4, 0) \therefore 3^2 + 1^2 + \lambda(12 - 0 - 7) = 0 \Rightarrow \lambda = -2.$

Putting λ in (ii), we get $x^2 + y^2 - 8x + 10y + 16 = 0, r = 5.$

38.(A) $AC = AB = BC$

$a^2 + y^2 = (2a)^2$

$y = \sqrt{3}a$

$\therefore C(0, \sqrt{3}a), \text{Centroid} \left(\frac{-a + a + 0}{3}, \frac{0 + 0 + \sqrt{3}a}{3} \right)$

In equilateral Δ , circumcentre = centroid $\therefore$ Circumcentre $D\left(0, \frac{\sqrt{3}a}{3}\right)$

Circumradius = $AD = \sqrt{\frac{4a^2}{3}}$

Equation of circumcircle of ΔABC is $x^2 + \left(y - \frac{\sqrt{3}a}{3}\right)^2 = \frac{4a^2}{3} \Rightarrow 3x^2 + 3y^2 - 2\sqrt{3}ay = 3a^2.$

39.(B) Centre $(3, 4).$

Circle touches x-axis, so $r = 4$

But $r = \sqrt{3^2 + 4^2 - (25 - a^2)} \Rightarrow a = \pm 4$

40.(6) $\therefore CP = CR$

$\Rightarrow \frac{|p - q + 10|}{\sqrt{2}} = p$

or $p - q + 10 = p\sqrt{2} \dots\dots(i)$

and $CP = CQ$

$\frac{p - q + 10}{\sqrt{2}} = -\left(\frac{p - q - 6}{\sqrt{2}}\right)$ or $p - q = -2 \dots\dots(ii)$

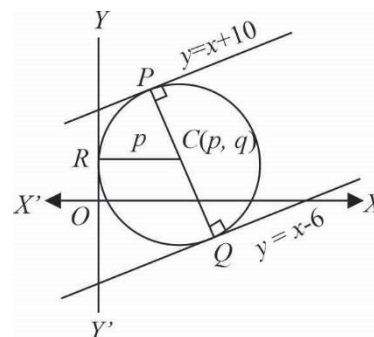
From Eqs. (i) and (ii), we get

$p = 4\sqrt{2}$ and $q = 4\sqrt{2} + 2$

Now, $p + q = 2 + 8\sqrt{2} = a + b\sqrt{2}$

$\therefore a = 2, b = 8$

Hence, $|a - b| = |2 - 8| = 6$



- 41.(1) Since, both the circles are passing through the origin $(0,0)$, the equation of tangent at $(0,0)$ of first circle will be same as that of the tangent at $(0,0)$ of second circle.

Equation of tangent at $(0,0)$ of first circle is

$$(3 + \sin \theta)x + (2 \cos \phi)y = 0 \quad \dots(i)$$

Equation of tangent at $(0,0)$ of second circle is

$$(2 \cos \phi)x + 2\lambda y = 0 \quad \dots(ii)$$

Therefore, Eqs. (i) and (ii) must be identical, then

$$\frac{3 + \sin \theta}{2 \cos \phi} = \frac{2 \cos \phi}{2\lambda}$$

$$\text{or } \lambda = \frac{2 \cos^2 \phi}{(3 + \sin \theta)}$$

$$\text{or } \lambda_{\max} = 1 \quad (\text{When } \sin \theta = -1 \text{ and } \cos \phi = 1)$$

- 42.(11) $\because C_1 : x^2 + y^2 - 2ax - 2ay + a^2 = 0$

Centre : (a, a) and radius : a

$$\text{and } C_2 : x^2 + y^2 - 2bx - 2by + b^2 = 0$$

Centre : (b, b) and radius : b

$\because C_1$ and C_2 touch each other, then

$$\sqrt{2}(b - a) = b + a \Rightarrow \frac{b}{a} = (\sqrt{2} + 1)^2 = 3 + \sqrt{8}$$

$$\Rightarrow \lambda = 3, \mu = 8$$

- 43.(2) The circles $S_1 : (x+1)^2 + \left(y + \frac{3}{2}\right)^2 = \left(\sqrt{\left(\frac{13}{4} - c\right)}\right)^2$

$$\text{Center } C_1 : \left(-1, -\frac{3}{2}\right), \text{ radius } r_1 : \sqrt{\left(\frac{13}{4} - c\right)}$$

$$\text{and } S_2 : \left(x - \frac{1}{2}\right)^2 + (y+1)^2 = \left(\sqrt{\left(\frac{5}{4} - c\right)}\right)^2$$

$$\text{centre } C_2 : \left(\frac{1}{2}, -1\right), \text{ radius } r_2 : \sqrt{\left(\frac{5}{4} - c\right)}$$

For intersect orthogonally

$$(C_1 C_2)^2 = r_1^2 + r_2^2$$

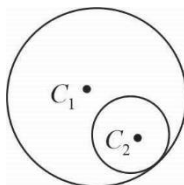
$$\Rightarrow \left(\frac{3}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{13}{4} - c + \frac{5}{4} - c$$

$$\text{or } 2c = 2$$

44.(15) Given pair of normal is $x^2 + 2xy + 3x + 6y = 0$

or $(x + 2y)(x + 3) = 0$

$\therefore$ Normals are $x + 2y = 0$ and $x + 3 = 0$ the point of intersection of normal $x + 2y = 0$ and $x + 3 = 0$ is the centre of required circle, we get centre $C_1 \equiv (-3, 3/2)$ and other circle is



$$x(x - 4) + y(y - 3) = 0$$

or $x^2 + y^2 - 4x - 3y = 0$... (i)

Its centre $C_2 \equiv (2, 3/2)$ and radius $r = \sqrt{4 + \frac{9}{4}} = \frac{5}{2}$

Since, the required circle just contains the given circle (i), the given circle should touch the required circle internally from inside.

$$\Rightarrow \lambda = \text{radius of the required circle} = |C_1 C_2| + r$$

$$= \sqrt{(-3 - 2)^2 + \left(\frac{3}{2} - \frac{3}{2}\right)^2} + \frac{5}{2}$$

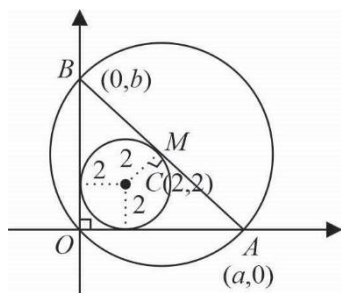
$$= 5 + \frac{5}{2} = \frac{15}{2}$$

$$\Rightarrow 2\lambda = 15$$

45.(1) The given circle is $x^2 + y^2 - 4x - 4y + 4 = 0$. This can be re-written as $(x - 2)^2 + (y - 2)^2 = 4$ which has centre $C(2, 2)$ and radius 2.

Let the equation of third side is

$$\frac{x}{a} + \frac{y}{b} = 1 \quad (\text{equation of } AB)$$



Length of perpendicular from $(2, 2)$ on $AB = \text{radius} = CM$

$$\therefore \frac{\left| \frac{2}{a} + \frac{2}{b} - 1 \right|}{\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2} \right)}} = 2$$

Since, origin and $(2, 2)$ lie on the same side of AB

$$\therefore -\frac{\left(\frac{2}{a} + \frac{2}{b} - 1\right)}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = 2$$

$$\text{or } \frac{2}{a} + \frac{2}{b} - 1 = -2\sqrt{\left(\frac{1}{a^2} + \frac{1}{b^2}\right)} \quad \dots\dots\dots(i)$$

Since, $\angle AOB = \frac{\pi}{2}$

Hence, AB is the diameter of the circle passing through ΔOAB , mid-point of AB is the centre of the circle i.e.

$$\left(\frac{a}{2}, \frac{b}{2}\right).$$

Let centre be $(h, k) \equiv \left(\frac{a}{2}, \frac{b}{2}\right)$ then $a = 2h$ and $b = 2k$.

Substituting the values of a and b in Eq. (i), then

$$\frac{2}{2h} + \frac{2}{2k} - 1 = -2\sqrt{\left(\frac{1}{4h^2} + \frac{1}{4k^2}\right)} \Rightarrow \frac{1}{h} + \frac{1}{k} - 1 = -\sqrt{\left(\frac{1}{h^2} + \frac{1}{k^2}\right)}$$

$$\text{or } h + k - hk + \sqrt{(h^2 + k^2)} = 0$$

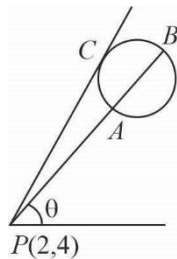
$\therefore$ Locus of $M(h, k)$ is

$$x + y - xy + \sqrt{(x^2 + y^2)} = 0$$

Hence, the required value of k is 1.

46.(16) Given, $(1 + \alpha x)^n = 1 + 8x + 24x^2 + \dots\dots\dots$

$$1 + n(\alpha \cdot x) + \frac{n(n-1)}{1 \cdot 2}(\alpha \cdot x)^2 + \dots\dots = 1 + 8x + 24x^2 + \dots\dots\dots$$



Equating the coefficients of x and x^2 , we get

$$n\alpha = 8, \frac{n\alpha(n\alpha - \alpha)}{1 \cdot 2} = 24$$

$$\text{or } \frac{8(8 - \alpha)}{2} = 24 \Rightarrow 8 - \alpha = 6$$

$$\therefore \alpha = 2 \text{ and } n = 4$$

Equation of line is $\frac{x-2}{\cos \theta} = \frac{y-4}{\sin \theta} = r$, then point

$(2+r \cos \theta, 4+r \sin \theta)$ lies on the circle $x^2 + y^2 = 4$,

Then, $(2+r \cos \theta)^2 + (4+r \sin \theta)^2 = 4$

or $r^2 + 4r(\cos \theta + 2 \sin \theta) + 16 = 0$

$$\therefore PA.PB = r_1 r_2 = \frac{16}{1} = 16$$

Aliter: $PA.PB = (PC)^2 = 2^2 + 4^2 - 4 = 16$

47.(35) Let $\phi(x, y) \equiv x^2 + y^2 + 2gx + 2fy + c = 0$

$$\therefore \phi(0, \lambda) = 0 + \lambda^2 + 0 + 2f\lambda + c = 0$$

Have equal roots,

$$\text{Then, } 2 + 2 = -\frac{2f}{1} \text{ and } 2.2 = \frac{c}{1}$$

$$\therefore f = -2 \text{ and } c = 4$$

and $\phi(\lambda, 0) \equiv \lambda^2 + 0 + 2g\lambda + 0 + c = 0$

$$\therefore \lambda^2 + 2g\lambda + 4 = 0$$

$$\therefore \frac{4}{5} + 5 = -2g$$

$$\Rightarrow g = -\frac{29}{10}$$

$$\therefore \text{Center} \equiv (-g, -f) = \left(\frac{29}{10}, 2\right) \equiv (\alpha, \beta)$$

$$\text{So } 10\alpha + 3\beta = 29 + 6 = 35$$

48.(4) The given is circles is

$$S : (x-3)^2 + (y-5)^2 = \left(\sqrt{(34-\lambda)}\right)^2$$

Since, point $P(1, 4)$ lies inside the circle

$$\therefore S_1 < 0$$

$$\Rightarrow 1 + 16 - 6 - 40 + \lambda < 0$$

$$\text{or } \lambda < 29 \quad \dots(i)$$

Also, circle neither touches nor cuts the axes, then

$$3 > \sqrt{(34-\lambda)} \text{ or } \lambda > 25 \quad \dots(ii)$$

$$\text{and } 5 > \sqrt{34-\lambda} \text{ or } \lambda > 9 \quad \dots(iii)$$

Form Eqs. (i), (ii) and (iii), we get

$$25 < \lambda < 29$$

Hence, difference = $29 - 25 = 4$

49.(6) $x^2 + y^2 - 2x - 2\lambda y - 8 = 0$

or $(x^2 + y^2 - 2x - 8) - 2\lambda y = 0$

Which is of the form $S + \lambda L = 0$

All the circles pass through the point of intersection of the circle $x^2 + y^2 - 2x - 8 = 0$ and $y = 0$

Solving, we get

$$x^2 - 2x - 8 = 0$$

or $(x-4)(x+2) = 0$

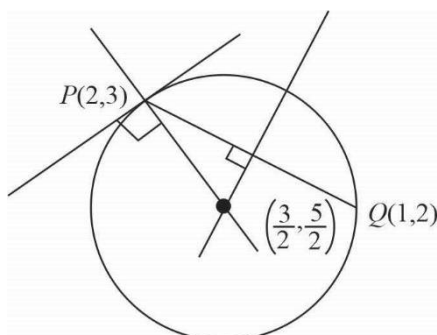
$\therefore A \equiv (-2, 0)$

and $B \equiv (4, 0)$

Hence, $|AB| = 6$

50.(4) Since, $x + y - 5 = 0$ is a tangent to the circle at $P(2, 3)$, so the centre of the circle lies on $x - y + 1 = 0$. Now $Q(1, 2)$ lies on the circle.

Equation of right bisector of PQ is $x + y - 4 = 0$



$\therefore$ Centre of circle is point of intersection of $x - y + 1 = 0$ and

$x + y - 4 = 0$ which is $\left(\frac{3}{2}, \frac{5}{2}\right)$

$$\therefore \text{Radius} = \frac{1}{\sqrt{\lambda}} = \frac{\left|\frac{3}{2} + \frac{5}{2} - 5\right|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

$\therefore \lambda = 2$

Hence, $\lambda^2 = 4$

51.(4) Let radius of given circle is a

$\therefore a = \sqrt{(4+2-c)} = \sqrt{(6-c)} \quad \dots(i)$

Now, radius of circle $C_1 = \frac{a}{\sqrt{2}}$ and radius of circle $C_2 = \frac{a}{(\sqrt{2})^2}$ and so on.

Also, $a + \frac{a}{\sqrt{2}} + \frac{a}{(\sqrt{2})^2} + \dots \infty = 2$ (given)

$$\text{Then, } \frac{a}{1 - \frac{1}{\sqrt{2}}} = 2$$

$$\text{or } a = 2 - \sqrt{2} = \sqrt{6-c} \quad [\text{From Eq. (i)}]$$

$$\text{or } 4 + 2 - 4\sqrt{2} = 6 - c$$

$$\therefore c = 4\sqrt{2}$$

$$\text{Hence, } \lambda = 4$$

52.(36) Let radii of circles be r_1 and r_2 and distance between centres is d , then

$$\sqrt{d^2 - (r_1 + r_2)^2} = 5 \quad \dots(i)$$

$$\text{and } \sqrt{d^2 - (r_1 - r_2)^2} = 13 \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$(r_1 + r_2)^2 - (r_1 - r_2)^2 = (13)^2 - (5)^2$$

$$\Rightarrow 4r_1r_2 = 144 \text{ or } \lambda = 36 \quad (\because r_1r_2 = \lambda)$$

53.(2) Since $(2, 3)$ lies on $ax + by - 5 = 0$

$$\therefore 2a + 3b - 5 = 0 \quad \dots(i)$$

Since, line is at greatest distance from centre

$$\Rightarrow \left(\frac{4-3}{3-2} \right) \left(-\frac{a}{b} \right) = -1 \text{ i.e. } a = b$$

$$\text{From Eq. (i), } a = b = 1 \Rightarrow a^{2021} + b^{2021} = 1$$

54.(1) Center of circles are $C_1 : (1, 2)$ and $C_2 : (5, -6)$

$$\text{Equation of } C_1C_2 \text{ is } y - 2 = \frac{-6-2}{5-1}(x-1)$$

$$\text{or } 2x + y - 4 = 0 \quad \dots(i)$$

$$\text{Equation of radical axis is } 8x - 16y - 56 = 0$$

$$\text{or } x - 2y - 7 = 0 \quad \dots(ii)$$

Point of intersection of Eqs. (i) and (ii) are $(3, -2)$

$$\therefore a = 3, b = -2$$

$$\text{or } a + b = 1$$

Complex Number

1.(B) $x^2 + x + 1 = (x - \omega)(x - \omega^2)$

$\Rightarrow$ If $P(x)$ is divisible by $x^2 + x + 1$, $P(\omega) = P(\omega^2) = 0 \Rightarrow P(\omega) = g(\omega^3) + \omega h(\omega^3) = g(1) + \omega h(1) = 0 \dots (i)$

and $P(\omega^2) = g(\omega^6) + \omega^2 h(\omega^6) = g(1) + \omega^2 h(1) = 0 \dots (ii)$

Solving (i) and (ii) we get : $g(1) = h(1) = 0 \Rightarrow g(1) = -h(1)$ and $g(1) + h(1) = 0 + 0 = 0$

2.(A) $\log_{1/2} |z - 2| > \log_{1/2} |z| \Rightarrow |z - 2| < |z|$

$\Rightarrow$ Distance of z from $(2, 0)$ is less than distance of z from $(0, 0) \Rightarrow \operatorname{Re}(z) > 1$

3.(A) $\frac{(\sqrt{5+12i} + \sqrt{5-12i})(\sqrt{5+12i} + \sqrt{5-12i})}{(\sqrt{5+12i} - \sqrt{5-12i})(\sqrt{5+12i} + \sqrt{5-12i})} = \frac{5+12i+5-12i+2\sqrt{5+12i}\sqrt{5-12i}}{5+12i-5+12i} = \frac{10+2 \times (+13)}{24i} = -\frac{3}{2}i$

4.(BC) Let $z_1 = a + bi$ and $z_2 = c + di$

$z_1 + z_2 = (a + c) + (b + d)i \Rightarrow \operatorname{Im}(z_1 + z_2) = 0 \Rightarrow (b + d) = 0$

$\operatorname{Im}(z_1 \cdot z_2) = 0 \Rightarrow ad + bc = 0$ or $ad - dc = 0$

or $d(a - c) = 0 \Rightarrow a = c$ or $d = 0$ In both cases we find that $z_1 = \bar{z}_2$

5.(B) $(\cos \theta + i \sin \theta)(\cos 3\theta + i \sin 3\theta)(\cos 5\theta + i \sin 5\theta) \dots (\cos(2n-1)\theta + i \sin(2n-1)\theta) = e^{i\theta} \cdot e^{i3\theta} \cdot e^{i5\theta} \dots e^{i(2n-1)\theta}$

6.(B) $|\lambda_1 a_1 + \lambda_2 a_2 + \dots + \lambda_n a_n| \leq |\lambda_1 a_1| + |\lambda_2 a_2| + \dots + |\lambda_n a_n|$

7.(A) For given series $T_r = \left(r + \frac{1}{\omega}\right) \left(r + \frac{1}{\omega^2}\right) = r^2 - r + 1$

$\Rightarrow \sum_{r=1}^n T_r = \sum_{r=1}^n (r^2 - r + 1) = \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + n = \frac{n(n^2 + 2)}{3}$

8.(C) $|a + b\omega + c\omega^2|^2 = (a + b\omega + c\omega^2) \cdot (\overline{a + b\omega + c\omega^2})$
 $= (a + b\omega + c\omega^2) \cdot (a + b\omega^2 + c\omega) = (a^2 + b^2 + c^2 - ab - bc - ca) = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2]$

Minimum value occurs when two integers are equal and third is 1 more than them.

$\Rightarrow |a + b\omega + c\omega^2|^2 = \frac{1}{2}[2] = 1$ or $|a + b\omega + c\omega^2| = 1$

9.(B) Let $z = x + iy$. Then $x^2 + y^2 = 1$

$\frac{z}{1 - z^2} = \frac{z}{z\bar{z} - z^2} = \frac{z}{z(\bar{z} - z)}$

We can see that $\operatorname{Re}\left(\frac{z}{1 - z^2}\right) = 0 \Rightarrow \frac{z}{1 - z^2}$ lies on y -axis.

10.(B) Let $a_k + ib_k = r_k (\cos \theta_k + i \sin \theta_k)$

$\Rightarrow r_1 \cdot r_2 \cdot r_3 \dots r_n e^{i(\theta_1 + \theta_2 + \theta_3 + \dots + \theta_n)} = A + iB \Rightarrow \tan(\theta_1 + \theta_2 + \theta_3 + \dots + \theta_n) = \frac{B}{A}$

$$\text{or} \quad \theta_1 + \theta_2 + \theta_3 + \dots + \theta_k = \tan^{-1} \left(\frac{B}{A} \right) \quad \text{or} \quad \sum_{k=1}^n \tan^{-1} \left(\frac{b_k}{a_k} \right) = \tan^{-1} \left(\frac{B}{A} \right)$$

11.(D) $z^{2m-1} = e^{i\theta(2m-1)}$

$$\sum_{m=1}^k \operatorname{Im}(z^{2m-1}) = \sin \theta + \sin 3\theta + \dots + \sin 29\theta = \frac{\sin(15\theta) \cdot \sin(15\theta)}{\sin(\theta)} = \frac{\sin^2(30^\circ)}{\sin(2^\circ)} = \frac{1}{4\sin(2^\circ)}$$

12.(B) Let us take $z_1 = \cos A + i \sin A$, $z_2 = \cos B + i \sin B$, $z_3 = \cos C + i \sin C$

$$\Rightarrow z_1 + z_2 + z_3 = 0 \quad \Rightarrow z_1^3 + z_2^3 + z_3^3 = 3z_1z_2z_3 = 3e^{iA} \cdot e^{iB} \cdot e^{iC} = 3e^{i(A+B+C)}$$

$$\Rightarrow \text{But } z_1^3 + z_2^3 + z_3^3 = e^{i3A} + e^{i3B} + e^{i3C} \Rightarrow 3e^{i(A+B+C)} = e^{i3A} + e^{i3B} + e^{i3C}$$

Equating real parts on both sides we get: $3\cos(A+B+C) = 3\cos(\pi) = -3 = \cos 3A + \cos 3B + \cos 3C$

13.(A) L.H.S. = $\frac{(\cos x + i \sin x)(\cos y + i \sin y)}{(\cos u + i \sin u)(\cos v + i \sin v)} \sin u \cos v = \sin u \cos v [\cos(x+y-u-v) + i \sin(x+y-u-v)]$

14.(D) $\log_{\tan 30^\circ} \left(\frac{2|z|^2 + 2|z| - 3}{|z| + 1} \right) < -2$

$$\text{or} \quad \frac{2|z|^2 + 2|z| - 3}{|z| + 1} > \left(\frac{1}{\sqrt{3}} \right)^{-2} = 3 \quad \text{or} \quad 2|z|^2 + 2|z| - 3 > 3|z| + 3$$

$$\text{or} \quad 2|z|^2 - |z| - 6 > 0 \quad \text{or} \quad (2|z| + 3)(|z| - 2) > 0$$

$$\Rightarrow |z| < \frac{-3}{2} \quad \text{or} \quad |z| > 2 \Rightarrow |z| > 2$$

15.(A) $z = \frac{\tan \alpha - i \left(\sin \frac{\alpha}{2} + \cos \frac{\alpha}{2} \right)}{1 + 2i \sin \frac{\alpha}{2}} = \frac{\left[\tan \alpha - i \left(\sin \frac{\alpha}{2} + \cos \frac{\alpha}{2} \right) \right] \left[1 - 2i \sin \frac{\alpha}{2} \right]}{1 + 4 \sin^2 \frac{\alpha}{2}}$ is purely imaginary

$$\Rightarrow \operatorname{Re}(z) = 0 \quad \text{or} \quad \tan \alpha - 2 \sin \frac{\alpha}{2} \left(\sin \frac{\alpha}{2} + \cos \frac{\alpha}{2} \right) = 0$$

$$\Rightarrow \tan \alpha - [1 - \cos \alpha + \sin \alpha] = 0 \quad \Rightarrow (\sin \alpha - \cos \alpha)(\sec \alpha - 1) = 0$$

$$\alpha = n\pi + \frac{\pi}{4}, \quad \text{or} \quad \alpha = 2n\pi$$

16.(C) $\sqrt{1-c^2} = nc - 1 \Rightarrow 1 - c^2 = n^2c^2 + 1 - 2nc \Rightarrow c = \frac{2n}{1+n^2}$

$$\begin{aligned} \text{Now, } \frac{c}{2n} \left(1 + \frac{n}{z} \right) &= \frac{1}{1+n^2} \left[(1 + ne^{i\theta})(1 + ne^{-i\theta}) \right] \\ &= \frac{1}{1+n^2} [1 + 2n \cos \theta + n^2] = 1 + \frac{2n}{1+n^2} \cos \theta = 1 + c \cos \theta \end{aligned}$$

17.(A) $z_1^3 - 3z_1z_2^2 = 2 \dots \dots (A)$ and $3z_1^2z_2 - z_2^3 = 11 \dots \dots (B)$

Operate $(A) + i(B)$ and $(A) - i(B)$

$$z_1^3 - 3z_1z_2^2 + i(3z_1^2z_2 - z_2^3) = 2 + 11i \Rightarrow (z_1 + iz_2)^3 = 2 + 11i$$

$$\text{and } z_1^3 - 3z_1z_2^2 - i(3z_1^2z_2 - z_2^3) = 2 - 11i \Rightarrow (z_1 - iz_2)^3 = 2 - 11i$$

$$\text{Now, } \{(z_1 + iz_2)(z_1 - iz_2)\}^3 = (2 + 11i)(2 - 11i) \Rightarrow (z_1^2 + z_2^2)^3 = 125 \Rightarrow z_1^2 + z_2^2 = 5$$

18.(C) $|z| - z = 1 + 2i$

Let $z = x + iy$, therefore $|x + iy| - (x + iy) = 1 + 2i$

Equating real and imaginary parts, we get : $\sqrt{x^2 + y^2} - x = 1$ and $y = -2 \Rightarrow x = \frac{3}{2}$

Hence complex number $z = \frac{3}{2} - 2i$.

Trick : Since $\left|\frac{3}{2} - 2i\right| - \left(\frac{3}{2} - 2i\right) = \sqrt{\frac{9}{4} + 4} - \frac{3}{2} + 2i = \frac{5}{2} - \frac{3}{2} + 2i = 1 + 2i$

19-21. 19.(B) 20.(B) 21.(C)

$$|z_1 + z_2|^2 = |z_1|^2 + |z_2|^2$$

$$\Rightarrow |z_1|^2 + |z_2|^2 + z_1\bar{z}_2 + \bar{z}_1z_2 = |z_1|^2 + |z_2|^2 \Rightarrow z_1\bar{z}_2 + \bar{z}_1z_2 = 0$$

$$\Rightarrow \operatorname{Re}(z_1\bar{z}_2) = 0 \quad (\because z_1\bar{z}_2 = \overline{\bar{z}_1z_2})$$

$\therefore z_1\bar{z}_2$ is purely imaginary

Hence $\arg\left(\frac{z_1}{z_2}\right) = \pm \frac{\pi}{2}$ Also, $i\left(\frac{z_1}{z_2}\right)$ is purely real.

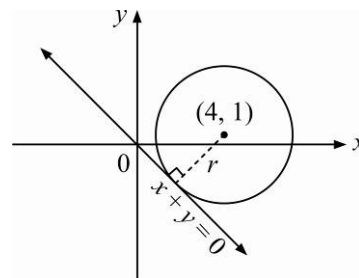
22.(D) Given $|\omega - (4 + i)| \leq r$ represents the interior region of the circle with centre $(4, 1)$ and radius r .

Also $|z - 1| \leq |z + 1| \Rightarrow |x + iy - 1| \leq |x + iy + 1|, z = x + iy$

$$(x - 1)^2 + y^2 \leq x^2 + (y + 1)^2$$

$$x + y \geq 0$$

$$r = \frac{|4 + i|}{\sqrt{2}} = \frac{5\sqrt{2}}{2} \quad \therefore \text{Largest value of } r \text{ is } \frac{5\sqrt{2}}{2}.$$



23.(D) $\frac{z_2}{z_1} = \lambda i, (\lambda \neq 0) \Rightarrow z_2 = z_1 \lambda i$

$$\left| \frac{2z_1 + 3z_2}{2z_1 - 3z_2} \right| = \left| \frac{2z_1 + 3z_1\lambda i}{2z_1 - 3z_1\lambda i} \right| = \frac{|z_1||2 + 3\lambda i|}{|z_1||2 - 3\lambda i|} = 1$$

24.(C) Let $z = x + iy$

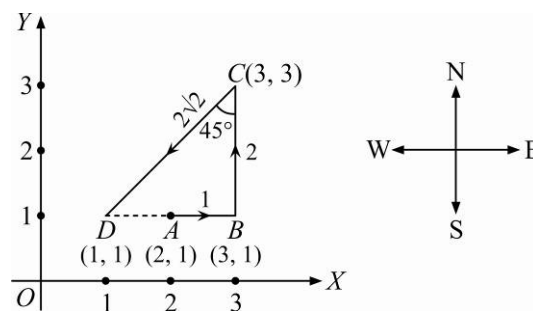
$$z + \sqrt{2}|z + 1| + i = 0 \Rightarrow x + i(y + 1) + \sqrt{2}[(x + 1) + iy] = 0.$$

Equating imaginary parts, $y + 1 = 0$ as $|z + 1|$ is real.

$$\therefore \sqrt{2}[(x + 1) + iy] = -x \Rightarrow 2[(x + 1)^2 + y^2] = x^2$$

$$\Rightarrow 2[(x + 1)^2 + (-1)^2] = x^2 \Rightarrow (x + 2)^2 = 0 \Rightarrow x = -2 \quad \therefore z = -2 - i \Rightarrow |z| = \sqrt{5}$$

- 25.(B) Now, point $2+i$ i.e. $A(2, 1)$ moves 1 unit eastwards and reaches $B(3, 1)$. Again, it moves 2 units northwards and reaches $C(3, 3)$. Finally, it moves $2\sqrt{2}$ units in south-westwards (i.e. at 45°) and reaches $D(1, 1)$. Hence, new position of $2+i$ is $1+i$.



- 26.(D) Now $z = 1+ai$ and z^3 is real

$$z^3 = 1+a^3i^3+3a^2i^2+3ai = (1-3a^2)+ia(3-a^2) \text{ is real}$$

$$\operatorname{Im}(z^3) = a(3-a^2) = 0$$

$$\Rightarrow a^2 = 3 \Rightarrow a = \sqrt{3} \quad \{\because a > 0\}$$

$$\therefore z = 1+\sqrt{3}i \Rightarrow z-1 = \sqrt{3}i \text{ and } z^3 = 1-3a^2 = 1-3 \times 3 = -8 \quad \{\because a = \sqrt{3}\}$$

$$1+z+z^2+\dots+z^{11} = \frac{1(z^{12}-1)}{z-1} = \frac{(z^3)^4-1}{\sqrt{3}i} = \frac{(-8)^4-1}{\sqrt{3}i} = \frac{4095}{\sqrt{3}i} \times \frac{\sqrt{3}i}{\sqrt{3}i} = -1365\sqrt{3}i.$$

- 27.(D) $w - \bar{w}z = k(1-z) \Rightarrow w-k = z(\bar{w}-k) \Rightarrow |w-k| = |z||\bar{w}-k|$

$$\Rightarrow |z| = 1 \text{ as } |w-k| \neq 0, \operatorname{Im}(w) \neq 0$$

$$\text{When } z = 1, \text{ then } w = \bar{w} \Rightarrow \operatorname{Im}(w) = 0, \text{ not possible} \quad \therefore |z| = 1 \text{ and } z \neq 1$$

- 28.(A) Let $z_1 = r_1(\cos \theta_1 + i \sin \theta_1) = r_1 e^{i\theta_1}$

$$\therefore z_2 = r_1(\cos \theta_1 - i \sin \theta_1) = r_1 e^{-i\theta_1} \text{ and } z_3 = r_2(\cos \theta_2 + i \sin \theta_2) = r_2 e^{i\theta_2}$$

$$\therefore z_4 = r_2(\cos \theta_2 - i \sin \theta_2) = r_2 e^{-i\theta_2}$$

$$\text{Now } \arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right) = \arg\left(\frac{r_1}{r_2} e^{i(\theta_1+\theta_2)}\right) + \arg\left(\frac{r_1}{r_2} e^{-i(\theta_1+\theta_2)}\right) = \theta_1 + \theta_2 - (\theta_1 + \theta_2) = 0$$

- 29.(B) $\frac{z-i}{z+i} = ki, k \neq 0 \quad (\because \text{purely imaginary})$

$$\Rightarrow z = \frac{k-i}{ki-1} \quad \therefore z + \frac{1}{z} = \frac{k-i}{ki-1} + \frac{ki-1}{k-i} = \frac{(k-i)^2 + (ki-1)^2}{(ki-1)(k-i)} = \frac{-4k}{k^2+1}$$

- 30.(B) $z = 1+\alpha i$ and $z^2 = x+iy$

$$\therefore 1-\alpha^2+2i\alpha = x+iy \Rightarrow x = 1-\alpha^2, y = 2\alpha \Rightarrow \alpha = y/2$$

$$\therefore x = 1 - (y^2/4) \Rightarrow y^2 + 4x - 4 = 0$$

- 31.(A) Given $|z| = 1$. Let $z = \cos \theta + i \sin \theta$.

$$\frac{1+z^2}{2iz} = \frac{1+\cos^2 \theta + i \sin 2\theta}{2i(\cos \theta + i \sin \theta)} = \frac{2\cos^2 \theta + 2i \sin \theta \cos \theta}{2i(\cos \theta + i \sin \theta)} = \frac{2\cos \theta (\cos \theta + i \sin \theta)}{2i(\cos \theta + i \sin \theta)} = \frac{\cos \theta}{i} = -i \cos \theta$$

$$\therefore a = \operatorname{Im}(-i \cos \theta). \text{ As } z \neq \pm 1, \text{ so } \theta \neq 0, \pi \Rightarrow a = -\cos \theta \Rightarrow a \in (-1, 1)$$

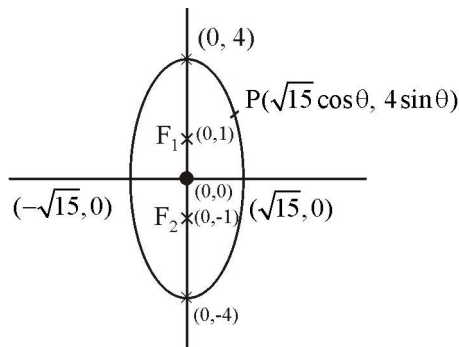
32.(7) z lies on ellipse $\frac{x^2}{15} + \frac{y^2}{16} = 1$

$$|z|^2 = 15\cos^2\theta + 16\sin^2\theta$$

$$= 15 + \sin^2\theta$$

$$m = \sqrt{15}, M = 4$$

$$\Rightarrow m + M = 4 + \sqrt{15}$$



33.(8) Let $\alpha = e^{i\theta}$, $z = ik$, $k, \theta \in R$

$$\Rightarrow -(\cos\theta + i\sin\theta)k^2 + ik + 1 = 0$$

$$\Rightarrow 1 - k^2\cos\theta = 0$$

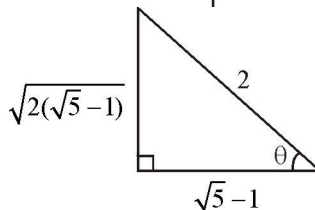
$$k(1 - k\sin\theta) = 0 \Rightarrow k\sin\theta = 1$$

$$\Rightarrow \frac{1}{k^2} = \cos\theta = \sin^2\theta$$

$$\Rightarrow \cos^2\theta + \cos\theta - 1 = 0$$

$$\Rightarrow \cos\theta = \frac{-1 + \sqrt{5}}{2}$$

$$\Rightarrow \tan^2\theta = \frac{2}{\sqrt{5}-1} = \frac{\sqrt{5}+1}{2}$$



34.(8) $\cos 17\theta + i\sin 17\theta = (\cos\theta + i\sin\theta)^{17}$

$$\sin 17\theta = {}^{17}C_1 \cos^{16}\theta \sin\theta - {}^{17}C_3 \cos^{14}\theta \sin^3\theta + {}^{17}C_5 \cos^{12}\theta \sin^5\theta$$

$$- {}^{17}C_7 \cos^{10}\theta \sin^7\theta + \dots - {}^{17}C_{15} \cos^2\theta \sin^{15}\theta + {}^{17}C_{17} \sin^{17}\theta$$

$$= \tan\theta \cos^{17}\theta ({}^{16}C_1 - {}^{16}C_3 \tan^2\theta + {}^{16}C_5 \tan^4\theta - {}^{16}C_7 \tan^6\theta + \dots - {}^{16}C_{15} \tan^{14}\theta + {}^{16}C_{17} \tan^{16}\theta)$$

If $\theta = \frac{r\pi}{17}$ $r = 1, 2, \dots, 8$

$$\Rightarrow 0 = {}^{17}C_1 - {}^{17}C_3 \tan^2\theta + {}^{17}C_5 \tan^4\theta - {}^{17}C_7 \tan^6\theta + \dots - {}^{17}C_{15} \tan^{14}\theta + {}^{17}C_{17} \tan^{16}\theta$$

Put $\tan^2\theta = x$

$${}^{17}C_1 - {}^{17}C_3 x + {}^{17}C_5 x^2 - {}^{17}C_7 x^3 + \dots - {}^{17}C_{15} x^7 + {}^{17}C_{17} x^8 = 0$$

$$\Rightarrow \sum_{r=1}^8 \tan^2 \frac{r\pi}{17} = \frac{{}^{17}C_{15}}{{}^{17}C_{17}} = \frac{17 \times 16}{2}$$

$$\Rightarrow \prod_{r=1}^8 \tan^2 \frac{r\pi}{17} = \frac{{}^{17}C_1}{{}^{17}C_{17}} = 17$$

$$\Rightarrow \frac{a}{b} = 8$$

$$\begin{matrix} \tan^2 \frac{\pi}{17} \\ \tan^2 \frac{2\pi}{17} \\ \vdots \\ \tan^2 \frac{8\pi}{17} \end{matrix}$$

35.(1) Let $\theta = y - 1 + yi$, $y \in R$

$$\Rightarrow |z| \leq |\omega|$$

$$\Rightarrow (x-1)^2 + x^2 \leq (y-1)^2 + y^2 \quad \forall y \in R$$

$$\Rightarrow (x-1)^2 + x^2 \leq \min \{(y-1)^2 + y^2, y \in R\}$$

$$\Rightarrow (x-1)^2 + x^2 \leq \frac{1}{2}$$

$$\Rightarrow (2x-1)^2 \leq 0$$

$$\Rightarrow x = \frac{1}{2}$$

$$\Rightarrow z = -\frac{1}{2} + \frac{i}{2}$$

36.(1) $\operatorname{Re}(z) > 1$

$\Rightarrow z$ lies in the right side of perpendicular bisector of points with complex representation 2 and 0.

$$\Rightarrow |z-2| < |z|$$

$$\Rightarrow \left| \frac{z-2}{2z} \right| < \frac{1}{2} \Rightarrow L = \frac{1}{2}$$

37.(8) $z^2 - \sqrt{3}z + 1 = 0 \Rightarrow z = \frac{\sqrt{3} \pm i}{2}$

$$\Rightarrow z = e^{i\pi/6} \text{ or } e^{-i\pi/6} \Rightarrow z^r + \frac{1}{z^r} = 2 \cos \frac{r\pi}{6}$$

$$\sum_{r=1}^5 \left(z^r + \frac{1}{z^r} \right)^2 = 4 \sum_{r=1}^5 \cos^2 \left(\frac{r\pi}{6} \right) = 2 \sum_{r=1}^5 \left(1 + \cos \frac{r\pi}{3} \right) = 8$$

38.(2) $\frac{z^4}{1+z^4} + \frac{z}{1+z} + \frac{z^2}{1+z^2} + \frac{z^4}{z+z^4} = \frac{z^5}{z+z^5} + \frac{z}{1+z} + \frac{z^2}{1+z^2} + \frac{z^5}{z^2+z^5} = \left(\frac{1}{1+z} + \frac{z}{1+z} \right) + \left(\frac{z^2}{1+z^2} + \frac{1}{z^2+1} \right) = 2$

39.(144) $z_{10} = e^{i\frac{2k_1\pi}{18}} e^{i\frac{2k_2\pi}{48}} = e^{i\frac{2\pi}{144}(8k_1+3k_2)} = e^{i\frac{2r\pi}{144}}$

r can take at most 144 distinct values and all values are attained using the periodicity of 2π

For example $3 \times 47 + 8 = 149$

$$\frac{149 \times 2\pi}{144} - 2\pi = \frac{2\pi \times 5}{144}$$

Which is not attained for $8k_1 + 3k_2$ between 0 to 143.

40.(259) $|f(z) - z| = |f(z)|$

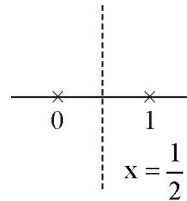
$$\Rightarrow |(a+bi)z - z| = |(a+bi)z| \quad \forall z \in \mathbb{C}$$

$$\Rightarrow |(a-1)+bi| = |a+bi|$$

$$\Rightarrow a = \frac{1}{2}$$

$$|a+ib| = 8$$

$$\Rightarrow \frac{1}{4} + b^2 = 64 \Rightarrow b^2 = \frac{255}{4}$$



41.(697) $1 + \frac{n}{z} = -\frac{i}{4} \Rightarrow z = \frac{-n}{1 + \frac{i}{4}} = \frac{-n \left(1 - \frac{i}{4} \right)}{\frac{17}{16}}$

$$z = \frac{16}{17} n \left(1 - \frac{i}{4} \right)$$

$$\therefore 164 = \frac{4n}{17} \Rightarrow n = 697$$

42.(53) $f(-\omega^2) = a\omega^{4036} - b\omega^{4034} + c\omega^{4032}$

$$\Rightarrow 2015 + 2019\sqrt{3}i = a\omega - b\omega^2 + c \Rightarrow 2015 + 2019\sqrt{3}i = \frac{b-a}{2} + c + i(a+b)\frac{\sqrt{3}}{2}$$

$$\Rightarrow a+b=4038 \Rightarrow a=b=2019 \text{ and } \frac{b-a}{2} + c = 2015 \Rightarrow c = 2015$$

$$\therefore f(1) = a+b+c = 4038 + 2015 = 6053$$

43.(19) $|z|^{2017} = |p||\bar{z}|$

$$\Rightarrow |z| = 0 \text{ or } |z| = |p|^{\frac{1}{2016}}$$

$$\therefore z^{2018} = p|z|^2$$

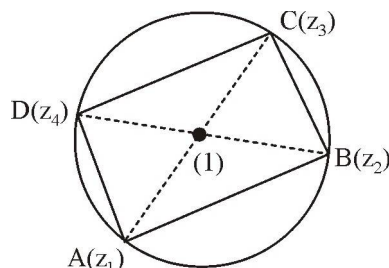
$$\Rightarrow z^{2018} = 0 \text{ or } |p|^{\frac{1}{1008}} p$$

$$\Rightarrow \text{No. of solutions} = 1 + 2018 = 2019.$$

44.(4) $\frac{z_1 + z_3}{2} = 1$

$$\frac{z_2 + z_4}{2} = 1$$

$$\Rightarrow z_1 + z_2 + z_3 + z_4 = 4$$



45.(245) $z = \left(\frac{1-i}{1+i}\right)^n = \left(\frac{-2i}{2}\right)^n = (-i)^n$

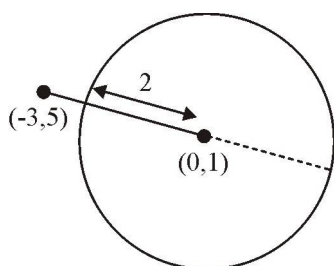
$$I_m(z) < 0 \text{ for } n = 4k+1 \quad k \in I$$

$$\text{Sum} = 1 + 5 + 9 + \dots + 97 = 1225$$

46.(10) $|z - iz_0| = |z + 3 - 5i|$

$$5 - 2 \leq |z + 3 - 5i| \leq 5 + 2$$

$$\Rightarrow 3 \leq |z + 3 - 5i| \leq 7$$



Permutations & Combinations

1.(A) Using de-arrangement theorem,

Number of ways to place 6 letters in 6 envelopes such that all are placed in wrong envelopes.

$$= \left[6 - \frac{1}{1} + \frac{1}{2} - \frac{1}{3} + \dots + \frac{1}{6} \right] = 360 - 120 + 30 - 6 + 1 = 265$$

2.(A) Required sum of number = [Sum of four digit numbers using 0, 1, 2, 3, 4, allowing 0 in first place] - [Sum of three digit numbers using 1, 2, 3, 4].

$$= \frac{5!}{5} [0 + 1 + 2 + 3 + 4] [1 + 10 + 10^2 + 10^3] - \frac{4!}{4} (1 + 2 + 3 + 4) (1 + 10 + 10^2) \\ = 24 \times 10 \times 1111 - 6 \times 10 \times 111 = 259980$$

3.(C) Refer to Solution Illustration-38

4.(B) Refer to Solution Illustration-41

5.(B) It is possible in two mutually exclusive cases

Case 1 : 2 children get none, one child gets three and all remaining 7 children get one each.

Case 2 : 2 children get none, 2 children get 2 each and all remaining 6 children get one each.

Using formula given in section (6.3), we get :

Case I : Number of ways = $\left(\frac{10!}{3!7!} \frac{1}{2!} \right) (10!)$

Case II : Number of ways = $\left(\frac{10!}{(2!)^2 \cdot 6!} \frac{1}{2!} \frac{1}{2!} \right) 10!$ Thus total ways = $(10!)^2 \left(\frac{1}{3!7!2!} + \frac{1}{(2!)^4 6!} \right)$

6.(B) Refer to Solution Illustrating the concepts (ii) : Page-46

7.(A) The 4 men can be seated at the circular table such that there is a vacant seat between every pair of men in $(4-1)! = 3!$ ways. Now, 4 vacant seats can be occupied by 4 women in $4!$ ways. Hence, the required number of seating arrangements = $3! \times 4! = 144$.

8.(B) Refer to Solution Illustration-48

9.(A) Refer to Solution Illustration-48

10.(B) A is not included, then others can be selected in $2^k - 1$ ways.
T is not included, then others can be selected in $2^k - 1$ ways.
M is not included, then others can be selected in $2^k - 1$ ways.
 $3(2^k - 1) = 93 \Rightarrow 2^k = 32 \Rightarrow k = 5$

11.(D) Refer to Solution OWS/Question-39

12.(B) Total no. of words = $\frac{9!}{2!}$

No. of words having "HIN" together = $7!$

No. of words having "DUS" together = $7!/2!$

No. of words having "TAN" together = $7!$

No. of words having "HIN" and "DUS" together = $5!$

No. of words having "HIN" and TAN" together = $5!$

No. of words having "DUS" and "TAN" together = $5!$

No. of words having HIN, DUS and TAN together = $3!$

No. of words required $\frac{9!}{2!} - (7! + \frac{7!}{2!} + 7!) + (5! + 5! + 5!) - 3! = 169194$

13.(B) Refer to Solution INE-E/Question-10

14.(C) Refer to Solution OWS/Question-13

15.(C) Refer to Solution OWS/Question-25

16.(A) For each of the first $(n-1)$ elements $a_1, a_2, \dots, a_{n-1}$ we have two choices: either a_i ($1 \leq i \leq n-1$) lies in the subset or a_i doesn't lie in the subset. For the last element we have just one choice. If even number of elements have already been selected, we do not include a_n in the subset, otherwise (when odd number of elements have been selected), we include it in the subset.

Thus, the number of subsets of $A = \{a_1, a_2, \dots, a_n\}$ which contain even number of elements is equal to 2^{n-1} .

17.(A) A number between 100 and 1000 has three digits.

So, we have to form all possible 3-digit numbers with distinct digits.

We cannot have 0 at the hundred's place. So, the hundred's place can be filled with any of the 9 digits 1, 2, 3, $\dots$, 9. So, there are 9 ways of filling the hundred's place.

Now, 9 digits are left including 0. So, ten's place can be filled with any of the remaining 9 digits in 9 ways. Now, the unit's place can be filled with in any of the remaining 8 digits. So, there are 8 ways of filling the unit's place.

Hence, the total number of required numbers = $9 \times 9 \times 8 = 648$.

18.(B) This can be done in four mutually exclusive ways as follows :

	Row R_1	Row R_2	Row R_3	Number of ways
I.	1	3	2	$({}^2C_1)({}^4C_3)({}^2C_2) = 8$
II.	1	4	1	$({}^2C_1)({}^4C_4)({}^2C_1) = 4$
III.	2	2	2	$({}^2C_2)({}^4C_2)({}^2C_2) = 6$
IV.	2	3	1	$({}^2C_2)({}^4C_3)({}^2C_1) = 8$
Total				26

Alternate Sol :

There are only two ways in which a row will not contain any X

$$\therefore \text{Total number of ways} = {}^8C_2 - 2 = 26$$

19.(A) In dictionary the words at each stage are arranged in alphabetical order. Starting with the letter A, and arranging the other four letters GAIN, we obtain $4! = 24$ words.

Thus, there are 24 words which start with A. These are the first 24 words.

Then, starting with G, and arranging the other four letters A, A, I, N in different ways, we obtain $\frac{4!}{2!} = \frac{24}{2} = 12$ words.

Thus, there are 12 words, which start with G.

Now, we start with I. The remaining 4 letters A, G, A, N can be arranged in $\frac{4!}{2!} = 12$ ways. So, there are 12 words, which start with I. Thus, we have so far constructed 48 words. The 49th word is NAAGI and hence the 50th word is NAAIG.

20.(C) ${}^{7-x}P_{x-3}$ is defined when $x-3 \leq 7-x \Rightarrow x \leq 5$ and $x-3 \geq 0 \Rightarrow x \geq 3$.

$\Rightarrow$ The possible values of ${}^{7-x}P_{x-3}$ are when $x = 3, 4$ or 5 .

$\Rightarrow$ The range of values of ${}^{7-x}P_{x-3}$ is $\{ {}^4P_0, {}^3P_1, {}^2P_2 \}$ or $\{1, 3, 2\}$.

21.(B) 18 correct results can be predicted in ${}^{21}C_{18}$ ways and 3 wrong results in 2^3 ways. Thus, required number of ways is ${}^{21}C_{18} 2^3$.

22.(C) A, I, I, I, O can occur at odd places in $\frac{5!}{3!}$ ways, and the remaining letters N, D, L, L can be arranged at the remaining places in $\frac{4!}{2!}$ ways.

23.(A) Number of ways $= {}^5C_1 + {}^5C_2 + {}^5C_3 = 25$

24.(D) As $0 + 1 + 2 + 3 + 4 + 5 = 15$, for a five digit number either do not use 0 or 3.

When 0 is not used, the number of such numbers is $5!$ And when 3 is not used, the number of such number is $5! - 4!$.

$\therefore$ Required number of ways $= 5! + (5! - 4!) = 216$

25.(A) We can choose 10 players from $(22 - 2)$ players in ${}^{20}C_{10}$ ways and one player from 2 players in 2C_1 ways.

$\therefore$ Required number of ways $= \frac{20!}{10!10!} \times 2 = 369512$

26.(B) Total number of books $= a + 2b + 3c + d$

Since there are b copies of each of two books, c copies of each of three books and single copies of d books.

Therefore the total number of arrangements is $\frac{(a + 2b + 3c + d)!}{a!(b!)^2(c!)^3}$.

27.(A) Let $n = 2m + 1$. Then, the common difference of the A.P. can be $1, 2, 3, \dots, m$. The number of AP's with $1, 2, 3, \dots, m$ common differences are $(2m - 1), (2m - 3), \dots, 1$ respectively

So, total number of AP's $= (2m - 1) + \dots + 1 = m^2 = \left(\frac{n-1}{2}\right)^2$. Hence, total number of ways $= \left(\frac{n-1}{2}\right)^2$

28.(B) Consider the host and the two particular persons as a single unit. So, there are effectively 19 persons. These 19 persons can be arranged in $18!$ ways and there are $2!$ ways of arranging the two persons on either side of the host.

29.(C) Digit at the extreme left can be chosen in 9 ways. (0 is not a possibility). Now the next digit can be chosen in 9 ways as consecutive digits are not same and so on.

$\Rightarrow$ Required number of numbers $= 9 \times 9 \times \dots \times 9 = 9^n$
 n times

30.(A) Required no. of ways $= {}^{22}C_{19} = 1540$

31.(B) $\frac{\frac{2n+1}{n+2} \times \frac{n-1}{2n-1}}{\frac{3}{5}} = \frac{(2n+1)(2n)}{(n+2)(n+1)(n)} = \frac{3}{5} \Rightarrow n = 4$

32.(B) The number of points of intersection

$$= \binom{n}{C_2}(2) + \binom{m}{C_2}(1) + \binom{n}{C_1}\binom{m}{C_1}(2) = n(n-1) + \frac{1}{2}m(m-1) + 2mn$$

33.(B) Number of ways is ${}^{12}C_6$ for M, 6C_4 for P and 2C_2 for C.

Thus, the required number of ways $= \binom{12}{C_6} \binom{6}{C_4} \binom{2}{C_2}$

34.(A) The total number of points is 15. From these 15 points we can obtain $^{15}C_3$ triangles. However, if all the 3 points are chosen on the same straight line, we do not get a triangle. Therefore, the required number of triangles $= ^{15}C_3 - 3(^5C_3) = 425$

35.(C) The required number of triangles is $^{15}C_3 - ^4C_3 = 455 - 4 = 451$

36.(D) M E D I T E R R A N E A N

R $\rightarrow$ 2, M $\rightarrow$ 1, D $\rightarrow$ 1, I $\rightarrow$ 1, T $\rightarrow$ 1, A $\rightarrow$ 2, N $\rightarrow$ 2, E $\rightarrow$ 3

R _ _ E

For letter words beginning with R (first letter) and ending with E (fourth letter) can be formed by two ways.

- (i) All two letters between R and E are distinct.
- (ii) All two letters between R and E are identical.
- (i) Excluding one R and one E, total distinct letters are 8 (R, M, D, I, T, A, N, E).
- (ii) Excluding one R and E, total identical pairs are 3 (AA, NN, EE).
- $\therefore$ Total such words = $56 + 3 = 59$.

37.(A) ${}^nC_r = \frac{n!}{r!(n-r)!}$ & ${}^nP_r = \frac{n!}{(n-r)!}$

$$\frac{{}^{n+2}C_6}{{}^{n-2}P_2} = 11$$

$$\Rightarrow \frac{(n+2)!}{6!(n+2-6)!} \bigg/ \frac{(n-2)!}{(n-2-2)!} = 11 \Rightarrow \frac{(n+2)!}{(n-2)!} = 11 \times 6! \Rightarrow (n+2)(n+1)n(n-1) = 11 \times 10 \times 9 \times 8$$

Equation on the both sides, $n = 9$

Clearly, $n = 9$ satisfies (A) $\therefore n^2 + 3n - 108 = 0 \Rightarrow (n+12)(n-9) = 0$

38.(B) Required numbers of ways

$$= {}^{15}C_1 \times {}^{15}C_1 + {}^{14}C_1 \times {}^{14}C_1 + {}^{13}C_1 \times {}^{13}C_1 + \dots + {}^1C_1 \times {}^1C_1 = 15^2 + 14^2 + 13^2 + \dots + 1^2$$

$$= \frac{15(15+1)(2 \times 15 + 1)}{6} = \frac{15 \times 16 \times 31}{6} = 1240$$

39.(D) Here $A = \{x_1, x_2, \dots, x_7\}$; $B = \{y_1, y_2, y_3\}$
 $f: A \rightarrow B$

There are three elements in A such that $f(x) = y_2$. This can be done by 7C_3 ways.

Out of other 4 elements of A , each one can be mapped with B except y_2 by 2 ways.

So, 4 elements of A can be mapped with 2 elements of B by 2^4 ways.

This contains 2 mapping such that all 4 element of A can be mapped with one element of B such that $f(x) = y_1$ or $f(x) = y_3$, i.e. 2 ways. So, for onto functions, it must be $2^4 - 2 = 14$.

$\therefore$ Total number of onto functions $f: A \rightarrow B$ is $14 \times {}^7C_3$.

40.(B) Sum of the digits in the unit's place of all the 4 digit numbers = $(4-1)!$ (sum of all four digits) = $3!(3+4+5+6) = 108$

41.(D) A number is divisible by 9 if its sum of digits is divisible by 9.

$$1+2+3+\dots+8 = \frac{8 \times 9}{2} = 36 \text{ is div. by } 9$$

Possible numbers with no repetitions are

- (i) 8 digits number excluding 0 & 9 = $8!$ (ii) 8 digits number excluding 1 & 8 = $7 \times 7!$
 (iii) 8 digits number excluding 2 & 7 = $7 \times 7!$ (iv) 8 digits number excluding 3 & 6 = $7 \times 7!$
 (v) 8 digits number excluding 4 & 5 = $7 \times 7!$ $\therefore$ Total number = $8! + 4 \times 7 \times 7! = 36(7!)$

42.(B) Number of games that n men played between themselves = $2 \cdot {}^nC_2$

Number of games that n men played with 2 women = $2 \cdot 2n$

$$\therefore 2 \cdot {}^nC_2 - 2 \cdot 2n = 66 \Rightarrow n(n-1) - 4n = 66$$

$$\Rightarrow (n-11)(n+6) = 0 \therefore n = 11$$

43.(C) $x_1 + x_2 + x_3 + \dots + x_8 = 30, \quad x_i \geq 2$

Let $x_i = a_i + 2$ then $a_1 + a_2 + a_3 + \dots + a_8 + 16 = 30$

$$a_1 + a_2 + a_3 + \dots + a_8 = 14, \quad a_i \geq 0$$

Total number of ways of dividing n identical items among r persons, each of whom, can receive 0, 1, 2, 3.... or more items ($\leq n$) = ${}^{n+r-1}C_{r-1}$

$$\therefore \text{Required number of ways} = {}^{14+8-1}C_{8-1} = {}^{21}C_7$$

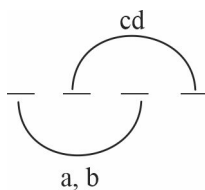
44.(365) $\frac{5}{5 \text{ options}} \frac{7}{X \text{ (not even)}} + \frac{5}{(not \text{ even})} \frac{7}{(not \text{ even})} = 5$
 (0, 2, 4, 6, 8)

Numbers not containing 5

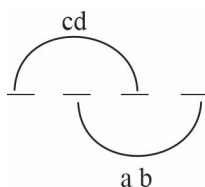
$$\frac{8}{8} \times \frac{9}{9} \times \frac{5}{5} = 360$$

(First digit should not be 0 or 5, 2nd digit should not be 5, 3rd digit should be even))

45.(8) a & b should both be at odd places or even places.



$$2 \times (2!) (2!) = 8$$



46.(162)

$\begin{array}{c} \text{---} \\ | \\ \text{---} \end{array}$ 1st, 2nd identical.
 identical 3rd should not be identical $9 \times 9 = 81$
 2nd, 3rd, identical
 $\begin{array}{c} \text{---} \\ | \\ \text{---} \end{array}$ 1st not identical & non-zero. $9 \times 9 = 81$
 identical Total = $81 + 81 = 162$

47.(41) Let two particular persons who either serve together or not at all are A and B . Also, let two particular persons who refuse to serve with each other are C and D .

Case 1: AB in $= {}^7C_3 - {}^5C_1 = 35 - 5 = 30$

AB in $- AB$ in and CD also in

Case 2: AB out $= {}^7C_5 - {}^5C_3 = 11$

AB out $- AB$ out and CD in

48.(126) There is only one arrangement of 5 different numbers in ascending or descending order. For a number to be a 5 digit zero should not be there while arranging in ascending order.

$$m = {}^{10}C_5, \quad n = {}^9C_5$$

49.(80) Each should belong to different pair and each shoe can have 2 options left or right shoe.

$${}^5C_4 \times 2^4 = 5 \times 16 = 80$$

50.(240)

$$\overbrace{1, 2, 3, 3}^{4 \text{ or } 5} \quad \overbrace{\text{Sum} = 9}$$

$$\frac{4!}{2!} \times 2 \times 10 \times 1 = 240$$

51.(52) (1) Exactly one 2's & one 3's
(3rd digit has 8 options) $-$ (3rd digit is a zero and no. begins with 0)

$${}^8C_1 \times 3! - 2!$$

$$= 8 \times 6 - 2 = 46$$

(2) Exactly two 2's & one 3's or one 2's or two 3's.

$$2 \times \left(\frac{3!}{2!} \right) = 6$$

$$46 + 6 = 52$$

52.(126) 5 match series $\text{I I I I I} = 1 = {}^4C_4$

(Last match won by India and 4 out of 1st 5 won by India)

[Here I at any position denotes that particular match is won by India]

6 match series $\text{-----I} \quad {}^5C_4$

7 match series $\text{-----I} \quad {}^6C_4$

8 match series $\text{-----I} \quad {}^7C_4$

9 match series $\text{-----I} \quad {}^8C_4$

$$\text{Total } {}^4C_4 + {}^5C_4 + {}^6C_4 + {}^7C_4 + {}^8C_4 = 126$$

53.(8) $\overline{3} \times \overline{2} \times \overline{2} \times \overline{2} \times \overline{2}$

$$3 \times 2^{n-1} = 384$$

$$2^{n-1} = 128 = 2^7$$

$$n-1 = 7 \Rightarrow n = 8$$

- 54.(10) Let $n(A), n(B), n(C)$ denote number of people possessing TV, VCR and tape recorder respectively & A, B, C be respectively set of teachers possessing them.

$$n(A) = 22, n(B) = 15, n(C) = 14$$

$$n(A \cap B \cap C) = 1$$

$$\text{Exactly two} = n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$$

$$9 = n(A \cap B) + n(B \cap C) + n(C \cap A) - 3 \times 1$$

$$\therefore n(A \cap B) + n(B \cap C) + n(C \cap A) = 12$$

Number of teachers possessing at least one

$$= n(A \cup B \cup C) = n(A) + n(B) + n(C) - (n(A \cap B) + n(B \cap C) + n(C \cap A)) + n(A \cap B \cap C)$$

$$= 22 + 15 + 14 - 12 + 1 = 40$$

$$\text{Number of teachers possessing none} = 50 - 40 = 10$$

- 55.(144) Total ways in AB are adjacent - Total ways in which AB are adjacent and CD are adjacent.

$$AB, C, D, E, F$$

$$AB, CD, E, F$$

$$5! \times 2! = 240$$

$$- 4! \times 2! \times 2! = 96$$

- 56.(30) last 2 digits should be 04, 12, 20, 24, 32, 40

$$\text{Ending with 04, 20, 40} = 3 \times (3!) = 18$$

$$\text{Ending with 12, 24, 32} = 3 \times (3! - 2!) = 6$$

Starting with O.

$$3 \times (6 - 2) = 3 \times 4 = 12$$

- 57.(50) Let A, B, C be the sets of students who read business India, business World and business Today.

$$n(A) = 80, n(B) = 50, n(C) = 30, n(A \cap B \cap C) = 5$$

$$n(A \cup B \cup C) = 100$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - [n(A \cap B) + n(B \cap C) + n(C \cap A)] + n(A \cap B \cap C)$$

$$100 = 80 + 50 + 30 - x + 5$$

$$x = 65$$

$$n(A \cap B) + n(B \cap C) + n(C \cap A) = \text{Number of students who read exactly two} + 3 \times n(A \cap B \cap C)$$

$$65 = y + 3 \times 5; y = 65 - 15 = 50$$

- 58.(420) $x_1 + x_2 + x_3 = 11$ or $x_1 + x_2 + x_3 = 7$

$$y_1 + y_2 = 7$$

$$y_1 + y_2 = 11$$

$${}^{10}C_2 \times {}^6C_1 + {}^6C_2 \times {}^{10}C_1$$

$$45 \times 6 + 15 \times 10$$

$$= 270 + 150 = 420$$

Binomial

1.(B) Expression = $(1+x^2)^{40} \cdot \left(x + \frac{1}{x}\right)^{-10} = (1+x^2)^{30} \cdot x^{10}$

$\Rightarrow$ The coefficient of x^{20} in $x^{10}(1+x^2)^{30}$ = the coefficient of x^{10} in $(1+x^2)^{30} = {}^{30}C_5 = {}^{30}C_{25}$

2.(D)

3.(C)

4.(C) $R = (\sqrt{2}+1)^{2n+1} = [R] + f$ (i)

$(\sqrt{2}-1)^{2n+1} = f' \in (0,1)$

$\therefore [R] + f - f' = (\sqrt{2}+1)^{2n+1} - (\sqrt{2}-1)^{2n+1} = \text{integer}$

$\Rightarrow f - f' = \text{integer}$

Since $f \in (0,1)$ and $f' \in (0,1) \Rightarrow -1 < f - f' < 1 \Rightarrow f - f' = 0 \Rightarrow f' = f$

$\therefore f = (\sqrt{2}-1)^{2n+1} = \frac{1}{(\sqrt{2}+1)^{2n+1}}$ From (i), $[R] = (\sqrt{2}+1)^{2n+1} - f = \frac{1}{f} - f$

5.(C) Given, $R = (7+4\sqrt{3})^{2n} = I + f$ (i) Let, $(7-4\sqrt{3})^{2n} = f'$

$\therefore (7+4\sqrt{3})^{2n} + (7-4\sqrt{3})^{2n} = \text{integer}$

$\Rightarrow I + f + f' = \text{integer} \Rightarrow f + f' = \text{integer} \Rightarrow f' = 1 - f = (7-4\sqrt{3})^{2n}$

$\therefore R(1-f) = (7+4\sqrt{3})^{2n} (7-4\sqrt{3})^{2n} = 1$

6.(B) Given expression, $(1+x)^n = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

$\Rightarrow {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$

Equating the coefficient of various powers of x , we get : $a_0 = {}^nC_0$, $a_1 = {}^nC_1$, $a_2 = {}^nC_2$, ..., $a_n = {}^nC_n$

$$\begin{aligned} \left(1 + \frac{a_1}{a_0}\right) \left(1 + \frac{a_2}{a_1}\right) \dots \left(1 + \frac{a_n}{a_{n-1}}\right) &= \frac{{}^nC_0 + {}^nC_1}{{}^nC_0} \cdot \frac{{}^nC_1 + {}^nC_2}{{}^nC_1} \dots \frac{{}^nC_{n-1} + {}^nC_n}{{}^nC_{n-1}} \\ &= \frac{{}^{n+1}C_1}{{}^nC_0} \cdot \frac{{}^{n+1}C_2}{{}^nC_1} \dots \frac{{}^{n+1}C_n}{{}^nC_{n-1}} \quad (\text{using } {}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r) \\ &= \left(\frac{1 \cdot {}^{n+1}C_1}{{}^nC_0} \cdot \frac{2 \cdot {}^{n+1}C_2}{{}^nC_1} \dots \frac{n \cdot {}^{n+1}C_n}{{}^nC_{n-1}} \right) \cdot \frac{1}{n!} \\ &= \frac{(n+1) \cdot {}^nC_0}{{}^nC_0} \cdot \frac{(n+1) \cdot {}^nC_1}{{}^nC_1} \dots \frac{(n+1) \cdot {}^nC_{n-1}}{{}^nC_{n-1}} \cdot \frac{1}{n!} = \frac{(n+1)^n}{n!} \end{aligned}$$

7.(D)

$$8.(A) \quad \sum_{r=0}^n \frac{(-1)^r}{{}^nC_r} = \frac{1}{{}^nC_0} - \frac{1}{{}^nC_1} + \frac{1}{{}^nC_2} - \frac{1}{{}^nC_3} + \dots - \frac{1}{{}^nC_n} = 0$$

$$9.(C) \quad \text{For } r \geq 0, \text{ we have } \frac{{}^nC_r}{r+2} = \frac{2}{(n+1)(n+2)} \times {}^{n+2}C_{r+2}$$

$$\begin{aligned} \therefore \sum_{r=0}^n (-2)^r \left(\frac{{}^nC_r}{r+2} \right) &= \frac{2}{(n+1)(n+2)} \times {}^{n+2}C_{r+2} (-2)^r = \frac{1}{2(n+1)(n+2)} \sum_{r=0}^n {}^{n+2}C_{r+2} (-2)^{r+2} \\ &= \frac{1}{2(n+1)(n+2)} \sum_{s=2}^{n+2} {}^{n+2}C_s (-2)^s \quad [\text{Putting } r+2 = s] \\ &= \frac{1}{2(n+1)(n+2)} \times \left[\sum_{s=0}^{n+2} {}^{n+2}C_s (-2)^s - {}^{n+2}C_0 (-2)^0 - {}^{n+2}C_1 (-2)^1 \right] \\ &= \frac{1}{2(n+1)(n+2)} \left[(1-2)^{n+2} - 1 + 2(n+2) \right] \\ &= \begin{cases} \frac{1}{2(n+1)(n+2)} (2n+4), & \text{if } n \text{ is even} \\ \frac{1}{2(n+1)(n+2)} (2n+3-1), & \text{if } n \text{ is odd} \end{cases} = \begin{cases} \frac{1}{n+1}, & \text{if } n \text{ is even} \\ \frac{1}{n+2}, & \text{if } n \text{ is odd} \end{cases} \end{aligned}$$

$$\begin{aligned} 10.(B) \quad \text{Given that } T_5 + T_6 = 0 &\Rightarrow {}^nC_4 a^{n-4} (-b)^4 + {}^nC_5 a^{n-5} (-b)^5 = 0 \\ &\Rightarrow {}^nC_4 a - {}^nC_5 b = 0 \Rightarrow \frac{a}{b} = \frac{{}^nC_5}{{}^nC_4} = \frac{n-4}{5} \end{aligned}$$

$$11.(C) \quad \sum_{r=0}^{100} {}^{100}C_r (x-3)^{100-r} 2^r = ((x-3)+2)^{100} = (x-1)^{100} = (1-x)^{100}$$

$$\therefore \text{Coefficient of } x^{53} = (-1)^{53} {}^{100}C_{53} = -{}^{100}C_{53}$$

$$\begin{aligned} 12.(C) \quad \text{Let, } X &= \frac{1}{57} {}^{57}C_4 \left[{}^{47}C_4 + \sum_{j=0}^3 {}^{50-j}C_3 + \sum_{j=0}^5 {}^{56-k}C_{53-k} \right] \\ &= \frac{1}{57} {}^{57}C_4 \left[{}^{47}C_4 + ({}^{47}C_3 + {}^{48}C_3 + {}^{49}C_3 + {}^{50}C_3) + ({}^{51}C_3 + \dots + {}^{56}C_3) \right] = \frac{{}^{57}C_4}{{}^{57}C_4} = 1 \end{aligned}$$

$$13.(D) \quad \frac{1}{81^n} (1 - 10 \cdot {}^{2n}C_1 + 10^2 \cdot {}^{2n}C_2 + \dots + 10^{2n}) = \frac{1}{81^n} (1-10)^{2n} = 1$$

$$14.(A) \quad \text{Let the term containing } x^7 \text{ in the expansion of } \left(ax^2 + \frac{1}{bx} \right)^8 \text{ is } T_{r+1}.$$

$$\therefore T_{r+1} = {}^8C_r (ax^2)^{8-r} \left(\frac{1}{bx} \right)^r = {}^8C_r \frac{a^{8-r}}{b^r} x^{16-3r}$$

$$\text{Since, this term contains } x^7 \Rightarrow 16-3r = 7 \Rightarrow r = 3$$

$$\therefore \text{Coefficient of } x^7 \text{ in the expansion of } \left(ax^2 + \frac{1}{bx}\right)^8 = {}^8C_3 \cdot \frac{a^5}{b^3} \quad \dots\dots (i)$$

Also, the term containing x^{-7} in the expansion of $\left(ax - \frac{1}{bx^2}\right)^8$ is T_{R+1}

$$T_{R+1} = {}^8C_R (ax)^{8-R} \left(-\frac{1}{bx^2}\right)^R = (-1)^R {}^8C_R \frac{a^{8-R}}{b^R} \cdot x^{8-3R}$$

$$\text{Since, this term contains } x^{-7} \Rightarrow 8 - 3R = -7 \Rightarrow R = 5$$

$$\therefore \text{Coefficient of } x^{-7} \text{ in the expansion of } \left(ax - \frac{1}{bx^2}\right)^8 = (-1)^5 {}^8C_5 \cdot \frac{a^3}{b^5} \quad \dots\dots (ii)$$

$$\text{From (i) and (ii) also from according to the question : } \left| {}^8C_3 \cdot \frac{a^5}{b^3} \right| = \left| {}^8C_5 \cdot \frac{a^3}{b^5} \right| \Rightarrow a^2 b^2 = 1 \Rightarrow ab = 1$$

$$\begin{aligned} 15.(A) \text{ We have, } (1+t^2)^{12} (1+t^{12}) (1+t^{24}) \\ = (1 + {}^{12}C_1 t^2 + {}^{12}C_2 t^4 + {}^{12}C_3 t^6 + \dots + {}^{12}C_6 t^{12} + \dots + {}^{12}C_{12} t^{24}) (1+t^{12} + t^{24} + t^{36}) \end{aligned}$$

$$\therefore \text{Coefficient of } t^{24} \text{ in } (1+t^2)^{12} (1+t^{12}) (1+t^{24}) = {}^{12}C_6 + 2$$

16.(C) Terms greater than 5!

$$i.e. (5!)^2, (6!)^2, \dots, (100!)^2 \text{ is divisible by } 100. \quad \therefore \text{For terms } (5!)^2, (6!)^2, \dots, (100!)^2$$

Remainder is 0.

$$\text{Now, consider } (1!)^2 + (2!)^2 + (3!)^2 + (4!)^2 = 1 + 4 + 36 + 576 = 617$$

When 617 is divided by 100, its remainder is 17 $\therefore$ Required remainder is 17.

$$\begin{aligned} 17.(D) \quad ({}^7C_0 + {}^7C_1) + ({}^7C_1 + {}^7C_2) + \dots + ({}^7C_6 + {}^7C_7) &= {}^8C_1 + {}^8C_2 + \dots + {}^8C_7 + ({}^8C_0 + {}^8C_8) - ({}^8C_0 + {}^8C_8) \\ &= (1+1)^8 - 2 = 2^8 - 2 \end{aligned}$$

$$18.(C) \quad (1+x)^{15} = a_0 + a_1 x + a_2 x^2 + \dots + a_{15} x^{15}$$

$$\Rightarrow {}^{15}C_0 + {}^{15}C_1 x + {}^{15}C_2 x^2 + \dots + {}^{15}C_{15} x^{15} = a_0 + a_1 x + a_2 x^2 + \dots + a_{15} x^{15}$$

Equating the coefficient of various powers of x , we get : $a_0 = {}^{15}C_0$, $a_1 = {}^{15}C_1$, $a_2 = {}^{15}C_2$, ..., $a_{15} = {}^{15}C_{15}$

$$\therefore \sum_{r=1}^{15} r \frac{a_r}{a_{r-1}} = \sum_{r=1}^{15} r \frac{{}^{15}C_r}{{}^{15}C_{r-1}} = \sum_{r=1}^{15} 15 - r + 1 = 120$$

$$\begin{aligned} 19.(B) \quad {}^{n-2}C_r + 2 \cdot {}^{n-2}C_{r-1} + {}^{n-2}C_{r-2} &= ({}^{n-2}C_r + {}^{n-2}C_{r-1}) + ({}^{n-2}C_{r-1} + {}^{n-2}C_{r-2}) \\ &= {}^{n-1}C_r + {}^{n-1}C_{r-1} \quad \left(\because {}^{n-2}C_{r-1} + {}^{n-2}C_r = {}^{n-1}C_{r-2} \right) \\ &= {}^nC_r \end{aligned}$$

$$20.(B) \quad {}^{15}C_8 + {}^{15}C_9 - {}^{15}C_6 - {}^{15}C_7 = {}^{15}C_8 + {}^{15}C_9 - {}^{15}C_9 - {}^{15}C_8 = 0$$

$$\begin{aligned}
 21.(B) \quad & {}^{18}C_{15} + 2\left({}^{18}C_{16}\right) + {}^{17}C_{16} + 1 = {}^nC_3 \\
 \Rightarrow & {}^{18}C_{15} + {}^{18}C_{16} + {}^{18}C_{16} + {}^{17}C_{16} + {}^{17}C_{17} = {}^nC_3 \quad \Rightarrow \quad {}^{19}C_{16} + {}^{18}C_{16} + {}^{18}C_{17} = {}^nC_3 \\
 \Rightarrow & {}^{19}C_{16} + {}^{19}C_{17} = {}^nC_3 \quad \Rightarrow \quad {}^{20}C_{17} = {}^nC_3 \quad \Rightarrow \quad {}^{20}C_3 = {}^nC_3 \quad \Rightarrow \quad n = 20
 \end{aligned}$$

22.(A) Given series is G.P. with $a = (1+x)^{2016}$, $r = \frac{x}{1+x}$ and $n = 2017$.

$$\begin{aligned}
 \therefore \quad & \frac{(1+x)^{2016} \left[1 - \left(\frac{x}{1+x} \right)^{2017} \right]}{1 - \frac{x}{1+x}} = \sum_{i=0}^{2016} a_i x^i \quad \Rightarrow \quad (1+x)^{2016} \left[1 - \left(\frac{x}{1+x} \right)^{2017} \right] (1+x) = \sum_{i=0}^{2016} a_i x^{i+1} \\
 & (1+x)^{2017} - x^{2017} = a_0 + a_1 x + a_2 x^2 + \dots + a_{2016} \cdot x^{2016}
 \end{aligned}$$

Equating, $a_{17} = \text{coeff. of } x^{17} = \frac{2017!}{17! 2000!}$.

$$23.(B) \quad \sum_{r=1}^{15} r^2 \left(\frac{{}^{15}C_r}{{}^{15}C_{r-1}} \right) = \sum_{r=1}^{15} r^2 \left(\frac{16-r}{r} \right) = \sum_{r=1}^{15} (16r - r^2) = 16 \sum_{r=1}^{15} r - \sum_{r=1}^{15} r^2 = 16 \frac{15 \cdot 16}{2} - \frac{15 \cdot 16 \cdot 31}{6} = 680$$

$$24.(B) \quad \frac{{}^nC_r}{{}^nC_{r+1}} = \frac{r+1}{n-r}$$

Let three successive terms in the expansion of $(1+x)^n$ be r th, $(r+1)$ th and $(r+2)$ th terms.

$$T_r = {}^nC_{r-1} x^{r-1}, T_{r+1} = {}^nC_r x^r, T_{r+2} = {}^nC_{r+1} x^{r+1}$$

Given ${}^nC_{r-1} : {}^nC_r : {}^nC_{r+1} = 1 : 7 : 42$

$$\frac{{}^nC_{r-1}}{{}^nC_r} = \frac{1}{7} \quad \Rightarrow \quad \frac{r}{n-(r-1)} = \frac{1}{7} \quad \Rightarrow \quad 8r = n+1 \quad \dots\dots (i)$$

$$\frac{{}^nC_r}{{}^nC_{r+1}} = \frac{7}{42} \quad \Rightarrow \quad \frac{r+1}{n-r} = \frac{1}{6} \quad 7r+6 = n \quad \dots\dots (ii)$$

Solve (i) and (ii), $r = 7$ First of these terms is 7^{th}

25.(C) Number of terms in $(1+x)^n$ is $n+1$.

$$(1+x)^{101} (1+x^2-x)^{100} = (1+x)(1+x)^{100} (1-x+x^2)^{100} = (1+x)(1+x^3)^{100}$$

After expansion, all the terms are distinct.

$\therefore$ Number of terms in the expansion $= 2(100+1) = 202$

26.(B) Let coeff. of r th term = coeff. of $(r+1)$ th term

$${}^{55}C_{r-1} 2^{55-(r-1)} \cdot \frac{1}{3^{r-1}} = {}^{55}C_r 2^{55-r} \cdot \frac{1}{3^r} \quad \Rightarrow \quad 6 = \frac{{}^{55}C_r}{{}^{55}C_{r-1}} = \frac{55-(r-1)}{r} \quad \Rightarrow \quad r = 8$$

$\therefore$ 8^{th} & 9^{th} terms

$$27.(D) \quad T_{r+1} = {}^{15}C_r \left(x^2\right)^{15-r} \left(\frac{2}{x}\right)^r = {}^{15}C_r \cdot 2^r \cdot x^{30-3r}$$

$$\text{For coeff. of } x^{15}, 30-3r = 15 \Rightarrow r = 5$$

$$\text{For independent of } x, 30-3r = 0 \Rightarrow r = 10$$

$$\text{Req. ratio} = \frac{{}^{15}C_5 \cdot 2^5}{{}^{15}C_{10} \cdot 2^{10}} = \frac{1}{2^5} = \frac{1}{32}$$

$$28.(B) \quad T_7 = 729 \Rightarrow {}^9C_6 \left(\frac{3}{\sqrt[3]{84}}\right)^{9-6} (\sqrt{3} \ln x)^6 = 729 \Rightarrow {}^9C_6 \cdot \frac{3^3}{84} \times 3^3 (\ln x)^6 = 729$$

$$\Rightarrow (\ln x)^6 = 1 \Rightarrow \ln x = \pm 1 \Rightarrow x = e \text{ or } x = e^{-1}$$

$$29.(D) \quad T_{r+1} = {}^{10}C_r 2^{\frac{10-r}{2}} 3^{\frac{r}{5}}$$

Terms to be rational, $\frac{10-r}{2}$ and $\frac{r}{5}$ must be rational.

$$\frac{10-r}{2} \text{ to be rational, } r = 0, 2, 4, 6, 8, 10$$

$$\frac{r}{5} \text{ to be rational, } r = 0, 5, 10$$

$$\text{Common values of } r = 0, 10 \quad \therefore \text{Sum} = {}^{10}C_0 2^5 + {}^{10}C_{10} 3^2 = 32 + 9 = 41$$

$$30.(A) \quad T_{r+1} = {}^{18}C_r \left(x^{\frac{1}{3}}\right)^{18-r} \left(\frac{1}{2x^{\frac{1}{3}}}\right)^r = {}^{18}C_r x^{\frac{18-2r}{3}} \cdot \frac{1}{2^r}$$

$$\text{For coeff. of } x^{-2}, \frac{18-2r}{3} = -2 \Rightarrow r = 12; \text{ For coeff. of } x^{-4}, \frac{18-2r}{3} = -4 \Rightarrow r = 15$$

$$m = \text{coeff. of } x^{-2} = {}^{18}C_{12} \cdot \frac{1}{2^{12}}; \quad n = \text{coeff. of } x^{-4} = {}^{18}C_{15} \cdot \frac{1}{2^{15}}$$

$$\frac{m}{n} = \frac{{}^{18}C_{12} \cdot 2^{15}}{{}^{18}C_{15} \cdot 2^{12}} = 182.$$

$$31.(A) \quad \text{Given } \left(1 - \frac{1}{x} + 3x^5\right) \left(2x^2 - \frac{1}{x}\right)^8 \quad \text{Now consider } \left(2x^2 - \frac{1}{x}\right)^8$$

$$T_{r+1} = {}^8C_r \left(2x^2\right)^{8-r} \left(-\frac{1}{x}\right)^r = {}^8C_r 2^{8-r} (-1)^r x^{16-2r-r}$$

$$\text{For independent of } x, 16-3r = 0 \Rightarrow r = 16/3. \text{ Not possible.}$$

$$\text{For coeff. of } x, 16-3r = 1 \Rightarrow r = 5$$

$$\text{For coeff. of } x^{-5}, 16-3r = -5 \Rightarrow r = 7$$

The term independent of x in the binomial expansion of $\left(1 - \frac{1}{x} + 3x^5\right)\left(2x^2 - \frac{1}{x}\right)^8$ is obtained when we multiply the coefficient of x in $\left(2x^2 - \frac{1}{x}\right)^8$ with $\left(-\frac{1}{x}\right)$ in $\left(1 - \frac{1}{x} + 3x^5\right)$ and multiply the coefficient of x^{-5} in $\left(2x^2 - \frac{1}{x}\right)^5$ with $3x^5$ in $\left(1 - \frac{1}{x} + 3x^5\right)$.

$\therefore$ Term independent of x

$$= {}^8C_5 \cdot 2^{8-5} \cdot (-1)^5 \cdot (-1) + {}^8C_7 \cdot 2^{8-7} \cdot (-1)^7 \cdot 3 = 8 \times 7 \times 8 - 8 \times 6 = 400$$

32.(D) In $(1+x)^n$, $T_{r+1} = {}^nC_r x^r$

Coeff. of x^r in $(1+x)^n = {}^nC_r$

Given series is in G.P. with $a = (1+x)^{1000}$, $r = \frac{x}{1+x}$ and $n = 1001$

$$\therefore \text{Sum} = \frac{a(1-r^n)}{1-r} = \frac{(1+x)^{1000} \left[1 - \left(\frac{x}{1+x} \right)^{1001} \right]}{1 - \frac{x}{1+x}} = \left[(1+x)^{1000} - \frac{x^{1001}}{1+x} \right] (x+1) = (1+x)^{1001} - x^{1001}$$

$$\therefore \text{Coeff. of } x^{50} \text{ in } (1+x)^{1001} - x^{1001} = {}^{1001}C_{50} = \frac{(1001)!}{(50)!(951)!}$$

33.(C) $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$

$$\text{and } 1+x^2+x^5 = \sum_{i=0}^5 a_i (1+x)^i = a_0 + a_1(1+x) + a_2(1+x)^2 + a_3(1+x)^3 + a_4(1+x)^4 + a_5(1+x)^5$$

Equation coeff. of x^5 , x^4 , x^3 and x^2 , we get :

$$a_5 = 1, a_5 \cdot {}^5C_4 + a_4 \cdot {}^4C_4 = 1,$$

$$a_5 \cdot {}^5C_3 + a_4 \cdot {}^4C_3 + a_3 \cdot {}^3C_3 = 0 \text{ and } a_5 \cdot {}^5C_2 + a_4 \cdot {}^4C_2 + a_3 \cdot {}^3C_2 + a_2 \cdot {}^2C_2 = 0 \text{ resp.}$$

$$\therefore a_5 = 1, 5a_5 + a_4 = 1 \Rightarrow a_4 = -4,$$

$$10a_5 + 4a_4 + a_3 = 0 \Rightarrow a_3 = 6, 10a_5 + 6a_4 + 3a_3 + a_2 = 0 \Rightarrow a_2 = -4$$

34.(6) The general term is

$$T_{r+1} = {}^nC_r (x)^{n-r} (a)^r ; T_{2r+4} = {}^{18}C_{2r+3} (1)^{18-2r-3} (x)^{2r+3}$$

$$T_{r-2} = {}^{18}C_{r-3} (1)^{18-r+3} (x)^{r-3}$$

Given that coefficient are equal then

$${}^{18}C_{2r+3} = {}^{18}C_{r-3}$$

{when ${}^nC_x = {}^nC_y$ then $x = y$ or $x + y = n$ }

$$\text{So, } 2r + 3 + r - 3 = 18$$

$$r = 6$$

35.(60) Coefficient of x^5 in $(1+x^2)^5 \cdot (1+x)^4$

$$(1 + {}^5C_1x^2 + {}^5C_2x^4 + {}^5C_3x^6 + \dots) (1+x)^4$$

$$(1 + 5x^2 + 10x^4 + \dots) (1+x)^4$$

$$\begin{aligned} \text{Coefficient of } x^5 &= 5 \cdot {}^4C_3 + 10 \cdot {}^4C_1 \\ &= 5 \cdot 4 + 10 \cdot 4 \\ &= 20 + 40 = 60 \end{aligned}$$

36.(990) $(1+x+x^2+x^3)^{11}$

$$\Rightarrow ((1+x) + x^2(1+x))^{11}$$

$$\Rightarrow (1+x)^{11}(1+x^2)^{11}$$

$$\begin{aligned} \Rightarrow (1 + {}^{11}C_1x + {}^{11}C_2x^2 + {}^{11}C_3x^3 + {}^{11}C_4x^4 + \dots) \\ (1 + {}^{11}C_1x^2 + {}^{11}C_2x^4 + \dots) \end{aligned}$$

$$\Rightarrow (1 \cdot {}^{11}C_2x^4 + {}^{11}C_2x^2 \cdot {}^{11}C_1x^2 + {}^{11}C_4x^4 \cdot 1)$$

$$\Rightarrow ({}^{11}C_2 + {}^{11}C_4 + {}^{11}C_2 \cdot {}^{11}C_1) = \text{coefficient of } x^4$$

$$= 990$$

37.(9) $(1+x^2)^2 \cdot (1+x)^n = \sum_{k=0}^{n+4} a_k \cdot x^k$

$$= a_0x^0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots$$

a_1, a_2, a_3 are the coefficient of x^1, x^2, x^3 in the expansion of $(1+x^2)^2(1+x)^n$

$$\text{Now, } (1+x^2)^2(1+x)^n$$

$$(1+2x^2+x^4)(1+x)^n$$

$$a_1 = \text{coefficient of } x^1 = {}^nC_1 = n$$

$$a_2 = \text{coefficient of } x^2 = {}^nC_2 + 2 \cdot {}^nC_0 = \frac{n(n-1)}{2} + 2 = \frac{n^2 - n + 4}{2}$$

$$a_3 = \text{coefficient of } x^3 = {}^nC_3 + 2 \cdot {}^nC_1 = \frac{n(n-1)(n-2)}{6} + 2n = \frac{n(n^2 - 3n + 14)}{6}$$

a_1, a_2, a_3 are in A.P.

$$2a_2 = a_1 + a_3$$

$$2 \left(\frac{n^2 - n + 4}{2} \right) = n + n \left(\frac{n^2 - 3n + 14}{6} \right)$$

$$n^2 - n + 4 = n \left(\frac{6 + n^2 - 3n + 14}{6} \right)$$

$$6n^2 - 6n + 24 = n(n^2 - 3n + 20)$$

$$6n^2 - 6n + 24 = n^3 - 3n^2 + 20n$$

$$n^3 - 9n^2 + 26n - 24 = 0$$

Putting $n = 2$

$$8 - 36 + 52 - 24 = 0$$

$$60 - 60 = 0$$

$\Rightarrow n = 2$ is a root

$(n - 2)$ is a factor

$$\begin{array}{r} n^2 - 7n + 12 \\ n - 2 \overline{) n^3 - 9n^2 + 26n - 24} \end{array}$$

$$\begin{array}{r} n^3 - 2n^2 \\ - \quad + \\ \hline -7n^2 + 26n \end{array}$$

$$\begin{array}{r} -7n^2 + 14n \\ + \quad - \\ \hline 12n - 24 \end{array}$$

$$\begin{array}{r} 12n - 24 \\ - \quad + \\ \hline \times \end{array}$$

$$(n - 2)(n^2 - 7n + 12) = 0$$

$$(n - 2)(n - 3)(n - 4) = 0$$

$$n = 2, 3, 4 \quad \text{sum} = 2 + 3 + 4 = 9$$

38.(1) $(7 + 4\sqrt{3})^n = p + \beta$

$$G = (7 - 4\sqrt{3})^n$$

$$\beta + G = 1$$

$$G(P + \beta) = (7 - 4\sqrt{3})^n (7 + 4\sqrt{3})^n = (49 - 48)^n = 1$$

39.(243) $f(n) = \frac{1}{a} ({}^nC_0 a^n - {}^nC_1 a^{n-1} + {}^nC_2 a^{n-2} + \dots + (-1)^{n-1} {}^nC_{n-1} a)$

$$= \frac{1}{a} ((a-1)^n - (-1)^n {}^nC_n) = \frac{1}{a} (3^{n/223} - (-1)^n)$$

$$f(n) = \frac{3^{n/223} - (-1)^n}{(3^{1/223} + 1)}$$

$$f(2007) = \frac{3^{2007/223} + 1}{3^{1/223} + 1}$$

$$f(2008) = \frac{3^{2008/223} - 1}{3^{1/223} + 1}$$

$$f(2007) + f(2008) = \frac{3^{2007/223} + 3^{2008/223}}{3^{1/223} + 1}$$

$$= \frac{3^9 + 3^{9 + \frac{1}{223}}}{3^{1/223} + 1} = 3^9 \frac{(1 + 3^{1/223})}{(1 + 3^{1/223})} = 3^9$$

$$3^9 = 81k \quad k = 3^5 = 243$$

40.(40) Let $\binom{30}{0}\binom{30}{10} - \binom{30}{1}\binom{30}{11} + \binom{30}{2}\binom{30}{12} - \dots + \binom{30}{20}\binom{30}{30}$

$$\therefore A = {}^{30}C_0 \cdot {}^{30}C_{10} - {}^{30}C_1 \cdot {}^{30}C_{11} + {}^{30}C_2 \cdot {}^{30}C_{12} - \dots + {}^{30}C_{20} \cdot {}^{30}C_{30}$$

$$= \text{coefficient of } x^{20} \text{ in } (1+x)^{30}(1-x)^{30}$$

$$= \text{coefficient of } x^{20} \text{ in } (1-x^2)^{30}$$

$$= (-1)^{10} {}^{30}C_{10}$$

$$= {}^{30}C_{10} = x_{C_y} \Rightarrow x = 30, y = 10$$

41.(0) t_{r+1} in $\left(ax + \frac{b}{x^2}\right)^9$

$$t_{r+1} = {}^9C_r (ax)^{9-r} \left(\frac{b}{x^2}\right)^r$$

$$= {}^9C_r a^{9-r} b^r \frac{x^{9-r}}{x^{2r}}$$

$$= {}^9C_r a^{9-r} b^r x^{9-3r}$$

For x^3 , $9 - 3r = 3$ so $r = 2$

$$\text{Coefficient} = {}^9C_r a^{9-r} b^r = {}^9C_2 a^7 b^2$$

Similarly, t_{r+1} in $\left(ax^2 - \frac{b}{x}\right)^9$

$$t_{r+1} = {}^9C_r (ax^2)^{9-r} \left(-\frac{b}{x}\right)^r$$

$$= {}^9C_r a^{9-r} (-b)^r x^{18-3r}$$

For x^{-3} ; $18 - 3r = -3$ so $r = 7$

$$\text{Coefficient} = {}^9C_r a^{9-r} (-b)^r = {}^9C_7 a^2 (-b)^7$$

Comparing both coefficients

$${}^9C_2 a^7 b^2 = {}^9C_7 a^2 (-b)^7$$

$$a^7 b^2 = -a^2 b^7 \quad ({}^9C_2 = {}^9C_7)$$

$$a^5 = -b^5 = (-b)^5$$

So, $a = -b$
 $a + b = 0$

42.(98) $t_5 = {}^nC_4$

$$t_6 = {}^nC_5$$

$$t_7 = {}^nC_6$$

Given, t_5, t_6, t_7 are in A.P. so $2t_6 = t_5 + t_7$

$$2 {}^nC_5 = {}^nC_4 + {}^nC_6$$

$$\frac{2n!}{5!(n-5)!} = \frac{n!}{4!(n-4)!} + \frac{n!}{6!(n-6)!}$$

$$\frac{2}{5!(n-5)!} = \frac{1}{4!(n-4)!} + \frac{1}{6!(n-6)!}$$

$$2 = \frac{5!(n-5)!}{4!(n-4)!} + \frac{5!(n-5)!}{6!(n-6)!}$$

$$2 = \frac{5}{(n-4)} + \frac{(n-5)}{6}$$

$$2 = \frac{30 + (n-4)(n-5)}{6(n-4)}$$

$$12(n-4) = 30 + (n-4)(n-5)$$

$$n^2 - 21n + 98 = 0$$

$$(n-7)(n-14) = 0 \text{ thus, } n = 7, 14$$

$$43.(17) \quad t_{r+1} = {}^{100}C_r x^{\frac{(100-r)}{2}} (y^{1/3})^r$$

So, $\frac{100-r}{2}$ should be integer as well as $\frac{r}{3}$ should be integer, $r = 0, 6, 12, 18, \dots, 96$

Thus, r can assume total 17 terms.

$$44.(6) \quad (1+ax)^n = {}^nC_0 + {}^nC_1 ax + {}^nC_2 a^2 x^2 + \dots$$

$$= 1 + na x + \frac{n(n-1)}{1.2} a^2 x^2 + \dots$$

$$= 1 + 8x + 24x^2 + \dots$$

$$\text{So, } na = 8$$

$$\frac{n(n-1)}{2} a^2 = 24$$

$$\text{Solving } n = 4, a = 2, \quad n + a = 6$$

$$45.(33) \quad t_{r+1} = {}^{256}C_r (\sqrt{3})^{256-r} (\sqrt[8]{5})^r = {}^{256}C_r 3^{\frac{256-r}{2}} 5^{r/8}$$

$$\frac{256-r}{2} \in I \text{ as well as } \frac{r}{8} \in I$$

$$r = 0, 8, 16, \dots, 256$$

Total terms = 33

$$46.(10) \quad \text{Let } S = 1 + 2x + 3x^2 + 4x^3 + \dots \infty$$

$$xS = x + 2x^2 + 3x^3 + \dots \infty$$

$$(1-x)S = 1 + x + x^2 + x^3 + \dots \infty = \frac{1}{(1-x)}$$

$$S = \frac{1}{(1-x)^2}$$

$$\text{Let } Z = [(1-x)^{-2}]^2 = (1-x)^{-4}$$

$$Z = 1 + 4x - \frac{4(-4-1)}{2!} x^2$$

$$\text{Coefficient of } x^2 = \frac{-4(-4-1)}{2!} = \frac{20}{2} = 10$$

47.(2) Middle term of $(1+kx)^6$ is $\left(\frac{6}{2}+1\right)^{\text{th}}$ term $t_4 = {}^6C_3 (kx)^3$

Middle term of $(1-kx)^8$ is $\left(\frac{8}{2}+1\right)^{\text{th}}$ term

$$t_5 = {}^8C_4 (-kx)^4$$

Since coefficient are equal so compare the values of coefficients

$${}^6C_3 k^3 = {}^8C_4 (-k)^4 \quad (\text{free from } x)$$

$$\frac{6 \cdot 5 \cdot 4}{1 \cdot 2 \cdot 3} k^3 = \frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1} k^4$$

$$4k^3 = \frac{8 \cdot 7}{4} k^4$$

$$k = \frac{2}{7}$$

48.(7) $\left[\frac{1}{2}x^{1/3} + x^{-1/5}\right]^8$

The general term

$$T_{r+1} = {}^8C_r \left(\frac{1}{2}x^{1/3}\right)^{8-r} (x^{-1/5})^r$$

$$= {}^8C_r \frac{1}{2^{8-r}} x^{\frac{8-r}{3} - \frac{r}{5}}$$

For term independent of x

$$\frac{8-r}{3} - \frac{r}{5} = 0$$

$$r = 5$$

$$T_6 = {}^8C_5 \frac{1}{2^3} = 7$$

Trigonometric Identities & Equations

$$1.(B) \quad a \sin x + 1 - 2 \sin^2 x = 2a - 7 \quad \Rightarrow \quad 2 \sin^2 x - a \sin x + 2a - 8 = 0 \quad \Rightarrow \quad \sin x = \frac{a \pm a - 8}{4}$$

$$\Rightarrow \quad \sin x = \frac{a - 4}{2}, \quad [\text{Reject } \sin x = 2]$$

$$\Rightarrow \quad \sin x = \frac{a - 4}{2} \Rightarrow -1 \leq \frac{a - 4}{2} \leq 1 \Rightarrow 2 \leq a \leq 6$$

$$2.(B) \quad \tan^2 x + \sec x = a \quad \Rightarrow \quad \sec^2 x + \sec x - 1 = a$$

$$t^2 + t - 1 = a$$

$$t \in (-\infty, -1] \cup [1, \infty)$$

Check from the graph to get : $a \in [-1, \infty)$

$$3.(A) \quad \sin(\theta - x) = a \quad \cos(\theta - y) = b$$

$$\cos(\theta - x) = \sqrt{1 - a^2} \quad \sin(\theta - y) = \sqrt{1 - b^2}$$

$$\cos(x - y) = \cos((\theta - y) - (\theta - x)) = \cos(\theta - y)\cos(\theta - x) + \sin(\theta - x)\sin(\theta - y) = b\sqrt{1 - a^2} + a\sqrt{1 - b^2}$$

$$4.(D) \quad \cot 123^\circ \cot 133^\circ \cot 137^\circ \cot 147^\circ \\ = \cot(90 + 33) \cot(90 + 43) \cot(180 - 43) \cot(180 - 33) = (-\tan 33)(-\tan 43)(-\cot 43)(-\cot 33) = 1$$

$$5.(B) \quad \text{LHS} = (1 + \tan 1^\circ)(1 + \tan 44^\circ)(1 + \tan 2^\circ)(1 + \tan 43^\circ) \dots (1 + \tan 45^\circ) \\ = (1 + \tan 1^\circ + \tan 44^\circ + \tan 1^\circ \tan 44^\circ) + (1 + \tan 2^\circ + \tan 43^\circ + \tan 2^\circ \tan 43^\circ) \dots (1 + \tan 45^\circ) \\ = (1 + 1)(1 + 1) \dots (1 + 1) \text{ 23 times} = 2^{23} \\ n = 23$$

$$6.(A) \quad 4^{\sin 2x + 2 \cos^2 x} + 4^{1 - \sin 2x + 2 \sin^2 x} = 65 \quad \Rightarrow \quad 4^{\sin 2x + \cos 2x + 1} + 4^{1 - \sin 2x - \cos 2x + 1} = 65$$

$$4^{\sin 2x + \cos 2x + 1} + 4^{3 - (1 + \sin 2x + \cos 2x)} = 65 \quad \text{Let,} \quad 4^{\sin 2x + \cos 2x + 1} = y$$

$$y + \frac{64}{y} = 65 \quad \Rightarrow \quad y^2 - 65y + 64 = 0$$

$$y = 1, y = 64$$

$$4^{\sin 2x + \cos 2x + 1} = 4^0 \text{ or } 4^{\sin 2x + \cos 2x + 1} = 4^3$$

$$\sin 2x + \cos 2x = -1$$

$$7.(C) \quad P = \cos \frac{\pi}{20} \cos \frac{3\pi}{20} \cos \frac{7\pi}{20} \cos \frac{9\pi}{20} \Rightarrow P = \cos \frac{\pi}{20} \sin \frac{\pi}{20} \cos \frac{3\pi}{20} \sin \frac{3\pi}{20}$$

$$P = \frac{1}{4} \sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4} \frac{\sqrt{5} - 1}{4} \times \frac{\sqrt{5} + 1}{4} = \frac{1}{16}$$

$$Q = \cos \frac{\pi}{11} \cos \frac{2\pi}{11} \cos \frac{4\pi}{11} \cos \frac{8\pi}{11} \cos \frac{16\pi}{11} ; \quad Q = \frac{1}{2^5} \frac{\sin 2^5 \frac{\pi}{11}}{\sin \frac{\pi}{11}} = \frac{1}{32} \frac{\sin \left(3\pi - \frac{\pi}{11} \right)}{\sin \frac{\pi}{11}} = \frac{1}{32} \Rightarrow \frac{P}{Q} = \frac{\frac{1}{16}}{\frac{1}{32}} = 2$$

8.(C) Given, $4\sin^4 x + \cos^4 x - 1 = 0$
 $\Rightarrow 4\sin^4 x + (\cos^2 x - 1)(\cos^2 x + 1) = 0 \Rightarrow 4\sin^4 x - \sin^2 x(1 - \sin^2 x + 1) = 0$
 $\Rightarrow \sin^2 x(5\sin^2 x - 2) = 0 \Rightarrow \sin x = 0 \text{ or } \pm\sqrt{\frac{2}{5}}$

Hence, $x = n\pi$ is the required solution.

9.(C) ABCD is a cyclic quadrilateral $\Rightarrow A + C = 180^\circ, B + D = 180^\circ \Rightarrow C = 180^\circ - A, D = 180^\circ - B$
 $\Rightarrow \cos C = \cos(180^\circ - A) = -\cos A, \cos D = \cos(180^\circ - B) = -\cos B$
 $\Rightarrow \cos A + \cos C = 0, \cos B + \cos D = 0 \Rightarrow \cos A + \cos B + \cos C + \cos D = 0$

10.(D) $3\sin P + 4\cos Q = 6 \dots (i)$ and $4\sin Q + 3\cos P = 1 \dots (ii)$
 Squaring and adding (i) and (ii), we get : $9 + 16 + 24(\sin P \cos Q + \cos P \sin Q) = 37$
 $\Rightarrow \sin(P + Q) = \frac{1}{2} \Rightarrow \sin R = \frac{1}{2} \Rightarrow R = \frac{\pi}{6}$

11.(A) Given that $\tan \theta_1 \cdot \tan \theta_2 = k$ $\frac{\cos(\theta_1 - \theta_2)}{\cos(\theta_1 + \theta_2)} = \frac{\cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2}{\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2} = \frac{1 + \tan \theta_1 \tan \theta_2}{1 - \tan \theta_1 \tan \theta_2} = \frac{1 + k}{1 - k}$

12.(D) $\frac{1}{\cos \alpha + \cos(2n+1)\alpha} = \frac{1}{2\cos(n+1)\alpha \cdot \cos n\alpha} = \frac{\sin \alpha}{2\sin \alpha \cdot \cos(n+1)\alpha \cdot \cos n\alpha}$
 $= \frac{\sin[(n+1)\alpha - n\alpha]}{2\sin \alpha \cdot \cos(n+1)\alpha \cdot \cos n\alpha} = \frac{\sin(n+1)\alpha \cdot \cos n\alpha - \cos(n+1)\alpha \cdot \sin n\alpha}{2\sin \alpha \cdot \cos(n+1)\alpha \cdot \cos n\alpha}$
 $= \frac{1}{2} \operatorname{cosec} \alpha \left[\frac{\sin(n+1)\alpha \cdot \cos n\alpha}{\cos(n+1)\alpha \cdot \cos n\alpha} - \frac{\cos(n+1)\alpha \cdot \sin n\alpha}{\cos(n+1)\alpha \cdot \cos n\alpha} \right] = \frac{1}{2} \operatorname{cosec} \alpha [\tan(n+1)\alpha - \tan n\alpha]$
 $\therefore \frac{1}{\cos \alpha + \cos(2n+1)\alpha} = \frac{1}{2} \operatorname{cosec} \alpha [\tan(n+1)\alpha - \tan n\alpha]$

Put $n = 1, 2, \dots$. Then $\frac{1}{\cos \alpha + \cos 3\alpha} = \frac{1}{2} \operatorname{cosec} \alpha [\tan 2\alpha - \tan \alpha],$

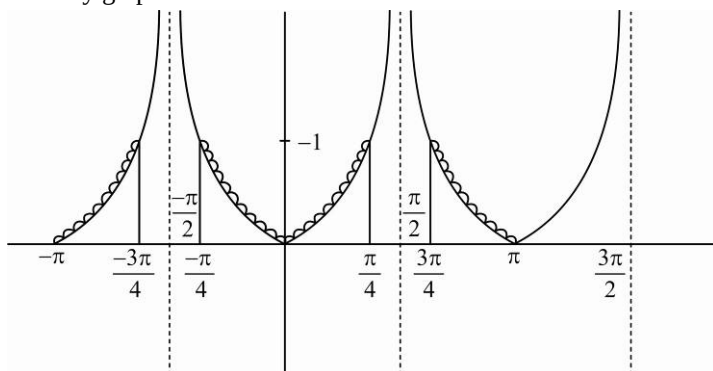
$\frac{1}{\cos \alpha + \cos 5\alpha} = \frac{1}{2} \operatorname{cosec} \alpha [\tan 3\alpha - \tan 2\alpha], \dots$

$\frac{1}{\cos \alpha + \cos 3\alpha} + \frac{1}{\cos \alpha + \cos 5\alpha} + \dots + \frac{1}{\cos \alpha + \cos(2n+1)\alpha}$
 $= \frac{1}{2} \operatorname{cosec} \alpha [(\tan 2\alpha - \tan \alpha) + (\tan 3\alpha - \tan 2\alpha) + \dots + \{\tan(n+1)\alpha - \tan n\alpha\}]$
 $= \frac{1}{2} \operatorname{cosec} \alpha [\tan(n+1)\alpha - \tan \alpha]$

13.(A) $-\sqrt{2} \leq \sin x - \cos x \leq \sqrt{2} \Rightarrow -2 \leq \sqrt{2}(\sin x - \cos x) \leq 2 \Rightarrow 1 \leq \sqrt{2}(\sin x - \cos x) + 3 \leq 5$
 $\Rightarrow 0 \leq \log_{\sqrt{5}} [\sqrt{2}(\sin x - \cos x) + 3] \leq 2$

14.(C) $\cos \frac{2\pi}{15} \cos \frac{4\pi}{15} \cos \frac{8\pi}{15} \cos \frac{14\pi}{15} = -\cos\left(\frac{\pi}{15}\right) \cos\left(\frac{2\pi}{15}\right) \cos\left(\frac{4\pi}{15}\right) \cos\left(\frac{8\pi}{15}\right) = \frac{1}{16}$

15.(A) Do it by graph



16.(B) $\cos \theta = \frac{1}{2} \left(x + \frac{1}{x} \right)$

Let $x + \frac{1}{x} = t \Rightarrow \frac{1}{2} \left(x^2 + \frac{1}{x^2} \right) = \frac{1}{2} (t^2 - 2) = \frac{1}{2} ((2 \cos \theta)^2 - 2) = 2 \cos^2 \theta - 1 = \cos 2\theta$

17.(B) $\sqrt{\frac{1+\frac{b}{a}}{1-\frac{b}{a}}} + \sqrt{\frac{1-\frac{b}{a}}{1+\frac{b}{a}}} = \sqrt{\frac{1+\tan x}{1-\tan x}} + \sqrt{\frac{1-\tan x}{1+\tan x}} = \frac{2}{\sqrt{1-\tan^2 x}} = \frac{2\sqrt{\cos^2 x}}{\sqrt{\cos^2 x - \sin^2 x}} = \frac{2 \cos x}{\sqrt{\cos 2x}}$

18.(AD) $\sec \theta + \tan \theta = 1$

and $\sec^2 \theta - \tan^2 \theta = 1 \Rightarrow (\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1 \Rightarrow \sec \theta - \tan \theta = 1 \Rightarrow \sec \theta = 1$ and $\tan \theta = 0$

Roots of the given equation are 1 and $\frac{c-2a+b}{a-2b+c} \Rightarrow \sec \theta$ and $\cos \theta$ are its roots ($\because$ both are 1).

19.(D) $\tan x + \cot x = 2$

We know that $\tan x + \cot x \leq -2$ and $\tan x + \cot x \geq 2$

Limiting condition is when $\tan x = \cot x = 1 \therefore x = \frac{\pi}{4} \Rightarrow \sin^{2n} x + \cos^{2n} x = \frac{1}{2^n} + \frac{1}{2^n} = \frac{1}{2^{n-1}}$

20.(D)

Function	Period
$\sin \frac{\pi}{2} x$	4
$\cos \frac{\pi}{3} x$	6
$\tan \frac{\pi}{3} x$	3

Take LCM of (4, 6, 3) = 12

21.(B) $P_n = \cos^n \theta + \sin^n \theta$

$$\begin{aligned} P_n - P_{n-2} &= \cos^n \theta + \sin^n \theta - \cos^{n-2} \theta - \sin^{n-2} \theta \\ &= \cos^{n-2} \theta (\cos^2 \theta - 1) + \sin^{n-2} \theta (\sin^2 \theta - 1) = \cos^{n-2} \theta (-\sin^2 \theta) + \sin^{n-2} \theta (-\cos^2 \theta) \\ &= -\sin^2 \theta \cos^2 \theta (\cos^{n-4} \theta + \sin^{n-4} \theta) = -\sin^2 \theta \cos^2 \theta P_{n-4} \end{aligned}$$

22.(A) $\sin \theta + \operatorname{cosec} \theta \geq 2$ and $\sin \theta + \operatorname{cosec} \theta \leq -2$

For $\sin \theta + \operatorname{cosec} \theta = 2$; $\sin \theta = 1 = \operatorname{cosec} \theta \Rightarrow \sin^8 \theta + \operatorname{cosec}^8 \theta = 1^8 + 1^8 = 2$

23.(C) $\cos 2001\pi + \cot 2001\frac{\pi}{2} + \sec \frac{2001\pi}{3} + \tan \frac{2001\pi}{4} + \operatorname{cosec} \frac{2001\pi}{6}$

$$= \cos \pi + \cot \frac{\pi}{2} + \sec 667\pi + \tan \frac{\pi}{4} + \operatorname{cosec} \frac{667\pi}{2} = -1 + 0 - 1 + 1 - 1 = -2$$

24.(A) $2 \tan \frac{\alpha}{2} = \tan \frac{\beta}{2}$; $\frac{3+5\cos \beta}{5+3\cos \beta} = \frac{3+5 \left(\frac{1-\tan^2 \frac{\beta}{2}}{1+\tan^2 \frac{\beta}{2}} \right)}{5+3 \left(\frac{1-\tan^2 \frac{\beta}{2}}{1+\tan^2 \frac{\beta}{2}} \right)} = \frac{3 \left(1+\tan^2 \frac{\beta}{2} \right) + 5 \left(1-\tan^2 \frac{\beta}{2} \right)}{5 \left(1+\tan^2 \frac{\beta}{2} \right) + 3 \left(1-\tan^2 \frac{\beta}{2} \right)}$

$$= \frac{8-2\tan^2 \frac{\beta}{2}}{8+2\tan^2 \frac{\beta}{2}} = \frac{8-8\tan^2 \frac{\alpha}{2}}{8+8\tan^2 \frac{\alpha}{2}} = \frac{1-\tan^2 \frac{\alpha}{2}}{1+\tan^2 \frac{\alpha}{2}} = \cos \alpha$$

25.(B) $(2\cos^2 x)^2 - 2\cos 2x - \frac{1}{2}\cos 4x$

$$= (1+\cos 2x)^2 - 2\cos 2x - \frac{1}{2}(2\cos^2 2x - 1) = 1 + \cos^2 2x + 2\cos 2x - 2\cos 2x - \cos^2 2x + \frac{1}{2} = \frac{3}{2}$$

26.(AD) $\sin \theta + \sin \phi = a \dots (i)$ & $\cos \theta + \cos \phi = b \dots (ii)$

Squaring and adding

$$\sin^2 \theta + \sin^2 \phi + 2\sin \theta \sin \phi = a^2 \Rightarrow \cos^2 \theta + \cos^2 \phi + 2\cos \theta \cos \phi = b^2$$

$$2 + 2(\cos \theta \cos \phi + \sin \phi \sin \theta) = a^2 + b^2 \Rightarrow \cos(\theta - \phi) = \frac{a^2 + b^2 - 2}{2}$$

$$2\cos^2\left(\frac{\theta - \phi}{2}\right) - 1 = \frac{a^2 + b^2}{2} - 1 \Rightarrow \cos^2\left(\frac{\theta - \phi}{2}\right) = \frac{a^2 + b^2}{4} \Rightarrow \cos\left(\frac{\theta - \phi}{2}\right) = \pm \frac{1}{2}\sqrt{a^2 + b^2}$$

27.(BD) $0 < x < \pi$

$0 < y < \pi$

$-\pi < -y < 0$

$-\pi < x - y < \pi$

$0 < x + y < 2\pi$

$\sin(x - y) = \frac{1}{2}$

$$x - y = \frac{\pi}{6} \text{ and } x - y = \frac{5\pi}{6}$$

$$x + y = \frac{\pi}{3} \text{ and } x + y = \frac{5\pi}{3}$$

$$x = \frac{\pi}{4} \quad y = \frac{\pi}{12}$$

$$x = \frac{11\pi}{12} \quad y = \frac{3\pi}{4}$$

$$\begin{aligned} 28.(D) \quad (a - b \cos 2\theta)(a - b \cos 2\varphi) &= \left[a - b \cdot \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} \right] \left[a - b \cdot \frac{1 - \tan^2 \varphi}{1 + \tan^2 \varphi} \right] \\ &= \left[\frac{a(1 + \tan^2 \theta) - b(1 - \tan^2 \theta)}{1 + \tan^2 \theta} \right] \left[\frac{a(1 + \tan^2 \varphi) - b(1 - \tan^2 \varphi)}{1 + \tan^2 \varphi} \right] \\ &= \frac{(a - b) + (a + b)\tan^2 \theta}{1 + \tan^2 \theta} \times \frac{(a - b) + (a + b)\tan^2 \varphi}{1 + \tan^2 \varphi} \\ &= \frac{(a - b)^2 + (a^2 - b^2)[\tan^2 \theta + \tan^2 \varphi] + (a + b)^2 \tan^2 \theta \tan^2 \varphi}{1 + (\tan^2 \theta + \tan^2 \varphi) + \tan^2 \theta \cdot \tan^2 \varphi} \\ &= \frac{(a - b)^2 + (a^2 - b^2)(\tan^2 \theta + \tan^2 \varphi) + (a + b)^2 \left(\frac{a - b}{a + b} \right)}{1 + (\tan^2 \theta + \tan^2 \varphi) + \frac{a - b}{a + b}} = \frac{(a - b) \left[a - b + (a + b)(\tan^2 \theta + \tan^2 \varphi) + a + b \right]}{\left[\frac{a + b + (a + b)(\tan^2 \theta + \tan^2 \varphi) + a - b}{(a + b)} \right]} \\ &= \frac{(a + b)(a - b) \left[2a + (a + b)(\tan^2 \theta + \tan^2 \varphi) \right]}{\left[2a + (a + b)(\tan^2 \theta + \tan^2 \varphi) \right]} = a^2 - b^2 \end{aligned}$$

*For alternate solution refer ICE-B of "Trigonometric Identities and Equations" Q. No: 15.

$$29.(C) \quad \sin x + \cos x + \tan x + \cot x + \sec x + \operatorname{cosec} x = 7$$

$$\Rightarrow (\sin x + \cos x) + \left(\frac{1}{\sin x \cdot \cos x} \right) + \frac{(\sin x + \cos x)}{\sin x \cdot \cos x} = 7 \Rightarrow (\sin x + \cos x) \left(1 + \frac{2}{\sin 2x} \right) = \left(7 - \frac{2}{\sin 2x} \right)$$

$$\text{Squaring, we have } (1 + \sin 2x) \left(1 + \frac{4}{\sin^2 2x} + \frac{4}{\sin 2x} \right) = 49 + \frac{4}{\sin^2 2x} - \frac{28}{\sin 2x}$$

$$\Rightarrow 1 + \frac{4}{\sin^2 2x} + \frac{4}{\sin 2x} + \sin 2x + \frac{4}{\sin 2x} + 4 = 49 + \frac{4}{\sin^2 2x} - \frac{28}{\sin 2x} \Rightarrow \sin 2x - 44 + \frac{36}{\sin 2x} = 0$$

$$\Rightarrow \sin^2 2x - 44 \sin 2x + 36 = 0 \Rightarrow \sin 2x = \frac{44 \pm \sqrt{1936 - 144}}{2} = \frac{44 \pm \sqrt{1792}}{2} = 22 \pm 8\sqrt{7}$$

$$\sin 2x \leq 1 \Rightarrow \sin 2x = 22 - 8\sqrt{7} \quad \therefore a = 22, b = 8$$

$$30.(A) \text{ Sum of the roots } = \sin^2 18^\circ + \cos^2 36^\circ = \left[\frac{\sqrt{5}-1}{4} \right]^2 + \left[\frac{\sqrt{5}+1}{4} \right]^2 = \frac{1}{16} \left[(\sqrt{5}+1)^2 + (\sqrt{5}-1)^2 \right] = \left(\frac{1}{16} \right) [2(5+1)] = \frac{3}{4}$$

$$\text{Product of the roots} = \sin^2 18^\circ \cos^2 36^\circ = \left[\frac{\sqrt{5}-1}{4} \right]^2 \left[\frac{\sqrt{5}+1}{4} \right]^2 = \left[\frac{5-1}{4 \cdot 4} \right]^2 = \frac{1}{16}$$

$$\Rightarrow \text{Required quadratic equation is } x^2 - \frac{3x}{4} + \frac{1}{16} = 0 \quad \text{or} \quad 16x^2 - 12x + 1 = 0$$

$$31.(C) \text{ Expanding D, we get } D = 1 + \sin \theta \cos \theta - \cos \theta (-\sin \theta - \cos \theta) + (-\sin^2 \theta + 1) \\ = 2 + 2 \sin \theta \cos \theta + \cos 2\theta = 2 + \sin 2\theta + \cos 2\theta = 2 + \sqrt{2} \cos \left(2\theta - \frac{\pi}{4} \right)$$

$$\text{As } -1 \leq \cos \left(2\theta - \frac{\pi}{4} \right) \leq 1, 2 - \sqrt{2} \leq D \leq 2 + \sqrt{2}$$

$$32.(C) \text{ From the given relation we have } \sin x - \sin 3x + 2 \sin 2x = 3 \Rightarrow 2 \sin x \cos 2x - 2 \sin 2x + 3 = 0$$

$$\Rightarrow (\sin x + \cos 2x)^2 + (\sin 2x - 1)^2 + 3 = \sin^2 x + \cos^2 2x + \sin^2 2x + 1$$

$$\Rightarrow (\sin x + \cos 2x)^2 + (\sin 2x - 1)^2 + \cos^2 x = 0 \text{ which is possible only if } \\ \sin x + \cos 2x = 0, \sin 2x = 1 \text{ and } \cos x = 0 \text{ which is not possible for any value of } x.$$

$$33.(D) (\sin \theta + \csc \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta \quad \therefore \text{ Minimum value} = 9$$

$$34.(B) \frac{1}{2} \tan \frac{\alpha}{2} + \frac{1}{4} \tan \frac{\alpha}{4} + \frac{1}{8} \tan \frac{\alpha}{8} + \dots + \frac{1}{2^n} \tan \frac{\alpha}{2^n} + \cot \alpha$$

$$\cot \alpha + \frac{1}{2} \tan \frac{\alpha}{2} = \frac{1 - \tan^2 \frac{\alpha}{2}}{2 \tan \frac{\alpha}{2}} + \frac{1}{2} \tan \frac{\alpha}{2} = \frac{1}{2 \tan \frac{\alpha}{2}} = \frac{1}{2} \cot \frac{\alpha}{2}$$

$$\text{Similarly } \frac{1}{4} \tan \frac{\alpha}{4} + \frac{1}{2} \cot \frac{\alpha}{2} = \frac{\cot \frac{\alpha}{2}}{4} \quad \therefore \text{ equation reduces to } \frac{\cot \frac{\alpha}{2^n}}{2^n}$$

$$35.(B) 25 \cos^2 \theta + 5 \cos \theta - 12 = 0$$

$$\Rightarrow \cos \theta = \frac{-5 \pm \sqrt{25 + 4 \times 25 \times 12}}{50} = \frac{-5 \pm 35}{50} = \frac{-4}{5}, \frac{3}{5}$$

$$\cos \theta = \frac{-4}{5} \text{ as } \frac{\pi}{2} < \alpha < \pi \Rightarrow \sin \theta = \frac{3}{5} \Rightarrow \sin 2\theta = 2 \sin \theta \cos \theta = \frac{-24}{25}$$

$$36.(B) x = \frac{-y}{2} = \frac{-z}{2} = k; \quad xy + yz + zx = -2k^2 + 4k^2 + (-2k^2) = 0$$

$$37.(B) \cos \left(x - \frac{3\pi}{2} \right) + \sin \left(\frac{3\pi}{2} + x \right) + \sin(32\pi + x) - 18 \cos(19\pi - x) + \cos(56\pi + x) - 9 \sin(x + 17\pi) \\ = -\sin x - \cos x + \sin x + 18 \cos x + \cos x + 9 \sin x = 18 \cos x + 9 \sin x \\ a + b = 27$$

$$38.(B) \quad [(\cos \beta \cos \gamma + \sin \beta \sin \gamma) + (\cos \gamma \cos \alpha + \sin \gamma \sin \alpha) + \cos \alpha \cos \beta + \sin \alpha \sin \beta] = -\frac{3}{2}$$

$$1+1+1+2[\sum \cos \alpha \cos \beta + \sum \sin \alpha \sin \beta] = 0$$

$$\cos^2 \alpha + \sin^2 \alpha + \cos^2 \beta + \sin^2 \beta + \cos^2 \gamma + \sin^2 \gamma + 2[\sum \cos \alpha \cos \beta + \sum \sin \alpha \sin \beta] = 0$$

$$(\cos \alpha + \cos \beta + \cos \gamma)^2 + (\sin \alpha + \sin \beta + \sin \gamma)^2 = 0$$

$$\Rightarrow \quad \cos \alpha + \cos \beta + \cos \gamma = 0 \quad \text{and} \quad \sin \alpha + \sin \beta + \sin \gamma = 0$$

$$39.(D) \quad \left| \sqrt{2 \sin^4 x + 18 \cos^2 x} - \sqrt{2 \cos^4 x + 18 \sin^2 x} \right| = 1$$

$$\sqrt{2 \sin^4 x + 18 \cos^2 x} - \sqrt{2 \cos^4 x + 18 \sin^2 x} = \pm 1$$

$$\sqrt{2 \sin^4 x + 18 \cos^2 x} = \sqrt{2 \cos^4 x + 18 \sin^2 x} \pm 1$$

$$\text{Squaring on both sides } 2 \sin^4 x + 18 \cos^2 x = 2 \cos^4 x + 18 \sin^2 x + 1 \pm 2 \sqrt{2 \cos^4 x + 18 \sin^2 x}$$

$$\Rightarrow \quad 2(\sin^4 x - \cos^4 x) + 18(\cos^2 x - \sin^2 x) - 1 = \pm 2 \sqrt{2 \cos^4 x + 18 \sin^2 x}$$

$$\Rightarrow \quad 16 \cos 2x - 1 = \pm 2 \sqrt{2 \cos^4 x + 18 \sin^2 x}$$

$$\Rightarrow \quad (16 \cos 2x - 1)^2 = 4 \left[2 \left(\frac{1 + \cos 2x}{2} \right)^2 + 9(1 - \cos 2x) \right]$$

$$\Rightarrow \quad \cos^2 2x = \frac{37}{254} \quad \Rightarrow \quad \cos 2x = \pm \sqrt{\frac{37}{254}}$$

$$\text{But } x \in [0, 2\pi] \quad \Rightarrow \quad 2x \in [0, 4\pi].$$

Clearly, the number of solutions is 8.

$$40.(B) \quad f'(x) = 4 \sin^3 x \cos x - 4 \cos^3 x \sin x = 4 \sin x \cos x (\sin^2 x - \cos^2 x) = -2 \sin 2x \cos 2x = -4 \sin 4x$$

For increasing function, $f'(x) > 0$

$$\therefore \quad -\sin 4x > 0 \quad \Rightarrow \quad \sin 4x < 0 \quad \Rightarrow \quad \pi < 4x < 2\pi \quad \Rightarrow \quad \frac{\pi}{4} < x < \frac{\pi}{2}$$

$$\therefore \quad \text{Increasing in } \left] \frac{\pi}{4}, \frac{\pi}{2} \right[$$

$$41.(C) \quad f(A) = \tan A + \tan B$$

$$f(A) = \tan A + \tan \left(\frac{\pi}{6} - A \right) \quad \left\{ \because A + B = \frac{\pi}{6} \right\}$$

$$f'(A) = \sec^2 A - \sec^2 \left(\frac{\pi}{6} - A \right)$$

$$f'(A) = 0 \quad \Rightarrow \quad \sec^2 A = \sec^2 \left(\frac{\pi}{6} - A \right) \quad \Rightarrow \quad A = \frac{\pi}{6} - A \quad \Rightarrow \quad A = \frac{\pi}{12}$$

$$f''(A) > 0 \quad \text{at } A = \frac{\pi}{12} = 15^\circ$$

$$\text{Min. value} = \tan 15^\circ + \tan 15^\circ = 2(2 - \sqrt{3}) = 4 - 2\sqrt{3}$$

42.(B) Let OP be a tower, due east of tower is A and due south of A is B .

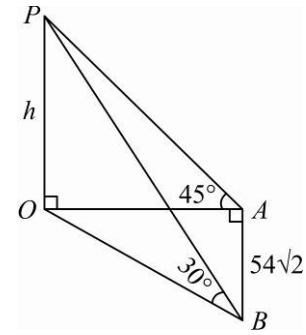
$$\text{In } \triangle POA, \tan 45^\circ = h/OA \Rightarrow OA = h$$

$$\text{In } \triangle POB, PO \perp OB$$

$$\tan 30^\circ = h/OB \Rightarrow OB = \sqrt{3}h$$

$$\text{In } \triangle AOB, OB^2 = OA^2 + AB^2$$

$$\Rightarrow (\sqrt{3}h)^2 = h^2 + (54\sqrt{2})^2 \quad \left\{ \because AB = 54\sqrt{2} \right\} \Rightarrow h = 54$$



43.(B) In $\triangle ABC$, $\angle C = 60^\circ \Rightarrow A + B = 120^\circ$

$$\text{Sine formula } \frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow \frac{a}{b} = \frac{\sin A}{\sin B} = \frac{\sin(120^\circ - B)}{\sin B}$$

$$2 + \sqrt{3} = \frac{\sin 120^\circ \cos B - \cos 120^\circ \sin B}{\sin B} \Rightarrow 2 + \sqrt{3} = \frac{\sqrt{3}}{2} \cot B + \frac{1}{2}$$

$$\cot B = 2 + \sqrt{3} = \cot 15^\circ \quad \therefore \angle B = 15^\circ \text{ and } \angle A = 105^\circ$$

$$44.(B) \quad 2 \tan^{-1} x = \begin{cases} \sin^{-1} \frac{2x}{1+x^2}, & |x| < 1 \\ \pi - \sin^{-1} \frac{2x}{1+x^2}, & x > 1 \end{cases}$$

$$\therefore f(x) = 2 \tan^{-1} x + \sin^{-1} \frac{2x}{1+x^2} = \pi - \sin^{-1} \frac{2x}{1+x^2} + \sin^{-1} \frac{2x}{1+x^2} = \pi \quad \therefore f(5) = \pi$$

45.(B) $2 \cos \theta + \sin \theta = 1 \dots\dots (i)$

$$(2 \cos \theta)^2 = (1 - \sin \theta)^2 \Rightarrow 4(1 - \sin^2 \theta) = 1 - 2 \sin \theta + \sin^2 \theta$$

$$5 \sin^2 \theta - 2 \sin \theta - 3 = 0 \Rightarrow (\sin \theta - 1)(5 \sin \theta + 3) = 0$$

$$\sin \theta = -\frac{3}{5} \quad \therefore \theta \neq \frac{\pi}{2}$$

$$\text{From (i), } \cos \theta = \frac{4}{5} \quad \therefore 7 \cos \theta + 6 \sin \theta = 7 \cdot \frac{4}{5} + 6 \cdot \left(-\frac{3}{5}\right) = 2$$

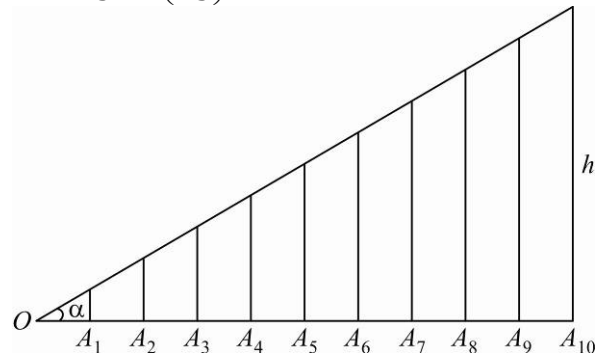
46.(C) $OA_1 = a, A_1 A_2 = A_2 A_3 = \dots = A_9 A_{10} = x$ (let)

$$\therefore A_1 A_{10} = 9x \text{ and } OA_{10} = a + 9x$$

$$\therefore \tan \alpha = \frac{h}{OA_{10}} = \frac{h}{a + 9x}$$

$$\Rightarrow a + 9x = h \cot \alpha$$

$$\Rightarrow x = \frac{h \cot \alpha - a}{9} = \frac{h \cos \alpha - a \sin \alpha}{9 \sin \alpha}$$



47.(C) $f(x) = 3\sin x + 4\cos x$ for $x \geq 0$

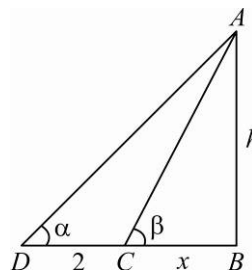
But $f(x)$ is an odd function $\therefore f\left(-\frac{11\pi}{6}\right) = -f\left(\frac{11\pi}{6}\right) = -\left[3\sin\frac{11\pi}{6} + 4\cos\frac{11\pi}{6}\right] = -\left[-\frac{3}{2} + 4 \cdot \frac{\sqrt{3}}{2}\right] = \frac{3}{2} - 2\sqrt{3}$

48.(A) In $\triangle ABC$: $h = x \tan B$

In $\triangle ABD$: $h = (x+2) \tan \alpha$

$$h = \left(\frac{h}{\tan \beta} + 2\right) \tan \alpha \Rightarrow h \left(1 - \frac{\tan \alpha}{\tan \beta}\right) = 2 \tan \alpha$$

$$\Rightarrow h = \frac{2 \tan \alpha \tan \beta}{\tan \beta - \tan \alpha} \Rightarrow h = \frac{2 \sin \alpha \sin \alpha}{\sin(\beta - \alpha)}$$



49.(A) $\sin 2x - 2\cos x + 4\sin x = 4$

$$\Rightarrow 2\sin x \cos x - 2\cos x + 4\sin x = 4 \Rightarrow 2\cos x(\sin x - 1) + 4(\sin x - 1) = 0$$

$$\Rightarrow (\sin x - 1)(2\cos x + 4) = 0 \Rightarrow \sin x = 1 \text{ as } \cos x + 2 \neq 0$$

$$\therefore x = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2} \text{ in } [0, 5\pi]$$

50.(A) **Statement I :** $-\frac{\pi}{2} \leq \sin^{-1} x \leq \frac{\pi}{2}$ and $0 \leq \cos^{-1} x \leq \pi$ $\therefore (\sin^{-1} x)^3 + (\cos^{-1} x)^3$ has some finite value.

Obviously, $(\sin^{-1} x)^3 + (\cos^{-1} x)^3 - a\pi^3 = 0$ has a solution for $\frac{1}{32} \leq a \leq \frac{7}{8}$.

For example, if $a = 100$, then $(\sin^{-1} x)^3 + (\cos^{-1} x)^3 \neq 100\pi^3$ for any x . $\therefore$ Statement I is false

Statement II : $x \in \mathbb{R}$, $\sin^{-1}(x)$ is undefined $\Rightarrow$ Statement II is false.

51.(C) Period of $|\sin x| = \pi$ and period of $|\cos x| = \pi$ $\therefore$ Period of $|\sin 4x| = \pi/4$ and period of $|\cos 2x| = \pi/2$

LMC of $\pi/4$ and $\pi/2 = \pi/2$

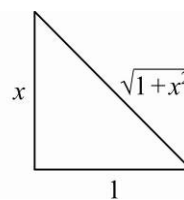
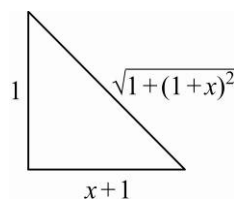
$\therefore$ Period of $f(x) = \pi/2$

52.(C) $\tan^{-1}\left(\cot\left(\frac{40\pi + 3\pi}{4}\right)\right) = \tan^{-1}\left(\cot\left(10\pi + \frac{3\pi}{4}\right)\right) = \tan^{-1}\left(\cot\frac{3\pi}{4}\right) = \tan^{-1}\left(\tan\left(\frac{\pi}{2} - \frac{3\pi}{4}\right)\right) = \tan^{-1}\left(\tan\left(-\frac{\pi}{4}\right)\right) = -\frac{\pi}{4}$

53.(A) $\sin(\cot^{-1}(1+x)) = \cos(\tan^{-1} x)$

$$\Rightarrow \sin\left(\sin^{-1}\frac{1}{\sqrt{1+(1+x)^2}}\right) = \cos\left(\cos^{-1}\frac{1}{\sqrt{1+x^2}}\right)$$

$$\Rightarrow \frac{1}{\sqrt{1+(1+x)^2}} = \frac{1}{\sqrt{1+x^2}} \Rightarrow 1+(1+x)^2 = 1+x^2 \Rightarrow x = -1/2$$



54.(D) $\sin^{-1} x = 2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2} \Rightarrow x = \frac{2x}{1+x^2} \Rightarrow x = 0, 1, -1$

All the three values of x satisfy the equation.

55.(B) **Statement 1 :** $2\sin^2 \theta - \cos 2\theta = 0$

$$\Rightarrow 2\sin^2 \theta - (1 - 2\sin^2 \theta) = 0 \Rightarrow \sin^2 \theta = \frac{1}{4}$$

$$\Rightarrow \sin \theta = \pm \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

$$2\cos^2 \theta - 3\sin \theta = 0$$

$$\Rightarrow 2(1 - \sin^2 \theta) - 3\sin \theta = 0 \Rightarrow (2\sin \theta - 1)(\sin \theta + 2) = 0$$

$$\Rightarrow \sin \theta = 1/2 \text{ as } \sin \theta + 2 \neq 0 \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6} \text{ in } [0, 2\pi]$$

$\therefore$ Number of common solutions is 2.

Statement 2 : $2\cos^2 \theta - 3\sin \theta = 0$

$$\Rightarrow \sin \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6} \text{ in } [0, \pi]$$

Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation for Statement 1.

$$\begin{aligned} 56.(C) \quad \tan^{-1}\left(\frac{1}{n^2 + n + 1}\right) &= \tan^{-1}\left(\frac{(n+1) - n}{1 + n(n+1)}\right) = \tan^{-1}(n+1) - \tan^{-1} n; \tan^{-1}\left(\frac{1}{n^2 + 3n + 3}\right) \\ &= \tan^{-1}\left(\frac{(n+2) - (n+1)}{1 + (n+2)(n+1)}\right) = \tan^{-1}(n+2) - \tan^{-1}(n+1); \dots, \tan^{-1}\left(\frac{1}{1 + (n+19)(n+20)}\right) \\ &= \tan^{-1}(n+20) - \tan^{-1}(n+19) \\ \therefore S &= \tan^{-1}(n+20) - \tan^{-1} n = \tan^{-1}\left(\frac{n+20-n}{1+n(n+20)}\right) = \tan^{-1}\left(\frac{20}{n^2 + 20n + 1}\right) \Rightarrow \tan S = \frac{20}{n^2 + 20n + 1} \end{aligned}$$

$$\begin{aligned} 57.(B) \quad 4 + \frac{1}{2}\sin^2 2x - 2(\cos^2 x)^2 &= 4 + \frac{1}{2}\sin^2 2x - 2\left(\frac{1 + \cos 2x}{2}\right)^2 = 4 + \frac{1}{2}(1 - \cos^2 2x) - \frac{1}{2}(1 + 2\cos 2x + \cos^2 2x) \\ &= 4 - \cos^2 2x - \cos 2x = \frac{17}{4} - \left(\cos 2x + \frac{1}{2}\right)^2 \quad \therefore \text{Max. value} = M = \frac{17}{4}, \text{ when } \cos 2x + \frac{1}{2} = 0. \\ \text{Min. value} = m &= \frac{17}{4} - \left(1 + \frac{1}{2}\right)^2 = 2, \text{ when } \cos 2x = 1 \quad \therefore M - m = \frac{17}{4} - 2 = \frac{9}{4}. \end{aligned}$$

$$\begin{aligned} 58.(D) \quad P &= \{\theta : \sin \theta - \cos \theta = \sqrt{2} \cos \theta\} = \sin \theta - \cos \theta = \sqrt{2} \cos \theta \\ \Rightarrow \sin \theta &= (\sqrt{2} + 1)\cos \theta \Rightarrow \tan \theta = \sqrt{2} + 1 \quad \therefore P = \{\theta : \tan \theta = \sqrt{2} + 1\} \\ Q &= \{\theta : \sin \theta + \cos \theta = \sqrt{2} + 1\} = \sin \theta + \cos \theta = \sqrt{2} \sin \theta = \cos \theta = (\sqrt{2} - 1)\sin \theta \\ \Rightarrow \tan \theta &= \sqrt{2} + 1 \quad \therefore Q = \{\theta : \tan \theta = \sqrt{2} + 1\} \quad \text{Hence, } P = Q. \end{aligned}$$

59.(C) θ is the arithmetic mean of α and β

$$\therefore \theta = \frac{\alpha + \beta}{2}$$

$$\text{Now, } \cos \alpha + \cos \beta = \frac{3}{2} \Rightarrow 2\cos \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2} = \frac{3}{2} \quad \dots\dots (i)$$

$$\text{and } \sin \alpha + \sin \beta = \frac{1}{2} \Rightarrow 2\sin \frac{\alpha + \beta}{2} \cos \frac{\alpha - \beta}{2} = \frac{1}{2} \quad \dots\dots (ii)$$

Dividing (ii) by (i), we get : $\tan \frac{\alpha + \beta}{2} = \frac{1}{3} \Rightarrow \tan \theta = \frac{1}{3}$

$$\therefore \sin 2\theta + \cos 2\theta = \frac{2 \tan \theta}{1 + \tan^2 \theta} + \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \frac{\frac{2}{3}}{1 + \frac{1}{9}} + \frac{1 - \frac{1}{9}}{1 + \frac{1}{9}} = \frac{2}{3} \times \frac{9}{10} + \frac{8}{9} \times \frac{9}{10} = \frac{3}{5} + \frac{4}{5} = \frac{7}{5}$$

60.(C) $2 \sin^3 \alpha - 7 \sin^2 \alpha + 7 \sin \alpha - 2 = 0$

$$(\sin \alpha - 1)(2 \sin^2 \alpha - 5 \sin \alpha + 2) = 0$$

$$(\sin \alpha - 1)(2 \sin \alpha - 1)(\sin \alpha - 2) = 0$$

$$\sin \alpha = 1, \sin \alpha = 1/2, \sin \alpha \neq 2$$

$$\alpha = \frac{\pi}{2}, \alpha = \frac{\pi}{6}, \frac{5\pi}{6} \text{ in } [0, 2\pi]$$

$\therefore$ Number of values of α is 3

61.(B) $\operatorname{cosec} \theta = \frac{p+q}{p-q} \Rightarrow \sin \theta = \frac{p-q}{p+q}$

$$\frac{1 + \cos \theta}{1 - \cos \theta} = \frac{2 \cos^2 \theta / 2}{2 \sin^2 \theta / 2} = \cot^2 \frac{\theta}{2}$$

$$\therefore \cot^2 \left(\frac{\pi}{4} + \frac{\theta}{2} \right) = \frac{1 + \cos \left(\frac{\pi}{2} + \theta \right)}{1 - \cos \left(\frac{\pi}{2} + \theta \right)} = \frac{1 - \sin \theta}{1 + \sin \theta} = \frac{1 - \frac{p-q}{p+q}}{1 + \frac{p-q}{p+q}} = \frac{q}{p} \quad \therefore \left| \cot \left(\frac{\pi}{4} + \frac{\theta}{2} \right) \right| = \sqrt{\frac{q}{p}}$$

62.(23) $\tan 1^\circ = \tan(45^\circ - 44^\circ) = \frac{\tan 45^\circ - \tan 44^\circ}{1 + \tan 45^\circ \tan 44^\circ} = \frac{1 - \tan 44^\circ}{1 + \tan 44^\circ}$

$$\text{Now, } 1 + \tan 1^\circ = 1 + \frac{1 - \tan 44^\circ}{1 + \tan 44^\circ}$$

$$1 + \tan 1^\circ = \frac{2}{1 + \tan 44^\circ}$$

$$(1 + \tan 1^\circ)(1 + \tan 44^\circ) = 2$$

$$\text{Similarly, } (1 + \tan 1^\circ)(1 + \tan 44^\circ) = 2$$

$$(1 + \tan 22^\circ)(1 + \tan 23^\circ) = 2$$

$$(1 + \tan 1^\circ)(1 + \tan 44^\circ)(1 + \tan 2^\circ)(1 + \tan 43^\circ) \dots (1 + \tan 22^\circ)(1 + \tan 23^\circ) \times (1 + \tan 45^\circ) = 2^n$$

$$2^{23} = 2^n$$

$$n = 23$$

63.(2) $P = \cos \frac{\pi}{20} \cos \frac{3\pi}{20} \cos \frac{7\pi}{20} \cos \frac{9\pi}{20} = \cos \frac{\pi}{20} \cos \frac{3\pi}{20} \cos \left(\frac{\pi}{2} - \frac{3\pi}{20} \right) \cos \left(\frac{\pi}{2} - \frac{\pi}{20} \right)$

$$= \cos \frac{\pi}{20} \sin \frac{\pi}{20} \cos \frac{3\pi}{20} \sin \frac{3\pi}{20} = \frac{1}{4} \sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4} \times \frac{\sqrt{5}-1}{4} \times \frac{\sqrt{5}+1}{4} = \frac{1}{16} \quad \therefore \sin \frac{\pi}{18} = \frac{\sqrt{5}-1}{4}; \sin \frac{3\pi}{18} = \frac{\sqrt{5}-1}{4}$$

$$Q = \cos \frac{\pi}{11} \cos \frac{2\pi}{11} \cos \frac{4\pi}{11} \cos \frac{8\pi}{11} \cos \frac{16\pi}{11} = \frac{1}{2^5 \sin \frac{\pi}{11}} \sin \frac{32\pi}{11} = \frac{1}{32} \frac{\sin \pi / 11}{\sin \pi / 11} = \frac{1}{32}$$

$$\frac{P}{Q} = 2$$

64.(0) In a cyclic quadrilateral

$$A + C = 180^\circ \text{ \& } B + D = 180^\circ$$

$$\cos A = -\cos C \text{ \& } \cos B = -\cos D$$

$$\cos A + \cos C = 0 \text{ and } \cos B + \cos D = 0$$

$$\cos A + \cos B + \cos C + \cos D = 0$$

65.(12) Period of function $f(ax)$ is $\frac{T}{|a|}$ if $f(x)$ is

Periodic with T.

$$f(x) = \sin\left(\frac{\pi x}{2}\right) + 2\cos\left(\frac{\pi x}{3}\right) - \tan\left(\frac{\pi x}{3}\right)$$

$$\text{Period} \quad \frac{2\pi}{\pi/2} \quad \frac{2\pi}{\pi/3} \quad \frac{\pi}{\pi/3}$$

$$\text{Period of } f(x) \text{ L.C.M.} \quad (4, 6, 3) \\ = 12$$

66.(2) $\sin \theta + \operatorname{cosec} \theta = 2$ If $t + \frac{1}{t} = 2 \Rightarrow t = 1$

$$\sin \theta = \operatorname{cosec} \theta = 1$$

$$\sin^8 \theta + \operatorname{cosec}^8 \theta = 2$$

67.(9) $f(\theta) = (\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = \sin^2 \theta + \operatorname{cosec}^2 \theta + 2 + \cos^2 \theta + \sec^2 \theta + 2$
 $= 5 + \frac{1}{\sin^2 \theta} + \frac{1}{\cos^2 \theta} = 5 + \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta} = 5 + \frac{4}{\sin^2 2\theta}$
 $(f(\theta))_{\min} = 9$

68.(0) $x = y \cos \frac{2\pi}{3} = z \cos \frac{4\pi}{3} = k$

$$\frac{1}{x} = \frac{1}{k}, \frac{1}{y} = \frac{\cos \frac{2\pi}{3}}{k}, \frac{1}{z} = \frac{\cos \frac{4\pi}{3}}{k}$$

$$xy + yz + zx = xyz \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right) = \frac{xyz}{k} \left[1 + \cos \frac{2\pi}{3} + \cos \frac{4\pi}{3} \right] = \frac{xyz}{k} \times 0 = 0$$

69.(27) $f(x) = \cos\left(x - \frac{3\pi}{2}\right) + \sin\left(\frac{3\pi}{2} + x\right) + \sin(32\pi + x) - 18\cos(19\pi - x) + \cos(56\pi + x) - 9\sin(x + 17\pi)$

$$f(x) = -\sin x - \cos x + \sin x + 18\cos x + \cos x + 9\sin x = 9\sin x + 18\cos x = a \sin x + b \cos x$$

$$a + b = 9 + 18 = 27$$

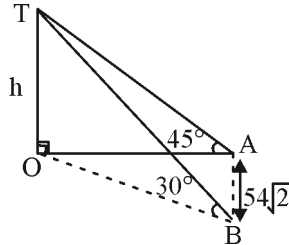
70.(2) Range of $a \sin \theta + b \cos \theta$ is $\left[-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}\right]$

Range of $(\sin x - \cos x)$ is $[-\sqrt{2}, \sqrt{2}]$

$\log_{\sqrt{5}} \left[\sqrt{2}(\sin x - \cos x) + 3 \right]$ varies for $[0, 2]$

Hence $a + b = 2$

71.(54)



$$\tan 45^\circ = \frac{h}{OA} \Rightarrow OA = h$$

$$\tan 30^\circ = \frac{h}{OB} \Rightarrow OB = h\sqrt{3}$$

$$\text{Now, } OA^2 + AB^2 = OB^2$$

$$h^2 + (54\sqrt{2})^2 = 3h^2$$

$$h = 54m$$

72.(2) $2 \cos \theta + \sin \theta = 1$

$$2 \left(\frac{1 - \tan^2 \theta / 2}{1 + \tan^2 \theta / 2} \right) + \left(\frac{2 \tan \theta / 2}{1 + \tan^2 \theta / 2} \right) = 1$$

$$\Rightarrow \tan \frac{\theta}{2} = 1, -\frac{1}{3} \quad \text{but } \tan \frac{\theta}{2} \neq 1 \left(\text{as } \theta \neq \frac{\pi}{2} \right)$$

$$\text{Now, } 7 \cos \theta + 6 \sin \theta = 7 \left(\frac{1 - \tan^2 \theta / 2}{1 + \tan^2 \theta / 2} \right) + 6 \left(\frac{2 \tan \theta / 2}{1 + \tan^2 \theta / 2} \right) = \frac{7 \tan^2 \theta / 2 + 12 \tan \theta / 2 + 7}{1 + \tan^2 \theta / 2} = 2 \text{ at } \tan \theta / 2 = -\frac{1}{3}$$

73.(3) $\sin 2x - 2 \cos x + 4 \sin x = 4$

$$2 \sin x \cos x - 2 \cos x + 4 \sin x - 4 = 0$$

$$2 \cos x (\sin x - 1) + 4 (\sin x - 1) = 0$$

$$(\sin x - 1)(2 \cos x + 4) = 0$$

$$\sin x = 1$$

x has 3 values in $[0, 5\pi]$

74.(3) at $x = 0, 1, -1$

$$75.(7) \quad \cos \alpha + \cos \beta = \frac{3}{2}$$

$$2 \cos \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) = \frac{3}{2} \dots\dots(1)$$

$$\sin \alpha + \sin \beta = \frac{1}{2}$$

$$2 \sin \left(\frac{\alpha + \beta}{2} \right) \cos \left(\frac{\alpha - \beta}{2} \right) = \frac{1}{2} \dots\dots(2)$$

$$\frac{(1)}{(2)}, \quad \cot \left(\frac{\alpha - \beta}{2} \right) = 3 \quad \therefore \frac{\alpha + \beta}{2} = \theta$$

$$\cot \theta = 3, \tan \theta = \frac{1}{3}$$

$$\sin 2\theta = \frac{2 \left(\frac{1}{3} \right)}{1 + \left(\frac{1}{3} \right)^2} = \frac{3}{5}$$

$$\cos 2\theta = \frac{1 - \frac{1}{9}}{1 + \frac{1}{9}} = \frac{4}{5}$$

$$5(\sin 2\theta + \cos \theta) = 7$$

$$76.(3) \quad 2 \sin^3 \alpha - 7 \sin^2 \alpha + 7 \sin \alpha - 2 = 0$$

$\sin \alpha = 1$ satisfies the equation

$$(\sin \alpha - 1)(2 \sin^2 \alpha - 5 \sin \alpha + 2) = 0$$

$$(\sin \alpha - 1)(2 \sin \alpha - 1)(\sin \alpha - 2) = 0$$

$$\sin \alpha = 1, \frac{1}{2} \Rightarrow \alpha = \frac{\pi}{2}, \frac{\pi}{6}, \frac{5\pi}{6}$$

Conic Section

- 1.(B) We have, $x^2 - 2 = -2\cos t$ and $y = 4\cos^2 \frac{t}{2}$
 $\Rightarrow x^2 - 2 = -2\cos t$ and $y = 4\left[\frac{1+\cos t}{2}\right] \Rightarrow x^2 - 2 = -2\cos t$ and $y = 2(1+\cos t)$
 $\Rightarrow y = 2\left[1 + \frac{x^2 - 2}{-2}\right] \Rightarrow y = 2 - x^2 + 2 \Rightarrow x^2 = 4 - y \quad \therefore x^2 = -(y - 4)$
 which is an equation of parabola having vertex at (0, 4).

- 2.(D) The equation of parabola is $\sqrt{(x-2)^2 + (y-3)^2} = \left| \frac{3x-4y+7}{\sqrt{(3)^2 + (-4)^2}} \right|$
 Clearly, its focus is (2, 3) and the equation of directrix is $3x - 4y + 7 = 0$
 $\therefore$ Length of the latusrectum = $2 \times$ Length of perpendicular from the focus (2, 3) on the directrix
 $\therefore$ Length of the latusrectum = $2 \left| \frac{3 \times 2 - 4 \times 3 + 7}{\sqrt{(3)^2 + (-4)^2}} \right| = \frac{2}{5}$

3-6. 3.(B) 4.(A) 5.(C) 6.(A)

- 7.(A) For $y^2 = 32x$, $a = 8$
 Then, any point on the parabola is $(8t^2, 16t)$, then the point $P(2, -8)$, we get : $t = -\frac{1}{2}$.
 Now, if PQ is the focal chord and point θ is $(8t_1^2, 16t_1)$ then $t_1 = -\frac{1}{t} = 2$. Hence, the point θ has co-ordinate (32, 32).

- 8.(A) Here, $M_{OA} \cdot M_{OB} = -1$
 $\Rightarrow \frac{2}{t_1} \cdot \frac{2}{t_2} = -1 \Rightarrow t_1 t_2 = -4 \quad \dots (i)$
 $M_{OP} \cdot M_{AB} = -1 \Rightarrow \frac{k}{h} \cdot \frac{2}{t_1 + t_2} = -1 \Rightarrow t_1 + t_2 = \frac{-2k}{h} \quad \dots (ii)$

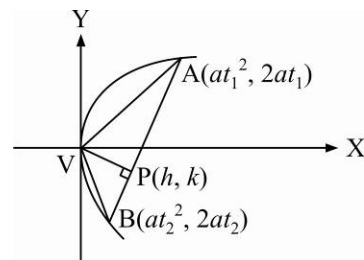
Equation of chord AB, is $y(t_1 + t_2) = 2x + 2at_1 t_2$, passes through (h, k)

$$\Rightarrow k(t_1 + t_2) = 2h + 2at_1 t_2 \quad \dots (iii)$$

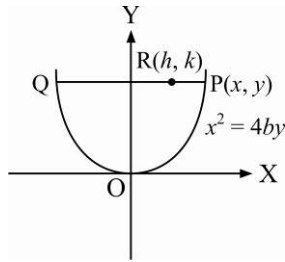
From equations (i), (ii) and (iii), we get :

$$k \cdot \left(\frac{-2k}{h} \right) = 2h - 8a$$

$$\Rightarrow -y^2 = x^2 - 4ax \quad \text{or} \quad x^2 + y^2 - 4ax = 0 \text{ is required locus.}$$



- 9.(C) Let PQ be a double ordinate of $x^2 = 4by$ and let R(h, k) be a point of trisection. Let the coordinates of P be (x, y). Then,
 $x = 3h$; $y = k$



But (x, y) lies on $x^2 = 4by$ $\therefore 9h^2 = 4bk$

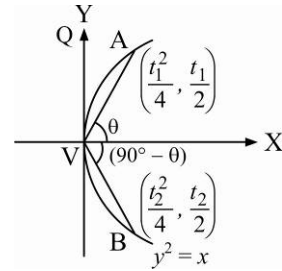
Hence, the locus of (h, k) is $9x^2 = 4by$

10.(D) $\tan \theta = \frac{t_1/2}{t_1^2/4} = \frac{2}{t_1}$

$\tan(90^\circ - \theta) = \frac{2}{t_2}$

$\therefore t_1 = 2 \cot \theta$ and $t_2 = 2 \tan \theta$

Now, $\left| \frac{VA}{VB} \right| = \frac{\sqrt{\frac{t_1^4}{16} + \frac{t_1^2}{4}}}{\sqrt{\frac{t_2^4}{16} + \frac{t_2^2}{4}}} = \frac{\sqrt{\cot^4 \theta + \cot^2 \theta}}{\sqrt{\tan^4 \theta + \tan^2 \theta}} = \frac{\cot \theta \cdot \operatorname{cosec} \theta}{\tan \theta \cdot \sec \theta} = \cot^3 \theta$



- 11.(D) We know that the vertex of the parabola given by $y = ax^2 + bx + c$ is at $\left(\frac{-b}{2a}, \frac{-D}{4a} \right)$.

So, the coordinates of the vertex of the parabola $y = \frac{a^3 x^2}{2} + \frac{a^2 x}{3} - 2a$ are $\left(\frac{-3}{4a}, \frac{-35a}{16} \right)$

Then, if (h, k) are the coordinates of the vertex.

Then, $h = -\frac{3}{4a}$ and $k = -\frac{35a}{16} \Rightarrow hk = \frac{105}{64}$

Hence, the locus of (h, k) is $xy = \frac{105}{64}$

- 12.(C) Equation of chord of contact of tangents from T (h, k) is $yk = 2(x + h)$ (i)

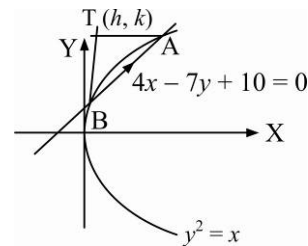
Also, equation of chord is $4x - 7y + 10 = 0$ (ii)

Since, equations (i) and (ii) are same,

$\therefore \frac{2}{4} = \frac{-k}{-7} = \frac{2h}{10}$

$\therefore h = \frac{5}{2}, k = \frac{7}{2}$

$\Rightarrow$ Point of intersection point of tangents is $\left(\frac{5}{2}, \frac{7}{2} \right)$



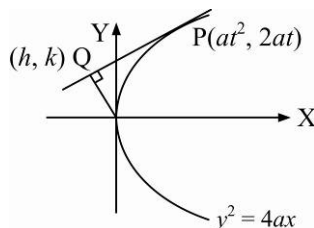
13.(A) Equation of tangent at P

$$\Rightarrow yt = x + at^2 \Rightarrow x - yt + at^2 = 0$$

As (h, k) is foot of perpendicular from $(0, 0)$

$$\Rightarrow \frac{h-0}{1} = \frac{k-0}{-t} = \frac{-(at^2)}{1+t^2}$$

$$\therefore t = -\frac{k}{h} \text{ and } h = -\frac{at^2}{1+t^2} \Rightarrow h = -\frac{a \cdot \frac{k^2}{h^2}}{1 + \frac{k^2}{h^2}} \Rightarrow x(x^2 + y^2) = -ay^2$$



14.(C) Equation of chord AB, is $y(t_1 + t_2) = 2x + 2at_1t_2$ passing through $(-1, -1)$

$$\Rightarrow (t_1 + t_2) = 2 + 2t_1t_2$$

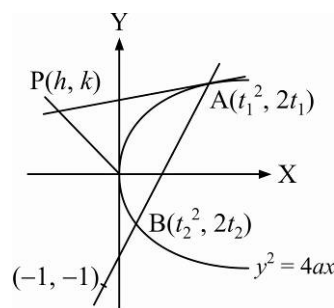
$$\Rightarrow (t_1 + t_2) = 2t_1t_2 + 2 \quad \dots (i)$$

As $P(h, k) = P(t_1t_2, t_1 + t_2)$

$$\therefore h = t_1t_2 \text{ and } k = t_1 + t_2 \quad \dots (ii)$$

From equations (i) and (ii), we get :

$$\Rightarrow y = 2(x+1)$$



15.(B) Let the coordinates of B and C be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$, respectively.

Then, the coordinates of A are $[at_1t_2, a(t_1 + t_2)]$

The equation of any tangent of $y^2 = 4ax$ is $ty = x + at^2$

$$\therefore p_1 = \frac{at_1t_2 - a(t_1 + t_2)t + at^2}{\sqrt{1+t^2}}, p_2 = \frac{at_1^2 - 2att_1 + at^2}{\sqrt{1+t^2}} \text{ and } p_3 = \frac{at_2^2 - 2at_1t_2 + at^2}{\sqrt{1+t^2}}$$

Clearly, $p_2p_3 = p_1^2$ Hence, p_2, p_1, p_3 are in G.P.

16.(B) The parabola is $(y-1)^2 = 4(x-1)$ whose directrix is the Y-axis or $x = 0$.

Any point on this parabola is $P(t^2 + 1, 2t + 1), t \in \mathbb{R}$.

$\therefore$ The equation of tangent at $P(t^2 + 1, 2t + 1)$ is $t(y-1) = (x-1) + t^2$

$$\text{or } x = ty - t^2 - t + 1 \quad \dots (i)$$

Tangent meets the directrix at $Q\left(0, \frac{t^2 + t - 1}{t}\right)$

Now, Q is the mid-point of PR. Therefore,

$$\begin{aligned} \frac{h+t^2+1}{2} &= 0 \quad \text{and} \quad \frac{k+2t+1}{2} = \frac{t^2+t-1}{t} \\ \Rightarrow t^2 &= -1-h \quad \text{and} \quad kt+2t^2+t = 2t^2+2t-2 \\ \Rightarrow t^2 &= -1-h \quad \text{and} \quad 2 = t(1-k) \end{aligned}$$

$$\text{On eliminating } t, \text{ we have } 4 = (-1-h)(1-k)^2 \Rightarrow (x+1)(y-1)^2 + 4 = 0$$

17.(B) Since, the normal at $(ap^2, 2ap)$ to $y^2 = 4ax$ meets the parabola at $(aq^2, 2aq)$.

Therefore, $q = -p - \frac{2}{p} \dots (i)$

Since, $OP \perp OQ$, therefore $\frac{2ap-0}{ap^2-0} \times \frac{2aq-0}{aq^2-0} = -1$

$\Rightarrow pq = -4 \Rightarrow p\left(-p - \frac{2}{p}\right) = -4$ [By equation (i)]

$\therefore p^2 = 2$

18.(A) Given, $\frac{2at_1}{2at_2} = \frac{1}{2} \Rightarrow 2t_1 = t_2, y^2 = 12x \therefore a = 3$

Now, $h = 2a + a(t_1^2 + t_2^2 + t_1t_2) = 6 + 3 \cdot 7t_1^2$ [$\because t_2 = 2t_1$]

and $k = -at_1t_2(t_1 + t_2) = -6t_1^2 \cdot 3t_1 = 18t_1^3 \Rightarrow \left(\frac{h-6}{21}\right)^3 = t_1^6 = \left(\frac{k}{18}\right)^2$

$\therefore$ Locus of (h, k) is $y^2 = \frac{12}{343}(x-6)^3$ i.e. $x = 3\left[7\left(\frac{y}{18}\right)^{2/3} + 2\right]$

19.(B) Let $(at^2, 2at)$ be a point on the parabola $y^2 = 4ax$ such that it cuts the parabola again at $(at_1^2, 2at_1)$. Then,

$t_1 = -t - \frac{2}{t} \Rightarrow |t_1| = \left|t + \frac{2}{t}\right| \Rightarrow |t_1| \geq \left|2\sqrt{t \times \frac{2}{t}}\right|$ [$\because AM \geq GM$]

$\Rightarrow |t_1| \geq 2\sqrt{2}$

Now, $OQ = \sqrt{(at_1^2 - 0)^2 + (2at_1 - 0)^2} \Rightarrow OQ = at_1\sqrt{t_1^2 + 4} \Rightarrow OQ \geq 2a\sqrt{2}\sqrt{8+4}$

$\therefore OQ \geq 4\sqrt{6}a$

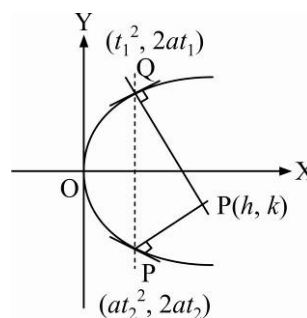
20.(A) Here, $m = \frac{2at_2 - 2at_1}{at_2^2 - at_1^2} = \frac{2}{t_2 + t_1} \dots (i)$

and $h = 2a + a(t_1^2 + t_2^2 + t_1t_2) \dots (ii)$

$k = -at_1t_2(t_1 + t_2) \dots (iii)$

From equations (i), (ii) and (iii), we get :

$h = 2a + a\left[\frac{4}{m^2} + \frac{km}{2a}\right]$ or $2m^2x = 4am^2 + 8a + ym^3$



21.(C)

22.(A) The two parabolas are given by

$y^2 = 4ax \dots (i)$

and $y^2 = 4c(x-b) \dots (ii)$

Equation to any normal of equation (i) is

$y = mx - 2am - am^3 \dots (iii)$

$y = m(x-b) - 2cm - cm^3 \dots (iv)$

If there is any common normal, then equations (iii) and (iv) must be identical. As the coefficients of x and y are equal, so the constant terms will be equal, hence $-2am - am^3 = -bm - 2mc - cm^3$

$$\Rightarrow m[m^2(c-a) - 2a + b + 2c] = 0$$

So, either $m = 0$ or $m^2 = \frac{2a-b-2c}{c-a}$

If $m = 0$, the common normal is the X -axis.

If $m^2 = \frac{2a-b-2c}{c-a}$

$$m = \sqrt{\frac{2(a-c)-b}{c-a}} = \sqrt{\left(-2 - \frac{b}{c-a}\right)}$$

then, $-2 - \frac{b}{c-a} > 0, \frac{-b}{c-a} > 2 \Rightarrow \frac{b}{a-c} > 2$

- 23.(A)** Let the tangents intersect at $P(h, k)$. Then, $lx + my + n = 0$ will be chord of contact of P . That means $lx + my + n = 0$ and $yk - 2ax - 2ah = 0$ will represent the same line.

Thus, $\frac{k}{m} = \frac{-2a}{l} = \frac{-2ah}{n} \Rightarrow h = \frac{n}{l}, k = \frac{-2am}{l}$

- 24.(C)** Let $P(h, k)$ be the pole of a focal chord of the parabola $y^2 = 4ax$. Then, the equation of chord is $ky - 2a(x + h) = 0$.

It passes through $(a, 0)$

$\therefore a + h = 0$

Hence, the locus of (h, k) is $x + a = 0$, i.e. $x = -a$

Clearly, it is the directrix of the parabola

- 25.(D)** The equation of the parabolas are

$$y^2 - 4x - 8y + 40 = 0 \quad \dots(i) \quad \text{and} \quad x^2 - 8x - 4y + 40 = 0 \quad \dots(ii)$$

We observe that, if we interchange x and y in equation (i), we obtain equation (ii). So, the two parabolas are symmetric about $y = x$.

If the two parabolas intersect on $y = x$, then the minimum distance between them is zero.

On solving $y = x$ and $y^2 - 4x - 8y + 40 = 0$, we get : $x^2 - 12x + 40 = 0$ which has imaginary roots.

So, the parabolas do not intersect. Consequently, distance between them is not zero.

Minimum distance between the two parabolas is the distance between tangents to the two parabolas which are parallel to $y = x$.

On differentiating equation (ii) w.r.t. x , we get :

$$2x - 8 - 4\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{x-4}{2}$$

If the tangent is parallel to $y = x$, then $\frac{x-4}{2} = 1 \Rightarrow x = 6$

On putting $x = 6$ in equation (ii), we get : $y = 7$

Thus, the coordinates of a point on parabola (ii) are $P(6, 7)$. The corresponding point on parabola (i) is $Q(7, 6)$.

Hence, required distance = $PQ = \sqrt{1+1} = \sqrt{2}$.

- 26.(A) The equation of tangent of slope m to the parabola $y^2 = 4x$ is $y = mx + \frac{1}{m}$. This will be a chord of the circle $x^2 + y^2 = 4$, if length of the perpendicular from the centre $(0, 0)$ is less than the radius.

$$\begin{aligned} \therefore \left| \frac{1}{m\sqrt{m^2+1}} \right| < 2 & \Rightarrow 4m^4 + 4m^2 - 1 > 0 \\ \Rightarrow \left(m^2 - \frac{\sqrt{2}-1}{2} \right) \left(m^2 + \frac{\sqrt{2}+1}{2} \right) > 0 & \Rightarrow m^2 - \frac{\sqrt{2}-1}{2} > 0 \\ \Rightarrow \left(m - \sqrt{\frac{\sqrt{2}-1}{2}} \right) \left(m + \sqrt{\frac{\sqrt{2}-1}{2}} \right) > 0 & \Rightarrow m \in \left(-\infty, -\sqrt{\frac{\sqrt{2}-1}{2}} \right) \cup \left(\sqrt{\frac{\sqrt{2}-1}{2}}, \infty \right) \end{aligned}$$

- 27.(D) The equation of any normal to $(x-2)^2 = 4(y-3)$ is $(x-2) = m(y-3) - 2m - m^3$

$$\text{If it passes through } \left(1, \frac{h}{2} \right) \Rightarrow 1-2 = m\left(\frac{h}{2} - 3 \right) - 2m - m^3$$

$$\Rightarrow 2m^3 + m(10-h) - 2 = 0$$

This equation will give three distinct values of m , if $f'(m) = 0$ has two distinct roots, where

$$f(m) = 2m^3 + m(10-h) - 2 \Rightarrow f'(m) = 6m^2 + (10-h)$$

$$\therefore f'(m) = 0 \Rightarrow m = \pm \sqrt{\frac{h-10}{6}}$$

The value of ' m ' are real and distinct if $h > 0$ i.e. $n \in (10, \infty)$

- 28.(C) Given equation is $2x^2 + 3y^2 - 8x - 18y + 35 = k$

$$(\sqrt{2}x - 2\sqrt{2})^2 + (\sqrt{3}y - 3\sqrt{3})^2 = k$$

So, no locus if $k < 0$, an ellipse for $k > 0$ and a point for $k = 0$.

- 29.(A) The coordinates of any point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, whose eccentric angle is θ are $(a \cos \theta, b \sin \theta)$.

The coordinates of end points of latusrectum are $\left(ae, \pm \frac{b^2}{a} \right)$

$$\therefore a \cos \theta = ae \quad \text{and} \quad b \sin \theta = \pm \frac{b^2}{a} \Rightarrow \tan \theta = \pm \frac{b}{ae} \quad \therefore \theta = \tan^{-1} \left(\pm \frac{b}{ae} \right)$$

- 30.(B) Let PQ and RS be two chords of the ellipse such that the eccentric angles of P, Q, R and S are α, β, γ and δ respectively. Suppose, PQ and RS meet major axis at $(C, 0)$ and $(-C, 0)$ respectively. Then,

$$\tan \frac{\alpha}{2} \times \tan \frac{\beta}{2} = \frac{c-a}{c+a} \quad \text{and} \quad \tan \frac{\gamma}{2} \tan \frac{\delta}{2} = \frac{-c-a}{-c+a} = \frac{c+a}{c-a}$$

$$\therefore \tan \frac{\alpha}{2} \tan \frac{\beta}{2} \tan \frac{\gamma}{2} \tan \frac{\delta}{2} = \frac{c-a}{c+a} \times \frac{c+a}{c-a} = 1$$

- 31.(A) The equation of tangents to $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are given by $y = mx \pm \sqrt{a^2 m^2 + b^2}$, where m is the slope of the tangent.

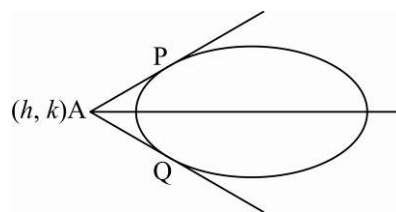
$$\text{For the ellipse } 4x^2 + 3y^2 = 5, \text{ we have } a = \frac{\sqrt{5}}{2}, b = \frac{\sqrt{5}}{\sqrt{3}}, m = \tan 60^\circ = \sqrt{3} \quad [\text{Given}]$$

$$\text{So, the equations of tangents are } y = \sqrt{3}x \pm \sqrt{\frac{5}{4} \times 3 + \frac{5}{3}} \Rightarrow y = \sqrt{3}x \pm \sqrt{\frac{65}{12}}$$

32.(D) Let PQ be any tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ which cuts the

$$\text{ellipse } \frac{x^2}{a(a+b)} + \frac{y^2}{b(a+b)} = 1 \quad \dots (i)$$

At P and Q.



Assume that tangents P and Q are at right angles, then the points A(h, k) will lie on the director circle by 2nd ellipse.

$$\therefore (h, k) \text{ will lie on } x^2 + y^2 = A^2 + B^2$$

$$x^2 + y^2 = a(a+b) + b(a+b) = (a+b)2$$

Since (h, k) lie on it.

$$\therefore h^2 + k^2 = (a+b)^2 \quad \dots (ii)$$

Also PQ is chord of contact of (h, k) w.r.t. equation (i)

$$\therefore \frac{hx}{a(a+b)} + \frac{ky}{b(a+b)} = 1$$

$$\text{It will be tangent to } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, \text{ if } a^2 l^2 + b^2 m^2 = n^2 \quad \Rightarrow \quad h^2 + k^2 = (a+b)^2$$

which is correct by equation (ii). Therefore, assumption that tangents subtends an angle of 90° is corrects.

33.(C) We know that, $PG = \frac{b}{a} \sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta} \quad \dots (i)$

Let S and S' be the foci and P be any point (x, y), then

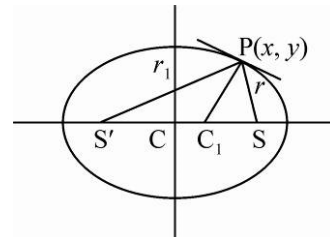
$$n_1 = SP \cdot S'P$$

$$= (a - ex)(a + ex) = a^2 - e^2 x^2$$

$$= a^2 - e^2 (a^2 \cos^2 \theta) \quad [\because x = a \cos \theta \text{ for point P}]$$

$$= a^2 + (b^2 - a^2) \cos^2 \theta \quad [\because b^2 = a^2 - a^2 e^2]$$

$$= a^2 (1 - \cos^2 \theta) + b^2 \cos^2 \theta = a^2 \sin^2 \theta + b^2 \cos^2 \theta \quad \therefore \sqrt{n_1} = \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$



Hence, from equation (i), we get : $PG = \frac{b}{a} \sqrt{r_1}$

34.(C) The equation of normals to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the points whose eccentric angles are θ and $\frac{\pi}{2} + \theta$ are respectively.

$$ax \sec \theta - by \operatorname{cosec} \theta = a^2 - b^2 \quad \text{and} \quad -ax \operatorname{cosec} \theta - by \sec \theta = a^2 - b^2$$

Since, ω is the angle between these two normals, we have

$$\tan \omega = \left| \frac{\frac{a}{b} \tan \theta + \frac{a}{b} \cot \theta}{1 - \frac{a^2}{b^2}} \right| = \left| \frac{ab(\tan \theta + \cot \theta)}{b^2 - a^2} \right| = \left| \frac{2ab}{\sin 2\theta (b^2 - a^2)} \right| = \frac{2ab}{(a^2 - b^2) \sin 2\theta}$$

$$\Rightarrow \frac{2 \cot \omega}{\sin 2\theta} = \frac{a^2 - b^2}{ab} = \frac{e^2}{\sqrt{1 - e^2}}$$

35.(C) Normal is given by $4x \sec \phi - 2y \operatorname{cosec} \phi = 12$

Coordinates of $Q = (3 \cos \phi, 0)$

Let $M \equiv (\alpha, \beta)$

$$\therefore \alpha = \frac{3 \cos \phi + 4 \cos \phi}{2} = \frac{7 \cos \phi}{2}$$

$$\Rightarrow \cos \phi = \frac{2}{7} \alpha \quad \text{and} \quad \beta = \frac{2 \sin \phi}{2} \Rightarrow \sin \phi = \beta$$

Now, $\cos^2 \phi + \sin^2 \phi = 1$

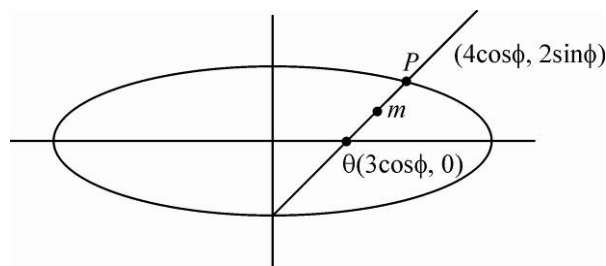
$$\Rightarrow \frac{4}{49} \alpha^2 + \beta^2 = 1 \Rightarrow \frac{4}{49} x^2 + y^2 = 1$$

Now, the latusrectum of given ellipse $x^2 + 4y^2 = 16$ is $x \pm 2\sqrt{3}$.

Solving $x = \pm 2\sqrt{3}$ and $\frac{4}{49} x^2 + y^2 = 1$.

$$\text{We have, } \frac{48}{49} + y^2 = 1 \Rightarrow y^2 = \frac{1}{49} \Rightarrow y = \pm \frac{1}{7}$$

$\therefore$ The point of intersection are $\left(\pm 2\sqrt{3}, \pm \frac{1}{7} \right)$



36.(B) The coordinates of extremity of the latusrectum which lies in the first quadrant are $\left(ae, \frac{b^2}{a} \right)$.

The equation of normal at (x_1, y_1) is $\frac{a^2 x}{x_1} - \frac{b^2 y}{y_1} = (a^2 - b^2)$

Therefore, the equation of normal at $\left(ae, \frac{b^2}{a} \right)$ is $\frac{a^2 x}{ae} - \frac{b^2 y}{b^2/a} = a^2 - b^2$

$$\Rightarrow ax - aey = e(a^2 - b^2) \Rightarrow ax - aey = ea^2 e^2 \Rightarrow x - ey = ae^3$$

This passes through the extremity of the minor axis, i.e. $(0, -b)$

$$\therefore 0 + eb - ae^3 = 0 \Rightarrow 0 = ae^2 \Rightarrow b^2 = a^2 e^4 \Rightarrow a^2(1 - e^2) = a^2 e^4$$

$$\Rightarrow e^4 + e^2 - 1 = 0$$

37.(A) Let $P(4 \cos \theta, 3 \sin \theta)$ be a point on the given ellipse such that its distance from the centre $(0, 0)$ of the ellipse is equal

to the mean of the lengths of the semi-major and semi-minor axes, i.e. $OP = \frac{4+3}{2}$

$$\Rightarrow \sqrt{16 \cos^2 \theta + 9 \sin^2 \theta} = \frac{7}{2} \Rightarrow 7 \cos^2 \theta + 9 = \frac{49}{4}$$

$$\Rightarrow \cos^2 \theta = \frac{13}{28} \Rightarrow \cos \theta = \pm \sqrt{\frac{13}{28}} \quad \text{and} \quad \sin \theta = \pm \sqrt{\frac{15}{28}}$$

$$\Rightarrow \cos \theta = \pm \frac{\sqrt{91}}{14} \quad \text{and} \quad \sin \theta = \pm \frac{\sqrt{105}}{14} \quad \text{Hence, the required points are given by } P\left(\pm \frac{2\sqrt{91}}{7}, \pm \frac{3\sqrt{105}}{14} \right)$$

38.(A) Let $P(3 \cos \theta, 2 \sin \theta)$ be a point on $4x^2 + 9y^2 = 36$. The equation of the tangent at P is $2x \cos \theta + 3y \sin \theta = 6$

This meets the tangents at extremities of major axis at T & T' . The equation of the circle with TT' as diameter is $x^2 + y^2 - 4y \operatorname{cosec} \theta - 5 = 0$

Clearly, it passes through $(-\sqrt{5}, 0)$

- 39.(B) Given, eccentricity of an ellipse = $\frac{2}{3}$
and length of latusrectum = 5

We know that, length of latusrectum of an ellipse is $\frac{2b^2}{a}$

$$\therefore \frac{2b^2}{a} = 5 \text{ and } e = \frac{2}{3} \Rightarrow 2b^2 = 5a \text{ and } e^2 = \frac{4}{9}$$

$$\Rightarrow 2b^2 = 5a \text{ and } \frac{a^2 - b^2}{a^2} = \frac{4}{9} \quad \left[\because e = \sqrt{\frac{a^2 - b^2}{a^2}} \right]$$

$$\Rightarrow 9a^2 - 9b^2 = 4a^2 \Rightarrow 5a^2 = 9b^2 \Rightarrow 5a^2 = \frac{9 \times 5a}{2}$$

$$\text{Thus, } a = \frac{9}{2} \text{ and } b = \frac{\sqrt{45}}{2}$$

$$\therefore \text{Equation of ellipse is } \frac{x^2}{\left(\frac{9}{2}\right)^2} + \frac{y^2}{\left(\frac{\sqrt{45}}{2}\right)^2} = 1 \quad \text{i.e.} \quad \frac{4x^2}{81} + \frac{4y^2}{45} = 1$$

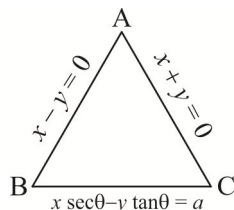
40.(C)

- 41.(A) The line $y = mx + 6$ is a tangent to the hyperbola

$$\frac{x^2}{100} - \frac{y^2}{49} = 1$$

$$\therefore 6^2 = 100m^2 - 49 \Rightarrow 36 = 100m^2 - 49 \Rightarrow m = \sqrt{\frac{17}{20}}$$

- 42.(A) Let equation of tangent of hyperbola $x^2 - y^2 = a^2$ at point $(a \sec \theta, a \tan \theta)$ is $x \sec \theta - y \tan \theta = a$.



Area of triangle formed by line $x - y = 0$, $x + y = 0$ and $x \sec \theta - y \tan \theta = a$.

Coordinates of A, B and C are $(0, 0)$, $\{a(\sec \theta + \tan \theta), a(\sec \theta + \tan \theta)\}$, $\{a(\sec \theta - \tan \theta), -a(\sec \theta - \tan \theta)\}$

$$\begin{aligned} \therefore \text{Area of } \triangle ABC &= \frac{1}{2} \begin{vmatrix} 0 & 0 & 1 \\ a(\sec \theta + \tan \theta) & a(\sec \theta + \tan \theta) & 1 \\ a(\sec \theta - \tan \theta) & -a(\sec \theta - \tan \theta) & 1 \end{vmatrix} \\ &= \frac{1}{2} \left| -a^2(\sec^2 \theta - \tan^2 \theta) - a^2(\sec^2 \theta - \tan^2 \theta) \right| = \frac{1}{2} \left| -2a^2 \right| = a^2 \text{ sq. units} \end{aligned}$$

- 43.(A) We have, $P(a \sec \alpha, b \tan \alpha)$ and $\theta(a \sec \beta, b \tan \beta)$ and points lie on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

$$\therefore \text{PQ is a chord of hyperbola } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Equation of chord of hyperbola joining $P(a \sec \alpha, b \tan \alpha)$ and $\theta(a \sec \beta, b \tan \beta)$ is

$$\frac{x}{a} \cos\left(\frac{\alpha - \beta}{2}\right) - \frac{y}{b} \sin\left(\frac{\alpha + \beta}{2}\right) = \cos\left(\frac{\alpha + \beta}{2}\right) \Rightarrow \frac{x \sec\left(\frac{\alpha + \beta}{2}\right)}{a \sec\left(\frac{\alpha - \beta}{2}\right)} - \frac{y}{b} \tan\left(\frac{\alpha + \beta}{2}\right) = 1$$

$$\Rightarrow \frac{x \sec\left(\frac{\alpha + \beta}{2}\right)}{a \sec \theta} - \frac{y}{b} \tan\left(\frac{\alpha + \beta}{2}\right) = 1 \quad [\because \alpha - \beta = 2\theta]$$

The equation of the touches the hyperbola $\frac{x^2}{a^2 \sec^2 \theta} - \frac{y^2}{b^2} = 1$

44.(C) Let $P(a \sec \theta, b \tan \theta)$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, where $e = \sqrt{\frac{a^2 + b^2}{a^2}}$

Equation of normal is $ax \cos \theta + by \cot \theta = a^2 + b^2$

$$\therefore \text{X-intercept i.e. } L = \left(\frac{a^2 + b^2}{a \cos \theta}, 0\right) \text{ and Y-intercept i.e. } M = \left(0, \frac{a^2 + b^2}{b \cot \theta}\right)$$

Let mid-point of L and M be Q(h, k).

$$\therefore h = \frac{a^2 + b^2}{2a \cos \theta} \quad \text{and} \quad k = \frac{a^2 + b^2}{2b \cot \theta} \quad \text{or} \quad \frac{2ha}{a^2 + b^2} = \sec \theta \quad \text{and} \quad \frac{2bk}{a^2 + b^2} = \tan \theta$$

$$\therefore \sec^2 \theta - \tan^2 \theta = 1 \quad \Rightarrow \quad \frac{4h^2 a^2}{(a^2 + b^2)^2} - \frac{4b^2 k^2}{(a^2 + b^2)^2} = 1$$

On taking locus of Q(h, k), we get :

$$\frac{4x^2 a^2}{(a^2 + b^2)^2} - \frac{4b^2 y^2}{(a^2 + b^2)^2} = 1 \quad \Rightarrow \quad \frac{x^2}{\left(\frac{a^2 + b^2}{2a}\right)^2} - \frac{y^2}{\left(\frac{a^2 + b^2}{2b}\right)^2} = 1$$

which is a hyperbola

$$\text{And its eccentricity is } e_1^2 = \frac{\left(\frac{a^2 + b^2}{2a}\right)^2 + \left(\frac{a^2 + b^2}{2b}\right)^2}{\left(\frac{a^2 + b^2}{2a}\right)^2}$$

$$e_1^2 = \frac{\frac{1}{4a^2} + \frac{1}{4b^2}}{\frac{1}{4a^2}} = \frac{b^2 + a^2}{b^2}$$

$$\therefore e_1^2 = \frac{a^2 + b^2}{b^2} = \frac{a^2 e^2}{b^2}$$

$$\therefore e_1^2 = e^2 \cdot \frac{1}{e^2 - 1} \quad \left[\because \frac{a^2}{b^2} = \frac{1}{e^2 - 1} \right] \quad \therefore e_1 = \frac{e}{\sqrt{e^2 - 1}}$$

45.(C) We have, $lx + my + n = 0$

$$\text{or } y = -\frac{lx}{m} - \frac{n}{m}$$

Now, slope $m' = \frac{l}{m}$ and $c = -\frac{n}{m}$

Then, condition of normal is $c^2(a^2 - b^2 m'^2) = m'^2(a^2 + b^2)^2$

$$\frac{n^2}{m^2} \left(a^2 - b^2 \frac{l^2}{m^2} \right) = \frac{l^2}{m^2} (a^2 + b^2)^2$$

$$\Rightarrow \frac{n^2}{m^2} (a^2 m^2 - b^2 l^2) = l^2 (a^2 + b^2)^2$$

$$\therefore \frac{a^2}{l^2} - \frac{b^2}{m^2} = \frac{(a^2 + b^2)^2}{n^2}$$

46.(C) Let the normal points be $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and (x_4, y_4)

Equation of normal is $\frac{a^2(x - x_i)}{x_i} = \frac{b^2(y - y_i)}{y_i}$ [where, $i = 1, 2, 3, 4$]

Let these normal passes through the point (h, k)

$$\text{So, } \frac{a^2(h - x)}{x} = \frac{b^2(k - y)}{y} \Rightarrow a^2 y(h - x) = b^2 x(k - y)$$

$$\text{i.e. } a^2 y(x - h) + b^2 x(y - k) = 0$$

47.(C) The equation of normal at $\left(ct, \frac{c}{t}\right)$ to the hyperbola $xy = c^2$ is

$$xt^3 - yt - ct^4 + c = 0 \quad \dots \text{(i)}$$

Since, equation (i) intersect at hyperbola $\left(ct_1, \frac{c}{t_1}\right)$

$$ct_1 t^3 - \frac{c}{t_1} t - ct^4 + c = 0$$

$$\Rightarrow t^3 t_1 = \frac{t}{t_1} + t^4 - 1 \Rightarrow t^3 t_1^2 - t - t^4 t_1 + t_1 = 0$$

$$\Rightarrow (t^3 t_1^2 - t^4 t_1) + (t_1 - t) = 0 \Rightarrow t^3 t_1 (t_1 - t) + t(t_1 - t) = 0$$

$$\Rightarrow (t_1 - t)(t^3 t_1 + 1) = 0 \Rightarrow t_1 \neq t$$

$$\text{or } t^3 t_1 + 1 = 0 \quad \therefore t^3 t_1 = -1$$

48.(C) Let (h, k) be the point of intersection of tangent at the end of normal chord. The equation of normal to the hyperbola $x^2 - y^2 = a^2$ at the point $(a \sec \theta, a \tan \theta)$

$$x \cos \theta + y \cot \theta = 2a \quad \dots \text{(i)}$$

Equation of chord of contact from the point (h, k) is

$$xh - yk = a^2 \quad \dots \text{(ii)}$$

Since, equations (i) and (ii) are identical, then

$$\frac{\cos \theta}{h} = \frac{\cot \theta}{-k} = \frac{2}{a} \quad \Rightarrow \quad \sec \theta = \frac{a}{2h} \text{ and } \tan \theta = \frac{a}{-2k}$$

$$\therefore \sec^2 \theta - \tan^2 \theta = 1 \quad \Rightarrow \quad \frac{a^2}{4h^2} - \frac{a^2}{4k^2} = 1$$

$$\Rightarrow \frac{k^2 - h^2}{h^2 k^2} = \frac{4}{a^2} \quad \Rightarrow \quad a^2 (k^2 - h^2) = 4h^2 k^2$$

On taking locus of (h, k) , we get : $a^2 (y^2 - x^2) = 4x^2 y^2$

49.(D) We have, $(x_i, y_i), i = 1, 2, 3, 4$ are the points on the rectangular hyperbola $xy = c^2$.

Equation of normal to the hyperbola $xy = c^2$ at point $\left(ct, \frac{c}{t}\right)$ is $ct^4 - t^3x + ty - c = 0$

It passes through (h, k) then $ct^4 - t^3h + tk - c = 0$

Which is a biquadratic in t .

Let its roots be t_1, t_2, t_3 and t_4 , then $\sum t_i = \frac{h}{c}, \sum t_i t_j = 0, \sum t_i t_j t_k = -\frac{k}{c}$ and $t_1 t_2 t_3 t_4 = -1$

Now, $\sum x_i = c \sum t_i = c \cdot \frac{h}{c} = h$

$$\begin{aligned} \text{and } \sum y_i &= c \left(\sum \frac{1}{t_i} \right) = c \left(\frac{\sum t_i t_j t_k}{t_1 t_2 t_3 t_4} \right) = \frac{c \cdot \left(-\frac{k}{c} \right)}{-1} = k \\ &= \frac{(x_1 + x_2 + x_3 + x_4)(y_1 + y_2 + y_3 + y_4)}{x_1 x_2 x_3 x_4} = \frac{hk}{-c^4} \quad \left[\text{using } x_1 x_2 x_3 x_4 = c^4 t_1 t_2 t_3 t_4 = -c^4 \right] \\ &= -\frac{hk}{c^4} \end{aligned}$$

50.(B) For hyperbola, $e > 1$.

$$9e^2 - 18e - 5 = 0 \quad \Rightarrow \quad (3e - 5)(3e - 1) = 0 \quad \Rightarrow \quad e = 5/3 \text{ as } e \neq 1/3.$$

Focus $(5, 0)$ and directrix $x = 9/5$.

$$5 - \frac{9}{5} = ae - \frac{a}{e} = a \left(e - \frac{1}{e} \right) \quad \Rightarrow \quad \frac{16}{5} = a \left(\frac{5}{3} - \frac{3}{5} \right) \quad \Rightarrow \quad a = 3$$

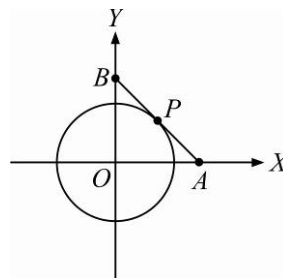
$$b^2 = a^2 (e^2 - 1) = 9 \left(\frac{25}{9} - 1 \right) = 16 \quad \therefore \quad a^2 - b^2 = 9 - 16 = -7.$$

51.(D) For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (i)

Tangent at (x_1, y_1) is $\frac{xx_1}{a^2} + \frac{yy_1}{b^2} = 1$

Any point on (i) is taken as $(a \cos \theta, b \sin \theta)$

$\therefore$ A point on $\frac{x^2}{27} + \frac{y^2}{3} = 1$ is $P(3\sqrt{3} \cos \theta, \sqrt{3} \sin \theta)$



$$\therefore \text{ Tangent at } P \text{ is } \frac{x \cdot 3\sqrt{3} \cos \theta}{27} + \frac{y \cdot \sqrt{3} \sin \theta}{3} = 1$$

Which intersect x -axis at $A\left(\frac{3\sqrt{3}}{\cos \theta}, 0\right)$ and y -axis at $B\left(0, \frac{\sqrt{3}}{\sin \theta}\right)$.

$$\therefore \text{ Area of } \triangle AOB = \frac{1}{2}(OA)(OB) = \frac{1}{2} \cdot \frac{3\sqrt{3}}{\cos \theta} \cdot \frac{\sqrt{3}}{\sin \theta} = \frac{9}{\sin 2\theta} \geq 9$$

$$\therefore \text{ Minimum area} = 9 \quad \{\text{Max. value of } \sin 2\theta = 1\}$$

52.(D) Equation of normal at $P(t)$ is $y + tx = 2at + at^3$.

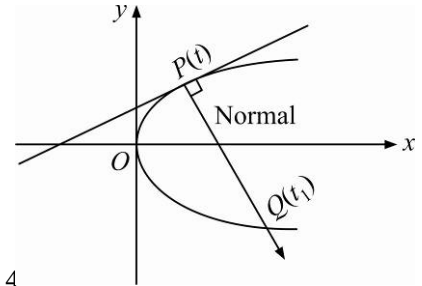
It passes through $Q(t_1)$ i.e. $(at_1^2, 2at_1)$, then $2at_1 + t \cdot at_1^2 = 2at + at^3$

$$\Rightarrow -2(t - t_1) = t(t - t_1)(t + t_1) \Rightarrow t_1 = -t - \frac{2}{t} = -\left(t + \frac{2}{t}\right)$$

$$\Rightarrow t_1^2 = t^2 + \frac{4}{t^2} + 4 \quad \dots\dots (i)$$

$$\therefore \text{ A.M.} \geq \text{G.M.} \Rightarrow \frac{t^2 + \frac{4}{t^2}}{2} \geq \sqrt{t^2 \cdot \frac{4}{t^2}} \Rightarrow t^2 + \frac{4}{t^2} \geq 4$$

$$\text{From (i), } t_1^2 \geq 4 + 4 \Rightarrow t_1^2 \geq 8 \quad \therefore \text{ Minimum value of } t_1^2 \text{ is 8.}$$



$$53.(D) \frac{x^2}{3} + \frac{y^2}{4} = 4 \Rightarrow \frac{x^2}{12} + \frac{y^2}{16} = 1$$

Its major axis is y -axis and foci $(0, \pm 2)$

$\therefore$ Vertices of hyperbola are $(0, \pm 2)$

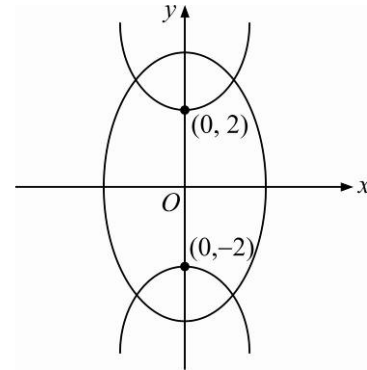
$$\text{Hyperbola, } \frac{y^2}{a^2} - \frac{x^2}{b^2} = 1$$

$$a = 2, e = \frac{3}{2}$$

$$b^2 = a^2(e^2 - 1) = 4\left(\frac{9}{4} - 1\right) = 5$$

$$\therefore \text{ Hyperbola is } \frac{y^2}{4} - \frac{x^2}{5} = 1$$

Clearly, $(5, 2\sqrt{3})$ does not lie on it.



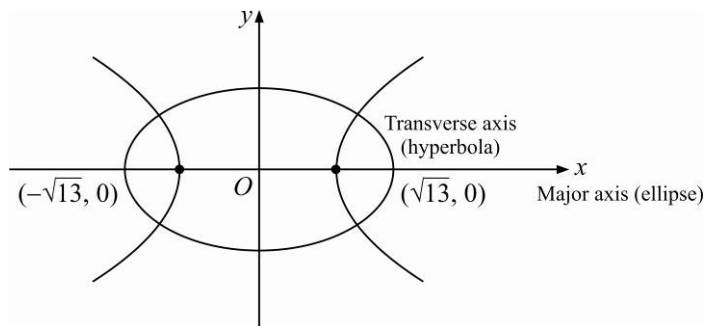
$$54.(C) \text{ Hyperbola } \frac{x^2}{4} - \frac{y^2}{9} = 1$$

$$c = \sqrt{a^2 + b^2} = \sqrt{4 + 9} = \sqrt{13}$$

$$\text{Foci } (\pm c, 0) = (\pm \sqrt{13}, 0)$$

$$\text{Eccentricity } e_1 = \frac{c}{a} = \frac{\sqrt{13}}{2}$$

$$\text{Given } e_1 e_2 = \frac{1}{2}$$



$$\Rightarrow \frac{\sqrt{13}}{2} \times e_2 = \frac{1}{2} \Rightarrow e_2 = \frac{1}{\sqrt{13}}$$

As ellipse passes through foci of the hyperbola, so for ellipse, $a = \sqrt{13}$

$$e_2^2 = 1 - \frac{b^2}{a^2} \Rightarrow \frac{1}{13} = 1 - \frac{b^2}{13} \Rightarrow b^2 = 12$$

$$\therefore \text{Ellipse is } \frac{x^2}{13} + \frac{y^2}{12} = 1$$

All the given points except $\left(\frac{1}{2}\sqrt{13}, \frac{\sqrt{3}}{2}\right)$ lie on this ellipse

55.(C) Ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, $b^2 = a^2(1 - e^2)$

Foci $F_1(ae, 0)$ and $F_2(-ae, 0)$

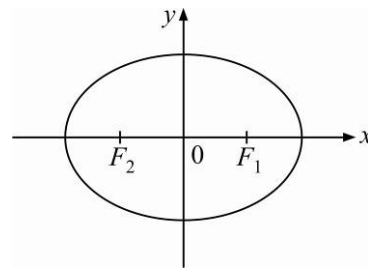
$$\text{Latus rectum} = \frac{2b^2}{a}$$

Given $F_1 F_2 = \frac{1}{2}$ (length of latus rectum)

$$\Rightarrow 2ae = \frac{1}{2} \left(\frac{2b^2}{a} \right) \Rightarrow 2a^2 e = b^2 = a^2(1 - e^2)$$

$$\Rightarrow e^2 + 2e - 1 = 0 \quad \therefore e = \frac{-2 \pm \sqrt{4 + 4}}{2 \times 1} \quad \left\{ \text{Use } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \right\}$$

$$\Rightarrow e = \frac{-2 \pm 2\sqrt{2}}{2} = -1 \pm \sqrt{2} \quad \therefore e = \sqrt{2} - 1, \quad \{ \because 0 < e < 1 \}$$



56.(B) Double ordinate PQ is a line perpendicular to the axis of the parabola.

Let P be $(-x_1, y_1)$, so Q is $(-x_1, -y_1)$,

To find locus of R , let R be (h, k)

Given $PR : RQ = 2 : 1$

$$\therefore h = \frac{-2x_1 - x_1}{3} = -x_1$$

$$k = \frac{-2y_1 + y_1}{3} = \frac{-y_1}{3}$$

$$\therefore x_1 = -h \text{ and } y_1 = 3k$$

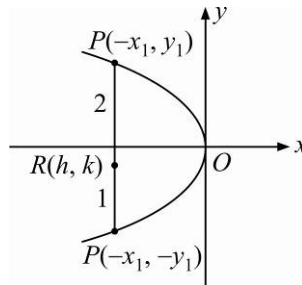
Since $(-x_1, y_1)$ lies on $y^2 = -4x$

$$\therefore y_1^2 = 4x_1$$

$$\Rightarrow (-3k)^2 = 4(-h) \Rightarrow 9k^2 = -4h$$

To find locus of R , replace h by x and k by y

$$\therefore \text{Locus of } R \text{ is } 9y^2 = -4x$$



7.(A) For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

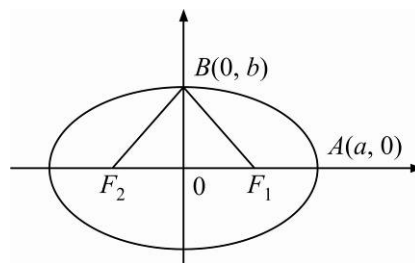
$\angle F_1BF_2 = 90^\circ$ (given), $F_1(ae, 0)$, $F_2(-ae, 0)$

Clearly, $F_1B = F_2B$

$\therefore \angle BF_1O = 45^\circ$

$\therefore OB = OF_1 \Rightarrow b = ae \Rightarrow e = b/a$

But $e^2 = 1 - \frac{b^2}{a^2} = 1 - e^2 \Rightarrow 2e^2 = 1 \Rightarrow e^2 = \frac{1}{2}$



58.(B) $AB = l$

$a^2 + b^2 = l^2$ (i)

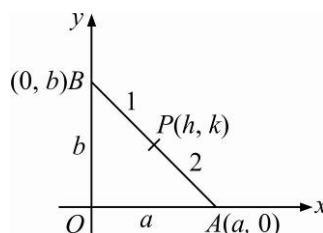
$h = \frac{1 \cdot a + 2 \cdot 0}{1 + 2} \Rightarrow a = 3h$

$k = \frac{1 \cdot 0 + 2 \cdot b}{1 + 2} \Rightarrow b = \frac{3k}{2}$

From (i), $9h^2 + \frac{9k^2}{4} = l^2$

Locus is $9x^2 + \frac{9y^2}{4} = l^2 \Rightarrow \frac{x^2}{1} + \frac{y^2}{4} = \frac{l^2}{9}$

$\therefore e = \sqrt{1 - \frac{b^2}{a^2}} = \frac{\sqrt{3}}{2}$



59.(C) AA' is the common chord. Points of intersections $x^2 + y^2 = 9$ and $y^2 = 8x$

$x^2 + 8x - 9 = 0$

$(x+9)(x-1) = 0$

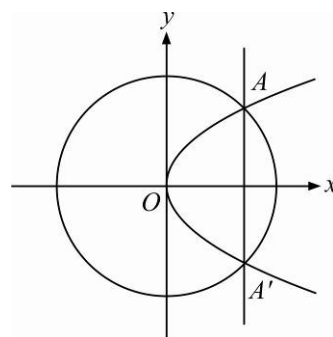
$x = 1, x = -9$ (rejected)

When $x = 1, y^2 = 8 \Rightarrow y = \pm 2\sqrt{2}$

$\therefore A(1, 2\sqrt{2})$ and $A'(1, -2\sqrt{2}) \Rightarrow AA' = 4\sqrt{2} = L_1$

Latus rectum $= 8 = L_2$

Clearly, $L_1 < L_2$



60.(D) For $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Normal at $(a \sec \theta, b \tan \theta)$ is

$ax \cos \theta + by \cot \theta = a^2 + b^2$

$\therefore$ Normal at $P(3 \sec \theta, 2 \tan \theta)$ is

$3x \cos \theta + 2y \cot \theta = 9 + 4 = 13$ (i)

and normal at $Q(3 \sec \phi, 2 \tan \phi)$ is

$3x \cos \phi + 2y \cot \phi = 13$

$$3x \sin \theta + 2y \tan \theta = 13 \quad \dots\dots (ii)$$

$$\therefore \phi = \pi/2 - \theta$$

$$(i) \sin \theta - (ii) \cos \theta$$

$$2y(\cos \theta - \sin \theta) = 13(\sin \theta - \cos \theta)$$

$$\Rightarrow y = -13/2$$

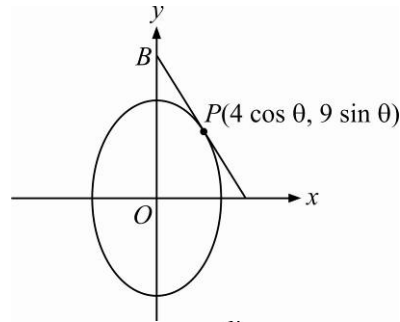
$$61.(C) \quad \frac{x^2}{4^2} + \frac{y^2}{9^2} = 1$$

Any point on ellipse be $P(4 \cos \theta, 9 \sin \theta)$

$$\therefore \text{Tangent at } P, \frac{x}{4} \cos \theta + \frac{y}{9} \sin \theta = 1$$

$$\Rightarrow A(4 \sec \theta, 0), B(0, 9 \operatorname{cosec} \theta)$$

$$\begin{aligned} \text{Req. area } (\triangle AOB) &= \left| \frac{1}{2} \cdot 4 \sec \theta \cdot 9 \operatorname{cosec} \theta \right| \\ &= \left| \frac{36}{\sin 2\theta} \right| \geq 36 \text{ as } |\sin 2\theta| \leq 1 \end{aligned}$$



62.(B) Any tangent to $y^2 = 4ax$ is

$$y = mx + \frac{a}{m}$$

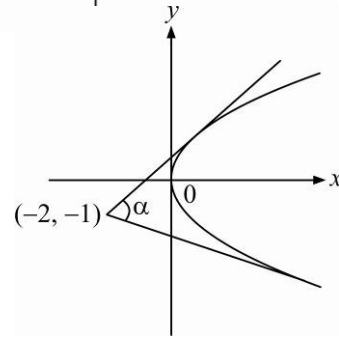
$\therefore$ A tangent to $y^2 = 4x$ is

$$y = mx + \frac{1}{m} \text{ as } a = 1$$

It passes through $(-2, -1)$

$$\therefore -1 = -2m + \frac{1}{m} \Rightarrow m = 1, m = -1/2$$

$$\text{Use } \tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \Rightarrow \tan \alpha = \left| \frac{1 + 1/2}{1 - 1/2} \right| = 3$$



63.(A) $y^2 = 6x \quad \therefore F(3/2, 0)$

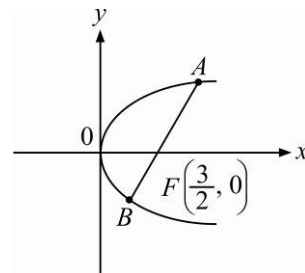
Equation of focal chord AB is

$$y - 0 = m \left(x - \frac{3}{2} \right)$$

$$mx - y - \frac{3m}{2} = 0$$

Its distance from vertex $(0, 0) = \frac{\sqrt{5}}{2}$

$$\Rightarrow \left| \frac{-3m/2}{\sqrt{1+m^2}} \right| = \frac{\sqrt{5}}{2} \Rightarrow m^2 = \frac{5}{4} \Rightarrow m = \pm \frac{\sqrt{5}}{2}$$



- 64.(A) Tangent at (x_1, y_1) to $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

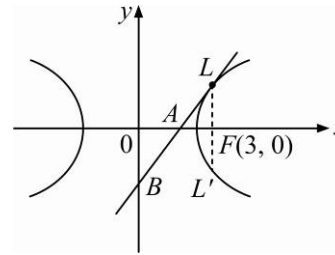
$$\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$$

LL' is latus rectum and $L\left(3, \frac{5}{2}\right)$

Tangent at $L\left(3, \frac{5}{2}\right)$ is

$$\frac{3x}{4} - \frac{y}{5} \cdot \frac{5}{2} = 1 \Rightarrow A\left(\frac{4}{3}, 0\right) \text{ and } (0, -2)$$

$$\therefore OA^2 - OB^2 = \frac{16}{9} - 4 = -\frac{20}{9}$$



- 65.(D) $x^2 = 8y, \frac{x^2}{3} + y^2 = 1$

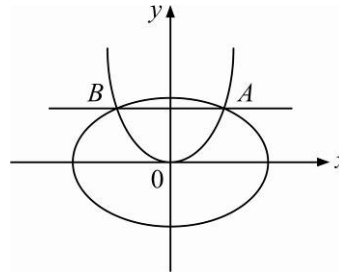
$$8y + 3y^2 = 3$$

$$\Rightarrow (3y-1)(y+3) = 3$$

$$\Rightarrow 3y-1=0, y+3=0$$

But $y+3=0$ does not satisfy $x^2 = 8y$

So, $3y-1=0$ is the required line AB



- 66.(C) Solving $\frac{x^2}{4c^2} + \frac{y^2}{c^2} = 1, x^2 + y^2 = 9a^2$, we get $x^2 = 36a^2 - 4c^2, y^2 = (4c^2 - 9a^2)/3$.

To intersect at four distinct points, $x^2 > 0, y^2 > 0$

$$36a^2 - 4c^2 > 0, 4c^2 - 9a^2 > 0$$

$$\Rightarrow 36a^2 > 4c^2, 4c^2 > 9a^2 \Rightarrow 3a > c, 2c > 3a \therefore a, c > 0$$

$$\Rightarrow (3a-c) > 0, 2c-3a > 0 \Rightarrow (3a-c)(2c-3a) > 0 \Rightarrow 9ac - 9a^2 - 2c^2 > 0$$

- 67.(B) **Statement 1 :**

$$x - 2y = 2 \Rightarrow x = 2y + 2$$

$$y^2 + 2x = 0 \Rightarrow y^2 + 2(2y + 2) = 0 \Rightarrow (y + 2)^2 = 0$$

This is a perfect square, so line is a tangent.

When $y = -2$, then $x = -2$

$\therefore$ Tangent at $(-2, -2)$. So, line touches parabola only at $(-2, -2)$.

Statement 2 :

For $y^2 = 4ax$, a general tangent is given by

$$y = mx + \frac{a}{m} \text{ at } \left(\frac{a}{m^2}, \frac{2a}{m}\right)$$

Comparing with $y^2 = -2x$, we get tangent $y = mx - \frac{1}{2m}$ at $\left(-\frac{1}{2m^2}, -\frac{1}{m}\right)$

Put $m = \frac{1}{2}$, we get the tangent $x - 2y = 2$

Statement 2 is a correct explanation for statement 1.

68.(C) E_1 : eccentricity $e_1 = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{2}{3}} = \frac{1}{3}$; E_2 : eccentricity $e_2 = \sqrt{1 - \frac{b^2}{16}}$

$$e_1 e_2 = \frac{1}{2} \text{ (given)} \Rightarrow b^2 = 4$$

Length of mirror axis $(L_2) = 2b = 4$

69.(B) Let the curves intersect at (x_1, y_1)

$$\frac{2x}{\alpha} + \frac{2y}{4} - \frac{dy}{dx} = 0$$

$$\Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = -\frac{x_1}{y_1} \cdot \frac{4}{\alpha} = m_1 \text{ and } 3y^2 \frac{dy}{dx} = 16$$

Now $m_1 m_2 = -1$

$$\Rightarrow -\frac{x_1}{y_1} \cdot \frac{4}{\alpha} \cdot \frac{16}{3y_1^2} = -1 \Rightarrow 64x_1 = 3\alpha y_1^3$$

$$\Rightarrow 64x_1 = 3\alpha \cdot 16x_1 \Rightarrow \alpha = 4/3 \quad \left[\because y_1^3 = 16x_1 \right]$$

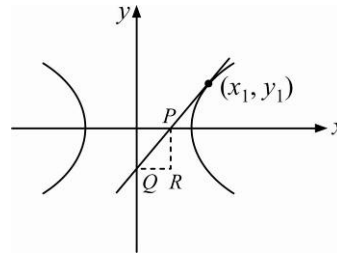
70.(D) A tangent at (x_1, y_1) on hyperbola $\frac{xx_1}{4} - \frac{yy_1}{2} = 1$

$$P\left(\frac{4}{x_1}, 0\right), Q\left(0, \frac{-2}{y_1}\right)$$

For rectangle OPRQ, $R\left(4/x_1, -2/y_1\right)$

Here R satisfies $\frac{4}{x^2} - \frac{2}{y^2} = 1$ as $\frac{x_1^2}{4} - \frac{y_1^2}{2} = 1$

Because (x_1, y_1) lies on the hyperbola.



71.(B) For $y^2 = 4ax$

Normal at $(at^2, 2at)$, $y + tx = 2at + at^3$

For $y^2 = 4x$, Ends of latus rectum are $(1, 2)$, $(1, -2)$

(A) For $(1, 2)$, $a = 1$, $t = 1$

Normal $y + x = 2 + 1$ (i)

(B) For $(1, -2)$, $a = 2$, $t = -1$

Normal $y - x = -2 - 1$ (ii)

Point of intersection of (i) and (ii) is $(3, 0)$.

72.(32) Let $P(at^2, 2at)$ be any point on the parabola $y^2 = 4ax$ and $S(a, 0)$ be the focus, then the other end of focal chord through P will be $Q\left(\frac{a}{t^2}, -\frac{2a}{t}\right)$.

Then, length of focal chord $PQ = a\left(t + \frac{1}{t}\right)^2$

$$\therefore t + \frac{1}{t} \geq 2 \Rightarrow a\left(t + \frac{1}{t}\right)^2 \geq 4a$$

or $PQ \geq 32$ $[\because 4a = 32]$

$\therefore \lambda = 32$

73.(16) Equation of tangent to parabola $y^2 = 4x$ at $P(t^2, 2t)$ is

$$ty = x + t^2$$

Given $t^2 \in [1, 4]$

$$\therefore \text{Area of } \triangle APN = \frac{1}{2}(AN)(PN)$$

$$= \frac{1}{2}(2t^2)(2t)$$

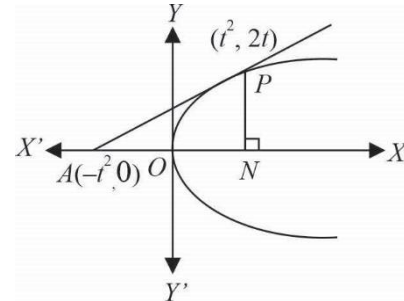
$$\Rightarrow \Delta = 2t^3 = 2(t^2)^{\frac{3}{2}}$$

$$\therefore t^2 \in [1, 4]$$

$$\Rightarrow \Delta_{\max} \text{ occurs, when } t^2 = 4$$

$$\Rightarrow \Delta_{\max} = 2(4)^{\frac{3}{2}} = 16 \text{ sq units}$$

$$\therefore \lambda = 16$$



74.(12) Given parabola $y^2 = 4x$. Now, then ends of latusrectum are $P(1, 2)$ and $Q(1, -2)$ or $P(t^2, 2t)$ and $Q(t_1^2, 2t_1)$, Where

$t = 1, t_1 = -1$. We know that the other end of normal is given by $t_2 = -t - \frac{2}{t}$

$$\Rightarrow A(t_2^2, 2t_2) \text{ and } A'(t_3^2, 2t_3)$$

Where $t_2 = -3, t_3 = 3$

or $A(9, -6)$ and $A'(9, 6)$

$$\therefore AA' = 12 \text{ units}$$

$$\therefore \lambda = 12$$

75.(5) $\therefore SA$ is perpendicular to BC [S is focus $(a, 0)$]

$$\Rightarrow \left(\frac{2at_1 - 0}{at_1^2 - a} \right) \left(\frac{2at_3 - 2at_2}{at_3^2 - at_2^2} \right) = -1$$

$$\Rightarrow \left(\frac{2t_1}{t_1^2 - 1} \right) \left(\frac{2}{t_3 + t_2} \right) = -1$$

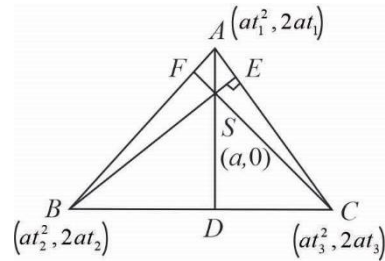
$$4t_1 + t_1^2 (t_2 + t_3) = t_2 + t_3 \quad \dots(i)$$

$$\text{Similarly, } 4t_2 + t_2^2 (t_3 + t_1) = t_3 + t_1 \quad \dots(ii)$$

On subtracting Eq. (ii) from Eq. (i), then

$$4 + (t_1 t_2 + t_2 t_3 + t_3 t_1) = -1 \Rightarrow t_1 t_2 + t_2 t_3 + t_3 t_1 = -5$$

$$\therefore |t_1 t_2 + t_2 t_3 + t_3 t_1| = 5$$



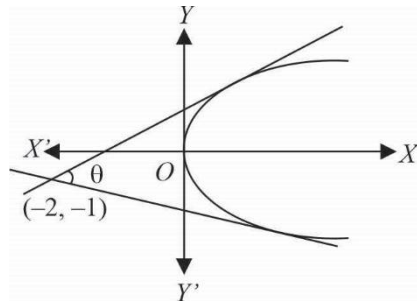
76.(3) Equation of tangent in terms of slope (m) of the parabola

$$y^2 = 4x \text{ is } y = mx + \frac{1}{m}.$$

$\therefore$ Point of intersection of tangents is $(-2, -1)$, then

$$-1 = -2m + \frac{1}{m}$$

$$2m^2 - m - 1 = 0$$



Let m_1, m_2 be slopes of the tangents, then

$$m_1 - m_2 = \frac{\sqrt{D}}{a} = \frac{\sqrt{(1+8)}}{2} = \frac{3}{2}$$

$$\text{and } m_1 m_2 = -\frac{1}{2}$$

$$\therefore \tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{3/2}{1 - 1/2} \right| = 3$$

77.(6) Let S be the focus and point of intersection of tangents at P and Q is R .

$$\therefore (SR)^2 = SP \cdot SQ = 4 \times 9 = 36$$

$$\therefore SR = 6$$

$$78.(12) \quad \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2)$$

$$\Rightarrow \quad \lim_{h \rightarrow 0} a^{2-h} = \lim_{h \rightarrow 0} \frac{b((2+h)^2 - b^2)}{2+h-2} = 8$$

$$\Rightarrow \quad a^2 = b \lim_{h \rightarrow 0} \frac{(2+h)^2 - b^2}{h} = 8$$

$$\text{at } h \rightarrow 0, (2+h)^2 - b^2 \rightarrow 0$$

$$\therefore \quad b^2 = 4 \text{ and } a^2 = 8$$

$$\therefore \text{ Locus of the pair of perpendicular tangents to the ellipse } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ is}$$

$$x^2 + y^2 = a^2 + b^2 = 8 + 4 = 12$$

$$\Rightarrow \quad r^2 = 12$$

$$79.(30) \quad \therefore \text{ Major axis on the line } y=2 \text{ and minor axis on the line } x=-1$$

$$\therefore \text{ Centre of ellipse is } (-1, 2)$$

$$\Rightarrow \quad h = -1, k = 2$$

$$\text{Also, } 2a = 10 \text{ and } 2b = 4$$

$$\therefore \quad M = a^2 = 25$$

$$\text{and } N = b^2 = 4$$

$$\text{Now, } h+k+M+N = -1+2+25+4 = 30$$

$$80.(10) \quad \text{Here, } a = 5, b = 4$$

$$\therefore \quad b^2 = a^2(1-e^2) \Rightarrow 16 = 25(1-e^2)$$

$$\therefore \quad e = \frac{3}{5}$$

$$\text{Foci } (\pm 3, 0)$$

$$\text{Here, } SA = 2 \quad [A \text{ and } A' \text{ are vertices}]$$

$$\text{Also gives } PS = 2$$

$$\therefore \quad P \text{ coincides with } A \text{ and } Q \text{ coincides with } A'$$

$$\therefore \quad PQ = 2a = 10$$

$$81.(4) \quad \text{For ellipse}$$

$$9x^2 + 16y^2 = 144 \text{ or } \frac{x^2}{16} + \frac{y^2}{9} = 1$$

$$\text{Equation of director circle is}$$

$$x^2 + y^2 = 25$$

$$\text{This director circle will cut the ellipse } \frac{x^2}{50} + \frac{y^2}{20} = 1 \text{ at four points.}$$

$$\text{Hence, number of points } = 4$$

82.(2) We know that, for $\theta = \frac{\pi}{2}$ the focal chord will be latusrectum with length $\frac{2b^2}{a}$.

$$\Rightarrow \frac{\lambda ab^2}{a^2 \sin^2\left(\frac{\pi}{2}\right) + b^2 \cos^2\left(\frac{\pi}{2}\right)} = \frac{2b^2}{a} \Rightarrow \frac{\lambda ab^2}{a^2 + 0} = \frac{2b^2}{a}$$

$$\therefore \lambda = 2$$

83.(2) The given lines are

$$\sqrt{3}x - y - 4\sqrt{3}t = 0 \quad \dots(i)$$

$$\text{and } \sqrt{3}tx + ty - 4\sqrt{3} = 0 \quad \dots(ii)$$

Eliminate 't' from Eqs. (i) and (ii), then

$$\frac{\sqrt{3}x - y}{4\sqrt{3}} = \frac{4\sqrt{3}}{\sqrt{3}x + y}$$

$$\Rightarrow 3x^2 - y^2 = 48$$

$$\text{or } \frac{x^2}{16} - \frac{y^2}{48} = 1$$

$$\text{or } 48 = 16(e^2 - 1)$$

$$e^2 = 4$$

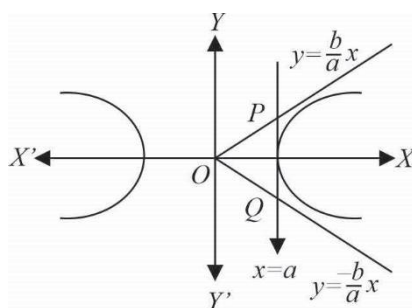
$$\therefore e = 2$$

84.(12)

Equation of tangent at $(a, 0)$ is $x = a$

Equation of asymptotes are $y = \pm \frac{b}{a}x$

$$\therefore P \equiv (a, b), Q \equiv (a, -b)$$



$$\therefore \text{Required area} = \frac{1}{2} \times a \times 2b$$

$$= ab$$

$$= (4)(3) = 12$$

(Here $a = 4, b = 3$)

- 85.(8) orthocentre of triangle formed by the points $\left(ct_1, \frac{c}{t_1}\right), \left(ct_2, \frac{c}{t_2}\right)$ and $\left(ct_3, \frac{c}{t_3}\right)$ on the rectangular hyperbola

$$xy = c^2 \text{ is } \left(-\frac{c}{t_1 t_2 t_3}, -ct_1 t_2 t_3\right)$$

Here, $c=2$ and $\frac{-c}{t_1 t_2 t_3} = \frac{1}{2}$ or $t_1 t_2 t_3 = -2c$

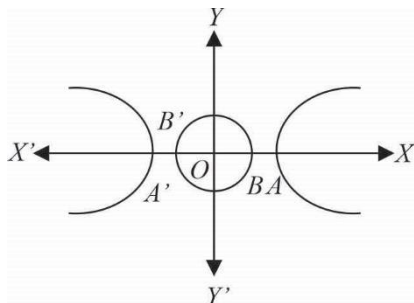
$\therefore$ Ordinate of orthocenter $= -c \ t_1 t_2 t_3$

$$= -c \times -2c$$

$$= 2c^2 = 2(2)^2$$

$$= 8$$

- 86.(3) $\because \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and $x^2 + y^2 = \frac{a^2}{4}$ having origin as their common centre.



So, vertex is the nearest point. Hence, shortest distance $= BA$

$$= a - \frac{a}{2} = \frac{a}{2}$$

$\therefore f(a, b) = \frac{a}{2}$

Hence, $f(4, 6) + f(2, 3) = \frac{4}{2} + \frac{2}{2} = 3$